



Submittal Review Response

Project Name: Hilo WWTP Rehabilitation and Replacement Project Phase 1
Submittal No.: 09672-001.0
Date: 9/26/2025

Client: County of Hawai'i Carollo Project No.: 203975
Contractor: Nan, Inc.
Submittal Name: Resinous Flooring
Reviewed By: Michele Hollenbaugh

SUBMITTAL REVIEW

Review is for general compliance with contract documents. No responsibility is assumed by Carollo for correctness of quantities, dimensions, and details. No deviation or variation is approved unless specifically addressed in these review comments. Refer to Section 01330 for additional requirements. The Contractor shall assume full responsibility for coordination with all other trades and deviations from contract requirements.

Approved	☐	No Exceptions
	☐	Make Corrections Noted - See Comments
	☐	Make Corrections Noted - Confirm
Not Approved	☒	Correct and Resubmit
	☐	Rejected - See Remarks
Receipt Acknowledged	☐	Filed for Record
	☐	With Comments - Resubmit

Review Comments:

1. Pre-installation conference shall be held in accordance with Section 09672.1.02.A.
2. Provide samples in accordance with Section 09672.1.03.B for Engineer's review.
3. Installer certificates shall be provided in accordance with Section 09672.1.03.C.
4. Final approval contingent upon receipt of warrant in accordance with Section 096752.1.03.F. Submitted warranties are in draft form and not complete.
5. Complete mock-up in accordance with Section 09672.1.04.C.1-4.
6. Final color selection shall be coordinated with Owner.

☐ High Priority

CONTRACTOR SUBMITTAL TRANSMITTAL FORM REV. A

Owner: County of Hawaii
Contractor: Nan, Inc. Project No.: WW-4705R
Project Name: Hilo WWTP Phase 1 Submittal Number: 09672.001.0
Submittal Title: RESINOUS FLOORING
TO: County OF Hawaii, DEM ☐ For Information Only
From: Nan Inc.

Specification No. and Subject of Submittal / Equipment Supplier	
Spec: 09672	Paragraph: 1.03A, 1.03D, 1.03E, 1.03F
Authorized By: ZELINSKY, Inc	Date Submitted: 09/11/2025

Submittal Certification	
Check Either (A) or (B):	
☐ (A)	We have verified that the equipment or material contained in this submittal meets all the requirements specified in the project manual or shown on the contract drawings with <u>no exceptions</u> .
☒ (B)	We have verified that the equipment or material contained in this submittal meets all the requirements specified in the project manual or shown on the contract drawings <u>except</u> for the deviations listed.
Certification Statement: By this submittal, I hereby represent that I have determined and verified all field measurements, field construction criteria, materials, dimensions, catalog numbers and similar data, and I have checked and coordinated each item with other applicable approved shop drawings and all Contract requirements.	
General Contractor's Reviewer's Signature: 	
Printed Name and Title: Krishna Dubbudu, Project Engineer	
In the event, Contractor believes the Submittal response does or will cause a change to the requirements of the Contract, Contractor shall immediately give written notice stating that Contractor considers the response to be a Change Order.	
Firm:	Signature: Date Returned:

PM/CM Office Use
Date Received GC to PM/CM:
Date Received PM/CM to Reviewer:
Date Received Reviewer to PM/CM:
Date Sent PM/CM to GC:

Nan, Inc

PROJECT: HILO WWTP REHABILITATION
AND REPLACEMENT PROJECT - PHASE 1

JOB NO. WW-4705R

THIS SUBMITTAL HAS BEEN CHECKED BY
THIS CONTRACTOR. IT IS CERTIFIED
CORRECT, COMPLETE, AND IN
COMPLIANCE WITH CONTRACT
DRAWINGS AND SPECIFICATIONS. ALL
AFFECTED CONTRACTORS AND
SUPPLIERS ARE AWARE OF, AND WILL
INTEGRATE THIS SUBMITTAL (UPON
APPROVAL) INTO THEIR OWN WORK.

DATE RECEIVED 9/12/2025
SPECIFICATION SECTION # 09672
SPECIFICATION RESINOUS FLOORING
PARAGRAPH 1.03A, 1.03D, 1.03E, 1.03F
DRAWING na
SUBCONTRACTOR ZELINSKY, Inc
SUPPLIER ZELINSKY, Inc
MANUFACTURER Green Chemistry Innovati

CERTIFIED BY CQCM or Designee : 

Hilo Wastewater Treatment Plant Phase 1
Contract No. WW-4705R

VARIANCE
Request Form

Variance Request 09672- RESINOUS FLOORING

The decision to grant or deny a variance is at the sole discretion of the relevant authority or entity responsible for reviewing such requests.

Specification section 09672- RESINOUS FLOORING

1. What are the benefits the variance has to the Government?
 - Product is a 100% solids epoxy resin that exhibits no amine blush when used in humid environments and excellent chemical and water resistant.
2. What are the positive and negative impacts to the project and Government if the variance is accepted?

Positive impacts:

 - Excellent Chemical and Water resistant.

Negative impacts:

 - This product is not readily available to the islands.
3. Discuss how the variance is "equal to" or "better than" the specification requirement.
 - Chemistry Innovations is no longer in business and there are no Product Data to compare.
4. Is there any cost Impact to the government if the variance product is approved?
 - No cost information available due to company out of business.

List of Variation[s]

Note: these numbers correspond to the spec section

1. Product **Green Chemistry Innovations** in lieu of **NeoGard**, refer to above.
2. Chemistry Innovations is no longer in business.

NEW SECTION

SECTION 09672^{AD3}

RESINOUS FLOORING

PART 1 GENERAL

1.01 SUMMARY

- A. Legal 3 Style (ALT 3) Section includes resinous flooring systems.

1.02 PREINSTALLATION MEETING

- A. Preinstallation Conference: Conduct conference at project location, exact location to be determined.
- B. Participants include but are not limited to Director or his/her Representative, General Contractor and Flooring Installation Contractor.

1.03 SUBMITTALS

- ✓ A. Product Data: For each type of product include manufacturer's technical data, safety data sheets, application instructions, and recommendations for each resinous flooring component required. ← Page- 13-27
- B. Samples for Verification: For each resinous flooring system required, 6-inches square, applied to a rigid backing by Installer for this Project. Samples must comply with specified system materials, application thickness, color and finish texture.
- C. Installer Certificates: Installer certification by material manufacturer for the system(s) specified. ← Provided After Approval
- ✓ D. Material Data Sheets, Safety Data Sheets (SDS) and Installation Instructions: For each resinous flooring component and system from manufacturer. ← Page- 35-145
- ✓ E. Maintenance Data: For each resinous flooring system maintenance instructions and period system evaluation from material manufacturer. ← Page- 164-165
- ✓ F. Warranty: Installation and material warranty from installer and material manufacturer. ← Page- 167-168

1.04 QUALITY ASSURANCE

- A. Installer Qualifications: A resinous flooring installer who has completed industry qualified training or experienced with similar flooring systems and project scope (references required).
- B. Engage an installer who is certified, in writing, by resinous flooring system manufacturer as qualified to apply the resinous flooring systems indicated.

- C. Mockups: Build mockups to verify selections made under Sample submittals and to demonstrate aesthetic effects and set quality standards for materials and execution.
 - 1. Apply full-thickness mockups on 10-foot square floor area selected by Director.
 - 2. Simulate finished lighting conditions for Director's review of mockups.
 - 3. Approval of mockups does not constitute approval of deviations from the Contract Documents contained in mockups unless Director specifically approves such deviations in writing.
 - 4. Subject to compliance with requirements, approved mockups may become part of the completed Work if undisturbed at time of Final Contract Acceptance.

1.05 DELIVERY, STORAGE, AND HANDLING

- A. Deliver materials in original packages and containers, with seals unbroken, bearing manufacturer's labels indicating brand name and directions for storage and mixing with other components.

1.06 FIELD CONDITIONS

- A. Environmental Limitations: Comply with resinous flooring system manufacturer's written instructions for substrate temperature, ambient temperature, moisture, ventilation, and other conditions affecting resinous flooring application.
- B. Lighting: Provide permanent lighting or, if permanent lighting is not in place, simulate permanent lighting conditions during resinous flooring application.
- C. Close spaces to traffic during resinous flooring application and for 24-hours after application unless manufacturer recommends a longer period.

PART 2 PRODUCTS

2.01 PERFORMANCE REQUIREMENTS

- A. Volatile Organic Content (VOC) of Liquid-Applied Flooring Coatings: Ultra-Low VOC (<23 g/L) when calculated according to 40 CFR 59, Subpart D (EPA Method 24) or ASTM D2369.
- B. Moisture Vapor Emissions: Flooring system must accommodate up to 10lbs/1000ft²/24hrs (4.5 kg/92.9m²/24hrs) vapor emissions as measured by ASTM F1869.
- C. Flammability: Self-extinguishing according to ASTM D635.

2.02 MANUFACTURERS

- A. Source Limitations: Obtain primary resinous flooring materials, including primers, resins, hardening agents, aggregates, grouting coats, and sealers, from a single manufacturer or the resinous flooring system manufacturer's approved supplier. Obtain secondary materials, including patching and fill material, joint sealant, divider

and termination strips, and repair materials recommended in writing by manufacturer of primary resinous flooring materials.

2.03 RESINOUS FLOORING

- A. Resinous Flooring System: Abrasion-, impact-, and chemical-resistant, aggregate-filled, and resin-based monolithic floor surfacing designed to produce a seamless floor.
- B. For reasons of convenience and to establish minimum standard of quality, character, design aesthetics, performance, functional and experiential quality for this project, the products of the following manufacturers are incorporated into these Specifications, along with general descriptions of the minimum resinous flooring components.
 - 1. Basis-of-Design Product: Subject to compliance with requirements, provided by Green Chemistry Innovations; GSPE - Textured Flooring System or comparable product approved via request for Substitution.
 - 2. The products of other manufacturers not listed shall be pre-qualified in accordance with paragraph 6.2 SUBSTITUTION OF MATERIALS AND EQUIPMENT of the County's General Requirements and Covenants as amended by the Special Provisions.
 - a. The burden of proof as to the comparative quality and suitability of any alternate brand or make shall be upon the bidder, and it shall furnish, at its own expense, all information relating thereto at the time of making such a request. The County shall be the sole judge as to the suitability of the alternate brand or make, and its decision shall be final. Additionally, the requestor shall identify the components it proposes to use in satisfaction of the specific resinous flooring component described in this Section. Approval of any substitution request does not inherently provide approval for deviations from any other requirement of the Section or of the project.
- C. System Characteristics:
 - 1. Color and Pattern: Colors are produced using combinations of aggregates, cement and resin. Standard color palette is provided by the manufacture.
 - a. To be selected by Director's representative from Manufacturer's standard Colored Quartz/Colored Quartz Blend:
 - 2. Wearing Surface: Finish texture/gloss as described by SSPC Concrete Coating Texture Standard (CTS):
 - a. To be selected by Director's representative from Manufacturer's standard concrete finished texture/gloss wearing surface.
 - 1) Roughness value (A-D): to be selected by Director's representative
 - 2) Gloss value: High Gloss (topcoat), BYK-Gardner Micro-gloss 60° >85 units.
 - 3) Overall System Thickness: 1/4-inch.
- D. Primer: Required when concrete substrate moisture vapor emission (MVE) rates are less than 10lb/1000ft² in-24 hours according to ASTM F1869.
 - 1. Basis-of-Design Product: Subject to compliance with requirements, provide by manufacturer Green Chemistry Innovations or comparable product compatible with resinous flooring system and approved via Request for Substitution.

2. Formulation Description:
 - a. Mixed solids: 98-100%
 - b. Volatile Content: Ultra-Low VOC (<23 g/L) when calculated according to 40 CFR 59, Subpart D (EPA Method 24) or ASTM D2369.
- E. Patching and Fill Material: Resinous product approved by resinous flooring system manufacturer and recommended by manufacturer for application indicated and approved via Request for Substitution.
- F. Body Coats:
 1. Manufacturers: Green Chemistry Innovations or comparable manufacturer approved via request for Substitution.
 2. Coating: GSPEC-FC Coating (GSPEC-FC) Clear/Pigmented as required to meet system specified color(s).
 - a. Formulation Description: High Solids, Ultra-Low VOC Epoxy-urethane resin containing zero isocyanates.
 3. Aggregates: GSPEC Bag Mix supplied by resinous flooring system manufacturer to match specified color(s).
 4. Application Method: Trowel or gauge rake.
 5. Number of Coats: Apply one coat to achieve specified system thickness.
- G. Broadcast:
 1. Manufacturers: Green Chemistry Innovations or manufacturer approved via Request for Substitution.
 2. Aggregate:
 - a. Colored quartz as recommended by coating manufacturer to meet specified system color to be selected by Director's representative from Manufacturer's standard products.
- H. Topcoat:
 1. Manufacturers: Subject to compliance with requirements, provide products by Green Chemistry Innovations or manufacturer approved via Request for Substitution.
 2. Coating: GSPEC-FC Coating (GSPEC-FC) Clear/Pigmented as required to meet system specified color(s).
 3. Formulation Description: High Solids, Ultra-Low VOC Epoxy-urethane resin containing zero isocyanates.
- I. System Physical Properties: Provide resinous flooring system with the following physical property requirements when tested according to test methods indicated:
 1. Tensile Strength: 5700 psi minimum according to ASTM D638.
 2. Water Absorption: 0.5 percent maximum according to ASTM D570.
 3. Abrasion Resistance: <40 mg maximum weight loss according to ASTM D4060 (Taber Wheel CS-17, 1000g).
 4. Hardness: 75-85 Shore D according to ASTM D2240.
 5. Bond Strength to Concrete: 100% Concrete Failure according to ASTM D4541.
 6. Elongation at break: 3-7% according to ASTM D638.
- J. System Chemical Resistance: Provide resinous flooring system with the following minimum chemical resistant property requirements:
 1. 10% Sulfuric Acid - <1.0% weight gain; minimum staining

2. 10% Sodium Hydroxide – No effect.
3. Gasoline – No effect.
4. Motor Oil – No effect.
5. Brake Fluid – No effect.
6. Vegetable Oil – No effect.
7. Bleach – No effect.
8. Ammonia – No effect.
9. Spray Paint – No effect.

PART 3 EXECUTION

3.01 PREPARATION

- A. Prepare and clean substrates according to resinous flooring system manufacturer's written instructions for substrate indicated. Provide clean, dry substrate for resinous flooring application.
- B. Concrete Substrates: Provide sound concrete surfaces free of laitance, glaze, efflorescence, curing compounds, form-release agents, dust, dirt, grease, oil, and other contaminants incompatible with resinous flooring.
 1. New Cast-in-Place Concrete must be placed over an effective moisture retarder >15 mils according to ACI 302.
 2. Roughen concrete substrates as follows:
 - a. Shotblast or diamond grind surfaces to Concrete Surface Profile (CSP) of 3-5 (ICRI Guideline 310.2R) with an apparatus that abrades the concrete surface, contains the dispensed shot within the apparatus, and recirculates the shot by vacuum pickup.
 - b. Comply with SSPC-SP 13/NACE No. 6, "SURFACE PREPARATION OF CONCRETE" requirements unless manufacturer's written instructions are more stringent.
 3. Repair damaged and deteriorated concrete according to resinous flooring system manufacturer's instructions.
 4. Verify that concrete substrates are dry and moisture-vapor emissions are within acceptable levels according to manufacturer's written instructions. Moisture evaluation and testing is to be conducted by an independent certified inspector to be retained by and paid for by the General Contractor.
 - a. Anhydrous Calcium Chloride Test: ASTM F1869. Proceed with application of resinous flooring without primer only after substrates have maximum moisture-vapor-emission rate of 10lb/1000ft² in 24-hours.
 - 1) If CaCl test (ASTM F1869) results are higher than 10lb/1000ft² in 24-hours, a primer application is required and the resinous flooring system manufacture must be consulted before proceeding.
 - b. Relative Humidity Test: Use in situ probes, ASTM F2170. Proceed with installation only after substrates have a maximum 80-percent relative humidity level measurement. If concrete humidity values exceed 80%, consult resinous flooring system manufacturer before applying system.
 5. Alkalinity and Adhesion Testing: Verify that concrete substrates have pH within acceptable range. Perform tests recommended by manufacturer. Proceed with application only after substrates pass testing.

- C. Patching and Filling: Use patching and fill material to fill holes and depressions in substrates according to manufacturer's written instructions.
- D. Control Joint Treatment: Treat control joints and other nonmoving substrate cracks to prevent cracks from reflecting through resinous flooring according to industry standards as detailed in Joint NACE Publication 02203/ICRI Technical Guideline 710.1/SSPC-TR 5 Technical Committee Report: "Design, Installation, and Maintenance of Protective Polymer Flooring Systems for Concrete".
- E. Resinous Materials: Mix components and prepare materials according to resinous flooring system manufacturer's written instructions.

3.02 APPLICATION

- A. Apply components of resinous flooring system according to manufacturer's written instructions to produce a uniform, monolithic wearing surface of thickness specified.
 - 1. Coordinate application of components to provide optimum adhesion of resinous flooring system to substrate, and optimum intercoat adhesion.
 - 2. Cure resinous flooring components according to manufacturer's written instructions. Prevent contamination during application and curing processes.
 - 3. Expansion and Isolation Joint Treatment: Honor all expansion/isolation joints located in the substrate utilize preformed joint strips or L-strips on both sides of the joint to terminate flooring system and fill joints with flexible sealant as detailed in Section 07900.
- B. Primer: Primer is only required if moisture vapor emissions exceed 10lb/1000ft² in 24-hours according to ASTM F1869. Apply primer over prepared substrate at manufacturer's recommended spreading rate.
- C. Body Coats: Apply self-leveling slurry body coats with a gauge rake or trowel in thickness indicated for flooring system. Allow the slurry to self-level prior to subsequent aggregate broadcast.
- D. Broadcast: Broadcast selected aggregate (silica sand/colored quartz/colored paint chip to refusal. Allow to cure and remove excess aggregate.
- E. Topcoat: Apply topcoat (Pigmented/Clear) using a squeegee, trowel and/or roller according to resinous flooring system manufacturer's instructions to achieve the desired texture. Allow to cure. (Cure time and open to service will depend upon temperature.)

3.03 FIELD QUALITY CONTROL

- A. Material Sampling: Director may, at any time and any number of times during resinous flooring application, require material samples for testing for compliance with requirements.
 - 1. General Contractor shall engage an independent testing agency to take samples of materials being used. Material samples will be taken, identified, sealed, and certified in presence of Contractor.

2. Testing agency will test samples for compliance with requirements, using applicable referenced testing procedures or, if not referenced, using testing procedures listed in manufacturer's product data.
 3. If test results show applied materials do not comply with specified requirements, pay for testing, remove noncomplying materials, prepare surfaces coated with unacceptable materials, and reapply flooring materials to comply with requirements.
- B. Core Sampling of Finished Floor System: At the direction of Director and at locations designated by Director, take one core sample per 1000-sq. ft. of resinous flooring, or portion of, to verify thickness. For each sample that fails to comply with requirements, take two additional samples. Repair damage caused by coring. Correct deficiencies in installed flooring as indicated by testing.

3.04 PROTECTION

- A. After application of the topcoat, the Work shall be ready for final inspection and acceptance by the Director or his agent.
- B. The General Contractor shall protect resinous flooring from damage and wear during the remainder of the construction period. Use protective methods and materials, including temporary covering, recommended in writing by resinous flooring system manufacturer.

END OF SECTION

^{AD3} Addendum No. 3 - June 2024



Painting, Wallcovering, & Industrial

91-310 Komohana Street
Kapolei, Hawaii 96707

Phone: (808) 682-1321
Fax: (808)-682-1535

www.zelinskyco.com
Contractor Lic. #C20440

To: Nan Inc. From: Alee Bowden Date: 9/11/2025

Job Site: Hilo Watewater Treatment Plant

ATTN: Makoa Ng

SUBJECT: 09672 Resinous Flooring Submittal

WE ARE SENDING YOU: ☒ Attached ☐ Sent Via Mail/UPS ☐ Hand Delivered

ITEM	NO.	Description
1	1	09672 Resinous Flooring Submittal

THESE ARE TRANSMITTED AS CHECKED BELOW:

- | | |
|--|---|
| ☒ For Approval | ☐ Disapproved |
| ☐ For your use | ☐ As Requested |
| ☐ Approved as Noted | ☐ Subject to Contract Requirements |
| ☐ Approved as Submitted | ☐ Color Selection |

REMARKS:

See Attached PDS and SDS Sheets
Preventive Maintenance
Sample Warranty Zelinsky
Sample Warranty Neogard

Neocrete

Table of Contents

70718/70719 Epoxy Joint Filler	Product Data Sheet	English	2
70991 Polyurethane Sealant	Product Data Sheet	English	4
70714/70715 Epoxy, High Performance	Product Data Sheet	English	6
70817/70818 Urethane, Chemical Resistant, Clear	Product Data Sheet	English	8
P1934 Fumed Silica	Product Data Sheet	English	10
Neocrete SL Mix	Product Data Sheet	English	11
Quartz Aggregate, UV-stable	Product Data Sheet	English	13
Neocrete V Cove Base	Product Data Sheet	English	15
Neocrete SL Quartz	Guide Specification	English	17

PRODUCT DATA

Product Data Sheet

70718/70719 — Flexible Epoxy

25000: Base 25009 : Curing Agent 95048



Description:	70718/70719 (Hempel 25000) is a two-component high solids, flexible, semi-rigid, stress-relieving epoxy hybrid joint filler.
Recommended use:	Applications include: Semi-rigid high strength joint compound; binder for patching spalled concrete to prevent further deterioration of the concrete; topping for high impact resistance..
Features:	Meets South Coast Air Quality Management District (SCAQMD) VOC requirements
Certificates/approvals:	LEED: MRc5, IEQc4.2
Availability:	Available in North America. Not included in Group Assortment; other regions must confirm.

Colors and packaging:

70718-02 (2500916650)	Gray	3-gallon kit
70719 (9504800000)	NA	
<i>Some standard colors and/or packaging may have extended lead times and/or minimum quantities. Contact Neogard for additional colors.</i>		

Physical constants:

Tensile strength	2,100 psi	ASTM D412
Elongation at break	90%	ASTM D412
Permanent set	<20%	ASTM D412
Tear resistance	100 pli	ASTM D1004
Water resistance	2% at 7 days	ASTM D471
MVT at 20 mils	0.2 English	ASTM E96
Shore A	95	ASTM D2240
Shore D	47	ASTM D2240
Adhesion	650 psi	ASTM D4541
Flammability	Pass	ASTM D635
Taber abrasion	15 mg/1,000 cycles CS-17	ASTM D4060
Weight/gallon (mixed)	9.05 lbs/gal	ASTM D1475
Weight solids (mixed)	99% at 12 hours	ASTM C1250
Volume solids (mixed)	99%	Calculated
Viscosity (mixed)	11,500 cps	ASTM D2196
Flash point	>250°F/122°C	ASTM D3278
VOC (mixed)	2.3 g/L	EPA Method 24
Shelf life	1 year	
Set time	6 hours	

The above tested results are typical values. Individual lots may vary up to 10% from the typical value. Further technical information can be found at www.neogard.com.

Application details:

Version, mixed product	25000
Mixing ratio:	Base 25009 : Curing Agent 95048 2 : 1 by volume
Mixing instructions:	Pre-mix base for 3–5 minutes before adding curing agent. Mix for a minimum of 5 minutes before applying. Jiffy Mixer paddle recommended.
Application method:	Roller
Thinner (max.vol.):	7055 Odorless Reducer (086JB) or Hempel's Thinner 08080 (xylene) (10%)
Pot life:	70 minutes at 75°F/23°C
Cleaning of tools:	7055 Odorless Reducer (086JB) or Hempel's Thinner 08080 (xylene)

Product Data Sheet

70718/70719 — Flexible Epoxy

25000: Base 25009 : Curing Agent 95048



Safety:	Handle with care. Use with adequate ventilation. Before and during use, observe all safety labels on packaging and paint containers, consult Neogard Safety Data Sheets and follow all local or national safety regulations.
Surface preparation:	According to Neogard Guide Specifications.
Application conditions:	Ambient temperature must be between 50°F/10°C and 85°F/29°C, and a minimum of 5°F/3°C above dew point.
Preceding coat:	According to Neogard Guide Specifications.
Subsequent coat:	Must be recoated within 24 hours at 70°F/21°C for subsequent coat to properly bond. If the 24-hour recoat window is missed, sand to dull finish before applying subsequent coat(s).
Storage temperature:	Store in a cool area to ensure full shelf life. Recommended temperature: 70°F/21°C.

Note: 70718/70719 is for professional use only.

Issued by: Hempel (USA) – 25000

This Product Data Sheet ("PDS") relates to the supplied product ("Product") and is subject to update from time-to-time. Accordingly, the buyer/applicator should refer to the PDS current as of the time of delivery. In addition to the PDS, the buyer/applicator may receive some or all of the specifications, statements and/or guidelines listed below or available at www.neogard.com (the "Additional Documents"):

No.	Document Description
1	PDS
2	Guide Specification
3	Application Manual
4	Other Technical Support Information (i.e. summary application tables, troubleshooting guides, maintenance manuals, chemical resistance charts and other technical information)

In the event of a conflict between this PDS and the Additional Documents, the conflict shall be resolved in accordance with the order of priority set forth above. In addition, the buyer/applicator should refer to the relevant Safety Data Sheet current as of the time of delivery and available at www.neogard.com. Buyer/applicator is responsible for determining the suitability of the intended use of the Product, and Neogard disclaims all responsibility for any use, handling and storage of the Product that is not in accordance with the requirements set forth in the relevant PDS and the Additional Documents. The terms and provisions hereof apply to this PDS, the Additional Documents and any other documents supplied by Neogard in respect of the Product. The Product is supplied and all technical assistance is given subject to the General Conditions of Sale of Hempel Products and/or Services available at www.hempel.com. NEOGARD MAKES NO OTHER WARRANTY THAT EXTENDS BEYOND THE WARRANTY REFERENCED THEREIN INCLUDING, WITHOUT LIMITATION, ANY IMPLIED WARRANTIES OR CONDITIONS OF MERCHANTABILITY OR FITNESS FOR A PARTICULAR PURPOSE. NEOGARD WILL NOT BE LIABLE FOR ANY INCIDENTAL OR CONSEQUENTIAL DAMAGES FOR BREACH OF ANY EXPRESS OR IMPLIED WARRANTY OR CONDITION, OR THAT IN ANY WAY ARISE IN RELATION TO THE PRODUCT. 70718-70719-PDS ksk 10272021.docx

Neogard®, a part of Hempel
2728 Empire Central - Dallas, Texas 75235 - Phone (214) 353-1600 - Fax (214) 357-7532 - www.neogard.com

Product Data Sheet

70991 (47XJB) Polyurethane Sealant



Description:	70991 (Hempel 47XJB) is a one-component, contractor/construction grade, smooth polyurethane sealant.
Recommended use:	70991 is used for sealing the following: expansion and control joints in pre-cast concrete panels; various roofing and siding applications; perimeters of doors, windows, and other wall penetrations.
Features:	May be used with most building materials, including stone, masonry, ceramic, marble, wood, steel, aluminum, fiber cement board and many other synthetic materials. Easy gunning; reduces installer's fatigue. Bonds well to most common building materials. Quicker skin and cure time; reduces jobsite dirt pickup.
Service temperatures:	-40°F–150°F (-40°C–66°C)
Certificates/approvals:	Type S, Grade NS, Class 25, Use NT, A and M. US Federal Specification TT-S 00230C (COMB-NBS) for one-component sealants as Class A, non-sag. Canadian Specification CAN/CGSB 19.13-M87. CARB and SCAQMD Compliant. Meets VOC requirements for OTC Regulation.
Availability:	Available in North America. Not included in Group Assortment; other regions must confirm.

Colors and packaging:

70991-4 (47XJB1L030)	Gray	10.1-oz cartridge
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Physical constants:

Hardness (Shore A)	42	ASTM D2240
Modulus at 100% Elongation	65 psi	ASTM D412
Modulus at 25% Elongation	45 psi	ASTM D412
Tensile Strength at Break	133 psi	ASTM D412
Elongation at Break	685%	ASTM D412
UV Resistance	Pass	ASTM C793
VOC	43 g/L (2.8%)	Calculated
Adhesion Peel	> 5 piw	TT-S-00230C/ASTM C794
Joint Movement Capability	+25%	TT-S-00230C/ASTM C719

Tool/Work Time	60 minutes
Skin Time	4 hours
Curing Time at 77°F/25°C	2–7 days, 1/16" per day
Flow, Sag or Slump	0.1 inch

The above tested results are typical values. Individual lots may vary up to 10% from the typical value. Further technical information can be found at www.neogard.com.

Application details:

Application method:	Caulking gun
Cleaning:	Clean tools with an aromatic solvent. Dry-wipe excess uncured sealant from surfaces, then follow with a solvent wipe. Cured sealant is difficult to remove without damaging the substrate.

Safety:	Handle with care. Before and during use, observe all safety labels on packaging and paint containers, consult Neogard Safety Data Sheets and follow all local or national safety regulations.
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Surface preparation:	Surface must be clean, sound, and dry. Do not apply over damp, contaminated, loose surfaces, old sealants, or other foreign substances that may impair adhesion. Pre-test substrates with a sample of 70991 before full application.
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Product Data Sheet

70991 (47XJB) Polyurethane Sealant



Application conditions:

Do not install when the dew point of the substrate is close to ambient temperature, or if moisture-vapor transmission (MVT) is present. Damp substrates with high moisture content will cause bubbling and foaming. Lower relative humidity and temperature will significantly extend curing time. Confined areas, deep joints, and moisture barrier substrates may also affect the full cure time and extend it by many days. High temperature/humidity can cause bubbles to develop during the curing process.

Subsequent coat:

According to Neogard system Guide Specifications. Test paints or coatings for adhesion before general application.

Remarks:

Do not apply to copper substrates. Will cause staining of porous substrates such as marble, limestone, and granite. Not for sealing narrow joints, fillet joints, and face nail holes. Not recommended for smearing and feathering over joints, or by itself joints where abrasion resistance is required (walkways, driveways, runways, etc.). Not for continuous immersion in water or any other fluid, or exterior or interior sealing below the waterline. Not for glazing applications. Bond line strength can be affected by UV rays through the clear material (glass, acrylic glass, polycarbonate). When fully cured, avoid exposure to fuels, or chlorinated, acid, or alkaline solutions. Contact with asphalts and other filler compounds impregnated with oil, asphalt, tar, etc., may deteriorate the cohesive strength of the substrate and ultimately compromise the seal. Lower relative humidity and temperature will significantly extend curing time. Confined areas, deep joints, and moisture barrier substrates may also affect the full cure time and extend it by many days. Exposure to UV/sunlight will alter original color or gloss. Effect is limited to the surface layer and will not compromise the sealing properties if joint dimensions are proper and the sealant is properly applied. Check shelf life prior to use. Do not use past shelf life. Shelf life of polyurethane sealants may be significantly reduced by high temperature and high relative humidity.

Application coverage rates (linear feet per 10.1 fl oz cartridge):

		Width							
		1/8"	1/4"	3/8"	1/2"	5/8"	3/4"	7/8"	1"
Depth	1/8"	99	49	33	24	20	16	14	12
	1/4"	NA	24	20	12	10	8	7	6
	3/8"	NA	NA	11	8	6	5	5	4
	1/2"	NA	NA	NA	6	5	4	3	3

Note:

70991 is for professional use only.

Issued by:

Hempel (USA) – 47XJB

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No.	Document Description
1	PDS
2	Guide Specification
3	Application Manual
4	Other Technical Support Information (i.e. summary application tables, troubleshooting guides, maintenance manuals, chemical resistance charts and other technical information)

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Product Data Sheet

70714/70715 (45060) – 2K Epoxy

Base 70714 (45069) : Curing Agent 70715 (95073)



Description:	70714/70715 (Hempel 45060) is a two-component, 100% solids epoxy resin.
Recommended use:	Applications include: High-gloss, high-build protective coating; binder for self-smoothing pigmented and color quartz flooring; low-viscosity binder for trowel-applied mortar flooring systems; high-quality primer for chemical resistant urethane top coats, and/or Neogard waterproofing and flooring systems; low-viscosity binder for trowel applied mortar and/or patching material; low viscosity binder for crack injection.
Features:	Formulated for use on damp or dry concrete. Near water-clear. Exhibits no amine blush when used in humid environments. Very low viscosity for deep penetration into concrete. Uniform wetting of filler aggregates. Excellent chemical and water resistance. Meets U.S.D.A. requirements for incidental food contact. Meets South Coast Air Quality Management District (SCAQMD) VOC requirements.
Certificates/approvals:	LEED: MRc5, IEQc4.2, BDc4, IDc4 MPI: MPI #212
Availability:	Available in North America. Not included in Group Assortment; other regions must confirm.

Colors and packaging:

70714 (4506900000)	Clear and Tintable	15- and 3-gallon kits
70714-01 (4506916640)	White	15- and 3-gallon kits
70714-02 (4506916650)	Gray	15- and 3-gallon kits
70714-03 (4506923820)	Tan	15- and 3-gallon kits
70714-04 (4506916670)	Dark Gray	15- and 3-gallon kits
70714-05 (4506957930)	Tile Red	15- and 3-gallon kits
70714-11 (4506916660)	Light Gray	15- and 3-gallon kits
70714-16 (4506919990)	Black	15- and 3-gallon kits
70714-06 (4506949890)	Green	3-gallon kits
70715 (9507300000)	NA	Curing agent, included in kits

Some standard colors and/or packaging may have extended lead times and/or minimum quantities. Contact Neogard for additional colors.

Physical constants:

Tensile strength	3,700 psi	ASTM D638
Elongation	25%	ASTM D638
Compressive strength	25,300 psi	ASTM D695
Flexural strength	3,180 psi	ASTM D790
Flexural modulus	57,700 psi	ASTM D790
Water resistance	0.21%	ASTM D570
MVT at 10 mils	0.16	ASTM E96
Taber abrasion	25 mg/1,000 CS-17	ASTM D4060
Shore d	78	ASTM D2240
Adhesion	350 psi	ASTM D4541
Flammability	Pass	ASTM D635
Weight/gallon (mixed)	8.8 lbs/gal	ASTM D1475
Weight solids (mixed)	100%	ASTM D4209
Volume solids (mixed)	100%	Calculated
VOC (mixed)	<10 g/L	EPA Method 24
Flash point	>200°F/93°C	ASTM D3278
Shelf life	1 year	
Thin film cure at 75°F/23°C	8–9 hours	
Full cure at 75°F/23°C	7 days	
Full chemical resistance	7 days	

The above tested results are typical values. Individual lots may vary up to 10% from the typical value. Further technical information can be found at www.neogard.com.

Product Data Sheet

70714/70715 (45060) – 2K Epoxy

Base 70714 (45069) : Curing Agent 70715 (95073)



Application details:

Version, mixed product	70714/70715 (45060)
Mixing ratio:	Base 70714 (45069) : Curing Agent 70715 (95073) 2 : 1 by volume
Mixing instructions:	Pre-mix base for 3–5 minutes before adding curing agent. Mix for a minimum of 5 minutes before applying. Jiffy Mixer paddle recommended.
Application method:	Roller or squeegee
Thinner (max.vol.):	7055 Odorless Reducer (086JB) or Hempel's Thinner 08080 (xylene) (10%)
Pot life:	30 minutes at 70°F/21°C
Cleaning of tools:	7055 Odorless Reducer (086JB) or Hempel's Thinner 08080 (xylene)
Safety:	Handle with care. Use with adequate ventilation. Before and during use, observe all safety labels on packaging and paint containers, consult Neogard Safety Data Sheets and follow all local or national safety regulations.
Surface preparation:	According to Neogard Guide Specifications.
Application conditions:	Ambient temperature must be between 40°F/4°C and 100°F/37°C, and a minimum of 5°F/3°C above dew point.
Preceding coat:	According to Neogard Guide Specifications.
Subsequent coat:	Must be recoated within 24 hours at 70°F/21°C for subsequent coat to properly bond. If the 24-hour recoat window is missed, sand to dull finish before applying subsequent coat(s).
Storage temperature:	Store in a cool area to ensure full shelf life. Recommended temperature: 70°F/21°C.

Note: **70714/70715 is for professional use only.**

Issued by: Hempel (USA) – 45060

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Product Data Sheet

70817/70818 — High-Solids Chemical Resistant Urethane
57070: Base 57079 : Curing Agent 95059



Description:	70817/70818 (Hempel 57070) is a two-component, ultra-high solids, chemical-resistant aliphatic polyurethane.
Recommended use:	As a topcoat for Neogard decorative flooring and waterproofing systems.
Features:	Meets South Coast Air Quality Management District (SCAQMD) VOC requirements.
Availability:	Available in North America. Not included in Group Assortment; other regions must confirm.

Colors and packaging:

70817 (5707900050)	Clear (Tintable)	
70817-01 (5707916640)	White	2-gallon kit (mixed)
70818 (9505900000)	N/A	
Contact Neogard for available colors.		

Physical constants:

	70817/70818	70817-01/70818	
Tensile strength	7,500 psi	5,800 psi	ASTM D2370
Elongation	12%	40%	ASTM D2370
Shore D	70	65	ASTM D2240
Water resistance	<1%, 7 days	<1%, 7 days	ASTM D471
Taber abrasion	23 mg/1,000 cycles CS-17	17 mg/1,000 cycles CS-17	ASTM D4060
Anti-microbial	Pass	Pass	JIS Z 2801-2010
Both versions			
Weight/gallon (mixed)	9.73 lbs/gal	ASTM D1475	
Weight solids (mixed)	92%	ASTM D5201	
Volume solids (mixed)	90%	ASTM D5201	
VOC (mixed)	< 100 g/L	ASTM D5201	
Flash point	70817: 167°F/75°C 70718: 316°F/157°C	ASTM D3278	
Working time at 75°F/23°C, 50% relative humidity	20–25 minutes		
Cure time at 77°F/24°C, 50% relative humidity	10–12 hours		
Recoat at 75°F/23°C, 50% relative humidity	Within 24 hours		
Shelf life	1 year		

The above tested results are typical values. Individual lots may vary up to 10% from the typical value. Further technical information can be found at www.neogard.com.

Application details:

Version, mixed product	57070
Mixing ratio:	Base 59079 : Curing Agent 95059 1 : 1 by volume
Mixing instructions:	Pre-mix base for 3 minutes. Add base to curing agent and mix for 3 more minutes. Jiffy Mixer recommended.
Application method:	Roller or squeegee
Application Rate:	140–200 ft ² /gallon
Indicated film thickness, dry:	12–8 mils
Thinner (max.vol.):	7055 Odorless Reducer (086JB) or Hempel's Thinner 08080 (xylene) (10%)
Pot life:	60–80 minutes at 75°F/23°C
Cleaning of tools:	7055 Odorless Reducer (086JB) or Hempel's Thinner 08080 (xylene)

Product Data Sheet

70817/70818 — High-Solids Chemical Resistant Urethane
57070: Base 57079 : Curing Agent 95059



Safety:	Handle with care. Use with adequate ventilation. Before and during use, observe all safety labels on packaging and paint containers, consult Neogard Safety Data Sheets and follow all local or national safety regulations.
Surface preparation:	According to Neogard Guide Specifications.
Application conditions:	Precondition material for 24 hours at 65°–75°F/18°C–23°C. Ambient temperature must 40°F–80°F/4°C–26°C. Substrate temperature must be 40°F–110°F/4°C–40°C and 5°F/3°C above the dew point. Relative humidity must be 30%–85%; may turn milky if applied in excess of 12 mils per coat or in areas of high humidity (>85%). Surface must be dry, clean, and without frost.
Preceding coat:	According to Neogard Guide Specifications.
Subsequent coat:	Must be recoated within 24 hours at 70°F/21°C for subsequent coat to properly bond. If the 24-hour recoat window is missed, sand to dull finish before applying subsequent coat(s).
Accelerators and additives:	7986 (994JB00000): Odor mask, synthetic vanilla.
Storage temperature:	Store in a cool area to ensure full shelf life. Recommended temperature: 75°F/23°C.
Remarks:	Open to foot traffic in 16 hours at 75°F/23°C, 50% relative humidity. Apply additional coats within 24 hours of initial application. If applying additional coats beyond 24 hours, lightly abrade surface with 80–100 grit sand paper and thoroughly clean surface before applying next coat. Do not apply directly to concrete.

Note: **70817/70818 is for professional use only.**

Issued by: Hempel (USA) – 57070

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Product Data Sheet

P1934 (D261) Fumed Silica



Description:	P1934 (Hempel D261) is a synthetic silica powder.	
Recommended use:	As a reinforcing filler and thickening agent for Neogard epoxies, in particular to create patching material for repair of concrete substrates.	
Availability:	Available in North America. Not included in Group Assortment. Availability subject to confirmation.	

Colors and packaging:

P1934 (D261)	White	20-lb bag
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Physical constants:

Loss on drying	< 1.5%	DIN EN ISO 787-2
Sieve residue	< 0.03%	DIN EN ISO 787-18
Tamped density	Approx. 40 g/l	DIN EN ISO 787-11
BET surface	175–225 m ² /g	DIN ISO 9277, DIN 66132

SiO ₂	> 99.8%	DIN EN ISO 3262-19
Density at 68°F/20°C	Approx. 1.27 oz/in ³ 2.2 g/cm ³	DIN 51757
Loss of weight	< 2%	DIN EN ISO 3262-19
Refraction index	1.46	
pH	3.8–4.3	DIN EN ISO 787-9

The above tested results are typical values. Individual lots may vary up to 10% from the typical value. Further technical information can be found at www.neogard.com.

Safety:	Handle with care. Before and during use, observe all safety labels on packaging and paint containers, consult Neogard Safety Data Sheets and follow all local or national safety regulations. Prevent inhalation. Use respirators and/or filter masks when handling. Beware of static discharge when pouring P1934.
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Storage:	Store in a cool area to ensure full shelf life. Recommended temperature: 75°F/23°C. Protect against humidity.
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Note: **P1934 is for professional use only.**

Issued by: Hempel (USA) – D261

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Product Data Sheet

Neocrete SL — 70800/70801/70804

48012: Resin 48019: Hardener 98010: Aggregate 66022



Description:	Neocrete SL (70800/70801/70804, Hempel 48012) is a self-leveling, water-dispersed polyurethane resin combined with a powder aggregate.
Recommended use:	As a surfacing/resurfacing system applied to the following substrates: concrete, polymer reinforced screeds, and properly supported wood floors.
Features:	High abrasion, impact, and chemical resistance. Resistant to thermal shock stresses such as hot water and daily steam cleaning. Resistant to acids, caustics, fuels, oils, and solvents. Steel troweled in one step; requires no primers or sealers. Meets U.S.D.A. requirements for incidental food contact. Meets South Coast Air Quality Management District (SCAQMD) VOC requirements.
Service temperatures:	-120°F–210°F/ -84°C–99°C
Certificates/approvals:	LEED: MRc5, IEQc4.2
Availability:	Available in North America. Not included in Group Assortment; other regions must confirm.

Colors and packaging:

(Hempel product codes in parentheses)

70800-02 (4801916650)	Gray	2-gallon can, 141 fluid oz. fill
70800-05 (4801957930)	Red	2-gallon can, 141 fluid oz. fill
70800-03 (4801923820)	Tan	2-gallon can, 141 fluid oz. fill
70801 (9801000000)	NA	1-gallon can, 90 fluid oz. fill
70804 (6602209990P038)	NA	38-lb bag

Some standard colors may have extended lead times and/or minimum quantities.

Physical constants:

Compressive strength	7,700 psi	ASTM C579
Tensile strength	712 psi	ASTM C307
Flexural strength	2,200 psi	ASTM C580
Modulus of elasticity	446,700 psi	ASTM C580
Shore D	84	ASTM D2240
Adhesion (concrete)	400 psi	ASTM D4541
Water resistance	0.42%	ASTM C413
Density	121 lbs/ft ³	ASTM C905-01
Coefficient of thermal expansion	2.4 x 10 ⁻⁵ in/in/°F	ASTM C531
Resistance to fungal growth	No Support of Growth	ASTM G21
Flammability	Pass	ASTM D635
Weight solids (mixed)	99.95%	ASTM D5201
Volume solids (mixed)	99.93%	ASTM D5201
Flash point	None	ASTM D3278
VOC (mixed)	8.4 g/L	ASTM D5201
Shelf life	1 year	
Typical thickness	3/16"	
Mixed unit coverage at 3/16"	32 square feet	
Initial join up time at 70°F/21°C per 3/16"	20 minutes	
Cure to foot traffic at 70°F/21°C per 3/16"	4–8 hours	
Cure to full use at 70°F/21°C per 3/16"	10–16 hours	

The above tested results are typical values. Individual lots may vary up to 10% from the typical value. Further technical information can be found at www.neogard.com.

Product Data Sheet

Neocrete SL — 70800/70801/70804

48012: Resin 48019: Hardener 98010: Aggregate 66022



Application details:

Version, mixed product

Mixing ratio:

Neocrete SL 48012

Base 70800 (48019) : Curing Agent : 70801 (98010) : Aggregate: 70806 (66022)

Mix 141 fluid ounces of 70800 (contents of one 2-gallon can) with 90 fluid ounces of 70801 (contents of one 1-gallon can) and one 38-pound bag of 70804 powder. Mix with a variable speed drill utilizing a Jiffy Mixer to suspend any settled pigment and attain a uniform color. Mix only until the material is smooth and has no clumps of aggregate. Over-mixing will shorten pot life, working time, and affect flow/leveling. Mixing time will vary depending on temperature conditions.

IMPORTANT: Pre-mix 70800 for a minimum of one minute before mixing with 70801 hardener. To avoid color variation from mix to mix, scrape all of the pre-mixed 70800 from the 70800 can into the mixing container.

Application method:

Steel trowel, screed box, gauge rake

Thinner (max.vol.):

Contact Neogard for thinning instructions

Pot life:

15 minutes at 70°F/21°C

Cleaning of tools:

7055 Odorless Reducer (086JB) or Hempel's Thinner 08080 (xylene)

Safety:

Handle with care. Use with adequate ventilation. Before and during use, observe all safety labels on packaging and paint containers, consult Neogard Safety Data Sheets and follow all local or national safety regulations.

Surface preparation:

According to Neogard Guide Specifications.

Application conditions:

Ambient temperature must be between 50°F/10°C and 85°F/29°C, and a minimum of 5°F/3°C above dew point.

Preceding coat:

According to Neogard Guide Specifications.

Subsequent coat:

According to Neogard Guide Specifications.

Storage temperature:

Store in a cool area to ensure full shelf life. Recommended temperature: 70°F/21°C.

Remarks:

For maximum adhesion, correct surface preparation is essential. Regular cleaning of applied surface is recommended in order to maintain slip resistance and aesthetics.

Note:

Neocrete SL is for professional use only.

Issued by:

Hempel (USA) – 48012

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Product Data Sheet

NeoQuartz (66GJB)

Crystalline Quartz Granules, Broadcast Medium, UV-Stable



Description:	NeoQuartz granules are a colored, crystalline quartz, UV-stable flooring and deck coating aggregate.
Recommended use:	As an aggregate for decorative and industrial flooring systems, and decorative deck coating systems.
Availability:	Available in North America. Not included in Group Assortment. Availability subject to confirmation.

Colors and packaging:

66GJB99980	Contact Neogard	50-pound bag (all colors)
	<i>Requires a minimum order of 8 bags. Some standard colors and/or packaging may have extended lead times and/or minimum quantities.</i>	

Physical constants:

Hardness (Moh's Scale)	6.5–7	
Bulk density	90–100 pounds/cubic foot	ASTM C29/C29-97 (2003)
Specific gravity	2.65	ASTM C128-01e1

Composition

Material	Weight %
Silica sand	97.5
Resin	2.65
Various colorants	0.5

Chemical analysis

SiO ₂	99.808%
Fe ₂	0.016%
Al ₂ O ₃	0.042%
CaO & MgO	0.200%
Loss on ignition	0.100

Sieve analysis

	Retained %
20	9.0
30	12.4
40	54.0
50	22.0
70	1.9
100	0.7

Application details:

Application method:	Broadcast; rate according to Neogard system Guide Specifications.
Safety:	Handle with care. Use with adequate ventilation. Before and during use, observe all safety labels on packaging and paint containers, consult Neogard Safety Data Sheets and follow all local or national safety regulations.
Storage temperature:	Store in a dry area.

Product Data Sheet

NeoQuartz (66GJB)

Crystalline Quartz Granules, Broadcast Medium, UV-Stable



Note: **NeoQuartz is for professional use only.**

Issued by: Hempel (USA) – 66GJB

This Product Data Sheet (“**PDS**”) relates to the supplied product (“**Product**”) and is subject to update from time-to-time. Accordingly, the buyer/applicator should refer to the PDS current as of the time of delivery. In addition to the PDS, the buyer/applicator may receive some or all of the specifications, statements and/or guidelines listed below or available at www.neogard.com (the “**Additional Documents**”):

No.	Document Description
1	PDS
2	Guide Specification
3	Application Manual
4	Other Technical Support Information (i.e. summary application tables, troubleshooting guides, maintenance manuals, chemical resistance charts and other technical information)

In the event of a conflict between this PDS and the Additional Documents, the conflict shall be resolved in accordance with the order of priority set forth above. In addition, the buyer/applicator should refer to the relevant Safety Data Sheet current as of the time of delivery and available at www.neogard.com. Buyer/applicator is responsible for determining the suitability of the intended use of the Product, and Neogard disclaims all responsibility for any use, handling and storage of the Product that is not in accordance with the requirements set forth in the relevant PDS and the Additional Documents. The terms and provisions hereof apply to this PDS, the Additional Documents and any other documents supplied by Neogard in respect of the Product. The Product is supplied and all technical assistance is given subject to the General Conditions of Sale of Hempel Products and/or Services available at www.hempel.com. NEOGARD MAKES NO OTHER WARRANTY THAT EXTENDS BEYOND THE WARRANTY REFERENCED THEREIN INCLUDING, WITHOUT LIMITATION, ANY IMPLIED WARRANTIES OR CONDITIONS OF MERCHANTABILITY OR FITNESS FOR A PARTICULAR PURPOSE. NEOGARD WILL NOT BE LIABLE FOR ANY INCIDENTAL OR CONSEQUENTIAL DAMAGES FOR BREACH OF ANY EXPRESS OR IMPLIED WARRANTY OR CONDITION, OR THAT IN ANY WAY ARISE IN RELATION TO THE PRODUCT. NeoQuartzUV-PDS ksk 04122021.docx

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Product Data Sheet

Neocrete V — 70800/70801/70803

48011: Base 48019 : Curing Agent 98010 : Aggregate 66021



Description:	Neocrete V (70800/70801/70803, Hempel 48011) is a heavy duty, water-dispersed polyurethane resin combined with a powder aggregate.
Recommended use:	As a cove base system applied to the following substrates: concrete, polymer reinforced screeds, and properly supported wood floors.
Features:	High abrasion, impact, and chemical resistance. Resistant to thermal shock stresses such as hot water and daily steam cleaning. Resistant to acids, caustics, fuels, oils, and solvents. Steel troweled in one step; requires no primers or sealers. Meets South Coast Air Quality Management District (SCAQMD) VOC requirements.
Service temperatures:	-120°F–210°F/ -84°C–99°C
Certificates/approvals:	LEED: MRc5, IEQc4.2
Availability:	Available in North America. Not included in Group Assortment; other regions must confirm.

Colors and packaging:

70800-02 (4801916650)	Gray	2-gallon can, 141 fluid oz. fill	Note: Two bags of 70803 required for one unit of Neocrete V mix.
70800-05 (4801957930)	Red	2-gallon can, 141 fluid oz. fill	
70800-03 (4801923820)	Tan	2-gallon can, 141 fluid oz. fill	
70801 (9801000000)	NA	1-gallon can, 90 fluid oz. fill	
70803 (6602109990)	NA	42-lb bags	

Some standard colors and may have extended lead times and/or minimum quantities.

Physical constants:

Compressive strength	6,800 psi	ASTM C579
Tensile strength	850 psi	ASTM C307
Flexural strength	1,800 psi	ASTM C580
Modulus of elasticity	167,000 psi	ASTM C580
Shore D	80	ASTM D2240
Adhesion (concrete)	400 psi	ASTM D4541
Water resistance	0.5%	ASTM C413
Density	132 lbs/ft ³	ASTM C905-01
Coefficient of thermal expansion	1.09 x 10 ⁻⁶ in/in°F	ASTM C531
Resistance to fungal growth	No support of growth	ASTM G21
Flammability	Pass	ASTM D635
Weight solids (mixed)	99.97%	ASTM D5201
Volume solids (mixed)	99.94%	ASTM D5201
Flash point	None	ASTM D3278
VOC (mixed)	8.4 g/L	ASTM D5201
Shelf life	1 year	
Typical thickness	1/8"	
Mixed unit coverage at 1/8"	84 linear feet	
Initial join up time at 70°F/21°C per 1/8"	20 minutes	
Cure to foot traffic at 70°F/21°C per 1/8"	4–8 hours	
Cure to full use at 70°F/21°C per 1/8"	10–16 hours	

The above tested results are typical values. Individual lots may vary up to 10% from the typical value. Further technical information can be found at www.neogard.com.

Product Data Sheet

Neocrete V — 70800/70801/70803

48011: Base 48019 : Curing Agent 98010 : Aggregate 66021



Application details:

Version, mixed product

Mixing instructions:

Neocrete V 48011

Base 70800 (48019) : Curing Agent 70801 (98010) : Aggregate 70803 (66021)

Mix 141 fluid ounces of 70800 (contents of one 2-gallon can) with 90 fluid ounces of 70801 (contents of one 1-gallon can) and two bags of 70803 powder. Mix with a variable speed drill utilizing a Jiffy Mixer to suspend any settled pigment and attain a uniform color. Mix only until the material is smooth and has no clumps of aggregate. Over-mixing will shorten pot life, working time, and affect flow/leveling. Mixing time will vary depending on temperature conditions. One mixed unit yields approximately 84 linear feet at 1/8" thickness and 4" in height.

IMPORTANT: Pre-mix 70800 for a minimum of one minute before mixing with 70801 hardener. To avoid color variation from mix to mix, scrape all of the pre-mixed 70800 from the 70800 can into the mixing container.

Application method:

Thinner (max.vol.):

Pot life:

Cleaning of tools:

Margin trowel and cove base trowel

Contact Neogard for thinning instructions

15 minutes at 70°F/21°C

7055 Odorless Reducer (086JB) or Hempel's Thinner 08080 (xylene)

Safety:

Handle with care. Use with adequate ventilation. Before and during use, observe all safety labels on packaging and paint containers, consult Neogard Safety Data Sheets and follow all local or national safety regulations.

Surface preparation:

According to Neogard Guide Specifications.

Application conditions:

Ambient temperature must be between 50°F/10°C and 85°F/29°C, and a minimum of 5°F/3°C above dew point.

Preceding coat:

According to Neogard Guide Specifications.

Subsequent coat:

According to Neogard Guide Specifications.

Storage temperature:

Store in a cool area to ensure full shelf life. Recommended temperature: 70°F/21°C.

Remarks:

For maximum adhesion, correct surface preparation is essential. Regular cleaning of applied surface is recommended in order to maintain slip resistance and aesthetics.

Note:

Neocrete V is for professional use only.

Issued by:

Hempel (USA) – 48011

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GUIDE SPECIFICATIONS

PART 1 GENERAL

1.1 SUMMARY

- A. Provide labor, materials, equipment and supervision necessary to install a heavy duty cementitious polyurethane floor system as outlined in this specification to new or existing concrete surfaces.
- B. The manufacturer's application instructions for each product used are considered part of this specification and should be followed at all times.
- C. Related Sections:
 - 1. Section 03 30 00: Cast-in-Place Concrete
 - 2. Section 07 92 00: Joint Sealants
 - 3. Section 07 95 00: Expansion Control

1.2 SYSTEM DESCRIPTION

- A. Neocrete SL Quartz shall be a complete system of compatible materials manufactured by Neogard to create a seamless flooring surface.
- B. Neocrete SL Quartz shall be designated for application on the specific type of substrate indicated on the drawings.

1.3 SUBMITTALS

- A. Technical Data: Submit manufacturer's product data, Safety Data Sheets (SDS) and installation instructions.
- B. Samples: Submit samples of Neocrete SL Quartz cementitious polyurethane flooring system. Samples shall be construed as examples of finished color and texture of the system only.
- C. Applicator Approval: Submit letter from manufacturer stating applicator is approved to install the Neocrete SL Quartz cementitious polyurethane flooring system.
- D. Warranty: Submit copy of manufacturer's standard sample warranty, identifying the terms and conditions stated in section 1.7 Warranty.

1.4 QUALITY ASSURANCE

- A. Supplier Qualifications: Neocrete SL Quartz, as manufactured by Neogard, is approved for use on this project.
- B. Applicator Qualifications: Applicators shall be approved to install specified system.
- C. Requirement of Regulatory Agencies: Specified materials shall meet existing Federal, State and local VOC regulations.
- D. Field Sample:
 - 1. Install a field sample of at least 100 square feet at the project site or pre-selected area as agreed to by owner's representative, applicator and manufacturer.
 - 2. Apply material in accordance with manufacturer's written application instructions.
 - 3. Field sample will be standard for judging color and texture on remainder of project.
 - 4. Maintain field sample during construction for workmanship comparison.
 - 5. Do not alter, move, or destroy field sample until work is completed and approved by Owner's representative.

1.5 DELIVERY, STORAGE AND HANDLING

- A. Delivery: Materials shall be delivered in original sealed containers, clearly marked with supplier's name, brand name and type of material.

- B. Storage and Handling: Recommended material storage temperature is 75°F/23°C. Handle products to prevent damage to container. All materials shall be stored in compliance with local fire and safety requirements. Do not store at high temperatures or in direct sunlight.

1.6 PROJECT CONDITIONS

- A. Read and follow the SDS and container labels for detailed health and safety information.
- B. Apply materials only when substrate temperature is 50°F/10°C or greater, and to a clean, dry surface. Do not apply if precipitation is imminent, or to a damp, unclean or frosty surface. Maintain a minimum substrate temperature of 50°F/10°C for a minimum of 48 hours before, during and after installation, or until cured.
- C. Apply materials only if ambient temperature between 50°F/10°C and 85°F/29°C. Ambient temperature must be a minimum of 5°F/3°C above dew point. Cure times, flow/leveling, cured physical properties, and overall appearance will be adversely affected if products are applied outside of these temperature ranges.
- D. Due to hydrostatic, capillary and moisture vapor pressure, substrates in contact with ground must have a properly installed, effective vapor barrier. Moisture vapor emission of concrete not to exceed 20 lbs/1,000 sq. ft./24 hours, when tested by the quantitative calcium chloride test method (ASTM F1869). Relative Humidity is not to exceed 80% when tested by In-situ Probe Test (ASTM F2170).
- E. Coordinate flooring work with other trades. Applicator shall have sole right of access to the specified area for the time needed to complete the application and allow the flooring system to cure adequately.
- F. Protect adjacent surfaces from damage resulting from installation of the system. If necessary, mask and/or cover adjacent surfaces, fixtures, equipment, and others by suitable means.
- G. Provide adequate ventilation.
- H. Provide a suitable work station to mix coating materials.
- I. Maintain work area in a neat and orderly condition, removing empty containers, rags and trash daily from the site.

1.7 WARRANTY

- A. Upon request, Neogard shall offer a manufacturer's standard warranty for institutional, commercial, industrial, and high-rise/multi-family residential projects only, upon substantial completion of the application and receipt of a properly executed warranty request form.

PART 2 MATERIALS

2.1 MANUFACTURER

- A. Neogard, a part of Hempel, 2728 Empire Central, Dallas, TX 75235, (214) 353-1600, www.neogard.com.

2.2 MATERIALS

- A. Neocrete SL Quartz (Hempel product numbers in parentheses):
 - 1. Crack and Joint Filler: 70718/70719 (25000) flexible epoxy.
 - 2. Sealant: 70991 (47XJB) or other polyurethane sealant approved by Neogard.
 - 3. Epoxy (100% Solids): 70714/70715 (45060) clear.
 - 4. Neocrete SL mix (48012):
 - a. Resin: 70800 (48019) series, gray, tan, or red in color.
 - b. Hardener: 70801 (98010).
 - c. Powder: Neocrete SL 70804 (6602209990P038, 38-lb bag).
 - 5. Fumed Silica: P1934 (D261).
 - 6. Seal Coats (two options):
 - a. 70817/70818 (57079/95059) clear Chemical Resistant Urethane (CRU).
 - b. 70869/70819 (57039/95055) clear Polyaspartic.

7. Aggregate: Blended Color Quartz.

2.3 MATERIAL PERFORMANCE CRITERIA

- A. Typical physical properties of cured Neocrete SL (70800/70801/70804) polyurethane used on this project are:
 1. Compressive Strength, ASTM C579, 7,700 psi
 2. Tensile Strength, ASTM C307, 712 psi
 3. Flexural Strength, ASTM C580, 2,200 psi
 4. Modulus of Elasticity, ASTM C580, 446,700 psi
 5. Shore D, ASTM D2240, 84
 6. Adhesion to Concrete, ASTM D4541, 400 psi
 7. Water Resistance, ASTM C413, 0.42%
 8. Density, ASTM C905-01, 121 lbs/ft³
 9. Coefficient of Thermal Expansion, ASTM C531, 2.4×10^{-5} in/in/°F
 10. Resistance to Fungal Growth, ASTM G21, No Support of Growth
 11. Flammability, ASTM D635, Pass
- B. Typical physical properties of cured 70817/70818 clear CRU used on this project are:
 1. Tensile Strength, ASTM D2370, 7,500 psi
 2. Elongation, ASTM D2370, 12%
 3. Shore D, ASTM D2240, 70
 4. Water Resistance, ASTM D471, < 1% (7 days)
 5. Taber Abrasion, ASTM D4060, 23 mg (1,000 CS-17)
 6. Anti-Microbial, JIS Z 2801-2010, Pass
- C. Typical physical properties of cured 70869/70819 clear polyaspartic used on this project are:
 1. Tensile Strength, ASTM D2370, 3,362 psi
 2. Elongation, ASTM D412, 63%
 3. Taber Abrasion, ASTM D4060, 55 mg (1,000 CS-17)
 4. Shore D, ASTM D2240, 70
 5. Anti-Microbial, JIS Z 2801-2010, Pass
- D. The above tested results are typical values. Individual lots may vary up to 10% from the typical value. Further technical information can be found at www.neogard.com.

2.4 ACCESSORIES

- A. Miscellaneous materials such as cleaning agents, adhesives, closed cell backer rod, deck drains, and others, shall be compatible with the specified Neocrete SL Quartz system.

2.5 MIXING

- A. Comply with manufacturer's instructions for mixing procedures.

PART 3 EXECUTION

3.1 EXAMINATION

- A. Verify that the work done under other sections meets the following requirements:
 1. That the concrete deck surface is free of ridges and sharp projections, sound and dry.
 2. That the concrete was cured for a minimum of 28 days. (Minimum of 3,500 psi compressive strength). The use of concrete curing agents, if any, shall be of the sodium silicate base only; others require written approval by Neogard.
 3. That damaged areas of the concrete substrate be restored to match adjacent areas. Use 70714/70715 epoxy and oven-dry silica aggregate approved by Neogard for filling and leveling at a ratio of one part epoxy mixed with four parts aggregate by volume.
 4. Due to hydrostatic, capillary and moisture vapor pressure, substrates in contact with ground must have a properly installed, effective vapor barrier. Moisture vapor emission of concrete not to exceed 20

lbs/1,000 sq. ft./24 hours, when tested by the quantitative calcium chloride test method (ASTM F1869). Relative Humidity is not to exceed 80% when tested by In-situ Probe Test (ASTM F2170).

3.2 PREPARATION

- A. Cleaning: Surfaces contaminated with oil or grease shall be vigorously scrubbed with a power broom and a strong non-sudsing detergent. Thoroughly wash, clean, and dry. Areas where oil or other contaminants penetrate deep into the concrete may require removal by mechanical methods. Do not apply materials unless surface is clean and dry.
- B. Shot-Blasting: Required surface preparation method for remedial construction is also the preferred method for new construction. Mechanically prepare surface by shot-blasting to industry standard surface texture (ICRI's CSP3-4) without causing additional surface defects in substrate. Shot-blasting does not remove deep penetrating oils, grease, tar or asphalt stains. Proper cleaning procedures should be followed to ensure proper bonding of the deck coating. Note: If shot-blasting is not practical, contact Neogard Technical Service.
- C. Cracks: After shot-blasting, fill all non-moving cracks with 70714/70715 epoxy, mixed with P1934 fumed silica to form a paste. The mix ratio is one part 70714/70715 epoxy to 3 parts P1934 by volume.
- D. Control and Cold Joints: Fill control and cold joints flush with 70718/70719 flexible epoxy at 3/4" depth. Install backer rod if necessary to limit depth to 3/4".
- E. Expansion and Isolation Joints: Expansion and isolation joints ≤ 1 " in width, shall be sealed with 70991 sealant. Sealant shall be applied to inside of joint only, not applied to floor surface.
- F. Key Cuts: Cut 1/8"–1/4" joints around perimeter of floor, drains, penetrations, doorways, and in field of floor to mechanically anchor floor system.

3.3 APPLICATION

- A. Factors That Affect Dry Film Thickness: Volume solids, thinning, surface profile, application technique and equipment, overspray, squeegee, brush and roller wet out, container residue, spills and other waste are among the many factors that affect the amount of wet coating required to yield proper dry film thickness. To ensure that specified dry film thickness is achieved, use a wet mil gauge to verify actual thickness of wet coating applied, adjusting as needed for those factors which directly affect the dry film build.
- B. Primer: Neocrete SL Quartz does not require a primer.
- C. Cementitious Polyurethane Mix:
 - 1. Pre-mix 70800 for a minimum of one minute before mixing with 70801 hardener.
 - 2. IMPORTANT: To avoid color variation from mix to mix, scrape all of the pre-mixed 70800 from the 70800 can into the mixing container.
 - 3. Mix 141 fluid ounces of 70800 resin (contents of one 2-gallon can) with 90 fluid ounces of 70801 hardener (contents of one 1-gallon can). Slowly add one 38-pound bag of 70804 powder to the resin mix. Mix with a variable speed drill utilizing a Jiffy Mixer to suspend any settled pigment and attain a uniform color. Continue mixing until the powder has been uniformly blended with the resin mix.
 - 4. Pour the cementitious polyurethane mix onto the floor and spread using a gauge rake. Immediately backroll with a spike roller to de-air and level the material.
 - 5. One unit of mixed material covers approximately 32 square feet at 3/16" thickness. Thickness and coverage rate can vary due to finish of substrate.
- D. Aggregate: Immediately broadcast aggregate (blended color quartz), evenly distributed, in wet cementitious polyurethane mix until refusal at a rate of approximately 40 pounds per 100 square feet. Make sure the aggregate is thrown up into the air so it will fall vertically into the wet cementitious polyurethane mix. Maintain a one to two foot wet edge without any aggregate to allow for a smooth transition to the next application of cementitious polyurethane mix.

Guide Specification

Neocrete SL Quartz

Section 09 67 23 Resinous Flooring



- E. Allow to cure 8–12 hours at 70°F/21°C. After curing, remove excess aggregate and lightly sand with a circular floor sander and #50 grit sandpaper to remove any rough spots. All debris from sanding must be removed to provide a clean, moisture-free surface.
- F. Seal Coats:
 - 1. CRU:
 - a. First Seal Coat: Pre-mix 70817 for 3 minutes. Add 70817 to 70818 and immediately mix for 3 minutes. Mix using a slow-speed drill with a Jiffy Mixer paddle. Apply mixed 70817/70818 at a rate of 160 square feet per gallon to achieve 10 mils DFT. Allow to cure 6 to 8 hours at 75°F/23°C.
 - b. Second Seal Coat: Pre-mix 70817 for 3 minutes. Add 70817 to 70818 and immediately mix for 3 minutes. Mix using a slow-speed drill with a Jiffy Mixer paddle. Apply mixed 70817/70818 at a rate of 200 square feet per gallon to achieve 8 mils DFT and allow to cure 8–12 hours at 75°F/23°C before allowing foot traffic.
 - 2. Polyaspartic: (NOTE: Do not split kits. Do not thin.)
 - a. First Seal Coat: Pre-mix 70869 for 3 minutes. Add entire contents of 70819 container to 70869 container and immediately mix for 3 minutes. Mix using a slow-speed drill with a Jiffy Mixer paddle. Apply mixed 70869/70819 at a rate of 160 square feet per gallon to achieve 10 mils DFT. Allow to cure 8–12 hours at 75°F/23°C before allowing foot traffic.
 - b. Second Seal Coat: Pre-mix 70869 for 3 minutes. Add entire contents of 70819 container to 70869 container and immediately mix for 3 minutes. Mix using a slow-speed drill with a Jiffy Mixer paddle. Apply mixed 70869/70819 at a rate of 200 square feet per gallon to achieve 8 mils DFT. Allow to cure 8–12 hours at 75°F/23°C before allowing foot traffic.
- G. Applicator is responsible for applying sufficient coating to the substrate.

3.4 CLEANING

- A. Remove debris resulting from completion of flooring operation from the project site.
- B. Refer to the Preventive Maintenance Manual for Neogard Floor Coating Systems for typical cleaning methods.

3.5 PROTECTION

- A. After completion of application, do not allow heavy traffic on coated surfaces for a period of at least 18 hours at 75°F/23°C.

END OF SECTION

Issued by: Hempel (USA) – Neogard Neocrete SL Quartz

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SAFETY DATA

Safety Data Sheet

NEOGARD

70714 EPOXY RESIN CLEAR



Conforms to ANSI Z400.1-2010 Standard - HCS 2012

Protective Clothing	General Hazard	DOT

SECTION 1: Identification of the substance/mixture and of the company/undertaking

1.1 Product identifier

Product name : NEOGARD
EPOXY RESIN CLEAR

Product identity : 4506900000, 70714

Product type : epoxy paint (base for multi-component product)

1.2 Relevant identified uses of the substance or mixture and uses advised against

Field of application : buildings and metal industry.

Ready-for-use mixture : Mixing Ratio:
45060 = 45069 2 Vol. / 95073 1 Vol.
45061 = 45069 2 Vol. / 95074 1 Vol.
45062 = 45069 2 Vol. / 95075 1 Vol.

Identified uses : Industrial/Professional use

TSCA : Unless otherwise stated. All components are listed or exempted.

1.3 Details of the supplier of the safety data sheet

Company details : NEOGARD, a Division of Hempel (USA), Inc.
2728 Empire Central
Dallas, TX 75235
Phone number: 1-214-353-1600
E-mail: hempel@hempel.com

1.4 Emergency telephone number (with hours of operation)

For Transportation Emergencies : CHEMTREC: **1-800-424-9300** (Toll-free in the U.S., Canada and the U.S. Virgin Islands) **703-527-3887**
(24 hours)
For calls originating elsewhere (Collect calls are accepted). Contract number: CCN10384
To preserve the effectiveness of arrangements for providing accurate and timely emergency response information, the basic identifying information (shipper name or contract number) must be included on shipping papers.
If the purchaser of this product is going to be shipping this product to other locations, the purchaser must arrange for its own Emergency Information Provider to respond to transport incidents. Hempel's 24 hour response contract does not cover non-Hempel shipments.

For all other information : In USA toll free calling available: 1-800- 678-6641 or (936)-523-6000
(8 AM - 5 PM CST) See Section 4 of the safety data sheet (first aid measures).

SECTION 2: Hazards identification

2.1 Classification of the substance or mixture

OSHA/HCS status : This material is considered hazardous by the OSHA Hazard Communication Standard (29 CFR 1910.1200).

GHS Classification : SKIN IRRITATION - Category 2
EYE IRRITATION - Category 2A
SKIN SENSITIZATION - Category 1
TOXIC TO REPRODUCTION (Fertility) - Category 2
TOXIC TO REPRODUCTION (Unborn child) - Category 2

2.2 Label elements

Safety Data Sheet

NEOGARD

70714 EPOXY RESIN CLEAR



SECTION 2: Hazards identification

Hazard pictograms :



Signal word :

Warning

Hazard statements :

H319 - Causes serious eye irritation.
H315 - Causes skin irritation.
H317 - May cause an allergic skin reaction.
H361 - Suspected of damaging fertility or the unborn child.

Precautionary statements :

Prevention :

Obtain special instructions before use. Do not handle until all safety precautions have been read and understood. Wear protective gloves. Wear eye or face protection. Wear protective clothing. Avoid breathing vapor. Wash hands thoroughly after handling. Contaminated work clothing must not be allowed out of the workplace.

Response :

IF exposed or concerned: Get medical attention. IF ON SKIN: Wash with plenty of soap and water. Wash contaminated clothing before reuse. If skin irritation or rash occurs: Get medical attention. IF IN EYES: Rinse cautiously with water for several minutes. Remove contact lenses, if present and easy to do. Continue rinsing. If eye irritation persists: Get medical attention.

Storage :

Store locked up.

Disposal :

Dispose of contents and container in accordance with all local, regional, national and international regulations.

Supplemental label elements :

None known.

2.3 Other hazards

Hazards not otherwise classified : None known.

SECTION 3: Composition/information on ingredients

Product definition :

Mixture

Physical state :

Liquid.

Product/ingredient name	Identifiers	%	GHS Classification
bisphenol A-(epichlorhydrin) epoxy resin MW =< 700	25068-38-6	≥50 - ≤75	SKIN IRRITATION - Category 2 EYE IRRITATION - Category 2A SKIN SENSITIZATION - Category 1
(C12-C14) Alkylglycidylether	68609-97-2	≥10 - ≤25	SKIN IRRITATION - Category 2 SKIN SENSITIZATION - Category 1
middle molecular epoxy resin MMW 700-1200	25068-38-6	≥10 - ≤25	SKIN IRRITATION - Category 2 EYE IRRITATION - Category 2A SKIN SENSITIZATION - Category 1
4-nonylphenol, branched	84852-15-3	<1	ACUTE TOXICITY (oral) - Category 4 SKIN CORROSION - Category 1B SERIOUS EYE DAMAGE - Category 1 TOXIC TO REPRODUCTION (Fertility) (oral) - Category 2 TOXIC TO REPRODUCTION (Unborn child) (oral) - Category 2

Any concentration shown as a range is to protect confidentiality or is due to batch variation.

There are no ingredients present which, within the current knowledge of the supplier and in the concentrations applicable, are classified as hazardous to health or the environment and hence require reporting in this section.

SECTION 4: First aid measures

4.1 Description of first aid measures

General :	In all cases of doubt, or when symptoms persist, seek medical attention. Never give anything by mouth to an unconscious person. If breathing is irregular, drowsiness, loss of consciousness or cramps: Call 911 and give immediate treatment (first aid).
Eye contact :	Check for and remove any contact lenses. Immediately flush eyes with plenty of water for at least 15 minutes, occasionally lifting the upper and lower eyelids. Seek immediate medical attention.
Inhalation :	Remove to fresh air. Keep person warm and at rest. If unconscious, place in recovery position and seek medical advice.
Skin contact :	Remove contaminated clothing and shoes. Wash skin thoroughly with soap and water or use recognized skin cleanser. Do NOT use solvents or thinners.
Ingestion :	If swallowed, seek medical advice immediately and show this container or label. Keep person warm and at rest. Do not induce vomiting unless directed to do so by medical personnel. Lower the head so that vomit will not re-enter the mouth and throat.
Protection of first-aiders :	No action shall be taken involving any personal risk or without suitable training. It may be dangerous to the person providing aid to give mouth-to-mouth resuscitation. Wash contaminated clothing thoroughly with water before removing it, or wear gloves.

4.2 Most important symptoms and effects, both acute and delayed

Potential acute health effects

Eye contact :	Causes serious eye irritation.
Inhalation :	No known significant effects or critical hazards.
Skin contact :	Causes skin irritation. May cause an allergic skin reaction.
Ingestion :	No known significant effects or critical hazards.

Over-exposure signs/symptoms

Eye contact :	Adverse symptoms may include the following: pain or irritation watering redness
Inhalation :	No specific data.
Skin contact :	Adverse symptoms may include the following: irritation redness
Ingestion :	No specific data.

4.3 Indication of any immediate medical attention and special treatment needed

Notes to physician :	Not applicable.
Specific treatments :	No specific treatment.

SECTION 5: Firefighting measures

5.1 Extinguishing media

Extinguishing media :	Recommended: alcohol resistant foam, CO ₂ , powders, water spray. Not to be used: waterjet.
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5.2 Special hazards arising from the substance or mixture

Hazards from the substance or mixture :	In a fire or if heated, a pressure increase will occur and the container may burst. This material is toxic to aquatic life with long lasting effects. Fire water contaminated with this material must be contained and prevented from being discharged to any waterway, sewer or drain.
Hazardous combustion products :	Decomposition products may include the following materials: carbon oxides halogenated compounds

5.3 Advice for firefighters

SECTION 5: Firefighting measures

Promptly isolate the scene by removing all persons from the vicinity of the incident if there is a fire. No action shall be taken involving any personal risk or without suitable training. Fire will produce dense black smoke. Exposure to decomposition products may cause a health hazard. Cool closed containers exposed to fire with water. Do not release runoff from fire to drains or watercourses. Fire-fighters should wear appropriate protective equipment and self-contained breathing apparatus (SCBA) with a full face-piece operated in positive pressure mode.

SECTION 6: Accidental release measures

6.1 Personal precautions, protective equipment and emergency procedures

Refer to protective measures listed in sections 7 and 8. No action shall be taken involving any personal risk or without suitable training.

6.2 Environmental precautions

Avoid dispersal of spilled material and runoff and contact with soil, waterways, drains and sewers. Inform the relevant authorities if the product has caused environmental pollution (sewers, waterways, soil or air). Water polluting material. May be harmful to the environment if released in large quantities.

6.3 Methods and materials for containment and cleaning up

Stop leak if without risk. Move containers from spill area. Approach release from upwind. Prevent entry into sewers, water courses, basements or confined areas. Wash spillages into an effluent treatment plant or proceed as follows. Contain and collect spillage with non-combustible, absorbent material e.g. sand, earth, vermiculite or diatomaceous earth and place in container for disposal according to local regulations (see Section 13). Contaminated absorbent material may pose the same hazard as the spilled product.

6.4 Reference to other sections

See Section 1 for emergency contact information.

See Section 8 for information on appropriate personal protective equipment.

See Section 13 for additional waste treatment information.

SECTION 7: Handling and storage

7.1 Precautions for safe handling

Contains epoxy constituents. Avoid all possible skin contact with epoxy and amine containing products, they may cause allergic reactions. Avoid inhalation of vapour, dust and spray mist. Avoid contact with skin and eyes. Eating, drinking and smoking should be prohibited in area where this material is handled, stored and processed. Appropriate personal protective equipment: see Section 8. Always keep in containers made from the same material as the original one.

7.2 Conditions for safe storage, including any incompatibilities

Store in accordance with local regulations. Store in a cool, well-ventilated area away from incompatible materials and ignition sources. Keep out of the reach of children. Keep away from: Oxidizing agents, strong alkalis, strong acids. No smoking. Prevent unauthorized access. Containers that are opened must be carefully resealed and kept upright to prevent leakage.

7.3 Specific end use(s)

See separate Product Data Sheet for recommendations or industrial sector specific solutions.

This product may be applied using several application techniques and methods of handling may be different for each. Application techniques include [but are not limited to] brushing, rolling, and spray application [conventional, HPLV, airless, pleural component or aerosol can]. Avoid the breathing of vapors and, if spraying, do not breath spray mist or aerosols.

SECTION 8: Exposure controls/personal protection

8.1 Control parameters

Product/ingredient name	Exposure limit values
No exposure limit value known.	

Recommended monitoring procedures

If this product contains ingredients with exposure limits, personal, workplace atmosphere or biological monitoring may be required to determine the effectiveness of the ventilation or other control measures and/or the necessity to use respiratory protective equipment. Reference should be made to appropriate monitoring standards. Reference to national guidance documents for methods for the determination of hazardous substances will also be required.

SECTION 8: Exposure controls/personal protection

8.2 Exposure controls

Appropriate engineering controls

Provide local exhaust and general ventilation systems to maintain airborne concentrations below OSHA, ACGIH, and manufacturer recommended exposure limits. Local exhaust ventilation is preferred because it prevents contaminant dispersion into work areas by controlling it at its source. Use local and general exhaust ventilation to effectively remove and prevent buildup of mists/vapors/fumes generated from the handling of this product.

Note: Local exhaust ventilation is designed to capture an emitted contaminant at or near its source, before the contaminant has a chance to disperse into the workplace air. General exhaust ventilation, also called dilution ventilation, is different from local exhaust ventilation because instead of capturing emissions at their source and removing them from the air, general exhaust ventilation allows the contaminant to be emitted into the workplace air and then dilutes the concentration of the contaminant to an acceptable level (e.g., to the PEL or below).

Individual protection measures

General :	Gloves must be worn for all work that may result in soiling. Apron/coveralls/protective clothing must be worn when soiling is so great that regular work clothes do not adequately protect skin against contact with the product. Safety eyewear should be used when there is a likelihood of exposure.
Hygiene measures :	Wash hands, forearms, and face thoroughly after handling compounds and before eating, smoking, using lavatory, and at the end of day.
Eye/face protection :	Safety eyewear complying with an approved standard should be used when a risk assessment indicates this is necessary to avoid exposure to liquid splashes, mists, gases or dusts. If contact is possible, the following protection should be worn, unless the assessment indicates a higher degree of protection: chemical splash goggles.
Hand protection :	Wear chemical-resistant gloves in combination with 'basic' employee training. The quality of the chemical-resistant protective gloves must be chosen as a function of the specific workplace concentrations and quantity of hazardous substances. Since the actual work situation is unknown. Supplier of gloves should be contacted in order to find the appropriate type. Below listed glove(s) should be regarded as generic advice: Recommended: Silver Shield / Barrier / 4H gloves, nitrile rubber, butyl rubber, Viton® Short term exposure: neoprene rubber, natural rubber (latex), polyvinyl alcohol (PVA), polyvinyl chloride (PVC)
Body protection :	Personal protective equipment for the body should be selected based on the task being performed and the risks involved handling this product. Wear suitable protective clothing. Always wear protective clothing when spraying.
Respiratory protection :	Wear appropriate respirator when ventilation is inadequate. Be sure to use approved/certified respirator or equivalent. It is not possible to specify precise filter type, since the actual work situation is unknown. Supplier of respirators should be contacted in order to find the appropriate filter.
Protective clothing (pictograms) :	

Note: Application of paint products by spraying requires additional safety precautions: Full body suit, Full face respirator with air supplied.

Environmental exposure controls

Emissions from ventilation or work process equipment should be checked to ensure they comply with the requirements of environmental protection legislation. In some cases, fume scrubbers, filters or engineering modifications to the process equipment will be necessary to reduce emissions to acceptable levels.

SECTION 9: Physical and chemical properties

9.1 Information on basic physical and chemical properties

Physical state :	Liquid.
Color :	Clear
Odor :	Amine-like.
pH :	Testing not relevant or not possible due to nature of the product.
Melting point/freezing point :	-16°C This is based on data for the following ingredient: bisphenol A-(epichlorhydrin) epoxy resin MW = < 700
Boiling point/boiling range :	Testing not relevant or not possible due to nature of the product.
Flash point :	Closed cup: 165°C (329°F)



SECTION 9: Physical and chemical properties

Evaporation rate :	Testing not relevant or not possible due to nature of the product.
Flammability :	Not available.
Upper/lower flammability or explosive limits :	No specific data.
Vapor pressure :	0 kPa This is based on data for the following ingredient: bisphenol A-(epichlorhydrin) epoxy resin MW = < 700
Vapor density :	Testing not relevant or not possible due to nature of the product.
Relative density :	1.105 g/cm ³
Solubility(ies) :	
Partition coefficient (LogKow) :	Testing not relevant or not possible due to nature of the product.
Auto-ignition temperature :	Testing not relevant or not possible due to nature of the product.
Decomposition temperature :	Testing not relevant or not possible due to nature of the product.
Viscosity :	Testing not relevant or not possible due to nature of the product.
Explosive properties :	Not available.
Oxidizing properties :	Testing not relevant or not possible due to nature of the product.

9.2 Other information

Solvent(s) % by weight (Included exempt solvent(s)):	0.73 % (w/w)
Water % by weight :	Weighted average: 0 %
VOC content (Coatings) :	0.0676 lbs/gal (8.1 g/l)
VOC content (Regulatory) :	0.0676 lbs/gal (8.1 g/l)
TOC Content (Volatile) :	Weighted average: 7 g/l
Solvent Gas :	Weighted average: 0.001 m ³ /l

SECTION 10: Stability and reactivity

10.1 Reactivity

No specific test data related to reactivity available for this product or its ingredients.

10.2 Chemical stability

The product is stable.

10.3 Possibility of hazardous reactions

Under normal conditions of storage and use, hazardous reactions will not occur.

10.4 Conditions to avoid

No specific data.

10.5 Incompatible materials

10.6 Hazardous decomposition products

When exposed to high temperatures (i.e. in case of fire) harmful decomposition products may be formed:

Decomposition products may include the following materials: carbon oxides halogenated compounds



SECTION 11: Toxicological information

11.1 Information on toxicological effects

Epoxy and amine containing products can cause skin disorders such as allergic eczema. The allergy may arise after only a short exposure period.

Acute toxicity

Product/ingredient name	Result	Species	Dose	Exposure
bisphenol A-(epichlorhydrin) epoxy resin MW =< 700	LD50 Dermal	Rabbit	>2000 mg/kg	-
(C12-C14) Alkylglycidylether	LD50 Dermal	Rat	>2000 mg/kg	-
	LD50 Oral	Rat	>2000 mg/kg	-
	LD50 Dermal	Rat	>4500 mg/kg	-
	LD50 Oral	Rat	>5000 mg/kg	-
middle molecular epoxy resin MMW 700-1200	LD50 Dermal	Rat	>2000 mg/kg	-
4-nonylphenol, branched	LD50 Dermal	Rabbit	>2000 mg/kg	-
	LD50 Oral	Rat	1246 mg/kg	-

Acute toxicity estimates

Route	ATE value
No known significant effects or critical hazards.	

Irritation/Corrosion

Product/ingredient name	Result	Species	Score	Exposure
bisphenol A-(epichlorhydrin) epoxy resin MW =< 700	Eyes - Mild irritant	Rabbit	-	-
(C12-C14) Alkylglycidylether	Skin - Mild irritant	Rabbit	-	-
	Skin - Moderate irritant	Rabbit	-	24 hours 500 microliters
	Eyes - Mild irritant	Rabbit	-	-
4-nonylphenol, branched	Eyes - Severe irritant	Rabbit	-	100 milligrams
	Skin - Severe irritant	Rabbit	-	24 hours 500 milligrams

Sensitizer

Product/ingredient name	Route of exposure	Species	Result
bisphenol A-(epichlorhydrin) epoxy resin MW =< 700	skin	Guinea pig	Sensitizing
(C12-C14) Alkylglycidylether	skin	Guinea pig	Sensitizing
middle molecular epoxy resin MMW 700-1200	skin	Guinea pig	Sensitizing

Information on the likely routes of exposure

Routes of entry anticipated: Oral, Dermal, Inhalation.

Potential chronic health effects

Sensitization : Contains bisphenol A-(epichlorhydrin) epoxy resin MW =< 700, (C12-C14) Alkylglycidylether, middle molecular epoxy resin MMW 700-1200. May produce an allergic reaction.

Other information : No additional known significant effects or critical hazards.

SECTION 12: Ecological information

12.1 Toxicity

Do not allow to enter drains or watercourses. Toxic to aquatic life with long lasting effects.

When spilled, this product may act as an oil, causing a film, sheen, emulsion, or sludge at or beneath the surface of a body of water. Oils of any kind can cause: (a) drowning of waterfowl due to lack of buoyancy, loss of insulating capacity of feathers, starvation and vulnerability to predators due to lack of mobility; (b) lethal effect on fish by coating gill surfaces, preventing respiration; (c) potential fish kills resulting from alteration in biochemical oxygen demand; (d) asphyxiation of benthic life forms when floating masses become engaged with surface debris and settle on the bottom; and (e) adverse aesthetic effects of fouled shoreline and beaches.



SECTION 12: Ecological information

Product/ingredient name	Result	Species	Exposure
bisphenol A-(epichlorhydrin) epoxy resin MW =< 700	Acute EC50 >11 mg/l	Algae	72 hours
	Acute EC50 1.4 - 1.7 mg/l	Daphnia - Daphnia magna	48 hours
	Acute LC50 3.1 mg/l	Fish - fathead minnow (Pimephales promelas)	96 hours
(C12-C14) Alkylglycidylether	Acute IC50 843.75 mg/l	Algae	72 hours
	Acute LC50 5000 mg/l	Fish	96 hours
middle molecular epoxy resin MMW 700-1200	Acute EC50 >100 mg/l	Daphnia	48 hours
	Acute LC50 >100 mg/l	Fish	96 hours
4-nonylphenol, branched	Acute EC50 0.03 mg/l Marine water	Algae - Skeletonema costatum	72 hours
	Acute EC50 137 µg/l Marine water	Crustaceans - Eohaustorius estuarii - Adult	48 hours
	Acute EC50 0.085 mg/l	Daphnia	48 hours
	Acute LC50 0.128 mg/l	Fish	96 hours
	Chronic NOEC 23 µg/l Fresh water	Fish - Pimephales promelas - Embryo	33 days

12.2 Persistence and degradability

Product/ingredient name	Test	Result	Dose	Inoculum
bisphenol A-(epichlorhydrin) epoxy resin MW =< 700	OECD 302B Inherent Biodegradability: Zahn-Wellens/EMPA Test	12 % - Not readily - 28 days	-	-
(C12-C14) Alkylglycidylether	-	87 % - Readily - 28 days	-	-

Product/ingredient name	Aquatic half-life	Photolysis	Biodegradability
bisphenol A-(epichlorhydrin) epoxy resin MW =< 700	-	-	Not readily
(C12-C14) Alkylglycidylether	-	-	Readily

12.3 Bioaccumulative potential

Product/ingredient name	LogP _{ow}	BCF	Potential
bisphenol A-(epichlorhydrin) epoxy resin MW =< 700	2.64 - 3.78	31	low
(C12-C14) Alkylglycidylether	3.77	160 - 263	low
middle molecular epoxy resin MMW 700-1200	2.64 - 3.78	31	low
4-nonylphenol, branched	5.4	740	high

12.4 Mobility in soil

Soil/water partition coefficient (K_{oc}) : No known data available in our database.

Mobility : No known data available in our database.

12.5 Other adverse effects

No known significant effects or critical hazards.

SECTION 13: Disposal considerations

13.1 Waste treatment methods

Disposal should be in accordance with applicable regional, national and local laws and regulations. Local regulations may be more stringent than regional or national requirements.

The information presented below only applies to the material as supplied. The identification based on characteristic(s) or listing may not apply if the material has been used or otherwise contaminated. It is the responsibility of the waste generator to determine the toxicity and physical properties of the material generated to determine the proper waste identification and disposal methods in compliance with applicable regulations.

Refer to Section 7 and Section 8 for additional handling information and protection of employees.



SECTION 13: Disposal considerations

The generation of waste should be avoided or minimized wherever possible. Disposal of this product, solutions and any by-products should at all times comply with the requirements of environmental protection and waste disposal legislation and any regional local authority requirements. Dispose of surplus and non-recyclable products via a licensed waste disposal contractor. Waste should not be disposed of untreated to the sewer unless fully compliant with the requirements of all authorities with jurisdiction. Waste packaging should be recycled. Incineration or landfill should only be considered when recycling is not feasible. This material and its container must be disposed of in a safe way. Care should be taken when handling emptied containers that have not been cleaned or rinsed out. Empty containers or liners may retain some product residues. Avoid dispersal of spilled material and runoff and contact with soil, waterways, drains and sewers.

SECTION 14: Transport information

Transport may take place according to national regulation or DOT for transport by road and by train, IMDG for transport by sea, IATA for Air shipment. Refer to specific Dangerous Goods Transport requirements under 49CFR, ICAO and IATA.

	14.1 UN no.	14.2 Proper shipping name	14.3 Transport hazard class(es)	14.4 PG*	14.5 Env*	Additional information
DOT Code	UN3082	ENVIRONMENTALLY HAZARDOUS SUBSTANCE, LIQUID, N.O.S.. (bisphenol A-(epichlorhydrin) epoxy resin MW =< 700)	9 -  	III	Yes.	The marine pollutant mark is not required when transported on inland waterways in sizes of ≤5 L or ≤5 kg or by road, rail, or inland air in non-bulk sizes. <u>Reportable quantity</u> (chlorine) 7635.2 lbs / 3466.4 kg [828.7 gal / 3137 L] Package sizes shipped in quantities less than the product reportable quantity are not subject to the RQ (reportable quantity) transportation requirements.
TDG Code	UN3082	ENVIRONMENTALLY HAZARDOUS SUBSTANCE, LIQUID, N.O.S.. (bisphenol A-(epichlorhydrin) epoxy resin MW =< 700)	9 -  	III	Yes.	Product classified as per the following sections of the Transportation of Dangerous Goods Regulations: 2.43-2.45 (Class 9), 2.7 (Marine pollutant mark). Non-bulk packages of this product are not regulated as dangerous goods when transported by road or rail.
SCT Code	UN3082	ENVIRONMENTALLY HAZARDOUS SUBSTANCE, LIQUID, N.O.S.. (bisphenol A-(epichlorhydrin) epoxy resin MW =< 700)	9 -  	III	Yes.	The environmentally hazardous substance mark is not required when transported in sizes of ≤5 L or ≤5 kg.
IMDG Code	UN3082	ENVIRONMENTALLY HAZARDOUS SUBSTANCE, LIQUID, N.O.S.. (bisphenol A-(epichlorhydrin) epoxy resin MW =< 700)	9 -  	III	Yes.	This product is not regulated as a dangerous good when transported in sizes of ≤5 L or ≤5 kg, provided the packagings meet the general provisions of 4.1.1.1, 4.1.1.2 and 4.1.1.4 to 4.1.1.8. <u>Emergency schedules</u> F-A, S-F
IATA Code	UN3082	ENVIRONMENTALLY HAZARDOUS SUBSTANCE, LIQUID, N.O.S.. (bisphenol A-(epichlorhydrin) epoxy resin MW =< 700)	9 -  	III	Yes.	This product is not regulated as a dangerous good when transported in sizes of ≤5 L or ≤5 kg, provided the packagings meet the general provisions of 5.0.2.4.1, 5.0.2.6.1.1 and 5.0.2.8.

Code : Classification

PG* : Packing group

Env.* : Environmental hazards

14.6 Special precautions for user

Transport within user's premises: always transport in closed containers that are upright and secure. Ensure that persons transporting the product know what to do in the event of an accident or spillage.

14.7 Transport in bulk according to Annex II of MARPOL and the IBC Code

Not applicable.

Safety Data Sheet

NEOGARD

70714 EPOXY RESIN CLEAR



SECTION 15: Regulatory information

15.1 Safety, health and environmental regulations/legislation specific for the substance or mixture

U.S. Federal regulations :

All components are listed or exempted.

TSCA 4(a) final test rules: 4-nonylphenol, branched

TSCA 5(a)2 final significant new use rules: branched 2-nonyl phenol; 4-nonylphenol, branched

TSCA 8(a) PAIR: branched 2-nonyl phenol; 4-nonylphenol, branched

TSCA 8(a) CDR Exempt/Partial exemption: Not determined

United States inventory (TSCA 8b): All components are listed or exempted.

Clean Water Act (CWA) 307: phenol

Clean Water Act (CWA) 311: epichlorohydrin; phenol

Clean Air Act Section 112(b) Hazardous Air Pollutants (HAPs) : Listed

Product/ingredient name	CAS number	Concentration
epichlorohydrin	106-89-8	0.018068
phenol	108-95-2	0.001236

Clean Air Act Section 602 Class I Substances : Not listed

Clean Air Act Section 602 Class II Substances : Not listed

DEA List I Chemicals (Precursor Chemicals) : Not listed

DEA List II Chemicals (Essential Chemicals) : Not listed

SARA 302/304 - SARA 311/312:

SARA 302/304: chlorine; epichlorohydrin; phenol

SARA 311/312 Hazards identification: Immediate (acute) health hazard

Product/ingredient name	%	Fire hazard	Sudden release of pressure	Reactive	Immediate (acute) health hazard	Delayed (chronic) health hazard
bisphenol A-(epichlorohydrin) epoxy resin MW =< 700	50 - 75	No.	No.	No.	Yes.	No.
(C12-C14) Alkylglycidylether	10 - 25	No.	No.	No.	Yes.	No.
middle molecular epoxy resin MMW 700-1200	10 - 25	No.	No.	No.	Yes.	No.
4-nonylphenol, branched	0.5 - 1	No.	No.	No.	Yes.	Yes.

SARA 313 :

SARA 313 notifications must not be detached from the MSDS and any copying and redistribution of the MSDS shall include copying and redistribution of the notice attached to copies of the MSDS subsequently redistributed.

Supplier notification :

Product/ingredient name	CAS number	Concentration
middle molecular epoxy resin MMW 700-1200	25068-38-6	10 - 20

State regulations :

Connecticut Carcinogen Reporting: None of the components are listed.

Connecticut Hazardous Material Survey: None of the components are listed.

Florida substances: None of the components are listed.

Illinois Chemical Safety Act: None of the components are listed.

Illinois Toxic Substances Disclosure to Employee Act: None of the components are listed.

Louisiana Reporting: None of the components are listed.

Louisiana Spill: None of the components are listed.

Massachusetts Spill: None of the components are listed.

Massachusetts Substances: None of the components are listed.

Michigan Critical Material: None of the components are listed.

Minnesota Hazardous Substances: None of the components are listed.

New Jersey Hazardous Substances: None of the components are listed.

New Jersey Spill: None of the components are listed.

New Jersey Toxic Catastrophe Prevention Act: None of the components are listed.

New York Acutely Hazardous Substances: None of the components are listed.

New York Toxic Chemical Release Reporting: None of the components are listed.

Pennsylvania RTK Hazardous Substances: None of the components are listed.

Rhode Island Hazardous Substances: None of the components are listed.

California Prop. 65 PFF :

WARNING: This product contains less than 0.1% of a chemical known to the State of California to cause cancer. **WARNING:** This product contains less than 1% of a chemical known to the State of California to cause birth defects or other reproductive harm.

Product/ingredient name	Cancer	Reproductive	No significant risk level	Maximum acceptable dosage level
epichlorohydrin	Yes.	Yes.	Yes.	

Safety Data Sheet

NEOGARD

70714 EPOXY RESIN CLEAR



SECTION 16: Other information

Remarks : Note: In USA, consult Code of Federal Regulations, Title 29, Labor, Parts 1910 and 1915 concerning occupational safety and health standards and regulations, as well as any other applicable Federal, State or local regulations that apply to safe practices in coating operations.

Warning! If you scrape, sand, or remove old paint, you may release lead dust. LEAD is TOXIC.

Validation : Validated by US - HSE Products Coordinator on 30 January 2018

GHS Classification

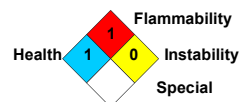
Procedure used to derive the classification.

Classification	Justification
SKIN IRRITATION - Category 2	Calculation method
EYE IRRITATION - Category 2A	Calculation method
SKIN SENSITIZATION - Category 1	Calculation method
TOXIC TO REPRODUCTION (Fertility) - Category 2	Calculation method
TOXIC TO REPRODUCTION (Unborn child) - Category 2	Calculation method

Hazardous Material Information System (U.S.A.)

Health	* 2
Fire hazard	1
Physical hazards	0
Personal protection	X

National Fire Protection Association (U.S.A.)



Personal Protective Equipment (PPE) shown in this section is a suggestion. Since conditions vary from one work location to another consult the facility safety & health program. Customer or end user is responsible to evaluate worker exposure conditions at the site of application and determine the appropriate PPE suitable for workers at that particular facility or location.

Abbreviations and acronyms :

ANSI = American National Standards Institute

HCS = Hazardous Communication System

TSCA = Toxic Substances Control Act

CFR = Code of federal Regulations

GHS = Globally Harmonized System of Classification and Labelling of Chemicals

OSHA = United States Occupational Health and Safety Administration

NIOSH = National Institute for Occupational Safety and Health

ACGIH = American Conference of Industrial Hygienists

IARC = International Agency for Research on Cancer.

NTP = National Toxicology Program

ATE = Acute Toxicity Estimate

OECD = Organisation for Economic Co-operation and Development

BCF = Bioconcentration Factor

DOT = United States Department of Transportation

ERG = Emergency Response Guide

TDG = Transport of Dangerous Goods, Canada

SCT = Transportation & Communications Ministry, Mexico

IMDG = International Maritime Dangerous Goods

IATA = International Air Transport Association

SARA = Superfund Amendments Reauthorization Act

EPCRA = Emergency Planning and Community Right to Know Act

Notice to reader

Indicates information that has changed from previously issued version.

To the best of our knowledge, the information contained herein is accurate. However, neither the above named supplier nor any of its subsidiaries assumes any liability whatsoever for the accuracy or completeness of the information contained herein. Final determination of suitability of any material is the sole responsibility of the user. All materials may present unknown hazards and should be used with caution. Although certain hazards are described herein, we cannot guarantee that these are the only hazards that exist.

Safety Data Sheet

NEOGARD EPOXY RESIN CURING AGENT 70715

Conforms to ANSI Z400.1-2010 Standard - HCS 2012

Protective Clothing	General Hazard	DOT
Consult your supervisor or S.O.P. for special handling	  	

SECTION 1: Identification of the substance/mixture and of the company/undertaking

1.1 Product identifier

Product name : NEOGARD EPOXY RESIN CURING AGENT
Product identity : 9507300000, 70715
Product type : Catalyst for epoxy resins.

1.2 Relevant identified uses of the substance or mixture and uses advised against

Field of application : used only as part of two- or multi component products.
Ready-for-use mixture : Used for: 17050, 251J1, 45060
Identified uses : Industrial/Professional use
TSCA : **Unless otherwise stated. All components are listed or exempted.**

1.3 Details of the supplier of the safety data sheet

Company details : NEOGARD, a Division of Hempel (USA), Inc.
2728 Empire Central
Dallas, TX 75235
Phone number: 1-214-353-1600
E-mail: hempel@hempel.com

1.4 Emergency telephone number (with hours of operation)

For Transportation Emergencies : CHEMTREC: **1-800-424-9300** (Toll-free in the U.S., Canada and the U.S. Virgin Islands) **703-527-3887**
(24 hours) For calls originating elsewhere (Collect calls are accepted). Contract number: CCN10384
To preserve the effectiveness of arrangements for providing accurate and timely emergency response information, the basic identifying information (shipper name or contract number) must be included on shipping papers.
If the purchaser of this product is going to be shipping this product to other locations, the purchaser must arrange for its own Emergency Information Provider to respond to transport incidents. Hempel's 24 hour response contract does not cover non-Hempel shipments.
For all other information : In USA toll free calling available: 1-800- 678-6641 or (936)-523-6000
(8 AM - 5 PM CST) See Section 4 of the safety data sheet (first aid measures).

SECTION 2: Hazards identification

2.1 Classification of the substance or mixture

OSHA/HCS status : This material is considered hazardous by the OSHA Hazard Communication Standard (29 CFR 1910.1200).
GHS Classification : FLAMMABLE LIQUIDS - Category 4
ACUTE TOXICITY (oral) - Category 4
ACUTE TOXICITY (inhalation) - Category 4
SKIN CORROSION - Category 1B
SERIOUS EYE DAMAGE - Category 1
SKIN SENSITIZATION - Category 1
TOXIC TO REPRODUCTION - Category 1B

2.2 Label elements

Hazard pictograms :



Safety Data Sheet

NEOGARD EPOXY RESIN CURING AGENT

70715



SECTION 2: Hazards identification

Signal word :	Danger
Hazard statements :	H227 - Combustible liquid. H302 + H332 - Harmful if swallowed or if inhaled. H314 - Causes severe skin burns and eye damage. H317 - May cause an allergic skin reaction. H360 - May damage fertility or the unborn child.
Precautionary statements :	
Prevention :	Obtain special instructions before use. Do not handle until all safety precautions have been read and understood. Wear protective gloves, protective clothing and eye or face protection. Keep away from flames and hot surfaces. No smoking. Use only outdoors or in a well-ventilated area. Avoid breathing vapor. Do not eat, drink or smoke when using this product. Wash thoroughly after handling. Contaminated work clothing must not be allowed out of the workplace.
Response :	IF exposed or concerned: Get medical advice or attention. IF INHALED: Remove person to fresh air and keep comfortable for breathing. Immediately call a POISON CENTER or doctor. Call a POISON CENTER or doctor if you feel unwell. IF SWALLOWED: Immediately call a POISON CENTER or doctor. Rinse mouth. Do NOT induce vomiting. IF ON SKIN (or hair): Take off immediately all contaminated clothing. Rinse skin with water. Immediately call a POISON CENTER or doctor. IF ON SKIN: Wash with plenty of water. If skin irritation or rash occurs: Get medical advice or attention. IF IN EYES: Rinse cautiously with water for several minutes. Remove contact lenses, if present and easy to do. Continue rinsing. Immediately call a POISON CENTER or doctor.
Storage :	Store locked up. Store in a well-ventilated place. Keep cool.
Disposal :	Dispose of contents and container in accordance with all local, regional, national and international regulations.
Supplemental label elements :	None known.

2.3 Other hazards

Hazards not otherwise classified : None known.

SECTION 3: Composition/information on ingredients

Product definition :	Mixture
Physical state :	Liquid.

Product/ingredient name	Identifiers	%	GHS Classification
benzyl alcohol	100-51-6	≥25 - ≤50	ACUTE TOXICITY (oral) - Category 4 ACUTE TOXICITY (inhalation) - Category 4 EYE IRRITATION - Category 2A
3-aminomethyl- 3,5,5-trimethylcyclohexylamine	2855-13-2	≥10 - ≤25	ACUTE TOXICITY (oral) - Category 4 ACUTE TOXICITY (dermal) - Category 4 SKIN CORROSION - Category 1B SERIOUS EYE DAMAGE - Category 1 SKIN SENSITIZATION - Category 1A
polyoxypropylenediamine	9046-10-0	≥10 - ≤25	SKIN CORROSION - Category 1C SERIOUS EYE DAMAGE - Category 1
Polymer of: Isophorone diamine and bisphenol A-(epichlorhydrin) epoxy resin salicylic acid	69-72-7	≥10 - ≤25	Not classified.
2,4,6-tris(dimethylaminomethyl)phenol	90-72-2	≥1 - ≤3	ACUTE TOXICITY (oral) - Category 4 SERIOUS EYE DAMAGE - Category 1 TOXIC TO REPRODUCTION - Category 2
4,4'-isopropylidenediphenol	80-05-7	≥1 - ≤3	ACUTE TOXICITY (oral) - Category 4 SKIN CORROSION - Category 1C SERIOUS EYE DAMAGE - Category 1 SKIN SENSITIZATION - Category 1
2,2'-iminodiethylamine	111-40-0	≥1 - ≤2.2	TOXIC TO REPRODUCTION - Category 1B SPECIFIC TARGET ORGAN TOXICITY (SINGLE EXPOSURE) (Respiratory tract irritation) - Category 3 ACUTE TOXICITY (oral) - Category 4 ACUTE TOXICITY (dermal) - Category 4 ACUTE TOXICITY (inhalation) - Category 2 SKIN CORROSION - Category 1B SERIOUS EYE DAMAGE - Category 1 SKIN SENSITIZATION - Category 1

Safety Data Sheet

NEOGARD EPOXY RESIN CURING AGENT

70715



SECTION 3: Composition/information on ingredients

3-aminopropyltriethoxysilane	919-30-2	≤0.3	SPECIFIC TARGET ORGAN TOXICITY (SINGLE EXPOSURE) (Respiratory tract irritation) - Category 3 ACUTE TOXICITY (oral) - Category 4 SKIN CORROSION - Category 1B SERIOUS EYE DAMAGE - Category 1 SKIN SENSITIZATION - Category 1
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Any concentration shown as a range is to protect confidentiality or is due to batch variation.

There are no ingredients present which, within the current knowledge of the supplier and in the concentrations applicable, are classified as hazardous to health or the environment and hence require reporting in this section.

SECTION 4: First aid measures

4.1 Description of first aid measures

General :	In all cases of doubt, or when symptoms persist, seek medical attention. Never give anything by mouth to an unconscious person. If breathing is irregular, drowsiness, loss of consciousness or cramps: Call 911 and give immediate treatment (first aid).
Eye contact :	Check for and remove any contact lenses. Immediately flush eyes with plenty of water for at least 15 minutes, occasionally lifting the upper and lower eyelids. Seek immediate medical attention.
Inhalation :	Remove to fresh air. Keep person warm and at rest. If not breathing, if breathing is irregular or if respiratory arrest occurs, provide artificial respiration or oxygen by trained personnel. Give nothing by mouth. If unconscious, place in recovery position and get medical attention immediately.
Skin contact :	Remove contaminated clothing and shoes. Wash skin thoroughly with soap and water or use recognized skin cleanser. Do NOT use solvents or thinners. In case of burns flush with water until the pain ceases. While flushing remove clothing from the affected area unless it is burnt into the skin. If hospital treatment is necessary flushing must continue during transfer and until the hospital staff takes over the treatment.
Ingestion :	If swallowed, seek medical advice immediately and show this container or label. Keep person warm and at rest. Do not induce vomiting unless directed to do so by medical personnel. Lower the head so that vomit will not re-enter the mouth and throat.
Protection of first-aiders :	No action shall be taken involving any personal risk or without suitable training. If it is suspected that fumes are still present, the rescuer should wear an appropriate mask or self-contained breathing apparatus. It may be dangerous to the person providing aid to give mouth-to-mouth resuscitation. Wash contaminated clothing thoroughly with water before removing it, or wear gloves.

4.2 Most important symptoms and effects, both acute and delayed

Potential acute health effects

Eye contact :	Causes serious eye damage.
Inhalation :	Harmful if inhaled.
Skin contact :	Causes severe burns. May cause an allergic skin reaction.
Ingestion :	Harmful if swallowed.

Over-exposure signs/symptoms

Eye contact :	Adverse symptoms may include the following: pain watering redness
Inhalation :	Adverse symptoms may include the following: reduced fetal weight increase in fetal deaths skeletal malformations
Skin contact :	Adverse symptoms may include the following: pain or irritation redness blistering may occur reduced fetal weight increase in fetal deaths skeletal malformations

Safety Data Sheet

NEOGARD EPOXY RESIN CURING AGENT 70715



SECTION 4: First aid measures

Ingestion : Adverse symptoms may include the following:
stomach pains
reduced fetal weight
increase in fetal deaths
skeletal malformations

4.3 Indication of any immediate medical attention and special treatment needed

Notes to physician : If gasses have been inhaled, from the decomposition of the product, symptoms may be delayed.
Specific treatments : No specific treatment.

SECTION 5: Firefighting measures

5.1 Extinguishing media

Extinguishing media : Recommended: alcohol resistant foam, CO₂, powders, water spray.
Not to be used: waterjet.

5.2 Special hazards arising from the substance or mixture

Hazards from the substance or mixture : In a fire or if heated, a pressure increase will occur and the container may burst. This material is harmful to aquatic life with long lasting effects. Fire water contaminated with this material must be contained and prevented from being discharged to any waterway, sewer or drain.

Hazardous combustion products : Decomposition products may include the following materials: carbon oxides nitrogen oxides

5.3 Advice for firefighters

Promptly isolate the scene by removing all persons from the vicinity of the incident if there is a fire. No action shall be taken involving any personal risk or without suitable training. Fire will produce dense black smoke. Exposure to decomposition products may cause a health hazard. Cool closed containers exposed to fire with water. Do not release runoff from fire to drains or watercourses. Fire-fighters should wear appropriate protective equipment and self-contained breathing apparatus (SCBA) with a full face-piece operated in positive pressure mode.

SECTION 6: Accidental release measures

6.1 Personal precautions, protective equipment and emergency procedures

Avoid all direct contact with the spilled material. Avoid breathing vapor or mist. Refer to protective measures listed in sections 7 and 8. No action shall be taken involving any personal risk or without suitable training. If the product contaminates lakes, rivers, or sewers, inform the appropriate authorities in accordance with local regulations.

6.2 Environmental precautions

Avoid dispersal of spilled material and runoff and contact with soil, waterways, drains and sewers. Inform the relevant authorities if the product has caused environmental pollution (sewers, waterways, soil or air). Water polluting material.

6.3 Methods and materials for containment and cleaning up

Stop leak if without risk. Move containers from spill area. Approach release from upwind. Prevent entry into sewers, water courses, basements or confined areas. Wash spillages into an effluent treatment plant or proceed as follows. Contain and collect spillage with non-combustible, absorbent material e.g. sand, earth, vermiculite or diatomaceous earth and place in container for disposal according to local regulations (see Section 13). Contaminated absorbent material may pose the same hazard as the spilled product.

6.4 Reference to other sections

See Section 1 for emergency contact information.
See Section 8 for information on appropriate personal protective equipment.
See Section 13 for additional waste treatment information.

Safety Data Sheet

NEOGARD EPOXY RESIN CURING AGENT 70715



SECTION 7: Handling and storage

7.1 Precautions for safe handling

Avoid inhalation of vapour, dust and spray mist. Avoid contact with skin and eyes. Eating, drinking and smoking should be prohibited in area where this material is handled, stored and processed. Appropriate personal protective equipment: see Section 8. Always keep in containers made from the same material as the original one.

7.2 Conditions for safe storage, including any incompatibilities

Store in accordance with local regulations. Store in a cool, well-ventilated area away from incompatible materials and ignition sources. Keep out of the reach of children. Keep away from: Oxidizing agents, strong alkalis, strong acids. No smoking. Prevent unauthorized access. Containers that are opened must be carefully resealed and kept upright to prevent leakage.

7.3 Specific end use(s)

See separate Product Data Sheet for recommendations or industrial sector specific solutions.

SECTION 8: Exposure controls/personal protection

8.1 Control parameters

Product/ingredient name	Exposure limit values
benzyl alcohol	AIHA WEEL (United States, 7/2018). TWA: 10 ppm 8 hours.
2,2'-iminodiethylamine	ACGIH TLV (United States, 3/2020). Absorbed through skin. TWA: 1 ppm 8 hours. TWA: 4.2 mg/m ³ 8 hours. NIOSH REL (United States, 10/2016). Absorbed through skin. TWA: 1 ppm 10 hours. TWA: 4 mg/m ³ 10 hours.

Recommended monitoring procedures

If this product contains ingredients with exposure limits, personal, workplace atmosphere or biological monitoring may be required to determine the effectiveness of the ventilation or other control measures and/or the necessity to use respiratory protective equipment. Reference should be made to appropriate monitoring standards. Reference to national guidance documents for methods for the determination of hazardous substances will also be required.

8.2 Exposure controls

Appropriate engineering controls

Provide local exhaust and general ventilation systems to maintain airborne concentrations below OSHA, ACGIH, and manufacturer recommended exposure limits. Local exhaust ventilation is preferred because it prevents contaminant dispersion into work areas by controlling it at its source. Use local and general exhaust ventilation to effectively remove and prevent buildup of mists/vapors/fumes generated from the handling of this product.

Note: Local exhaust ventilation is designed to capture an emitted contaminant at or near its source, before the contaminant has a chance to disperse into the workplace air. General exhaust ventilation, also called dilution ventilation, is different from local exhaust ventilation because instead of capturing emissions at their source and removing them from the air, general exhaust ventilation allows the contaminant to be emitted into the workplace air and then dilutes the concentration of the contaminant to an acceptable level (e.g., to the PEL or below).

Individual protection measures

General :	Gloves must be worn for all work that may result in soiling. Apron/coveralls/protective clothing must be worn when soiling is so great that regular work clothes do not adequately protect skin against contact with the product. Safety eyewear should be used when there is a likelihood of exposure.
Hygiene measures :	Wash hands, forearms, and face thoroughly after handling compounds and before eating, smoking, using lavatory, and at the end of day.
Eye/face protection :	Safety eyewear complying with an approved standard should be used when a risk assessment indicates this is necessary to avoid exposure to liquid splashes, mists, gases or dusts. If contact is possible, the following protection should be worn, unless the assessment indicates a higher degree of protection: chemical splash goggles and/or face shield. If inhalation hazards exist, a full-face respirator may be required instead.
Hand protection :	Wear chemical-resistant gloves in combination with 'basic' employee training. The quality of the chemical-resistant protective gloves must be chosen as a function of the specific workplace concentrations and quantity of hazardous substances.

Safety Data Sheet

NEOGARD EPOXY RESIN CURING AGENT

70715



SECTION 8: Exposure controls/personal protection

Since the actual work situation is unknown. Supplier of gloves should be contacted in order to find the appropriate type. Below listed glove(s) should be regarded as generic advice:

Short term exposure: neoprene rubber, butyl rubber, natural rubber (latex), polyvinyl chloride (PVC)
May be used: nitrile rubber
Recommended: polyvinyl alcohol (PVA), Silver Shield / Barrier / 4H gloves, Viton®

Body protection :	Personal protective equipment for the body should be selected based on the task being performed and the risks involved handling this product. Wear suitable protective clothing. Chemical-resistant apron.
Respiratory protection :	If working areas have insufficient ventilation, wear half or totally covering mask equipped with gas filter of type Organic Vapor, when grinding use particle filter of type P95, P99 or P100. When spraying use a combined filter (organic vapor / HEPA or organic vapor / P100 type). Be sure to use approved/certified respirator or equivalent. Always wear an air-fed respirator when spraying in a continuous and prolonged work situation (e.g. hood with supply of fresh or compressed air or a full face, powered air purifying filter).
Protective clothing (pictograms) :	

Consult your supervisor or S.O.P. for special handling

Note: Application of paint products by spraying requires additional safety precautions: Full body suit, Full face respirator with air supplied.

Environmental exposure controls

Emissions from ventilation or work process equipment should be checked to ensure they comply with the requirements of environmental protection legislation. In some cases, fume scrubbers, filters or engineering modifications to the process equipment will be necessary to reduce emissions to acceptable levels.

SECTION 9: Physical and chemical properties

9.1 Information on basic physical and chemical properties

Physical state :	Liquid.
Color :	Clear
Odor :	Solvent-like
pH :	Testing not relevant or not possible due to nature of the product.
Melting point/freezing point :	-15.4°C This is based on data for the following ingredient: benzyl alcohol
Boiling point/boiling range :	Testing not relevant or not possible due to nature of the product.
Flash point :	Closed cup: 93°C (199.4°F)
Evaporation rate :	Testing not relevant or not possible due to nature of the product.
Flammability :	Highly flammable in the presence of the following materials or conditions: open flames, sparks and static discharge. Flammable in the presence of the following materials or conditions: heat.
Upper/lower flammability or explosive limits :	1 - 14.3 vol %
Vapor pressure :	0.023 kPa This is based on data for the following ingredient: benzyl alcohol
Vapor density :	Testing not relevant or not possible due to nature of the product.
Relative density :	0.991 g/cm³
Solubility(ies) :	Partially soluble in the following materials: cold water and hot water.
Partition coefficient (LogKow) :	Testing not relevant or not possible due to nature of the product.
Auto-ignition temperature :	Testing not relevant or not possible due to nature of the product.
Decomposition temperature :	Testing not relevant or not possible due to nature of the product.
Viscosity :	Testing not relevant or not possible due to nature of the product.
Explosive properties :	Slightly explosive in the presence of the following materials or conditions: open flames, sparks and static discharge, heat and organic materials.
Oxidizing properties :	Testing not relevant or not possible due to nature of the product.

Safety Data Sheet

NEOGARD EPOXY RESIN CURING AGENT 70715



SECTION 9: Physical and chemical properties

9.2 Other information

Solvent(s) % by weight (Included exempt solvent(s)):	8.1 % (w/w)
Water % by weight :	Weighted average: 0 %
VOC content (Coatings) :	0.673 lbs/gal (80.7 g/l)
VOC content (Regulatory) :	0.673 lbs/gal (80.7 g/l)
TOC Content (Volatile) :	Weighted average: 65 g/l
Solvent Gas :	Weighted average: 0.073 m³/l

SECTION 10: Stability and reactivity

10.1 Reactivity

No specific test data related to reactivity available for this product or its ingredients.

10.2 Chemical stability

The product is stable.

10.3 Possibility of hazardous reactions

Under normal conditions of storage and use, hazardous reactions will not occur.

10.4 Conditions to avoid

No specific data.

10.5 Incompatible materials

Reactive or incompatible with the following materials: oxidizing materials.

Slightly reactive or incompatible with the following materials: reducing materials and organic materials.

10.6 Hazardous decomposition products

When exposed to high temperatures (i.e. in case of fire) harmful decomposition products may be formed:

Decomposition products may include the following materials: carbon oxides nitrogen oxides

SECTION 11: Toxicological information

11.1 Information on toxicological effects

Exposure to component solvent vapor concentrations may result in adverse health effects such as mucous membrane and respiratory system irritation and adverse effects on the kidneys, liver and central nervous system. Solvents may cause some of the above effects by absorption through the skin. Symptoms and signs include headaches, dizziness, fatigue, muscular weakness, drowsiness and, in extreme cases, loss of consciousness. Repeated or prolonged contact with the preparation may cause removal of natural fat from the skin, resulting in non-allergic contact dermatitis and absorption through the skin. If splashed in the eyes, the liquid may cause irritation and reversible damage. Accidental swallowing may cause stomach pain. Chemical lung inflammation may occur if the product is taken into the lungs via vomiting.

Inhalation of a corrosive substance may result in health effects such as stinging, coughing and in extreme cases, dyspnoea or loss of consciousness with a risk of lung damage, possibly lung oedema. Cauterization of skin and mucous membrane. If splashed in the eyes, the liquid may cause irreversible damage. Accidental swallowing may cause stinging and cauterization to mouth, oesophagus and stomach. Symptoms and signs include bloody vomiting, chock and loss of consciousness.

Acute toxicity

Safety Data Sheet

NEOGARD EPOXY RESIN CURING AGENT

70715

SECTION 11: Toxicological information

Product/ingredient name	Result	Species	Dose	Exposure
benzyl alcohol	LC50 Inhalation Dusts and mists	Rat	>4178 mg/m³	4 hours
3-aminomethyl- 3,5,5-trimethylcyclohexylamine	LD50 Oral	Rat	1230 mg/kg	-
	LC50 Inhalation Dusts and mists	Rat	>5.01 mg/l	4 hours
	LD50 Dermal	Rabbit	1840 mg/kg	-
polyoxypropylenediamine	LD50 Oral	Rat	1030 mg/kg	-
	LD50 Dermal	Rabbit	2980 mg/kg	-
	LD50 Oral	Rat	2880 mg/kg	-
salicylic acid	LC50 Inhalation Dusts and mists	Rat	>0.9 mg/l	1 hours
	LD50 Dermal	Rat	>2000 mg/kg	-
	LD50 Oral	Rat	891 mg/kg	-
2,4,6-tris(dimethylaminomethyl) phenol	LD50 Dermal	Rabbit	1465 mg/kg	-
	LD50 Oral	Rat	1200 mg/kg	-
	LD50 Oral	Rat	2169 mg/kg	-
4,4'-isopropylidenediphenol	LD50 Dermal	Rabbit	>2000 mg/kg	-
	LD50 Oral	Rat	3250 mg/kg	-
	LD50 Oral	Rat	3250 mg/kg	-
2,2'-iminodiethylamine	LC50 Inhalation Dusts and mists	Rat	0.07 mg/l	4 hours
	LD50 Dermal	Rabbit	1045 mg/kg	-
	LD50 Dermal	Rabbit	1090 mg/kg	-
	LD50 Oral	Rat	1080 mg/kg	-
3-aminopropyltriethoxysilane	LD50 Dermal	Rabbit	4.29 g/kg	-
	LD50 Oral	Rat	1.57 g/kg	-

Acute toxicity estimates

Route	ATE value
Oral	1927.65 mg/kg
Dermal	7177.96 mg/kg
Inhalation (vapors)	37.37 mg/l
Inhalation (dusts and mists)	4.84 mg/l

Irritation/Corrosion

Product/ingredient name	Result	Species	Score	Exposure
benzyl alcohol	Eyes - Visible necrosis	Rabbit	-	-
	Skin - Mild irritant	Rabbit	-	-
3-aminomethyl- 3,5,5-trimethylcyclohexylamine	Skin - Severe irritant	Rabbit	-	-
	Eyes - Severe irritant	Rabbit	-	-
polyoxypropylenediamine	Skin - Severe irritant	Rabbit	-	-
	Eyes - Severe irritant	Rabbit	-	-
salicylic acid	Eyes - Severe irritant	Rabbit	-	-
2,4,6-tris(dimethylaminomethyl) phenol	Eyes - Severe irritant	Rabbit	-	24 hours 50 Micrograms
	Skin - Severe irritant	Rabbit	-	24 hours 2 milligrams
4,4'-isopropylidenediphenol	Eyes - Severe irritant	Rabbit	-	24 hours 250 Micrograms
	Skin - Mild irritant	Rabbit	-	24 hours 500 milligrams
2,2'-iminodiethylamine	Skin - Severe irritant	Rabbit	-	-
	Eyes - Severe irritant	Rabbit	-	-
3-aminopropyltriethoxysilane	Eyes - Severe irritant	Rabbit	-	24 hours 750 Micrograms
	Skin - Severe irritant	Rabbit	-	24 hours 5 milligrams

Sensitizer

Product/ingredient name	Route of exposure	Species	Result
3-aminomethyl- 3,5,5-trimethylcyclohexylamine	skin	Guinea pig	Sensitizing
2,2'-iminodiethylamine	skin	Mouse	Sensitizing

Specific target organ toxicity (single exposure)

Safety Data Sheet

NEOGARD EPOXY RESIN CURING AGENT

70715

SECTION 11: Toxicological information

Product/ingredient name	Category	Route of exposure	Target organs
4,4'-isopropylidenediphenol	Category 3		Respiratory tract irritation
2,2'-iminodiethylamine	Category 3		Respiratory tract irritation

Information on the likely routes of exposure

Routes of entry anticipated: Oral, Dermal, Inhalation.

Potential chronic health effects

Sensitization : Contains 3-aminomethyl-3,5,5-trimethylcyclohexylamine, 4,4'-isopropylidenediphenol, 2,2'-iminodiethylamine. May produce an allergic reaction.

Other information : No additional known significant effects or critical hazards.

SECTION 12: Ecological information

12.1 Toxicity

Do not allow to enter drains or watercourses. Harmful to aquatic life with long lasting effects.

When spilled, this product may act as an oil, causing a film, sheen, emulsion, or sludge at or beneath the surface of a body of water. Oils of any kind can cause: (a) drowning of waterfowl due to lack of buoyancy, loss of insulating capacity of feathers, starvation and vulnerability to predators due to lack of mobility; (b) lethal effect on fish by coating gill surfaces, preventing respiration; (c) potential fish kills resulting from alteration in biochemical oxygen demand; (d) asphyxiation of benthic life forms when floating masses become engaged with surface debris and settle on the bottom; and (e) adverse aesthetic effects of fouled shoreline and beaches.

Product/ingredient name	Result	Species	Exposure
benzyl alcohol	Acute EC50 230 mg/l	Daphnia	48 hours
	Acute IC50 770 mg/l	Algae	72 hours
	Acute LC50 460 mg/l	Fish	96 hours
3-aminomethyl-3,5,5-trimethylcyclohexylamine	Acute EC50 37 mg/l	Algae	72 hours
	Acute EC50 >50 mg/l	Aquatic plants	72 hours
	Acute EC50 23 mg/l	Daphnia	48 hours
	Acute LC50 110 mg/l	Fish	96 hours
	Chronic NOEC 3 mg/l	Daphnia	21 days
polyoxypropylenediamine	Acute EC50 15 mg/l	Algae	72 hours
	Acute EC50 80 mg/l	Daphnia	48 hours
	Acute LC50 772 mg/l	Fish	96 hours
2,4,6-tris(dimethylaminomethyl)phenol	Acute EC50 84 mg/l	Algae	72 hours
	Acute LC50 175 mg/l	Fish	96 hours
4,4'-isopropylidenediphenol	Acute LC50 7.5 mg/l	Fish	96 hours
	Chronic NOEC 0.8 mg/l Fresh water	Daphnia - Daphnia magna - Neonate	21 days
	Chronic NOEC 0.2 - 20 ppb Fresh water	Fish - Xiphophorus helleri - Juvenile (Fledgling, Hatchling, Weanling)	60 days
2,2'-iminodiethylamine	Acute EC50 1164 mg/l	Algae	72 hours
	Acute EC50 16 mg/l	Daphnia	48 hours
	Acute LC50 430 mg/l	Fish	96 hours

12.2 Persistence and degradability

Product/ingredient name	Test	Result	Dose	Inoculum
benzyl alcohol	OECD 301A 301A Ready	95 - 97 % - Readily - 21 days	-	-
	Biodegradability - DOC Die-Away Test			
	OECD 301C 301C Ready	92 - 96 % - Readily - 14 days	-	-
	Biodegradability - Modified MITI Test (I)			
3-aminomethyl-3,5,5-trimethylcyclohexylamine	-	8 % - Not readily - 28 days	-	-
polyoxypropylenediamine	-	0 % - Not readily - 28 days	-	-
salicylic acid	-	100 % - Readily - 14 days	-	-
2,4,6-tris(dimethylaminomethyl)phenol	OECD 301D 301D Ready	4 % - Not readily - 28 days	-	-

Safety Data Sheet

NEOGARD EPOXY RESIN CURING AGENT

70715



SECTION 12: Ecological information

4,4'-isopropylidenediphenol 2,2'-iminodiethylamine	Biodegradability - Closed Bottle Test - OECD 301D Ready Biodegradability - Closed Bottle Test	1 - 2 % - Not readily - 28 days 87 % - Readily - 21 days	- -	- -
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Product/ingredient name	Aquatic half-life	Photolysis	Biodegradability
benzyl alcohol	-	-	Readily
3-aminomethyl-	-	-	Not readily
3,5,5-trimethylcyclohexylamine	-	-	
polyoxypropylenediamine	-	-	Not readily
salicylic acid	-	-	Readily
2,4,6-tris(dimethylaminomethyl)	-	-	Not readily
phenol	-	-	
4,4'-isopropylidenediphenol	-	-	Not readily
2,2'-iminodiethylamine	-	-	Readily

12.3 Bioaccumulative potential

Product/ingredient name	LogP _{ow}	BCF	Potential
benzyl alcohol	0.87	1.37	low
3-aminomethyl-3,5,5-trimethylcyclohexylamine	0.99	-	low
polyoxypropylenediamine	1.34	-	low
salicylic acid	2.21 - 2.26	-	low
2,4,6-tris(dimethylaminomethyl)phenol	0.219	-	low
4,4'-isopropylidenediphenol	3.4	20 - 67	low
2,2'-iminodiethylamine	-5.58	2.8 - 6.3	low
3-aminopropyltriethoxysilane	1.7	3.4	low

12.4 Mobility in soil

Soil/water partition coefficient (K_{oc}) : No known data available in our database.

Mobility : No known data available in our database.

12.5 Other adverse effects

No known significant effects or critical hazards.

SECTION 13: Disposal considerations

13.1 Waste treatment methods

Disposal should be in accordance with applicable regional, national and local laws and regulations. Local regulations may be more stringent than regional or national requirements.

The information presented below only applies to the material as supplied. The identification based on characteristic(s) or listing may not apply if the material has been used or otherwise contaminated. It is the responsibility of the waste generator to determine the toxicity and physical properties of the material generated to determine the proper waste identification and disposal methods in compliance with applicable regulations.

Refer to Section 7 and Section 8 for additional handling information and protection of employees.

The generation of waste should be avoided or minimized wherever possible. Disposal of this product, solutions and any by-products should at all times comply with the requirements of environmental protection and waste disposal legislation and any regional local authority requirements. Dispose of surplus and non-recyclable products via a licensed waste disposal contractor. Waste should not be disposed of untreated to the sewer unless fully compliant with the requirements of all authorities with jurisdiction. Waste packaging should be recycled. Incineration or landfill should only be considered when recycling is not feasible. This material and its container must be disposed of in a safe way. Care should be taken when handling emptied containers that have not been cleaned or rinsed out. Empty containers or liners may retain some product residues. Vapor from product residues may create a highly flammable or explosive atmosphere inside the container. Do not cut, weld or grind used containers unless they have been cleaned thoroughly internally. Avoid dispersal of spilled material and runoff and contact with soil, waterways, drains and sewers.

Safety Data Sheet

NEOGARD EPOXY RESIN CURING AGENT 70715



SECTION 14: Transport information

Transport may take place according to national regulation or DOT for transport by road and by train, IMDG for transport by sea, IATA for Air shipment. Refer to specific Dangerous Goods Transport requirements under 49CFR, ICAO and IATA.

	14.1 UN no.	14.2 Proper shipping name	14.3 Transport hazard class(es)	14.4 PG*	14.5 Env*	Additional information
DOT Code	UN3066	PAINT RELATED MATERIAL	8 - 	III	No.	
TDG Code	UN3066	PAINT RELATED MATERIAL	8 - 	III	No.	Product classified as per the following sections of the Transportation of Dangerous Goods Regulations: 2.40-2.42 (Class 8).
SCT Code	UN3066	PAINT RELATED MATERIAL	8 - 	III	No.	-
IMDG Code	UN3066	PAINT RELATED MATERIAL	8 - 	III	No.	<u>Emergency schedules</u> F-A, S-B
IATA Code	UN3066	PAINT RELATED MATERIAL	8 - 	III	No.	-

Code : Classification

PG* : Packing group

Env.* : Environmental hazards

14.6 Special precautions for user

Transport within user's premises: always transport in closed containers that are upright and secure. Ensure that persons transporting the product know what to do in the event of an accident or spillage.

14.7 Transport in bulk according to IMO instruments

Not applicable.

SECTION 15: Regulatory information

15.1 Safety, health and environmental regulations/legislation specific for the substance or mixture

U.S. Federal regulations :

All components are active or exempted.

TSCA 8(a) PAIR: benzaldehyde

TSCA 8(a) CDR Exempt/Partial exemption: Not determined

United States inventory (TSCA 8b): All components are active or exempted.

Clean Water Act (CWA) 307: phenol

Clean Water Act (CWA) 311: phenol

Clean Air Act Section 112(b) Hazardous Air Pollutants (HAPs) : Listed

Product/ingredient name	CAS number	Concentration
phenol	108-95-2	0.0047947

Clean Air Act Section 602 Class I Substances : Not listed

Clean Air Act Section 602 Class II Substances : Not listed

DEA List I Chemicals (Precursor Chemicals) : Not listed

DEA List II Chemicals (Essential Chemicals) : Not listed

SARA 302/304 :

Product/ingredient name	%	EHS	SARA 302 TPQ		SARA 304 RQ	
			(lbs)	(gallons)	(lbs)	(gallons)
phenol	≤0.1	Yes.	500 / 10000	-	1000	-

Safety Data Sheet

NEOGARD EPOXY RESIN CURING AGENT

70715



SECTION 15: Regulatory information

SARA 304 RQ : 20856537.1 lbs / 9468867.8 kg [2524127.4 gal / 9554861.6 L]

SARA 311/312 Classification : FLAMMABLE LIQUIDS - Category 4
ACUTE TOXICITY (oral) - Category 4
ACUTE TOXICITY (inhalation) - Category 4
SKIN CORROSION - Category 1B
SERIOUS EYE DAMAGE - Category 1
SKIN SENSITIZATION - Category 1
TOXIC TO REPRODUCTION - Category 1B

Product/ingredient name	%	Classification
benzyl alcohol	≥25 - ≤50	ACUTE TOXICITY (oral) - Category 4 ACUTE TOXICITY (inhalation) - Category 4 EYE IRRITATION - Category 2A
3-aminomethyl-3,5,5-trimethylcyclohexylamine	≥10 - ≤25	ACUTE TOXICITY (oral) - Category 4 ACUTE TOXICITY (dermal) - Category 4 SKIN CORROSION - Category 1B SERIOUS EYE DAMAGE - Category 1 SKIN SENSITIZATION - Category 1A
polyoxypropylenediamine	≥10 - ≤25	SKIN CORROSION - Category 1C SERIOUS EYE DAMAGE - Category 1
salicylic acid	≥1 - ≤3	ACUTE TOXICITY (oral) - Category 4 SERIOUS EYE DAMAGE - Category 1
2,4,6-tris(dimethylaminomethyl)phenol	≥1 - ≤3	TOXIC TO REPRODUCTION - Category 2 ACUTE TOXICITY (oral) - Category 4 SKIN CORROSION - Category 1C
4,4'-isopropylidenediphenol	≥1 - ≤3	SERIOUS EYE DAMAGE - Category 1 SKIN SENSITIZATION - Category 1 TOXIC TO REPRODUCTION - Category 1B
2,2'-iminodiethylamine	≥1 - ≤2.2	SPECIFIC TARGET ORGAN TOXICITY (SINGLE EXPOSURE) (Respiratory tract irritation) - Category 3 ACUTE TOXICITY (oral) - Category 4 ACUTE TOXICITY (dermal) - Category 4 ACUTE TOXICITY (inhalation) - Category 2 SKIN CORROSION - Category 1B SERIOUS EYE DAMAGE - Category 1 SKIN SENSITIZATION - Category 1
3-aminopropyltriethoxysilane	≤0.3	SPECIFIC TARGET ORGAN TOXICITY (SINGLE EXPOSURE) (Respiratory tract irritation) - Category 3 ACUTE TOXICITY (oral) - Category 4 SKIN CORROSION - Category 1B SERIOUS EYE DAMAGE - Category 1 SKIN SENSITIZATION - Category 1

SARA 313 : SARA 313 notifications must not be detached from the SDS and any copying and redistribution of the SDS shall include copying and redistribution of the notice attached to copies of the SDS subsequently redistributed.

Form R - Reporting requirements :

Product/ingredient name	CAS number	Concentration
4,4'-isopropylidenediphenol	80-05-7	1 - 3

Supplier notification :

Product/ingredient name	CAS number	Concentration
4,4'-isopropylidenediphenol	80-05-7	1 - 3

State regulations :

Connecticut Carcinogen Reporting: None of the components are listed.
Connecticut Hazardous Material Survey: None of the components are listed.
Florida substances: None of the components are listed.
Illinois Chemical Safety Act: None of the components are listed.
Illinois Toxic Substances Disclosure to Employee Act: None of the components are listed.
Louisiana Reporting: None of the components are listed.
Louisiana Spill: None of the components are listed.
Massachusetts Substances: The following components are listed: BENZYL ALCOHOL; 4,4'-ISOPROPYLDENEDIPHENOL; DIETHYLENE TRIAMINE
Massachusetts Spill: None of the components are listed.
Michigan Critical Material: None of the components are listed.
Minnesota Hazardous Substances: None of the components are listed.
New Jersey Spill: None of the components are listed.
New Jersey Toxic Catastrophe Prevention Act: None of the components are listed.
New Jersey Hazardous Substances: The following components are listed: ISOPHORONEDIAMINE; CYCLOHEXANEMETHANAMINE, 5-AMINO-1,3,3-TRIMETHYL-; BISPHENOL A; PHENOL, 4,4'-(1-METHYLETHYLIDENE)BIS-; 4,4'-ISOPROPYLDENEDIPHENOL; DIETHYLENE TRIAMINE; 1,2-ETHANEDIAMINE, N-(2-AMINOETHYL)-
New York Hazardous Substances: None of the components are listed.
New York Toxic Chemical Release Reporting: None of the components are listed.
Pennsylvania RTK Hazardous Substances: The following components are listed:

Safety Data Sheet

NEOGARD EPOXY RESIN CURING AGENT 70715



SECTION 15: Regulatory information

BENZENEMETHANOL; 4,4'-ISOPROPYLIDENEDIPHENOL; 1,2-ETHANEDIAMINE, N-(2-AMINOETHYL)-

Rhode Island Hazardous Substances: None of the components are listed.

California Prop. 65 PFF :

WARNING: This product can expose you to Bisphenol A, which is known to the State of California to cause birth defects or other reproductive harm. For more information go to www.P65Warnings.ca.gov.

Product/ingredient name	Cancer	Reproductive	No significant risk level	Maximum acceptable dosage level
4,4'-isopropylidenediphenol	No.	Yes.		Yes.

SECTION 16: Other information

Remarks :

Note: In USA, consult Code of Federal Regulations, Title 29, Labor, Parts 1910 and 1915 concerning occupational safety and health standards and regulations, as well as any other applicable Federal, State or local regulations that apply to safe practices in coating operations.

Warning! If you scrape, sand, or remove old paint, you may release lead dust. LEAD is TOXIC.

Validation :

Validated by US - HSE Products Coordinator on 10 February 2021

GHS Classification

Procedure used to derive the classification.

Classification	Justification
FLAMMABLE LIQUIDS - Category 4 ACUTE TOXICITY (oral) - Category 4 ACUTE TOXICITY (inhalation) - Category 4 SKIN CORROSION - Category 1B SERIOUS EYE DAMAGE - Category 1 SKIN SENSITIZATION - Category 1 TOXIC TO REPRODUCTION - Category 1B	On basis of test data Calculation method Calculation method Calculation method Calculation method Calculation method Calculation method

Hazardous Material Information System (U.S.A.)

Health	3
Fire hazard	2
Physical hazards	0
Personal protection	X

National Fire Protection Association (U.S.A.)

Health	3	2	0	Flammability
				Instability
				Special

Personal Protective Equipment (PPE) shown in this section is a suggestion. Since conditions vary from one work location to another consult the facility safety & health program. Customer or end user is responsible to evaluate worker exposure conditions at the site of application and determine the appropriate PPE suitable for workers at that particular facility or location.

Abbreviations and acronyms :

ANSI = American National Standards Institute
HCS = Hazardous Communication System
TSCA = Toxic Substances Control Act
CFR = Code of federal Regulations
GHS = Globally Harmonized System of Classification and Labelling of Chemicals
OSHA = United States Occupational Health and Safety Administration
NIOSH = National Institute for Occupational Safety and Health
ACGIH = American Conference of Industrial Hygienists
IARC = International Agency for Research on Cancer.
NTP = National Toxicology Program
ATE = Acute Toxicity Estimate

OECD = Organisation for Economic Co-operation and Development
BCF = Bioconcentration Factor
DOT = United States Department of Transportation
ERG = Emergency Response Guide
TDG = Transport of Dangerous Goods, Canada
SCT = Transportation & Communications Ministry, Mexico
IMDG = International Maritime Dangerous Goods
IATA = International Air Transport Association
SARA = Superfund Amendments Reauthorization Act
EPCRA = Emergency Planning and Community Right to Know Act

Notice to reader

Indicates information that has changed from previously issued version.

To the best of our knowledge, the information contained herein is accurate. However, neither the above named supplier nor any of its subsidiaries assumes any liability whatsoever for the accuracy or completeness of the information contained herein. Final determination of suitability of any material is the sole responsibility of the user. All materials may present unknown hazards and should be used with caution. Although certain hazards are described herein, we cannot guarantee that these are the only hazards that exist.

Safety Data Sheet

NEOGARD FLEXIBLE EPOXY JOINT COMPOUND

70718-02 GRAY



Conforms to ANSI Z400.1-2010 Standard - HCS 2012

Protective Clothing	General Hazard	DOT

SECTION 1: Identification of the substance/mixture and of the company/undertaking

1.1 Product identifier

Product name : NEOGARD FLEXIBLE EPOXY JOINT COMPOUND GRAY
Product identity : 2500916650, 70718-02
Product type : epoxy filler/putty (base for multi-component product)

1.2 Relevant identified uses of the substance or mixture and uses advised against

Field of application : buildings and metal industry.
Ready-for-use mixture : Mixing Ratio:
25000 = 25009 2 Vol. / 95048 1 Vol.
Identified uses : Industrial/Professional use
TSCA : Unless otherwise stated. All components are listed or exempted.

1.3 Details of the supplier of the safety data sheet

Company details : NEOGARD, a Division of Hempel (USA), Inc.
2728 Empire Central
Dallas, TX 75235
Phone number: 1-214-353-1600
E-mail: hempel@hempel.com

1.4 Emergency telephone number (with hours of operation)

For Transportation Emergencies : CHEMTREC: **1-800-424-9300** (Toll-free in the U.S., Canada and the U.S. Virgin Islands) **703-527-3887**
(24 hours) For calls originating elsewhere (Collect calls are accepted). Contract number: CCN10384
To preserve the effectiveness of arrangements for providing accurate and timely emergency response information, the basic identifying information (shipper name or contract number) must be included on shipping papers.
If the purchaser of this product is going to be shipping this product to other locations, the purchaser must arrange for its own Emergency Information Provider to respond to transport incidents. Hempel's 24 hour response contract does not cover non-Hempel shipments.
For all other information : In USA toll free calling available: 1-800- 678-6641 or (936)-523-6000
(8 AM - 5 PM CST) See Section 4 of the safety data sheet (first aid measures).

SECTION 2: Hazards identification

2.1 Classification of the substance or mixture

OSHA/HCS status : This material is considered hazardous by the OSHA Hazard Communication Standard (29 CFR 1910.1200).
GHS Classification : SKIN IRRITATION - Category 2
SERIOUS EYE DAMAGE - Category 1
SKIN SENSITIZATION - Category 1
CARCINOGENICITY - Category 2

2.2 Label elements

Hazard pictograms :



Safety Data Sheet

NEOGARD FLEXIBLE EPOXY JOINT COMPOUND

70718-02 GRAY



SECTION 2: Hazards identification

Signal word :	Danger
Hazard statements :	H315 - Causes skin irritation. H317 - May cause an allergic skin reaction. H318 - Causes serious eye damage. H351 - Suspected of causing cancer.
Precautionary statements :	
Prevention :	Obtain special instructions before use. Do not handle until all safety precautions have been read and understood. Wear protective gloves, protective clothing and eye or face protection. Avoid breathing vapor. Wash thoroughly after handling. Contaminated work clothing must not be allowed out of the workplace.
Response :	IF exposed or concerned: Get medical advice or attention. Take off contaminated clothing and wash it before reuse. Wash contaminated clothing before reuse. IF ON SKIN: Wash with plenty of water. If skin irritation or rash occurs: Get medical advice or attention. IF IN EYES: Rinse cautiously with water for several minutes. Remove contact lenses, if present and easy to do. Continue rinsing. Immediately call a POISON CENTER or doctor.
Storage :	Store locked up.
Disposal :	Dispose of contents and container in accordance with all local, regional, national and international regulations.
Supplemental label elements :	None known.

2.3 Other hazards

Hazards not otherwise classified : None known.

SECTION 3: Composition/information on ingredients

Product definition :	Mixture
Physical state :	Liquid.

Product/ingredient name	Identifiers	%	GHS Classification
bisphenol A-(epichlorhydrin) epoxy resin MW =< 700	25068-38-6	≥25 - ≤50	SKIN IRRITATION - Category 2 EYE IRRITATION - Category 2A SKIN SENSITIZATION - Category 1
blocked polyurethane oxirane, mono[(C12-14-alkyloxy)methyl] derivs.	2155840-39-2 68609-97-2	≥25 - ≤50 ≥5 - ≤10	SERIOUS EYE DAMAGE - Category 1 SKIN IRRITATION - Category 2 SKIN SENSITIZATION - Category 1
titanium dioxide	13463-67-7	≥5 - ≤10	Not classified.
cashew nutshell liquid	8007-24-7	≥1 - ≤3	ACUTE TOXICITY (oral) - Category 4 ACUTE TOXICITY (dermal) - Category 4 SKIN IRRITATION - Category 2 SERIOUS EYE DAMAGE - Category 1 SKIN SENSITIZATION - Category 1A
4-methylpentan-2-one	108-10-1	≤0.3	FLAMMABLE LIQUIDS - Category 2 ACUTE TOXICITY (inhalation) - Category 4 EYE IRRITATION - Category 2A CARCINOGENICITY - Category 2 SPECIFIC TARGET ORGAN TOXICITY (SINGLE EXPOSURE) (Respiratory tract irritation) - Category 3
maleic anhydride	108-31-6	<0.1	ACUTE TOXICITY (oral) - Category 4 SKIN CORROSION - Category 1B SERIOUS EYE DAMAGE - Category 1 RESPIRATORY SENSITIZATION - Category 1 SKIN SENSITIZATION - Category 1A SPECIFIC TARGET ORGAN TOXICITY (REPEATED EXPOSURE) - Category 1

Any concentration shown as a range is to protect confidentiality or is due to batch variation.

Occupational exposure limits, if available, are listed in Section 8.

There are no additional ingredients present which, within the current knowledge of the supplier and in the concentrations applicable, are classified as hazardous to health or the environment and hence require reporting in this section.

SECTION 4: First aid measures

4.1 Description of first aid measures

General :	In all cases of doubt, or when symptoms persist, seek medical attention. Never give anything by mouth to an unconscious person. If breathing is irregular, drowsiness, loss of consciousness or cramps: Call 911 and give immediate treatment (first aid).
Eye contact :	Check for and remove any contact lenses. Immediately flush eyes with plenty of water for at least 15 minutes, occasionally lifting the upper and lower eyelids. Seek immediate medical attention/advice.
Inhalation :	Remove to fresh air and keep at rest in a position comfortable for breathing. Give nothing by mouth. If not breathing, if breathing is irregular or if respiratory arrest occurs, provide artificial respiration or oxygen by trained personnel. If unconscious, place in recovery position and get medical attention immediately.
Skin contact :	Remove contaminated clothing and shoes. Wash skin thoroughly with soap and water or use recognized skin cleanser. Do NOT use solvents or thinners.
Ingestion :	If swallowed, seek medical advice immediately and show this container or label. Keep person warm and at rest. Do not induce vomiting unless directed to do so by medical personnel. Lower the head so that vomit will not re-enter the mouth and throat.
Protection of first-aiders :	No action shall be taken involving any personal risk or without suitable training. If it is suspected that fumes are still present, the rescuer should wear an appropriate mask or self-contained breathing apparatus. It may be dangerous to the person providing aid to give mouth-to-mouth resuscitation. Wash contaminated clothing thoroughly with water before removing it, or wear gloves.

4.2 Most important symptoms and effects, both acute and delayed

Potential acute health effects

Eye contact :	Causes serious eye damage.
Inhalation :	No known significant effects or critical hazards.
Skin contact :	Causes skin irritation. May cause an allergic skin reaction.
Ingestion :	No known significant effects or critical hazards.

Over-exposure signs/symptoms

Eye contact :	Adverse symptoms may include the following: pain watering redness
Inhalation :	No specific data.
Skin contact :	Adverse symptoms may include the following: pain or irritation redness blistering may occur
Ingestion :	Adverse symptoms may include the following: stomach pains

4.3 Indication of any immediate medical attention and special treatment needed

Notes to physician :	Not applicable.
Specific treatments :	No specific treatment.

SECTION 5: Firefighting measures

5.1 Extinguishing media

Extinguishing media :	Recommended: alcohol resistant foam, CO ₂ , powders, water spray. Not to be used: waterjet.
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5.2 Special hazards arising from the substance or mixture

Hazards from the substance or mixture :	In a fire or if heated, a pressure increase will occur and the container may burst. This material is toxic to aquatic life with long lasting effects. Fire water contaminated with this material must be contained and prevented from being discharged to any waterway, sewer or drain.
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SECTION 5: Firefighting measures

Hazardous combustion products : Decomposition products may include the following materials: carbon oxides halogenated compounds metal oxide/oxides

5.3 Advice for firefighters

Promptly isolate the scene by removing all persons from the vicinity of the incident if there is a fire. No action shall be taken involving any personal risk or without suitable training. Fire will produce dense black smoke. Exposure to decomposition products may cause a health hazard. Cool closed containers exposed to fire with water. Do not release runoff from fire to drains or watercourses. Fire-fighters should wear appropriate protective equipment and self-contained breathing apparatus (SCBA) with a full face-piece operated in positive pressure mode.

SECTION 6: Accidental release measures

6.1 Personal precautions, protective equipment and emergency procedures

Avoid all direct contact with the spilled material. Refer to protective measures listed in sections 7 and 8. No action shall be taken involving any personal risk or without suitable training. If the product contaminates lakes, rivers, or sewers, inform the appropriate authorities in accordance with local regulations.

6.2 Environmental precautions

Avoid dispersal of spilled material and runoff and contact with soil, waterways, drains and sewers. Inform the relevant authorities if the product has caused environmental pollution (sewers, waterways, soil or air). Water polluting material. May be harmful to the environment if released in large quantities.

6.3 Methods and materials for containment and cleaning up

Stop leak if without risk. Move containers from spill area. Approach release from upwind. Prevent entry into sewers, water courses, basements or confined areas. Wash spillages into an effluent treatment plant or proceed as follows. Contain and collect spillage with non-combustible, absorbent material e.g. sand, earth, vermiculite or diatomaceous earth and place in container for disposal according to local regulations (see Section 13). Contaminated absorbent material may pose the same hazard as the spilled product.

6.4 Reference to other sections

See Section 1 for emergency contact information.

See Section 8 for information on appropriate personal protective equipment.

See Section 13 for additional waste treatment information.

SECTION 7: Handling and storage

7.1 Precautions for safe handling

Contains epoxy constituents. Avoid all possible skin contact with epoxy and amine containing products, they may cause allergic reactions. Avoid inhalation of vapour, dust and spray mist. Avoid contact with skin and eyes. Eating, drinking and smoking should be prohibited in area where this material is handled, stored and processed. Appropriate personal protective equipment: see Section 8. Always keep in containers made from the same material as the original one.

7.2 Conditions for safe storage, including any incompatibilities

Store in accordance with local regulations. Store in a cool, well-ventilated area away from incompatible materials and ignition sources. Keep out of the reach of children. Keep away from: Oxidizing agents, strong alkalis, strong acids. No smoking. Prevent unauthorized access. Containers that are opened must be carefully resealed and kept upright to prevent leakage.

7.3 Specific end use(s)

See separate Product Data Sheet for recommendations or industrial sector specific solutions.

This product may be applied using several application techniques and methods of handling may be different for each. Application techniques include [but are not limited to] brushing, rolling, and spray application [conventional, HPLV, airless, pleural component or aerosol can]. Avoid the breathing of vapors and, if spraying, do not breath spray mist or aerosols.

SECTION 8: Exposure controls/personal protection

8.1 Control parameters

Product/ingredient name	Exposure limit values
titanium dioxide	OSHA PEL (United States, 5/2018). TWA: 15 mg/m ³ 8 hours. Form: Total dust ACGIH TLV (United States, 7/2023). TWA: 2.5 mg/m ³ 8 hours. Form: respirable fraction, finescale particles ACGIH TLV (United States, 7/2023). TWA: 20 ppm 8 hours. STEL: 75 ppm 15 minutes. NIOSH REL (United States, 10/2020). TWA: 50 ppm 10 hours. TWA: 205 mg/m ³ 10 hours. STEL: 75 ppm 15 minutes. STEL: 300 mg/m ³ 15 minutes. OSHA PEL (United States, 5/2018). TWA: 100 ppm 8 hours. TWA: 410 mg/m ³ 8 hours.
4-methylpentan-2-one	
maleic anhydride	ACGIH TLV (United States, 7/2023). Skin sensitizer. Inhalation sensitizer. TWA: 0.01 mg/m ³ 8 hours. Form: Inhalable fraction and vapor NIOSH REL (United States, 10/2020). TWA: 1 mg/m ³ 10 hours. TWA: 0.25 ppm 10 hours. OSHA PEL (United States, 5/2018). TWA: 1 mg/m ³ 8 hours. TWA: 0.25 ppm 8 hours.

Recommended monitoring procedures

If this product contains ingredients with exposure limits, personal, workplace atmosphere or biological monitoring may be required to determine the effectiveness of the ventilation or other control measures and/or the necessity to use respiratory protective equipment. Reference should be made to appropriate monitoring standards. Reference to national guidance documents for methods for the determination of hazardous substances will also be required.

8.2 Exposure controls

Appropriate engineering controls

Provide local exhaust and general ventilation systems to maintain airborne concentrations below OSHA, ACGIH, and manufacturer recommended exposure limits. Local exhaust ventilation is preferred because it prevents contaminant dispersion into work areas by controlling it at its source. Use local and general exhaust ventilation to effectively remove and prevent buildup of mists/vapors/fumes generated from the handling of this product.

Note: Local exhaust ventilation is designed to capture an emitted contaminant at or near its source, before the contaminant has a chance to disperse into the workplace air. General exhaust ventilation, also called dilution ventilation, is different from local exhaust ventilation because instead of capturing emissions at their source and removing them from the air, general exhaust ventilation allows the contaminant to be emitted into the workplace air and then dilutes the concentration of the contaminant to an acceptable level (e.g., to the PEL or below).

Individual protection measures

General :	Gloves must be worn for all work that may result in soiling. Apron/coveralls/protective clothing must be worn when soiling is so great that regular work clothes do not adequately protect skin against contact with the product. Safety eyewear should be used when there is a likelihood of exposure.
Hygiene measures :	Wash hands, forearms, and face thoroughly after handling compounds and before eating, smoking, using lavatory, and at the end of day.
Eye/face protection :	Safety eyewear complying with an approved standard should be used when a risk assessment indicates this is necessary to avoid exposure to liquid splashes, mists, gases or dusts. If contact is possible, the following protection should be worn, unless the assessment indicates a higher degree of protection: chemical splash goggles and/or face shield. If inhalation hazards exist, a full-face respirator may be required instead.
Hand protection :	Wear chemical-resistant gloves in combination with 'basic' employee training. The quality of the chemical-resistant protective gloves must be chosen as a function of the specific workplace concentrations and quantity of hazardous substances.

SECTION 8: Exposure controls/personal protection

Since the actual work situation is unknown. Supplier of gloves should be contacted in order to find the appropriate type. Below listed glove(s) should be regarded as generic advice:

Recommended: Silver Shield / Barrier / 4H gloves, nitrile rubber (>0.3 mm), butyl rubber (>0.5 mm), Viton®

May be used: nitrile rubber (>0.1 mm), butyl rubber (>0.3 mm)

Short term exposure: neoprene rubber (>0.1 mm), natural rubber (latex) (>0.4 mm), polyvinyl alcohol (PVA), polyvinyl chloride (PVC)

Body protection :

Personal protective equipment for the body should be selected based on the task being performed and the risks involved handling this product.

Wear suitable protective clothing. Always wear protective clothing when spraying.

Chemical-resistant apron.

Respiratory protection :

Wear appropriate respirator when ventilation is inadequate. Be sure to use approved/certified respirator or equivalent. It is not possible to specify precise filter type, since the actual work situation is unknown.

Supplier of respirators should be contacted in order to find the appropriate filter.

Protective clothing (pictograms) :



Note: Application of paint products by spraying requires additional safety precautions: Full body suit, Full face respirator with air supplied.

Environmental exposure controls

Emissions from ventilation or work process equipment should be checked to ensure they comply with the requirements of environmental protection legislation. In some cases, fume scrubbers, filters or engineering modifications to the process equipment will be necessary to reduce emissions to acceptable levels.

SECTION 9: Physical and chemical properties

9.1 Information on basic physical and chemical properties

Physical state :	Liquid.
Color :	Gray
Odor :	Amine-like.
pH :	Testing not relevant or not possible due to nature of the product.
Melting point/freezing point :	-16°C This is based on data for the following ingredient: bisphenol A-(epichlorhydrin) epoxy resin MW = < 700
Boiling point/boiling range :	Testing not relevant or not possible due to nature of the product.
Flash point :	Closed cup: 139°C (282.2°F)
Evaporation rate :	Testing not relevant or not possible due to nature of the product.
Flammability :	Not available.
Upper/lower flammability or explosive limits :	No specific data.
Vapor pressure :	0 kPa This is based on data for the following ingredient: bisphenol A-(epichlorhydrin) epoxy resin MW = < 700
Vapor density :	Testing not relevant or not possible due to nature of the product.
Relative density :	1.13 g/cm³
Partition coefficient (LogKow) :	Testing not relevant or not possible due to nature of the product.
Auto-ignition temperature :	Testing not relevant or not possible due to nature of the product.
Decomposition temperature :	Testing not relevant or not possible due to nature of the product.
Viscosity :	Testing not relevant or not possible due to nature of the product.
Explosive properties :	Not available.
Oxidizing properties :	Testing not relevant or not possible due to nature of the product.

9.2 Other information

Safety Data Sheet

NEOGARD FLEXIBLE EPOXY JOINT COMPOUND

70718-02 GRAY



SECTION 9: Physical and chemical properties

Solvent(s) % by weight (Included exempt solvent(s)):	0.47 % (w/w)
Water % by weight :	Weighted average: 0 %
VOC content (Coatings) :	0.0446 lbs/gal (5.3 g/l)
VOC content (Regulatory) :	0.0446 lbs/gal (5.3 g/l)
TOC Content (Volatile) :	Weighted average: 4 g/l
Solvent Gas :	Weighted average: 0.001 m³/l

SECTION 10: Stability and reactivity

10.1 Reactivity

No specific test data related to reactivity available for this product or its ingredients.

10.2 Chemical stability

The product is stable.

10.3 Possibility of hazardous reactions

Under normal conditions of storage and use, hazardous reactions will not occur.

10.4 Conditions to avoid

No specific data.

10.5 Incompatible materials

10.6 Hazardous decomposition products

When exposed to high temperatures (i.e. in case of fire) harmful decomposition products may be formed:

Decomposition products may include the following materials: carbon oxides halogenated compounds metal oxide/oxides

SECTION 11: Toxicological information

11.1 Information on toxicological effects

Epoxy and amine containing products can cause skin disorders such as allergic eczema. The allergy may arise after only a short exposure period.

Acute toxicity

Product/ingredient name	Result	Species	Dose	Exposure
bisphenol A-(epichlorhydrin) epoxy resin MW =< 700	LD50 Dermal	Rabbit	>2000 mg/kg	-
	LD50 Dermal	Rat	>2000 mg/kg	-
	LD50 Oral	Rat	>2000 mg/kg	-
oxirane, mono[(C12-14-alkyloxy) methyl] derivs.	LD50 Dermal	Rat	>4500 mg/kg	-
	LD50 Oral	Rat	>5000 mg/kg	-
titanium dioxide	LC50 Inhalation Dusts and mists	Rat	>6.8 mg/l	4 hours
	LD50 Dermal	Rabbit	>5000 mg/kg	-
	LD50 Oral	Rat	>5000 mg/kg	-
cashew nutshell liquid	LD50 Oral	Rat	500 mg/kg	-
	LC50 Inhalation Vapor	Rat	11 mg/l	4 hours
4-methylpentan-2-one	LD Dermal	Rabbit	>3 g/kg	-
maleic anhydride	LD50 Dermal	Rabbit	2620 mg/kg	-
	LD50 Oral	Rat	1090 mg/kg	-

Acute toxicity estimates

Route	ATE value
Oral	33076.99 mg/kg
Dermal	72769.38 mg/kg

Safety Data Sheet

NEOGARD FLEXIBLE EPOXY JOINT COMPOUND

70718-02 GRAY



SECTION 11: Toxicological information

Irritation/Corrosion

Product/ingredient name	Result	Species	Score	Exposure
bisphenol A-(epichlorhydrin) epoxy resin MW =< 700	Eyes - Mild irritant	Rabbit	-	-
oxirane, mono[(C12-14-alkyloxy)methyl] derivs.	Skin - Mild irritant	Rabbit	-	-
	Eyes - Mild irritant	Rabbit	-	-
titanium dioxide	Skin - Moderate irritant	Rabbit	-	-
4-methylpentan-2-one	Skin - Mild irritant	Human	-	72 hours 300 Micrograms Intermittent
	Eyes - Moderate irritant	Rabbit	-	24 hours 100 microliters
	Skin - Mild irritant	Rabbit	-	24 hours 500 milligrams
maleic anhydride	Eyes - Severe irritant	Rabbit	-	1 Percent
	Skin - Severe irritant	Rabbit	-	-

Sensitizer

Product/ingredient name	Route of exposure	Species	Result
bisphenol A-(epichlorhydrin) epoxy resin MW =< 700	skin	Guinea pig	Sensitizing
oxirane, mono[(C12-14-alkyloxy)methyl] derivs.	skin	Guinea pig	Sensitizing

Carcinogen Classification

Product/ingredient name	IARC	NTP	OSHA
titanium dioxide	2B	-	-
4-methylpentan-2-one	2B	-	-

Specific target organ toxicity (single exposure)

Product/ingredient name	Category	Route of exposure	Target organs
4-methylpentan-2-one	Category 3		Respiratory tract irritation

Specific target organ toxicity (repeated exposure)

Product/ingredient name	Category	Route of exposure	Target organs
maleic anhydride	Category 1	inhalation	respiratory system

Information on the likely routes of exposure

Routes of entry anticipated: Oral, Dermal, Inhalation.

Potential chronic health effects

Sensitization : Contains bisphenol A-(epichlorhydrin) epoxy resin MW =< 700, oxirane, mono[(C12-14-alkyloxy)methyl] derivs., cashew nutshell liquid. May produce an allergic reaction.

Other information : No additional known significant effects or critical hazards.

SECTION 12: Ecological information

12.1 Toxicity

Do not allow to enter drains or watercourses. Toxic to aquatic life with long lasting effects.

When spilled, this product may act as an oil, causing a film, sheen, emulsion, or sludge at or beneath the surface of a body of water. Oils of any kind can cause: (a) drowning of waterfowl due to lack of buoyancy, loss of insulating capacity of feathers, starvation and vulnerability to predators due to lack of mobility; (b) lethal effect on fish by coating gill surfaces, preventing respiration; (c) potential fish kills resulting from alteration in biochemical oxygen demand; (d) asphyxiation of benthic life forms when floating masses become engaged with surface debris and settle on the bottom; and (e) adverse aesthetic effects of fouled shoreline and beaches.

Safety Data Sheet

NEOGARD FLEXIBLE EPOXY JOINT COMPOUND

70718-02 GRAY



SECTION 12: Ecological information

Product/ingredient name	Result	Species	Exposure
bisphenol A-(epichlorhydrin) epoxy resin MW =< 700	Acute EC50 11 mg/l	Algae	72 hours
oxirane, mono[(C12-14-alkyloxy) methyl] derivs.	Acute EC50 1.8 mg/l Acute LC50 2 mg/l Acute IC50 843.75 mg/l	Daphnia Fish Algae	48 hours 96 hours 72 hours
titanium dioxide	Acute LC50 5000 mg/l Acute LC50 >100 mg/l Acute LC50 >100 mg/l	Fish Daphnia Fish	96 hours 48 hours 96 hours
4-methylpentan-2-one	Chronic NOEC 7800 - 39000 µg/l Fresh water Chronic NOEC 168 mg/l Fresh water	Daphnia - Daphnia magna Fish - Pimephales promelas - Embryo	21 days 33 days

12.2 Persistence and degradability

Product/ingredient name	Test	Result	Dose	Inoculum
bisphenol A-(epichlorhydrin) epoxy resin MW =< 700	OECD 302B Inherent Biodegradability: Zahn-Wellens/EMPA Test	12 % - Not readily - 28 days	-	-
oxirane, mono[(C12-14-alkyloxy) methyl] derivs.	-	87 % - Readily - 28 days	-	-
4-methylpentan-2-one	-	84 % - 14 days	100 mg/l	-

Product/ingredient name	Aquatic half-life	Photolysis	Biodegradability
bisphenol A-(epichlorhydrin) epoxy resin MW =< 700	-	-	Not readily
oxirane, mono[(C12-14-alkyloxy) methyl] derivs.	-	-	Readily
4-methylpentan-2-one	-	-	Readily

12.3 Bioaccumulative potential

Product/ingredient name	LogP _{ow}	BCF	Potential
bisphenol A-(epichlorhydrin) epoxy resin MW =< 700	2.64 - 3.78	31	low
oxirane, mono[(C12-14-alkyloxy)methyl] derivs.	3.77	160 - 263	low
cashew nutshell liquid	>4.78	-	high
4-methylpentan-2-one	1.31	2	low
maleic anhydride	-2.78	-	low

12.4 Mobility in soil

Soil/water partition coefficient (K_{oc}) : No known data available in our database.

Mobility : No known data available in our database.

12.5 Other adverse effects

No known significant effects or critical hazards.

SECTION 13: Disposal considerations

13.1 Waste treatment methods

Disposal should be in accordance with applicable regional, national and local laws and regulations. Local regulations may be more stringent than regional or national requirements.

The information presented below only applies to the material as supplied. The identification based on characteristic(s) or listing may not apply if the material has been used or otherwise contaminated. It is the responsibility of the waste generator to determine the toxicity and physical properties of the material generated to determine the proper waste identification and disposal methods in compliance with applicable regulations.

Refer to Section 7 and Section 8 for additional handling information and protection of employees.

SECTION 13: Disposal considerations

The generation of waste should be avoided or minimized wherever possible. Disposal of this product, solutions and any by-products should at all times comply with the requirements of environmental protection and waste disposal legislation and any regional local authority requirements. Dispose of surplus and non-recyclable products via a licensed waste disposal contractor. Waste should not be disposed of untreated to the sewer unless fully compliant with the requirements of all authorities with jurisdiction. Waste packaging should be recycled. Incineration or landfill should only be considered when recycling is not feasible. This material and its container must be disposed of in a safe way. Care should be taken when handling emptied containers that have not been cleaned or rinsed out. Empty containers or liners may retain some product residues. Avoid dispersal of spilled material and runoff and contact with soil, waterways, drains and sewers.

SECTION 14: Transport information

Transport may take place according to national regulation or DOT for transport by road and by train, IMDG for transport by sea, IATA for Air shipment. Refer to specific Dangerous Goods Transport requirements under 49CFR, ICAO and IATA.

	14.1 UN no.	14.2 Proper shipping name	14.3 Transport hazard class(es)	14.4 PG*	14.5 Env*	Additional information
DOT Code	UN3082	ENVIRONMENTALLY HAZARDOUS SUBSTANCE, LIQUID, N.O.S.. (bisphenol A-(epichlorhydrin) epoxy resin MW =< 700)	9 -  	III	Yes.	The marine pollutant mark is not required when transported on inland waterways in sizes of ≤5 L or ≤5 kg or by road, rail, or inland air in non-bulk sizes.
TDG Code	UN3082	ENVIRONMENTALLY HAZARDOUS SUBSTANCE, LIQUID, N.O.S.. (bisphenol A-(epichlorhydrin) epoxy resin MW =< 700)	9 -  	III	Yes.	Product classified as per the following sections of the Transportation of Dangerous Goods Regulations: 2.43-2.45 (Class 9), 2.7 (Marine pollutant mark). Non-bulk packages of this product are not regulated as dangerous goods when transported by road or rail.
SCT Code	UN3082	ENVIRONMENTALLY HAZARDOUS SUBSTANCE, LIQUID, N.O.S.. (bisphenol A-(epichlorhydrin) epoxy resin MW =< 700)	9 -  	III	Yes.	The environmentally hazardous substance mark is not required when transported in sizes of ≤5 L or ≤5 kg.
IMDG Code	UN3082	ENVIRONMENTALLY HAZARDOUS SUBSTANCE, LIQUID, N.O.S.. (bisphenol A-(epichlorhydrin) epoxy resin MW =< 700)	9 -  	III	Yes.	This product is not regulated as a dangerous good when transported in sizes of ≤5 L or ≤5 kg, provided the packagings meet the general provisions of 4.1.1.1, 4.1.1.2 and 4.1.1.4 to 4.1.1.8. Emergency schedules F-A, S-F
IATA Code	UN3082	ENVIRONMENTALLY HAZARDOUS SUBSTANCE, LIQUID, N.O.S.. (bisphenol A-(epichlorhydrin) epoxy resin MW =< 700)	9 -  	III	Yes.	This product is not regulated as a dangerous good when transported in sizes of ≤5 L or ≤5 kg, provided the packagings meet the general provisions of 5.0.2.4.1, 5.0.2.6.1.1 and 5.0.2.8.

Code : Classification

PG* : Packing group

Env.* : Environmental hazards

14.6 Special precautions for user

Transport within user's premises: always transport in closed containers that are upright and secure. Ensure that persons transporting the product know what to do in the event of an accident or spillage.

14.7 Transport in bulk according to IMO instruments

Not applicable.

Safety Data Sheet

NEOGARD FLEXIBLE EPOXY JOINT COMPOUND

70718-02 GRAY



SECTION 15: Regulatory information

15.1 Safety, health and environmental regulations/legislation specific for the substance or mixture

U.S. Federal regulations :

Not determined.

TSCA 8(a) PAIR: silica/polydimethylsiloxane reaction product

TSCA 8(a) CDR Exempt/Partial exemption: Not determined

United States inventory (TSCA 8b): Not determined.

Clean Water Act (CWA) 307: ethylbenzene

Clean Water Act (CWA) 311: xylene; ethylbenzene; 1-chloro-2,3-epoxypropane; maleic anhydride

Clean Air Act Section 112(b) Hazardous Air Pollutants (HAPs) : Listed

Product/ingredient name	CAS number	Concentration
4-methylpentan-2-one	108-10-1	0.26038
xylene	1330-20-7	0.06622
ethylbenzene	100-41-4	0.02896
1-chloro-2,3-epoxypropane	106-89-8	0.0058807
maleic anhydride	108-31-6	0.0010329

Clean Air Act Section 602 Class I Substances : Not listed

Clean Air Act Section 602 Class II Substances : Not listed

DEA List I Chemicals (Precursor Chemicals) : Not listed

DEA List II Chemicals (Essential Chemicals) : Not listed

SARA 302/304 :

Product/ingredient name	%	EHS	SARA 302 TPQ		SARA 304 RQ	
			(lbs)	(gallons)	(lbs)	(gallons)
1-chloro-2,3-epoxypropane	<0.1	Yes.	1000	101.6	100	10.2

SARA 304 RQ :

1700484 lbs / 772019.7 kg [180483.2 gal / 683203.3 L]

SARA 311/312 Classification :

SKIN IRRITATION - Category 2
 SERIOUS EYE DAMAGE - Category 1
 SKIN SENSITIZATION - Category 1
 CARCINOGENICITY - Category 2

Product/ingredient name	%	Classification
bisphenol A-(epichlorhydrin) epoxy resin MW = < 700	≥25 - ≤50	SKIN IRRITATION - Category 2 EYE IRRITATION - Category 2A SKIN SENSITIZATION - Category 1
blocked polyurethane oxirane, mono[(C12-14-alkyloxy)methyl] derivs.	≥25 - ≤50 ≥5 - ≤10	SERIOUS EYE DAMAGE - Category 1 SKIN IRRITATION - Category 2 SKIN SENSITIZATION - Category 1
cashew nutshell liquid	≥1 - ≤3	ACUTE TOXICITY (oral) - Category 4 ACUTE TOXICITY (dermal) - Category 4 SKIN IRRITATION - Category 2 SERIOUS EYE DAMAGE - Category 1
4-methylpentan-2-one	≤0.3	SKIN SENSITIZATION - Category 1A FLAMMABLE LIQUIDS - Category 2 ACUTE TOXICITY (inhalation) - Category 4 EYE IRRITATION - Category 2A CARCINOGENICITY - Category 2 SPECIFIC TARGET ORGAN TOXICITY (SINGLE EXPOSURE) (Respiratory tract irritation) - Category 3
maleic anhydride	<0.1	HNOC - Defatting irritant ACUTE TOXICITY (oral) - Category 4 SKIN CORROSION - Category 1B SERIOUS EYE DAMAGE - Category 1 RESPIRATORY SENSITIZATION - Category 1 SKIN SENSITIZATION - Category 1A SPECIFIC TARGET ORGAN TOXICITY (REPEATED EXPOSURE) - Category 1

State regulations :

Connecticut Carcinogen Reporting: None of the components are listed.

Connecticut Hazardous Material Survey: None of the components are listed.

Florida substances: None of the components are listed.

Illinois Chemical Safety Act: None of the components are listed.

Illinois Toxic Substances Disclosure to Employee Act: None of the components are listed.

Louisiana Reporting: None of the components are listed.

Louisiana Spill: None of the components are listed.

Massachusetts Substances: The following components are listed: TITANIUM DIOXIDE

Massachusetts Spill: None of the components are listed.

Michigan Critical Material: None of the components are listed.

Minnesota Hazardous Substances: None of the components are listed.

SECTION 15: Regulatory information

New Jersey Spill: None of the components are listed.

New Jersey Toxic Catastrophe Prevention Act: None of the components are listed.

New Jersey Hazardous Substances: The following components are listed: TITANIUM DIOXIDE

New York Hazardous Substances: None of the components are listed.

New York Toxic Chemical Release Reporting: None of the components are listed.

Pennsylvania RTK Hazardous Substances: The following components are listed: TITANIUM OXIDE

Rhode Island Hazardous Substances: None of the components are listed.

California Prop. 65 PFF :

WARNING: This product can expose you to chemicals including Methyl isobutyl ketone and Epichlorohydrin, which are known to the State of California to cause cancer and birth defects or other reproductive harm. This product can expose you to chemicals including Titanium dioxide, Carbon black and Ethylbenzene, which are known to the State of California to cause cancer, and Bisphenol A, which is known to the State of California to cause birth defects or other reproductive harm. For more information go to www.P65Warnings.ca.gov.

Product/ingredient name	Cancer	Reproductive	No significant risk level	Maximum acceptable dosage level
titanium dioxide	Yes.	No.		
4-methylpentan-2-one	Yes.	Yes.		
carbonblack	Yes.	No.		
4,4'-isopropylidenediphenol	No.	Yes.		
ethylbenzene	Yes.	No.	Yes.	Yes.
1-chloro-2,3-epoxypropane	Yes.	Yes.	Yes.	

SECTION 16: Other information

Remarks :

Note: In USA, consult Code of Federal Regulations, Title 29, Labor, Parts 1910 and 1915 concerning occupational safety and health standards and regulations, as well as any other applicable Federal, State or local regulations that apply to safe practices in coating operations.

Warning! If you scrape, sand, or remove old paint, you may release lead dust. LEAD is TOXIC.

Validation :

Validated by US - HSE Products Coordinator on 28 March 2024

GHS Classification

Procedure used to derive the classification.

Classification	Justification
SKIN IRRITATION - Category 2	Calculation method
SERIOUS EYE DAMAGE - Category 1	Calculation method
SKIN SENSITIZATION - Category 1	Calculation method
CARCINOGENICITY - Category 2	Calculation method

Hazardous Material Information System (U.S.A.)

Health	3
Fire hazard	1
Physical hazards	0
Personal protection	X

National Fire Protection Association (U.S.A.)

Health	1	0	Flammability
			Instability
			Special

Personal Protective Equipment (PPE) shown in this section is a suggestion. Since conditions vary from one work location to another consult the facility safety & health program. Customer or end user is responsible to evaluate worker exposure conditions at the site of application and determine the appropriate PPE suitable for workers at that particular facility or location.

Abbreviations and acronyms :

ANSI = American National Standards Institute
HCS = Hazardous Communication System
TSCA = Toxic Substances Control Act
CFR = Code of federal Regulations
GHS = Globally Harmonized System of Classification and Labelling of Chemicals
OSHA = United States Occupational Health and Safety Administration
NIOSH = National Institute for Occupational Safety and Health
ACGIH = American Conference of Industrial Hygienists
IARC = International Agency for Research on Cancer.
NTP = National Toxicology Program
ATE = Acute Toxicity Estimate

OECD = Organisation for Economic Co-operation and Development
BCF = Bioconcentration Factor
DOT = United States Department of Transportation
ERG = Emergency Response Guide
TDG = Transport of Dangerous Goods, Canada
SCT = Transportation & Communications Ministry, Mexico
IMDG = International Maritime Dangerous Goods
IATA = International Air Transport Association
SARA = Superfund Amendments Reauthorization Act
EPCRA = Emergency Planning and Community Right to Know Act

Notice to reader

 Indicates information that has changed from previously issued version.

To the best of our knowledge, the information contained herein is accurate. However, neither the above named supplier nor any of its subsidiaries assumes any liability whatsoever for the accuracy or completeness of the information contained herein. Final determination of suitability of any material is the sole responsibility of the user. All materials may present unknown hazards and should be used with caution. Although certain hazards are described herein, we cannot guarantee that these are the only hazards that exist.

Safety Data Sheet

NEOGARD FLEXIBLE EPOXY CURING AGENT 70719



Conforms to ANSI Z400.1-2010 Standard - HCS 2012

Protective Clothing	General Hazard	DOT
Consult your supervisor or S.O.P. for special handling	  	 

SECTION 1: Identification of the substance/mixture and of the company/undertaking

1.1 Product identifier

Product name : NEOGARD FLEXIBLE EPOXY CURING AGENT
Product identity : 9504800000, 70719
Product type : Catalyst.

1.2 Relevant identified uses of the substance or mixture and uses advised against

Field of application : used only as part of two- or multi component products.
Ready-for-use mixture : Used for: 25000
Identified uses : Industrial/Professional use
TSCA : **Unless otherwise stated. All components are listed or exempted.**

1.3 Details of the supplier of the safety data sheet

Company details : NEOGARD, a Division of Hempel (USA), Inc.
2728 Empire Central
Dallas, TX 75235
Phone number: 1-214-353-1600
E-mail: hempel@hempel.com

1.4 Emergency telephone number (with hours of operation)

For Transportation Emergencies : CHEMTREC: **1-800-424-9300** (Toll-free in the U.S., Canada and the U.S. Virgin Islands) **703-527-3887**
(24 hours) For calls originating elsewhere (Collect calls are accepted). Contract number: CCN10384
To preserve the effectiveness of arrangements for providing accurate and timely emergency response information, the basic identifying information (shipper name or contract number) must be included on shipping papers.
If the purchaser of this product is going to be shipping this product to other locations, the purchaser must arrange for its own Emergency Information Provider to respond to transport incidents. Hempel's 24 hour response contract does not cover non-Hempel shipments.
For all other information : In USA toll free calling available: 1-800- 678-6641 or (936)-523-6000
(8 AM - 5 PM CST) See Section 4 of the safety data sheet (first aid measures).

SECTION 2: Hazards identification

2.1 Classification of the substance or mixture

OSHA/HCS status : This material is considered hazardous by the OSHA Hazard Communication Standard (29 CFR 1910.1200).
GHS Classification : FLAMMABLE LIQUIDS - Category 4
ACUTE TOXICITY (inhalation) - Category 4
SKIN CORROSION - Category 1B
SERIOUS EYE DAMAGE - Category 1
SKIN SENSITIZATION - Category 1
TOXIC TO REPRODUCTION - Category 1B

2.2 Label elements

Hazard pictograms :



Safety Data Sheet

NEOGARD FLEXIBLE EPOXY CURING AGENT

70719



SECTION 2: Hazards identification

Signal word :	Danger
Hazard statements :	H227 - Combustible liquid. H314 - Causes severe skin burns and eye damage. H317 - May cause an allergic skin reaction. H332 - Harmful if inhaled. H360 - May damage fertility or the unborn child.
Precautionary statements :	
Prevention :	Obtain special instructions before use. Do not handle until all safety precautions have been read and understood. Wear protective gloves, protective clothing and eye or face protection. Keep away from flames and hot surfaces. No smoking. Use only outdoors or in a well-ventilated area. Avoid breathing vapor. Wash thoroughly after handling. Contaminated work clothing must not be allowed out of the workplace.
Response :	IF exposed or concerned: Get medical advice or attention. IF INHALED: Remove person to fresh air and keep comfortable for breathing. Immediately call a POISON CENTER or doctor. Call a POISON CENTER or doctor if you feel unwell. IF SWALLOWED: Immediately call a POISON CENTER or doctor. Rinse mouth. Do NOT induce vomiting. IF ON SKIN (or hair): Take off immediately all contaminated clothing. Rinse skin with water. Immediately call a POISON CENTER or doctor. IF ON SKIN: Wash with plenty of water. If skin irritation or rash occurs: Get medical advice or attention. IF IN EYES: Rinse cautiously with water for several minutes. Remove contact lenses, if present and easy to do. Continue rinsing. Immediately call a POISON CENTER or doctor.
Storage :	Store locked up. Store in a well-ventilated place. Keep cool.
Disposal :	Dispose of contents and container in accordance with all local, regional, national and international regulations.
Supplemental label elements :	None known.

2.3 Other hazards

Hazards not otherwise classified : None known.

SECTION 3: Composition/information on ingredients

Product definition :	Mixture
Physical state :	Liquid.

Product/ingredient name	Identifiers	%	GHS Classification
reactionproduct of dimer C18-fatty acids with polyethylenepolyamines benzyl alcohol	68410-23-1	≥25 - ≤50	SERIOUS EYE DAMAGE - Category 1
	100-51-6	≥10 - ≤25	ACUTE TOXICITY (oral) - Category 4 ACUTE TOXICITY (inhalation) - Category 4 EYE IRRITATION - Category 2A SKIN IRRITATION - Category 2 SKIN SENSITIZATION - Category 1A ACUTE TOXICITY (dermal) - Category 4 SKIN CORROSION - Category 1B SERIOUS EYE DAMAGE - Category 1 SKIN SENSITIZATION - Category 1 ACUTE TOXICITY (oral) - Category 4 SERIOUS EYE DAMAGE - Category 1 TOXIC TO REPRODUCTION - Category 2 SERIOUS EYE DAMAGE - Category 1 SKIN SENSITIZATION - Category 1 TOXIC TO REPRODUCTION - Category 1B SPECIFIC TARGET ORGAN TOXICITY (SINGLE EXPOSURE) (Respiratory tract irritation) - Category 3 ACUTE TOXICITY (oral) - Category 4 ACUTE TOXICITY (dermal) - Category 4 ACUTE TOXICITY (inhalation) - Category 2 SKIN CORROSION - Category 1B SERIOUS EYE DAMAGE - Category 1 SKIN SENSITIZATION - Category 1 SPECIFIC TARGET ORGAN TOXICITY (SINGLE EXPOSURE) (Respiratory tract irritation) - Category 3
2,4-bis(ethenyl)phenol	61788-44-1	≥10 - ≤25	
3,6-diazaoctanethylenediamin	112-24-3	≥5 - ≤10	
salicylic acid	69-72-7	≥1 - ≤3	
4,4'-isopropylidenediphenol	80-05-7	≥1 - ≤3	
2,2'-iminodiethylamine	111-40-0	≥1 - ≤3	

Any concentration shown as a range is to protect confidentiality or is due to batch variation.

Safety Data Sheet

NEOGARD FLEXIBLE EPOXY CURING AGENT

70719



SECTION 3: Composition/information on ingredients

Occupational exposure limits, if available, are listed in Section 8.

There are no additional ingredients present which, within the current knowledge of the supplier and in the concentrations applicable, are classified as hazardous to health or the environment and hence require reporting in this section.

SECTION 4: First aid measures

4.1 Description of first aid measures

General :	In all cases of doubt, or when symptoms persist, seek medical attention. Never give anything by mouth to an unconscious person. If breathing is irregular, drowsiness, loss of consciousness or cramps: Call 911 and give immediate treatment (first aid).
Eye contact :	Check for and remove any contact lenses. Immediately flush eyes with plenty of water for at least 15 minutes, occasionally lifting the upper and lower eyelids. Seek immediate medical attention/advice.
Inhalation :	Remove to fresh air and keep at rest in a position comfortable for breathing. Give nothing by mouth. If not breathing, if breathing is irregular or if respiratory arrest occurs, provide artificial respiration or oxygen by trained personnel. If unconscious, place in recovery position and get medical attention immediately.
Skin contact :	Remove contaminated clothing and shoes. Wash skin thoroughly with soap and water or use recognized skin cleanser. Do NOT use solvents or thinners. In case of burns flush with water until the pain ceases. While flushing remove clothing from the affected area unless it is burnt into the skin. If hospital treatment is necessary flushing must continue during transfer and until the hospital staff takes over the treatment.
Ingestion :	If swallowed, seek medical advice immediately and show this container or label. Keep person warm and at rest. Do not induce vomiting unless directed to do so by medical personnel. Lower the head so that vomit will not re-enter the mouth and throat.
Protection of first-aiders :	No action shall be taken involving any personal risk or without suitable training. If it is suspected that fumes are still present, the rescuer should wear an appropriate mask or self-contained breathing apparatus. It may be dangerous to the person providing aid to give mouth-to-mouth resuscitation. Wash contaminated clothing thoroughly with water before removing it, or wear gloves.

4.2 Most important symptoms and effects, both acute and delayed

Potential acute health effects

Eye contact :	Causes serious eye damage.
Inhalation :	Harmful if inhaled.
Skin contact :	Causes severe burns. May cause an allergic skin reaction.
Ingestion :	No known significant effects or critical hazards.

Over-exposure signs/symptoms

Eye contact :	Adverse symptoms may include the following: pain watering redness
Inhalation :	Adverse symptoms may include the following: reduced fetal weight increase in fetal deaths skeletal malformations
Skin contact :	Adverse symptoms may include the following: pain or irritation redness blistering may occur reduced fetal weight increase in fetal deaths skeletal malformations
Ingestion :	Adverse symptoms may include the following: stomach pains reduced fetal weight increase in fetal deaths skeletal malformations

SECTION 4: First aid measures

4.3 Indication of any immediate medical attention and special treatment needed

Notes to physician : If gasses have been inhaled, from the decomposition of the product, symptoms may be delayed.

Specific treatments : No specific treatment.

SECTION 5: Firefighting measures

5.1 Extinguishing media

Extinguishing media : Recommended: alcohol resistant foam, CO₂, powders, water spray.
Not to be used: waterjet.

5.2 Special hazards arising from the substance or mixture

Hazards from the substance or mixture : In a fire or if heated, a pressure increase will occur and the container may burst. This material is toxic to aquatic life with long lasting effects. Fire water contaminated with this material must be contained and prevented from being discharged to any waterway, sewer or drain.

Hazardous combustion products : Decomposition products may include the following materials: carbon oxides nitrogen oxides

5.3 Advice for firefighters

Promptly isolate the scene by removing all persons from the vicinity of the incident if there is a fire. No action shall be taken involving any personal risk or without suitable training. Fire will produce dense black smoke. Exposure to decomposition products may cause a health hazard. Cool closed containers exposed to fire with water. Do not release runoff from fire to drains or watercourses. Fire-fighters should wear appropriate protective equipment and self-contained breathing apparatus (SCBA) with a full face-piece operated in positive pressure mode.

SECTION 6: Accidental release measures

6.1 Personal precautions, protective equipment and emergency procedures

Avoid all direct contact with the spilled material. Exclude sources of ignition and be aware of explosion hazard. Ventilate the area. Avoid breathing vapor or mist. Refer to protective measures listed in sections 7 and 8. No action shall be taken involving any personal risk or without suitable training. If the product contaminates lakes, rivers, or sewers, inform the appropriate authorities in accordance with local regulations.

6.2 Environmental precautions

Avoid dispersal of spilled material and runoff and contact with soil, waterways, drains and sewers. Inform the relevant authorities if the product has caused environmental pollution (sewers, waterways, soil or air). Water polluting material. May be harmful to the environment if released in large quantities.

6.3 Methods and materials for containment and cleaning up

Stop leak if without risk. Move containers from spill area. Approach release from upwind. Prevent entry into sewers, water courses, basements or confined areas. Wash spillages into an effluent treatment plant or proceed as follows. Contain and collect spillage with non-combustible, absorbent material e.g. sand, earth, vermiculite or diatomaceous earth and place in container for disposal according to local regulations (see Section 13). Use spark-proof tools and explosion-proof equipment. Contaminated absorbent material may pose the same hazard as the spilled product.

6.4 Reference to other sections

See Section 1 for emergency contact information.
See Section 8 for information on appropriate personal protective equipment.
See Section 13 for additional waste treatment information.

SECTION 7: Handling and storage

7.1 Precautions for safe handling

Vapors are heavier than air and may spread along floors. Vapors may form explosive mixtures with air. Prevent the creation of flammable or explosive concentrations of vapors in air and avoid vapor concentrations higher than the occupational exposure limits. In addition, the product should be used only in areas from which all naked lights and other sources of ignition have been excluded. Electrical equipment should be protected to the appropriate standard. To dissipate static electricity during transfer, ground drum and connect to receiving container with bonding strap. No sparking tools should be used.

Avoid inhalation of vapour, dust and spray mist. Avoid contact with skin and eyes. Eating, drinking and smoking should be prohibited in area where this material is handled, stored and processed. Appropriate personal protective equipment: see Section 8. Always keep in containers made from the same material as the original one.

Safety Data Sheet

NEOGARD FLEXIBLE EPOXY CURING AGENT

70719



SECTION 7: Handling and storage

7.2 Conditions for safe storage, including any incompatibilities

Store in accordance with local regulations. Store in a cool, well-ventilated area away from incompatible materials and ignition sources. Keep out of the reach of children. Keep away from: Oxidizing agents, strong alkalis, strong acids. No smoking. Prevent unauthorized access. Containers that are opened must be carefully resealed and kept upright to prevent leakage.

7.3 Specific end use(s)

See separate Product Data Sheet for recommendations or industrial sector specific solutions.

SECTION 8: Exposure controls/personal protection

8.1 Control parameters

Product/ingredient name	Exposure limit values
benzyl alcohol	OARS WEEL (United States, 4/2022). TWA: 10 ppm 8 hours.
3,6-diazaoctanethylenediamin	OARS WEEL (United States, 4/2022). Absorbed through skin. TWA: 1 ppm 8 hours.
2,2'-iminodiethylamine	ACGIH TLV (United States, 1/2023). Absorbed through skin. TWA: 1 ppm 8 hours. TWA: 4.2 mg/m ³ 8 hours. NIOSH REL (United States, 10/2020). Absorbed through skin. TWA: 1 ppm 10 hours. TWA: 4 mg/m ³ 10 hours.

Recommended monitoring procedures

If this product contains ingredients with exposure limits, personal, workplace atmosphere or biological monitoring may be required to determine the effectiveness of the ventilation or other control measures and/or the necessity to use respiratory protective equipment. Reference should be made to appropriate monitoring standards. Reference to national guidance documents for methods for the determination of hazardous substances will also be required.

8.2 Exposure controls

Appropriate engineering controls

Provide local exhaust and general ventilation systems to maintain airborne concentrations below OSHA, ACGIH, and manufacturer recommended exposure limits. Local exhaust ventilation is preferred because it prevents contaminant dispersion into work areas by controlling it at its source. Use local and general exhaust ventilation to effectively remove and prevent buildup of mists/vapors/fumes generated from the handling of this product.

Note: Local exhaust ventilation is designed to capture an emitted contaminant at or near its source, before the contaminant has a chance to disperse into the workplace air. General exhaust ventilation, also called dilution ventilation, is different from local exhaust ventilation because instead of capturing emissions at their source and removing them from the air, general exhaust ventilation allows the contaminant to be emitted into the workplace air and then dilutes the concentration of the contaminant to an acceptable level (e.g., to the PEL or below).

Individual protection measures

General :	Gloves must be worn for all work that may result in soiling. Apron/coveralls/protective clothing must be worn when soiling is so great that regular work clothes do not adequately protect skin against contact with the product. Safety eyewear should be used when there is a likelihood of exposure.
Hygiene measures :	Wash hands, forearms, and face thoroughly after handling compounds and before eating, smoking, using lavatory, and at the end of day.
Eye/face protection :	Safety eyewear complying with an approved standard should be used when a risk assessment indicates this is necessary to avoid exposure to liquid splashes, mists, gases or dusts. If contact is possible, the following protection should be worn, unless the assessment indicates a higher degree of protection: chemical splash goggles and/or face shield. If inhalation hazards exist, a full-face respirator may be required instead.
Hand protection :	Wear chemical-resistant gloves in combination with 'basic' employee training. The quality of the chemical-resistant protective gloves must be chosen as a function of the specific workplace concentrations and quantity of hazardous substances.

Safety Data Sheet

NEOGARD FLEXIBLE EPOXY CURING AGENT

70719



SECTION 8: Exposure controls/personal protection

Since the actual work situation is unknown. Supplier of gloves should be contacted in order to find the appropriate type. Below listed glove(s) should be regarded as generic advice:

Short term exposure: neoprene rubber, natural rubber (latex), polyvinyl chloride (PVC)
Recommended: polyvinyl alcohol (PVA), Silver Shield / Barrier / 4H gloves, Viton®
May be used: butyl rubber, nitrile rubber

Body protection :	Personal protective equipment for the body should be selected based on the task being performed and the risks involved handling this product. Wear suitable protective clothing. Chemical-resistant apron.
Respiratory protection :	If working areas have insufficient ventilation, wear half or totally covering mask equipped with gas filter of type Organic Vapor, when grinding use particle filter of type P95, P99 or P100. When spraying use a combined filter (organic vapor / HEPA or organic vapor / P100 type). Be sure to use approved/certified respirator or equivalent. Always wear an air-fed respirator when spraying in a continuous and prolonged work situation (e.g. hood with supply of fresh or compressed air or a full face, powered air purifying filter).
Protective clothing (pictograms) :	

Consult your supervisor or S.O.P. for special handling

Note: Application of paint products by spraying requires additional safety precautions: Full body suit, Full face respirator with air supplied.

Environmental exposure controls

Emissions from ventilation or work process equipment should be checked to ensure they comply with the requirements of environmental protection legislation. In some cases, fume scrubbers, filters or engineering modifications to the process equipment will be necessary to reduce emissions to acceptable levels.

SECTION 9: Physical and chemical properties

9.1 Information on basic physical and chemical properties

Physical state :	Liquid.
Color :	Clear
Odor :	Solvent-like
pH :	Testing not relevant or not possible due to nature of the product.
Melting point/freezing point :	Testing not relevant or not possible due to nature of the product.
Boiling point/boiling range :	Testing not relevant or not possible due to nature of the product.
Flash point :	Closed cup: 93°C (199.4°F)
Evaporation rate :	Testing not relevant or not possible due to nature of the product.
Flammability :	Highly flammable in the presence of the following materials or conditions: open flames, sparks and static discharge. Flammable in the presence of the following materials or conditions: heat.
Upper/lower flammability or explosive limits :	1 - 13 vol %
Vapor pressure :	Testing not relevant or not possible due to nature of the product.
Vapor density :	Testing not relevant or not possible due to nature of the product.
Relative density :	1.02 g/cm ³
Partition coefficient (LogKow) :	Testing not relevant or not possible due to nature of the product.
Auto-ignition temperature :	Testing not relevant or not possible due to nature of the product.
Decomposition temperature :	Testing not relevant or not possible due to nature of the product.
Viscosity :	Testing not relevant or not possible due to nature of the product.
Explosive properties :	Slightly explosive in the presence of the following materials or conditions: open flames, sparks and static discharge, heat and organic materials.
Oxidizing properties :	Testing not relevant or not possible due to nature of the product.

9.2 Other information

Safety Data Sheet

NEOGARD FLEXIBLE EPOXY CURING AGENT

70719



SECTION 9: Physical and chemical properties

Solvent(s) % by weight (Included exempt solvent(s)):	6.6 % (w/w)
Water % by weight :	Weighted average: 0 %
VOC content (Coatings) :	0.559 lbs/gal (67 g/l)
VOC content (Regulatory) :	0.559 lbs/gal (67 g/l)
TOC Content (Volatile) :	Weighted average: 51 g/l
Solvent Gas :	Weighted average: 0.054 m ³ /l

SECTION 10: Stability and reactivity

10.1 Reactivity

No specific test data related to reactivity available for this product or its ingredients.

10.2 Chemical stability

The product is stable.

10.3 Possibility of hazardous reactions

Under normal conditions of storage and use, hazardous reactions will not occur.

10.4 Conditions to avoid

No specific data.

10.5 Incompatible materials

Extremely reactive or incompatible with the following materials: acids.
Reactive or incompatible with the following materials: oxidizing materials and organic materials.
Slightly reactive or incompatible with the following materials: reducing materials.

10.6 Hazardous decomposition products

When exposed to high temperatures (i.e. in case of fire) harmful decomposition products may be formed:
Decomposition products may include the following materials: carbon oxides nitrogen oxides

SECTION 11: Toxicological information

11.1 Information on toxicological effects

Exposure to component solvent vapor concentrations may result in adverse health effects such as mucous membrane and respiratory system irritation and adverse effects on the kidneys, liver and central nervous system. Solvents may cause some of the above effects by absorption through the skin. Symptoms and signs include headaches, dizziness, fatigue, muscular weakness, drowsiness and, in extreme cases, loss of consciousness. Repeated or prolonged contact with the preparation may cause removal of natural fat from the skin, resulting in non-allergic contact dermatitis and absorption through the skin. If splashed in the eyes, the liquid may cause irritation and reversible damage. Accidental swallowing may cause stomach pain. Chemical lung inflammation may occur if the product is taken into the lungs via vomiting.

Inhalation of a corrosive substance may result in health effects such as stinging, coughing and in extreme cases, dyspnoea or loss of consciousness with a risk of lung damage, possibly lung oedema. Cauterization of skin and mucous membrane. If splashed in the eyes, the liquid may cause irreversible damage. Accidental swallowing may cause stinging and cauterization to mouth, oesophagus and stomach. Symptoms and signs include bloody vomiting, choke and loss of consciousness.

Acute toxicity

Safety Data Sheet

NEOGARD FLEXIBLE EPOXY CURING AGENT

70719



SECTION 11: Toxicological information

Product/ingredient name	Result	Species	Dose	Exposure
benzyl alcohol	LC50 Inhalation Dusts and mists	Rat	>4178 mg/m ³	4 hours
	LD50 Oral	Rat	1230 mg/kg	-
2,4-bis(ethenyl)phenol	LD50 Dermal	Rabbit	>5010 mg/kg	-
	LD50 Oral	Rat	2500 mg/kg	-
3,6-diazaoctanethylenediamin	LD50 Dermal	Rabbit	550 mg/kg	-
	LD50 Oral	Rat	1716 mg/kg	-
salicylic acid	LC50 Inhalation Dusts and mists	Rat	>0.9 mg/l	1 hours
	LD50 Dermal	Rat	>2000 mg/kg	-
	LD50 Oral	Rat	891 mg/kg	-
4,4'-isopropylidenediphenol	LD50 Dermal	Rabbit	>2000 mg/kg	-
	LD50 Oral	Rat	3250 mg/kg	-
	LD50 Oral	Rat	3250 mg/kg	-
2,2'-iminodiethylamine	LC50 Inhalation Dusts and mists	Rat	0.07 mg/l	4 hours
	LD50 Dermal	Rabbit	1045 mg/kg	-
	LD50 Dermal	Rabbit	1090 mg/kg	-
	LD50 Oral	Rat	1080 mg/kg	-

Acute toxicity estimates

Route	ATE value
Oral	4549.03 mg/kg
Dermal	11303.07 mg/kg
Inhalation (vapors)	52.48 mg/l
Inhalation (dusts and mists)	3.32 mg/l

Irritation/Corrosion

Product/ingredient name	Result	Species	Score	Exposure
benzyl alcohol	Eyes - Visible necrosis	Rabbit	-	-
	Skin - Mild irritant	Rabbit	-	-
2,4-bis(ethenyl)phenol	Eyes - Irritant	Rabbit	-	-
	Skin - Irritant	Rabbit	-	-
3,6-diazaoctanethylenediamin	Eyes - Moderate irritant	Rabbit	-	24 hours 20 milligrams
	Skin - Severe irritant	Rabbit	-	24 hours 5 milligrams
salicylic acid	Eyes - Severe irritant	Rabbit	-	-
4,4'-isopropylidenediphenol	Eyes - Severe irritant	Rabbit	-	24 hours 250 Micrograms
	Skin - Mild irritant	Rabbit	-	24 hours 500 milligrams
2,2'-iminodiethylamine	Eyes - Severe irritant	Rabbit	-	-
	Skin - Severe irritant	Rabbit	-	-

Sensitizer

Product/ingredient name	Route of exposure	Species	Result
3,6-diazaoctanethylenediamin	skin	Guinea pig	Sensitizing
2,2'-iminodiethylamine	skin	Mouse	Sensitizing

Specific target organ toxicity (single exposure)

Product/ingredient name	Category	Route of exposure	Target organs
4,4'-isopropylidenediphenol	Category 3		Respiratory tract irritation
2,2'-iminodiethylamine	Category 3		Respiratory tract irritation

Information on the likely routes of exposure

Routes of entry anticipated: Oral, Dermal, Inhalation.

Potential chronic health effects

Sensitization : Contains 2,4-bis(ethenyl)phenol, 3,6-diazaoctanethylenediamin, 4,4'-isopropylidenediphenol, 2,2'-iminodiethylamine. May produce an allergic reaction.

Other information : No additional known significant effects or critical hazards.

Safety Data Sheet

NEOGARD FLEXIBLE EPOXY CURING AGENT

70719



SECTION 12: Ecological information

12.1 Toxicity

Do not allow to enter drains or watercourses. Toxic to aquatic life with long lasting effects.

When spilled, this product may act as an oil, causing a film, sheen, emulsion, or sludge at or beneath the surface of a body of water. Oils of any kind can cause: (a) drowning of waterfowl due to lack of buoyancy, loss of insulating capacity of feathers, starvation and vulnerability to predators due to lack of mobility; (b) lethal effect on fish by coating gill surfaces, preventing respiration; (c) potential fish kills resulting from alteration in biochemical oxygen demand; (d) asphyxiation of benthic life forms when floating masses become engaged with surface debris and settle on the bottom; and (e) adverse aesthetic effects of fouled shoreline and beaches.

Product/ingredient name	Result	Species	Exposure
reactionproduct of dimer C18-fatty acids with polyethylenepolyamines	Acute EC50 4.11 mg/l	Algae	72 hours
benzyl alcohol	Acute LC50 7.07 mg/l Acute EC50 230 mg/l Acute LC50 770 mg/l Acute LC50 460 mg/l	Fish Daphnia Algae Fish	96 hours 48 hours 72 hours 96 hours
2,4-bis(ethenyl)phenol	Acute EC50 3.14 mg/l Acute EC50 1 - 10 mg/l Acute EC50 14.8 mg/l Acute EC50 20 mg/l	Algae Daphnia Fish Algae	72 hours 48 hours 96 hours 72 hours
3,6-diazaoctanethylenediamin	Acute EC50 31.1 mg/l Acute LC50 330 mg/l Acute LC50 7.5 mg/l	Daphnia Fish Fish	48 hours 96 hours 96 hours
4,4'-isopropylidenediphenol	Chronic NOEC 0.8 mg/l Fresh water Chronic NOEC 0.2 - 20 ppb Fresh water	Daphnia - Daphnia magna - Neonate Fish - Xiphophorus helleri - Juvenile (Fledgling, Hatchling, Weanling)	21 days 60 days
2,2'-iminodiethylamine	Acute EC50 1164 mg/l Acute EC50 16 mg/l Acute LC50 430 mg/l	Algae Daphnia Fish	72 hours 48 hours 96 hours

12.2 Persistence and degradability

Product/ingredient name	Test	Result	Dose	Inoculum
reactionproduct of dimer C18-fatty acids with polyethylenepolyamines	OECD 301B 301B Ready Biodegradability - CO ₂ Evolution Test	100 % - Inherent - 74 days	-	-
benzyl alcohol	OECD 301A 301A Ready Biodegradability - DOC Die-Away Test	95 - 97 % - Readily - 21 days	-	-
salicylic acid	OECD 301C 301C Ready Biodegradability - Modified MITI Test (I)	92 - 96 % - Readily - 14 days	-	-
4,4'-isopropylidenediphenol	-	100 % - Readily - 14 days	-	-
2,2'-iminodiethylamine	OECD 301D Ready Biodegradability - Closed Bottle Test	1 - 2 % - Not readily - 28 days 87 % - Readily - 21 days	- -	- -

Product/ingredient name	Aquatic half-life	Photolysis	Biodegradability
benzyl alcohol	-	-	Readily
2,4-bis(ethenyl)phenol	-	-	Not readily
salicylic acid	-	-	Readily
4,4'-isopropylidenediphenol	-	-	Not readily
2,2'-iminodiethylamine	-	-	Readily

12.3 Bioaccumulative potential

Safety Data Sheet

NEOGARD FLEXIBLE EPOXY CURING AGENT

70719

SECTION 12: Ecological information

Product/ingredient name	LogP _{ow}	BCF	Potential
benzyl alcohol	0.87	1.37	low
3,6-diazaoctanethylenediamin	-1.66 - -1.4	-	low
salicylic acid	2.21 - 2.26	-	low
4,4'-isopropylidenediphenol	3.4	20 - 67	low
2,2'-iminodiethylamine	-5.58	2.8 - 6.3	low

12.4 Mobility in soil

Soil/water partition coefficient (K_{oc}) : No known data available in our database.

Mobility : No known data available in our database.

12.5 Other adverse effects

No known significant effects or critical hazards.

SECTION 13: Disposal considerations

13.1 Waste treatment methods

Disposal should be in accordance with applicable regional, national and local laws and regulations. Local regulations may be more stringent than regional or national requirements.

The information presented below only applies to the material as supplied. The identification based on characteristic(s) or listing may not apply if the material has been used or otherwise contaminated. It is the responsibility of the waste generator to determine the toxicity and physical properties of the material generated to determine the proper waste identification and disposal methods in compliance with applicable regulations.

Refer to Section 7 and Section 8 for additional handling information and protection of employees.

The generation of waste should be avoided or minimized wherever possible. Disposal of this product, solutions and any by-products should at all times comply with the requirements of environmental protection and waste disposal legislation and any regional local authority requirements. Dispose of surplus and non-recyclable products via a licensed waste disposal contractor. Waste should not be disposed of untreated to the sewer unless fully compliant with the requirements of all authorities with jurisdiction. Waste packaging should be recycled. Incineration or landfill should only be considered when recycling is not feasible. This material and its container must be disposed of in a safe way. Care should be taken when handling emptied containers that have not been cleaned or rinsed out. Empty containers or liners may retain some product residues. Vapor from product residues may create a highly flammable or explosive atmosphere inside the container. Do not cut, weld or grind used containers unless they have been cleaned thoroughly internally. Avoid dispersal of spilled material and runoff and contact with soil, waterways, drains and sewers.

SECTION 14: Transport information

Transport may take place according to national regulation or DOT for transport by road and by train, IMDG for transport by sea, IATA for Air shipment. Refer to specific Dangerous Goods Transport requirements under 49CFR, ICAO and IATA.

14.1 UN no.	14.2 Proper shipping name	14.3 Transport hazard class(es)	14.4 PG*	14.5 Env*	Additional information
DOT Code	UN3066	PAINT RELATED MATERIAL. (2,4-bis(ethenyl)phenol)	8 -	III	Yes. The marine pollutant mark is not required when transported on inland waterways in sizes of ≤5 L or ≤5 kg or by road, rail, or inland air in non-bulk sizes.
TDG Code	UN3066	PAINT RELATED MATERIAL. (2,4-bis(ethenyl)phenol)	8 -	III	Yes. Product classified as per the following sections of the Transportation of Dangerous Goods Regulations: 2.40-2.42 (Class 8), 2.7 (Marine pollutant mark). The marine pollutant mark is not required when transported by road or rail.
SCT Code	UN3066	PAINT RELATED MATERIAL	8 -	III	Yes. -

Safety Data Sheet

NEOGARD FLEXIBLE EPOXY CURING AGENT

70719



SECTION 14: Transport information

IMDG Code	UN3066	PAINT RELATED MATERIAL. (2,4-bis(ethenyl)phenol)	8 -	 	III	Yes. The marine pollutant mark is not required when transported in sizes of ≤5 L or ≤5 kg. Emergency schedules F-A, S-B
IATA Code	UN3066	PAINT RELATED MATERIAL	8 -		III	Yes. The environmentally hazardous substance mark may appear if required by other transportation regulations.

Code : Classification
PG* : Packing group
Env.* : Environmental hazards

14.6 Special precautions for user

Transport within user's premises: always transport in closed containers that are upright and secure. Ensure that persons transporting the product know what to do in the event of an accident or spillage.

14.7 Transport in bulk according to IMO instruments

Not applicable.

SECTION 15: Regulatory information

15.1 Safety, health and environmental regulations/legislation specific for the substance or mixture

U.S. Federal regulations :

All components are active or exempted.

TSCA 8(a) PAIR: 2,4-bis(ethenyl)phenol; benzaldehyde; α-chlorotoluene

TSCA 8(a) CDR Exempt/Partial exemption: Not determined

United States inventory (TSCA 8b): All components are active or exempted.

Clean Water Act (CWA) 307: phenol

Clean Water Act (CWA) 311: phenol; α-chlorotoluene; styrene

Clean Air Act Section 112(b) Hazardous Air Pollutants (HAPs) : Listed

Product/ingredient name	CAS number	Concentration
phenol	108-95-2	0.0059477
α-chlorotoluene	100-44-7	0.0021635
styrene	100-42-5	<0.001

Clean Air Act Section 602 Class I Substances : Not listed

Clean Air Act Section 602 Class II Substances : Not listed

DEA List I Chemicals (Precursor Chemicals) : Not listed

DEA List II Chemicals (Essential Chemicals) : Not listed

SARA 302/304 :

Product/ingredient name	%	EHS	SARA 302 TPQ		SARA 304 RQ	
			(lbs)	(gallons)	(lbs)	(gallons)
phenol	≤0.1	Yes.	500 / 10000	-	1000	-
α-chlorotoluene	<0.1	Yes.	500	54.4	100	10.9

SARA 304 RQ :

4622207.1 lbs / 2098482 kg [545630.2 gal / 2065435 L]

SARA 311/312 Classification :

FLAMMABLE LIQUIDS - Category 4
ACUTE TOXICITY (inhalation) - Category 4
SKIN CORROSION - Category 1B
SERIOUS EYE DAMAGE - Category 1
SKIN SENSITIZATION - Category 1
TOXIC TO REPRODUCTION - Category 1B

Safety Data Sheet

NEOGARD FLEXIBLE EPOXY CURING AGENT

70719



SECTION 15: Regulatory information

Product/ingredient name	%	Classification
reactionproduct of dimer C18-fatty acids with polyethylenepolyamines benzyl alcohol	≥25 - ≤50 ≥10 - ≤25	SERIOUS EYE DAMAGE - Category 1 ACUTE TOXICITY (oral) - Category 4 ACUTE TOXICITY (inhalation) - Category 4 EYE IRRITATION - Category 2A SKIN IRRITATION - Category 2
2,4-bis(ethenyl)phenol	≥10 - ≤25	SKIN SENSITIZATION - Category 1A ACUTE TOXICITY (dermal) - Category 4
3,6-diazaoctanethylenediamin	≥5 - ≤10	SKIN CORROSION - Category 1B SERIOUS EYE DAMAGE - Category 1 SKIN SENSITIZATION - Category 1
salicylic acid	≥1 - ≤3	ACUTE TOXICITY (oral) - Category 4 SERIOUS EYE DAMAGE - Category 1 TOXIC TO REPRODUCTION - Category 2
4,4'-isopropylidenediphenol	≥1 - ≤3	SERIOUS EYE DAMAGE - Category 1 SKIN SENSITIZATION - Category 1 TOXIC TO REPRODUCTION - Category 1B SPECIFIC TARGET ORGAN TOXICITY (SINGLE EXPOSURE) (Respiratory tract irritation) - Category 3
2,2'-iminodiethylamine	≥1 - ≤3	ACUTE TOXICITY (oral) - Category 4 ACUTE TOXICITY (dermal) - Category 4 ACUTE TOXICITY (inhalation) - Category 2 SKIN CORROSION - Category 1B SERIOUS EYE DAMAGE - Category 1 SKIN SENSITIZATION - Category 1 SPECIFIC TARGET ORGAN TOXICITY (SINGLE EXPOSURE) (Respiratory tract irritation) - Category 3

SARA 313 :

SARA 313 notifications must not be detached from the SDS and any copying and redistribution of the SDS shall include copying and redistribution of the notice attached to copies of the SDS subsequently redistributed.

Form R - Reporting requirements :

Product/ingredient name	CAS number	Concentration
4,4'-isopropylidenediphenol	80-05-7	1 - 3

Supplier notification :

Product/ingredient name	CAS number	Concentration
4,4'-isopropylidenediphenol	80-05-7	1 - 3

State regulations :

Connecticut Carcinogen Reporting: None of the components are listed.
Connecticut Hazardous Material Survey: None of the components are listed.
Florida substances: None of the components are listed.
Illinois Chemical Safety Act: None of the components are listed.
Illinois Toxic Substances Disclosure to Employee Act: None of the components are listed.
Louisiana Reporting: None of the components are listed.
Louisiana Spill: None of the components are listed.
Massachusetts Substances: The following components are listed: BENZYL ALCOHOL; TRIETHYLENETETRAMINE; 4,4'-ISOPROPYLIDENEDIPHENOL; DIETHYLENE TRIAMINE
Massachusetts Spill: None of the components are listed.
Michigan Critical Material: None of the components are listed.
Minnesota Hazardous Substances: None of the components are listed.
New Jersey Spill: None of the components are listed.
New Jersey Toxic Catastrophe Prevention Act: None of the components are listed.
New Jersey Hazardous Substances: The following components are listed: TRIETHYLENE TETRAMINE; BISPHENOL A; DIETHYLENE TRIAMINE
New York Hazardous Substances: None of the components are listed.
New York Toxic Chemical Release Reporting: None of the components are listed.
Pennsylvania RTK Hazardous Substances: The following components are listed: BENZENEMETHANOL; 1,2-ETHANEDIAMINE, N,N'-BIS(2-AMINOETHYL)-; 4,4'-ISOPROPYLIDENEDIPHENOL; 1,2-ETHANEDIAMINE, N-(2-AMINOETHYL)-
Rhode Island Hazardous Substances: None of the components are listed.

California Prop. 65 PFF :

WARNING: This product can expose you to chemicals including Benzyl chloride and Styrene, which are known to the State of California to cause cancer, and Bisphenol A, which is known to the State of California to cause birth defects or other reproductive harm. For more information go to www.P65Warnings.ca.gov.

Product/ingredient name	Cancer	Reproductive	No significant risk level	Maximum acceptable dosage level
4,4'-isopropylidenediphenol	No.	Yes.		Yes.
α-chlorotoluene	Yes.	No.	Yes.	
styrene	Yes.	No.	Yes.	

Safety Data Sheet

NEOGARD FLEXIBLE EPOXY CURING AGENT

70719



SECTION 16: Other information

Remarks : Note: In USA, consult Code of Federal Regulations, Title 29, Labor, Parts 1910 and 1915 concerning occupational safety and health standards and regulations, as well as any other applicable Federal, State or local regulations that apply to safe practices in coating operations.

Warning! If you scrape, sand, or remove old paint, you may release lead dust. LEAD is TOXIC.

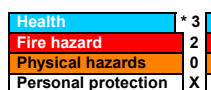
Validation : Validated by US - HSE Products Coordinator on 15 February 2024

GHS Classification

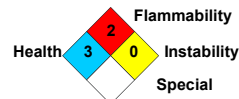
Procedure used to derive the classification.

Classification	Justification
FLAMMABLE LIQUIDS - Category 4 ACUTE TOXICITY (inhalation) - Category 4 SKIN CORROSION - Category 1B SERIOUS EYE DAMAGE - Category 1 SKIN SENSITIZATION - Category 1 TOXIC TO REPRODUCTION - Category 1B	On basis of test data Calculation method Calculation method Calculation method Calculation method Calculation method

Hazardous Material Information System (U.S.A.)



National Fire Protection Association (U.S.A.)



Personal Protective Equipment (PPE) shown in this section is a suggestion. Since conditions vary from one work location to another consult the facility safety & health program. Customer or end user is responsible to evaluate worker exposure conditions at the site of application and determine the appropriate PPE suitable for workers at that particular facility or location.

Abbreviations and acronyms :

ANSI = American National Standards Institute
HCS = Hazardous Communication System
TSCA = Toxic Substances Control Act
CFR = Code of federal Regulations
GHS = Globally Harmonized System of Classification and Labelling of Chemicals
OSHA = United States Occupational Health and Safety Administration
NIOSH = National Institute for Occupational Safety and Health
ACGIH = American Conference of Industrial Hygienists
IARC = International Agency for Research on Cancer.
NTP = National Toxicology Program
ATE = Acute Toxicity Estimate

OECD = Organisation for Economic Co-operation and Development
BCF = Bioconcentration Factor
DOT = United States Department of Transportation
ERG = Emergency Response Guide
TDG = Transport of Dangerous Goods, Canada
SCT = Transportation & Communications Ministry, Mexico
IMDG = International Maritime Dangerous Goods
IATA = International Air Transport Association
SARA = Superfund Amendments Reauthorization Act
EPCRA = Emergency Planning and Community Right to Know Act

Notice to reader

Indicates information that has changed from previously issued version.

To the best of our knowledge, the information contained herein is accurate. However, neither the above named supplier nor any of its subsidiaries assumes any liability whatsoever for the accuracy or completeness of the information contained herein. Final determination of suitability of any material is the sole responsibility of the user. All materials may present unknown hazards and should be used with caution. Although certain hazards are described herein, we cannot guarantee that these are the only hazards that exist.

Safety Data Sheet

NEOGARD

70801 NEOCRETE WATER BORNE POLYURETHANE RESIN
CURING AGENT



Conforms to ANSI Z400.1-2010 Standard - HCS 2012

Protective Clothing	General Hazard	DOT
Consult your supervisor or S.O.P. for special handling	 	

SECTION 1: Identification of the substance/mixture and of the company/undertaking

1.1 Product identifier

Product name : NEOGARD
NEOCRETE WATER BORNE POLYURETHANE RESIN CURING AGENT

Product identity : 9801000000, 70801

Product type : Resins.

1.2 Relevant identified uses of the substance or mixture and uses advised against

Field of application : used only as part of two- or multi component products.

Ready-for-use mixture : Used for: 48010 48011 48012

Identified uses : Industrial/Professional use

TSCA : Unless otherwise stated. All components are listed or exempted.

1.3 Details of the supplier of the safety data sheet

Company details : NEOGARD, a Division of Hempel (USA), Inc.
2728 Empire Central
Dallas, TX 75235
Phone number: 1-214-353-1600
E-mail: hempel@hempel.com

1.4 Emergency telephone number (with hours of operation)

For Transportation Emergencies : CHEMTREC: **1-800-424-9300** (Toll-free in the U.S., Canada and the U.S. Virgin Islands) **703-527-3887**
(24 hours) For calls originating elsewhere (Collect calls are accepted). Contract number: CCN10384
To preserve the effectiveness of arrangements for providing accurate and timely emergency response information, the basic identifying information (shipper name or contract number) must be included on shipping papers.
If the purchaser of this product is going to be shipping this product to other locations, the purchaser must arrange for its own Emergency Information Provider to respond to transport incidents. Hempel's 24 hour response contract does not cover non-Hempel shipments.

For all other information : In USA toll free calling available: 1-800- 678-6641 or (936)-523-6000
(8 AM - 5 PM CST) See Section 4 of the safety data sheet (first aid measures).

SECTION 2: Hazards identification

2.1 Classification of the substance or mixture

OSHA/HCS status : This material is considered hazardous by the OSHA Hazard Communication Standard (29 CFR 1910.1200).

GHS Classification : ACUTE TOXICITY (inhalation) - Category 4
SKIN IRRITATION - Category 2
EYE IRRITATION - Category 2A
RESPIRATORY SENSITIZATION - Category 1
SKIN SENSITIZATION - Category 1
CARCINOGENICITY - Category 2
SPECIFIC TARGET ORGAN TOXICITY (SINGLE EXPOSURE) (Respiratory tract irritation) - Category 3
SPECIFIC TARGET ORGAN TOXICITY (REPEATED EXPOSURE) (respiratory tract) - Category 2

2.2 Label elements

Safety Data Sheet

NEOGARD

70801 NEOCRETE WATER BORNE POLYURETHANE RESIN
CURING AGENT



SECTION 2: Hazards identification

Hazard pictograms :



Signal word :

Danger

Hazard statements :

H332 - Harmful if inhaled.
H319 - Causes serious eye irritation.
H315 - Causes skin irritation.
H334 - May cause allergy or asthma symptoms or breathing difficulties if inhaled.
H317 - May cause an allergic skin reaction.
H351 - Suspected of causing cancer.
H335 - May cause respiratory irritation.
H373 - May cause damage to organs through prolonged or repeated exposure. (respiratory tract)

Precautionary statements :

Prevention :

Obtain special instructions before use. Do not handle until all safety precautions have been read and understood. Wear protective gloves. Wear eye or face protection. Wear protective clothing. Wear respiratory protection. Use only outdoors or in a well-ventilated area. Do not breathe vapor. Wash hands thoroughly after handling. Contaminated work clothing must not be allowed out of the workplace.

Response :

Get medical attention if you feel unwell. IF exposed or concerned: Get medical attention. IF INHALED: If breathing is difficult, remove person to fresh air and keep comfortable for breathing. Call a POISON CENTER or physician if you feel unwell. If experiencing respiratory symptoms: Call a POISON CENTER or physician. IF ON SKIN: Wash with plenty of soap and water. Wash contaminated clothing before reuse. If skin irritation or rash occurs: Get medical attention. IF IN EYES: Rinse cautiously with water for several minutes. Remove contact lenses, if present and easy to do. Continue rinsing. If eye irritation persists: Get medical attention.

Storage :

Store locked up.

Disposal :

Dispose of contents and container in accordance with all local, regional, national and international regulations.

Supplemental label elements :

None known.

2.3 Other hazards

Hazards not otherwise classified : None known.

SECTION 3: Composition/information on ingredients

Product definition :

Mixture

Physical state :

Liquid.

Product/ingredient name	Identifiers	%	GHS Classification
4,4'-methylenebis(phenyl isocyanate) (isomers and homologues)	9016-87-9	≥25 - ≤50	ACUTE TOXICITY (inhalation) - Category 4 SKIN IRRITATION - Category 2 EYE IRRITATION - Category 2A RESPIRATORY SENSITIZATION - Category 1 SKIN SENSITIZATION - Category 1 CARCINOGENICITY - Category 2 SPECIFIC TARGET ORGAN TOXICITY (SINGLE EXPOSURE) (Respiratory tract irritation) - Category 3 SPECIFIC TARGET ORGAN TOXICITY (REPEATED EXPOSURE) (respiratory tract) (inhalation) - Category 2
4,4'-methylenediphenyl diisocyanate	101-68-8	≥25 - ≤50	ACUTE TOXICITY (inhalation) - Category 4 SKIN IRRITATION - Category 2 EYE IRRITATION - Category 2A RESPIRATORY SENSITIZATION - Category 1 SKIN SENSITIZATION - Category 1 CARCINOGENICITY (inhalation) - Category 2 SPECIFIC TARGET ORGAN TOXICITY (SINGLE EXPOSURE) (Respiratory tract irritation) - Category 3 SPECIFIC TARGET ORGAN TOXICITY (REPEATED EXPOSURE) (respiratory tract) (inhalation) - Category 2
o-(p-isocyanatobenzyl)phenyl isocyanate	5873-54-1	≥10 - ≤25	ACUTE TOXICITY (inhalation) - Category 4



SECTION 3: Composition/information on ingredients

			SKIN IRRITATION - Category 2 EYE IRRITATION - Category 2A RESPIRATORY SENSITIZATION - Category 1 SKIN SENSITIZATION - Category 1 CARCINOGENICITY (inhalation) - Category 2 SPECIFIC TARGET ORGAN TOXICITY (SINGLE EXPOSURE) (Respiratory tract irritation) - Category 3 SPECIFIC TARGET ORGAN TOXICITY (REPEATED EXPOSURE) (respiratory tract) (inhalation) - Category 2
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Any concentration shown as a range is to protect confidentiality or is due to batch variation.

There are no ingredients present which, within the current knowledge of the supplier and in the concentrations applicable, are classified as hazardous to health or the environment and hence require reporting in this section.

SECTION 4: First aid measures

4.1 Description of first aid measures

General :	In all cases of doubt, or when symptoms persist, seek medical attention. Never give anything by mouth to an unconscious person. If breathing is irregular, drowsiness, loss of consciousness or cramps: Call 911 and give immediate treatment (first aid).
Eye contact :	Check for and remove any contact lenses. Immediately flush eyes with plenty of water for at least 15 minutes, occasionally lifting the upper and lower eyelids. Seek immediate medical attention.
Inhalation :	Remove to fresh air. Keep person warm and at rest. If not breathing, if breathing is irregular or if respiratory arrest occurs, provide artificial respiration or oxygen by trained personnel. Give nothing by mouth. If unconscious, place in recovery position and get medical attention immediately.
Skin contact :	Remove contaminated clothing and shoes. Wash skin thoroughly with soap and water or use recognized skin cleanser. Do NOT use solvents or thinners.
Ingestion :	If swallowed, seek medical advice immediately and show this container or label. Keep person warm and at rest. Do not induce vomiting unless directed to do so by medical personnel. Lower the head so that vomit will not re-enter the mouth and throat.
Protection of first-aiders :	No action shall be taken involving any personal risk or without suitable training. If it is suspected that fumes are still present, the rescuer should wear an appropriate mask or self-contained breathing apparatus. It may be dangerous to the person providing aid to give mouth-to-mouth resuscitation. Wash contaminated clothing thoroughly with water before removing it, or wear gloves.

4.2 Most important symptoms and effects, both acute and delayed

Potential acute health effects

Eye contact :	Causes serious eye irritation.
Inhalation :	Harmful if inhaled. May cause respiratory irritation. May cause allergy or asthma symptoms or breathing difficulties if inhaled.
Skin contact :	Causes skin irritation. May cause an allergic skin reaction.
Ingestion :	No known significant effects or critical hazards.

Over-exposure signs/symptoms

Eye contact :	Adverse symptoms may include the following: pain or irritation watering redness
Inhalation :	Adverse symptoms may include the following: respiratory tract irritation coughing wheezing and breathing difficulties asthma
Skin contact :	Adverse symptoms may include the following: irritation redness
Ingestion :	No specific data.



SECTION 4: First aid measures

4.3 Indication of any immediate medical attention and special treatment needed

Notes to physician : If gasses have been inhaled, from the decomposition of the product, symptoms may be delayed.
Specific treatments : No specific treatment.

SECTION 5: Firefighting measures

5.1 Extinguishing media

Extinguishing media : Recommended: alcohol resistant foam, CO₂, powders, water spray.
Not to be used: waterjet.

5.2 Special hazards arising from the substance or mixture

Hazards from the substance or mixture : In a fire or if heated, a pressure increase will occur and the container may burst.
Hazardous combustion products : Decomposition products may include the following materials: carbon oxides nitrogen oxides

5.3 Advice for firefighters

Promptly isolate the scene by removing all persons from the vicinity of the incident if there is a fire. No action shall be taken involving any personal risk or without suitable training. Fire will produce dense black smoke. Exposure to decomposition products may cause a health hazard. Cool closed containers exposed to fire with water. Do not release runoff from fire to drains or watercourses. Fire-fighters should wear appropriate protective equipment and self-contained breathing apparatus (SCBA) with a full face-piece operated in positive pressure mode.

SECTION 6: Accidental release measures

6.1 Personal precautions, protective equipment and emergency procedures

Avoid all direct contact with the spilled material. Refer to protective measures listed in sections 7 and 8. No action shall be taken involving any personal risk or without suitable training. If the product contaminates lakes, rivers, or sewers, inform the appropriate authorities in accordance with local regulations.

6.2 Environmental precautions

Avoid dispersal of spilled material and runoff and contact with soil, waterways, drains and sewers. Inform the relevant authorities if the product has caused environmental pollution (sewers, waterways, soil or air).

6.3 Methods and materials for containment and cleaning up

Stop leak if without risk. Move containers from spill area. Approach release from upwind. Prevent entry into sewers, water courses, basements or confined areas. Wash spillages into an effluent treatment plant or proceed as follows. Contain and collect spillage with non-combustible, absorbent material e.g. sand, earth, vermiculite or diatomaceous earth and place in container for disposal according to local regulations (see Section 13). Contaminated absorbent material may pose the same hazard as the spilled product.

6.4 Reference to other sections

See Section 1 for emergency contact information.
See Section 8 for information on appropriate personal protective equipment.
See Section 13 for additional waste treatment information.

SECTION 7: Handling and storage

7.1 Precautions for safe handling

Contains isocyanates. Exposure to isocyanate may result in acute irritation and/or sensitisation when breathing.
Care should be taken when re-opening partly-used containers.

Avoid inhalation of vapour, dust and spray mist. Avoid contact with skin and eyes. Eating, drinking and smoking should be prohibited in area where this material is handled, stored and processed. Appropriate personal protective equipment: see Section 8. Always keep in containers made from the same material as the original one.

7.2 Conditions for safe storage, including any incompatibilities



SECTION 7: Handling and storage

Store in accordance with local regulations for flammable liquids. Store in a cool, well-ventilated area away from incompatible materials and ignition sources. Keep out of the reach of children. Keep away from: Oxidizing agents, strong alkalis, strong acids as well as of amines, alcohols and water. No smoking. Prevent unauthorized access. Containers that are opened must be carefully resealed and kept upright to prevent leakage.

7.3 Specific end use(s)

See separate Product Data Sheet for recommendations or industrial sector specific solutions.

SECTION 8: Exposure controls/personal protection

8.1 Control parameters

Product/ingredient name	Exposure limit values
4,4'-methylenediphenyl diisocyanate	ACGIH TLV (United States, 3/2017). TWA: 0.005 ppm 8 hours. NIOSH REL (United States, 10/2016). TWA: 0.05 mg/m ³ 10 hours. TWA: 0.005 ppm 10 hours. CEIL: 0.2 mg/m ³ 10 minutes. CEIL: 0.02 ppm 10 minutes. OSHA PEL (United States, 6/2016). CEIL: 0.02 ppm CEIL: 0.2 mg/m ³

Recommended monitoring procedures

If this product contains ingredients with exposure limits, personal, workplace atmosphere or biological monitoring may be required to determine the effectiveness of the ventilation or other control measures and/or the necessity to use respiratory protective equipment. Reference should be made to appropriate monitoring standards. Reference to national guidance documents for methods for the determination of hazardous substances will also be required.

8.2 Exposure controls

Appropriate engineering controls

Provide local exhaust and general ventilation systems to maintain airborne concentrations below OSHA, ACGIH, and manufacturer recommended exposure limits. Local exhaust ventilation is preferred because it prevents contaminant dispersion into work areas by controlling it at its source. Use local and general exhaust ventilation to effectively remove and prevent buildup of mists/vapors/fumes generated from the handling of this product.

Note: Local exhaust ventilation is designed to capture an emitted contaminant at or near its source, before the contaminant has a chance to disperse into the workplace air. General exhaust ventilation, also called dilution ventilation, is different from local exhaust ventilation because instead of capturing emissions at their source and removing them from the air, general exhaust ventilation allows the contaminant to be emitted into the workplace air and then dilutes the concentration of the contaminant to an acceptable level (e.g., to the PEL or below).

Individual protection measures

General :	Gloves must be worn for all work that may result in soiling. Apron/coveralls/protective clothing must be worn when soiling is so great that regular work clothes do not adequately protect skin against contact with the product. Safety eyewear should be used when there is a likelihood of exposure.
Hygiene measures :	Wash hands, forearms, and face thoroughly after handling compounds and before eating, smoking, using lavatory, and at the end of day.
Eye/face protection :	Safety eyewear complying with an approved standard should be used when a risk assessment indicates this is necessary to avoid exposure to liquid splashes, mists, gases or dusts. If contact is possible, the following protection should be worn, unless the assessment indicates a higher degree of protection: chemical splash goggles.
Hand protection :	Wear chemical-resistant gloves in combination with 'basic' employee training. The quality of the chemical-resistant protective gloves must be chosen as a function of the specific workplace concentrations and quantity of hazardous substances. Since the actual work situation is unknown. Supplier of gloves should be contacted in order to find the appropriate type.
Body protection :	Personal protective equipment for the body should be selected based on the task being performed and the risks involved handling this product.
Respiratory protection :	Use appropriate respiratory protection if there is a risk of exceeding any exposure limits.



SECTION 8: Exposure controls/personal protection

Protective clothing (pictograms) :

Consult your supervisor or S.O.P. for special handling

Note: Application of paint products by spraying requires additional safety precautions: Full body suit, Full face respirator with air supplied.

Environmental exposure controls

Emissions from ventilation or work process equipment should be checked to ensure they comply with the requirements of environmental protection legislation. In some cases, fume scrubbers, filters or engineering modifications to the process equipment will be necessary to reduce emissions to acceptable levels.

SECTION 9: Physical and chemical properties

9.1 Information on basic physical and chemical properties

Physical state :	Liquid.
Color :	Clear
Odor :	isocyanate
pH :	Testing not relevant or not possible due to nature of the product.
Melting point/freezing point :	37.28°C This is based on data for the following ingredient: diphenylmethane-diisocyanate (isomers and homologues)
Boiling point/boiling range :	Testing not relevant or not possible due to nature of the product.
Flash point :	Closed cup: 203°C (397.4°F)
Evaporation rate :	Testing not relevant or not possible due to nature of the product.
Flammability :	Not available.
Upper/lower flammability or explosive limits :	No specific data.
Vapor pressure :	0 kPa This is based on data for the following ingredient: 4,4'-methylenediphenyl diisocyanate
Vapor density :	Testing not relevant or not possible due to nature of the product.
Relative density :	1.23 g/cm ³
Solubility(ies) :	
Partition coefficient (LogKow) :	Testing not relevant or not possible due to nature of the product.
Auto-ignition temperature :	Testing not relevant or not possible due to nature of the product.
Decomposition temperature :	Testing not relevant or not possible due to nature of the product.
Viscosity :	Testing not relevant or not possible due to nature of the product.
Explosive properties :	Not available.
Oxidizing properties :	Testing not relevant or not possible due to nature of the product.

9.2 Other information

Solvent(s) % by weight (Included exempt solvent(s)):	0 % (w/w)
Water % by weight :	Weighted average: 0 %
VOC content (Coatings) :	0 lbs/gal (0 g/l)
VOC content (Regulatory) :	0 lbs/gal (0 g/l)
TOC Content (Volatile) :	Weighted average: 0 g/l
Solvent Gas :	Weighted average: 0 m ³ /l



SECTION 10: Stability and reactivity

10.1 Reactivity

No specific test data related to reactivity available for this product or its ingredients.

10.2 Chemical stability

The product is stable.

10.3 Possibility of hazardous reactions

Under normal conditions of storage and use, hazardous reactions will not occur.

10.4 Conditions to avoid

No specific data.

10.5 Incompatible materials

10.6 Hazardous decomposition products

When exposed to high temperatures (i.e. in case of fire) harmful decomposition products may be formed:

Decomposition products may include the following materials: carbon oxides nitrogen oxides

SECTION 11: Toxicological information

11.1 Information on toxicological effects

Isocyanate containing products have characteristics that include producing acute irritation and/or sensitisation when breathing, subsequent asthmatic problems and lung contractions. Sensitised people can, as a result from this, show asthmatic symptoms with exposure to atmospheric concentrations far below the TLV. Repeated exposures will lead to permanent damage to the respiratory system.

Acute toxicity

Product/ingredient name	Result	Species	Dose	Exposure
Diphenylmethane-diisocyanate (isomers and homologues)	LC50 Inhalation Vapor	Rat	490 mg/m ³	4 hours
	LD50 Dermal	Rabbit	>9400 mg/kg	-
4,4'-methylenediphenyl diisocyanate o-(p-isocyanatobenzyl)phenyl isocyanate	LD50 Oral	Rat	49 g/kg	-
	LD50 Oral	Rat	9200 mg/kg	-
	LD50 Dermal	Rabbit	>9400 mg/kg	-
	LD50 Oral	Rat	>2000 mg/kg	-

Acute toxicity estimates

Route	ATE value
Inhalation (vapors)	24.44 mg/l
Inhalation (dusts and mists)	2.727 mg/l

Irritation/Corrosion

Product/ingredient name	Result	Species	Score	Exposure
Diphenylmethane-diisocyanate (isomers and homologues)	Eyes - Mild irritant	Rabbit	-	100 milligrams
	Skin - Mild irritant	Rabbit	-	-
4,4'-methylenediphenyl diisocyanate	Eyes - Moderate irritant	Rabbit	-	100 milligrams

Sensitizer



SECTION 11: Toxicological information

Product/ingredient name	Route of exposure	Species	Result
diphenylmethane-diisocyanate (isomers and homologues)	skin	Mouse	Sensitizing
4,4'-methylenediphenyl diisocyanate	Respiratory skin	Rat	Sensitizing
	Respiratory skin	Mouse	Sensitizing
	Respiratory skin	Guinea pig	Sensitizing
o-(p-isocyanatobenzyl)phenyl isocyanate	Respiratory	Mouse	Sensitizing
	Respiratory	Guinea pig	Sensitizing

Carcinogen Classification

Product/ingredient name	IARC	NTP	OSHA
diphenylmethane-diisocyanate (isomers and homologues)	3	-	-
4,4'-methylenediphenyl diisocyanate	3	-	-

Specific target organ toxicity (single exposure)

Product/ingredient name	Category	Route of exposure	Target organs
diphenylmethane-diisocyanate (isomers and homologues)	Category 3	Not applicable.	Respiratory tract irritation
4,4'-methylenediphenyl diisocyanate	Category 3	Not applicable.	Respiratory tract irritation
o-(p-isocyanatobenzyl)phenyl isocyanate	Category 3	Not applicable.	Respiratory tract irritation

Specific target organ toxicity (repeated exposure)

Product/ingredient name	Category	Route of exposure	Target organs
diphenylmethane-diisocyanate (isomers and homologues)	Category 2	Inhalation	respiratory tract
4,4'-methylenediphenyl diisocyanate	Category 2	Inhalation	respiratory tract
o-(p-isocyanatobenzyl)phenyl isocyanate	Category 2	Inhalation	respiratory tract

Information on the likely routes of exposure

Routes of entry anticipated: Oral, Dermal, Inhalation.

Potential chronic health effects

Sensitization : Contains diphenylmethane-diisocyanate (isomers and homologues), 4,4'-methylenediphenyl diisocyanate, o-(p-isocyanatobenzyl)phenyl isocyanate. May produce an allergic reaction.

Other information : No additional known significant effects or critical hazards.

SECTION 12: Ecological information

12.1 Toxicity

Do not allow to enter drains or watercourses.

When spilled, this product may act as an oil, causing a film, sheen, emulsion, or sludge at or beneath the surface of a body of water. Oils of any kind can cause: (a) drowning of waterfowl due to lack of buoyancy, loss of insulating capacity of feathers, starvation and vulnerability to predators due to lack of mobility; (b) lethal effect on fish by coating gill surfaces, preventing respiration; (c) potential fish kills resulting from alteration in biochemical oxygen demand; (d) asphyxiation of benthic life forms when floating masses become engaged with surface debris and settle on the bottom; and (e) adverse aesthetic effects of fouled shoreline and beaches.

Product/ingredient name	Result	Species	Exposure
4,4'-methylenediphenyl diisocyanate	Acute EC50 >1640 mg/l	Algae	72 hours
	Acute LC50 >1000 mg/l	Fish	96 hours
o-(p-isocyanatobenzyl)phenyl isocyanate	Acute EC50 >1640 mg/l	Algae	72 hours
	Acute LC50 >1000 mg/l	Fish	96 hours

12.2 Persistence and degradability



SECTION 12: Ecological information

Product/ingredient name	Test	Result	Dose	Inoculum
Diphenylmethane-diisocyanate (isomers and homologues)	OECD 302C 302C Inherent Biodegradability: Modified MITI Test (II)	- % - Not readily - 28 days	-	-
4,4'-methylenediphenyl diisocyanate	OECD 302C Inherent Biodegradability: Modified MITI Test (II)	- % - Not readily - 28 days	-	-
o-(p-isocyanatobenzyl)phenyl isocyanate	OECD 302C Inherent Biodegradability: Modified MITI Test (II)	- % - Not readily - 28 days	-	-

Product/ingredient name	Aquatic half-life	Photolysis	Biodegradability
Diphenylmethane-diisocyanate (isomers and homologues)	-	-	Not readily
4,4'-methylenediphenyl diisocyanate	-	-	Not readily
o-(p-isocyanatobenzyl)phenyl isocyanate	-	-	Not readily

12.3 Bioaccumulative potential

Product/ingredient name	LogP _{ow}	BCF	Potential
Diphenylmethane-diisocyanate (isomers and homologues)	-	200	low
4,4'-methylenediphenyl diisocyanate	4.51	200	low
o-(p-isocyanatobenzyl)phenyl isocyanate	4.51	200	low

12.4 Mobility in soil

Soil/water partition coefficient (K_{oc}) : No known data available in our database.

Mobility : No known data available in our database.

12.5 Other adverse effects

No known significant effects or critical hazards.

SECTION 13: Disposal considerations

13.1 Waste treatment methods

Disposal should be in accordance with applicable regional, national and local laws and regulations. Local regulations may be more stringent than regional or national requirements.

The information presented below only applies to the material as supplied. The identification based on characteristic(s) or listing may not apply if the material has been used or otherwise contaminated. It is the responsibility of the waste generator to determine the toxicity and physical properties of the material generated to determine the proper waste identification and disposal methods in compliance with applicable regulations.

Refer to Section 7 and Section 8 for additional handling information and protection of employees.

The generation of waste should be avoided or minimized wherever possible. Disposal of this product, solutions and any by-products should at all times comply with the requirements of environmental protection and waste disposal legislation and any regional local authority requirements. Dispose of surplus and non-recyclable products via a licensed waste disposal contractor. Waste should not be disposed of untreated to the sewer unless fully compliant with the requirements of all authorities with jurisdiction. Waste packaging should be recycled. Incineration or landfill should only be considered when recycling is not feasible. This material and its container must be disposed of in a safe way. Care should be taken when handling emptied containers that have not been cleaned or rinsed out. Empty containers or liners may retain some product residues. Avoid dispersal of spilled material and runoff and contact with soil, waterways, drains and sewers.

Safety Data Sheet

NEOGARD

70801 NEOCRETE WATER BORNE POLYURETHANE RESIN
CURING AGENT



SECTION 14: Transport information

Transport may take place according to national regulation or DOT for transport by road and by train, IMDG for transport by sea, IATA for Air shipment. Refer to specific Dangerous Goods Transport requirements under 49CFR, ICAO and IATA.

14.1 UN no.	14.2 Proper shipping name	14.3 Transport hazard class(es)	14.4 PG*	14.5 Env*	Additional information
DOT Code	Not regulated.				Reportable quantity (4,4'-methylenediphenyl diisocyanate) 11111.1 lbs / 5044.4 kg [1083.4 gal / 4101.2 L] Package sizes shipped in quantities less than the product reportable quantity are not subject to the RQ (reportable quantity) transportation requirements.
TDG Code	Not regulated.				
SCT Code	Not regulated.				
IMDG Code	Not regulated.				
IATA Code	Not regulated.				

Code : Classification

PG* : Packing group

Env.* : Environmental hazards

14.6 Special precautions for user

Transport within user's premises: always transport in closed containers that are upright and secure. Ensure that persons transporting the product know what to do in the event of an accident or spillage.

14.7 Transport in bulk according to Annex II of MARPOL and the IBC Code

Not applicable.

SECTION 15: Regulatory information

15.1 Safety, health and environmental regulations/legislation specific for the substance or mixture

U.S. Federal regulations :

All components are listed or exempted.

TSCA 8(a) PAIR: 4,4'-methylenediphenyl diisocyanate; o-(p-isocyanatobenzyl)phenyl isocyanate

TSCA 8(a) CDR Exempt/Partial exemption: Not determined

United States inventory (TSCA 8b): All components are listed or exempted.

TSCA 8(c) calls for record of SAR: diphenylmethane-diisocyanate (isomers and homologues); 4,4'-methylenediphenyl diisocyanate; o-(p-isocyanatobenzyl)phenyl isocyanate

Clean Water Act (CWA) 307: 4,4'-methylenediphenyl diisocyanate

Clean Air Act Section 112(b) Hazardous Air Pollutants (HAPs) : Listed

Product/ingredient name	CAS number	Concentration
4,4'-methylenediphenyl diisocyanate	101-68-8	45

Clean Air Act Section 602 Class I Substances : Not listed

Clean Air Act Section 602 Class II Substances : Not listed

DEA List I Chemicals (Precursor Chemicals) : Not listed

DEA List II Chemicals (Essential Chemicals) : Not listed

SARA 302/304 - SARA 311/312:

SARA 302/304: No products were found.

SARA 311/312 Hazards identification: Immediate (acute) health hazard, Delayed (chronic) health hazard

Safety Data Sheet

NEOGARD

70801 NEOCRETE WATER BORNE POLYURETHANE RESIN

CURING AGENT



SECTION 15: Regulatory information

Product/ingredient name	%	Fire hazard	Sudden release of pressure	Reactive	Immediate (acute) health hazard	Delayed (chronic) health hazard
Diphenylmethane-diisocyanate (isomers and homologues)	25 - 50	No.	No.	No.	Yes.	Yes.
4,4'-methylenediphenyl diisocyanate	25 - 50	No.	No.	No.	Yes.	Yes.
o-(p-isocyanatobenzyl)phenyl isocyanate	5 - 10	No.	No.	No.	Yes.	Yes.

SARA 313 :

SARA 313 notifications must not be detached from the MSDS and any copying and redistribution of the MSDS shall include copying and redistribution of the notice attached to copies of the MSDS subsequently redistributed.

Form R - Reporting requirements :

Product/ingredient name	CAS number	Concentration
Diphenylmethane-diisocyanate (isomers and homologues)	9016-87-9	20 - 50
4,4'-methylenediphenyl diisocyanate	101-68-8	20 - 50

Supplier notification :

Product/ingredient name	CAS number	Concentration
Diphenylmethane-diisocyanate (isomers and homologues)	9016-87-9	20 - 50
4,4'-methylenediphenyl diisocyanate	101-68-8	20 - 50

State regulations :

Connecticut Carcinogen Reporting: None of the components are listed.
Connecticut Hazardous Material Survey: None of the components are listed.
Florida substances: None of the components are listed.
Illinois Chemical Safety Act: None of the components are listed.
Illinois Toxic Substances Disclosure to Employee Act: None of the components are listed.
Louisiana Reporting: None of the components are listed.
Louisiana Spill: None of the components are listed.
Massachusetts Spill: None of the components are listed.
Massachusetts Substances: The following components are listed: METHYLENE BISPHENYL ISOCYANATE; DIPHENYLMETHANE DIISOCYANATE; MDI
Michigan Critical Material: None of the components are listed.
Minnesota Hazardous Substances: None of the components are listed.
New Jersey Hazardous Substances: The following components are listed: METHYLENE DIPHENYL DIISOCYANATE (POLYMERIC); ISOCYANIC ACID, POLYMETHYLENEPOLYPHENYLENE ESTER; METHYLENE BISPHENYL ISOCYANATE; BENZENE, 1,1'-METHYLENEBIS[4-ISOCYANATO-; DIISOCYANATES
New Jersey Spill: None of the components are listed.
New Jersey Toxic Catastrophe Prevention Act: None of the components are listed.
New York Acutely Hazardous Substances: The following components are listed: Methylene diphenyl diisocyanate
New York Toxic Chemical Release Reporting: None of the components are listed.
Pennsylvania RTK Hazardous Substances: The following components are listed: BENZENE, 1,1'-METHYLENEBIS[4-ISOCYANATO-
Rhode Island Hazardous Substances: None of the components are listed.

SECTION 16: Other information

Remarks :

Note: In USA, consult Code of Federal Regulations, Title 29, Labor, Parts 1910 and 1915 concerning occupational safety and health standards and regulations, as well as any other applicable Federal, State or local regulations that apply to safe practices in coating operations.
Warning! If you scrape, sand, or remove old paint, you may release lead dust. LEAD is TOXIC.

Validation :

Validated by US - HSE Products Coordinator on 1 February 2018

GHS Classification

Procedure used to derive the classification.

Classification	Justification
ACUTE TOXICITY (inhalation) - Category 4	Calculation method
SKIN IRRITATION - Category 2	Calculation method
EYE IRRITATION - Category 2A	Calculation method
RESPIRATORY SENSITIZATION - Category 1	Calculation method
SKIN SENSITIZATION - Category 1	Calculation method
CARCINOGENICITY - Category 2	Calculation method
SPECIFIC TARGET ORGAN TOXICITY (SINGLE EXPOSURE) (Respiratory tract irritation) - Category 3	Calculation method
SPECIFIC TARGET ORGAN TOXICITY (REPEATED EXPOSURE) (respiratory tract) - Category 2	Calculation method

Safety Data Sheet

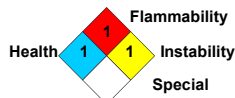
NEOGARD

70801 NEOCRETE WATER BORNE POLYURETHANE RESIN
CURING AGENT



SECTION 16: Other information

Health	* 2
Fire hazard	1
Physical hazards	0
Personal protection	X



Personal Protective Equipment (PPE) shown in this section is a suggestion. Since conditions vary from one work location to another consult the facility safety & health program. Customer or end user is responsible to evaluate worker exposure conditions at the site of application and determine the appropriate PPE suitable for workers at that particular facility or location.

Abbreviations and acronyms :

ANSI = American National Standards Institute

HCS = Hazardous Communication System

TSCA = Toxic Substances Control Act

CFR = Code of federal Regulations

GHS = Globally Harmonized System of Classification and Labelling of Chemicals

OSHA = United States Occupational Health and Safety Administration

NIOSH = National Institute for Occupational Safety and Health

ACGIH = American Conference of Industrial Hygienists

IARC = International Agency for Research on Cancer.

NTP = National Toxicology Program

ATE = Acute Toxicity Estimate

OECD = Organisation for Economic Co-operation and Development

BCF = Bioconcentration Factor

DOT = United States Department of Transportation

ERG = Emergency Response Guide

TDG = Transport of Dangerous Goods, Canada

SCT = Transportation & Communications Ministry, Mexico

IMDG = International Maritime Dangerous Goods

IATA = International Air Transport Association

SARA = Superfund Amendments Reauthorization Act

EPCRA = Emergency Planning and Community Right to Know Act

Notice to reader

Indicates information that has changed from previously issued version.

To the best of our knowledge, the information contained herein is accurate. However, neither the above named supplier nor any of its subsidiaries assumes any liability whatsoever for the accuracy or completeness of the information contained herein. Final determination of suitability of any material is the sole responsibility of the user. All materials may present unknown hazards and should be used with caution. Although certain hazards are described herein, we cannot guarantee that these are the only hazards that exist.



Conforms to ANSI Z400.1-2010 Standard - HCS 2012

Protective Clothing	General Hazard	DOT

SECTION 1: Identification of the substance/mixture and of the company/undertaking

1.1 Product identifier

Product name : NEOGARD
CHEMICAL RESISTANT URETHANE HS

Product identity : 5707900050, 70817

Product type : polyurethane paint (base for multi-component product)

1.2 Relevant identified uses of the substance or mixture and uses advised against

Field of application : buildings and metal industry.

Ready-for-use mixture : Mix ratio:
57070 = 57079 1 Vol. / 95059 1 Vol.

Identified uses : Industrial/Professional use

TSCA : **Unless otherwise stated. All components are listed or exempted.**

1.3 Details of the supplier of the safety data sheet

Company details : NEOGARD, a Division of Hempel (USA), Inc.
2728 Empire Central
Dallas, TX 75235
Phone number: 1-214-353-1600
E-mail: hempel@hempel.com

1.4 Emergency telephone number (with hours of operation)

For Transportation Emergencies : CHEMTREC: **1-800-424-9300** (Toll-free in the U.S., Canada and the U.S. Virgin Islands) **703-527-3887**
(24 hours) For calls originating elsewhere (Collect calls are accepted). Contract number: CCN10384
To preserve the effectiveness of arrangements for providing accurate and timely emergency response information, the basic identifying information (shipper name or contract number) must be included on shipping papers.
If the purchaser of this product is going to be shipping this product to other locations, the purchaser must arrange for its own Emergency Information Provider to respond to transport incidents. Hempel's 24 hour response contract does not cover non-Hempel shipments.

For all other information : In USA toll free calling available: 1-800- 678-6641 or (936)-523-6000
(8 AM - 5 PM CST) See Section 4 of the safety data sheet (first aid measures).

SECTION 2: Hazards identification

2.1 Classification of the substance or mixture

OSHA/HCS status : This material is considered hazardous by the OSHA Hazard Communication Standard (29 CFR 1910.1200).

GHS Classification : FLAMMABLE LIQUIDS - Category 4
SKIN IRRITATION - Category 2
EYE IRRITATION - Category 2A
SKIN SENSITIZATION - Category 1
TOXIC TO REPRODUCTION (Unborn child) - Category 2

2.2 Label elements

Hazard pictograms :





SECTION 2: Hazards identification

Signal word :	Warning
Hazard statements :	H227 - Combustible liquid. H319 - Causes serious eye irritation. H315 - Causes skin irritation. H317 - May cause an allergic skin reaction. H361 - Suspected of damaging the unborn child.
Precautionary statements :	
Prevention :	Obtain special instructions before use. Do not handle until all safety precautions have been read and understood. Wear protective gloves. Wear eye or face protection. Wear protective clothing. Keep away from flames and hot surfaces. - No smoking. Avoid breathing vapor. Wash hands thoroughly after handling. Contaminated work clothing must not be allowed out of the workplace.
Response :	IF exposed or concerned: Get medical attention. IF ON SKIN: Wash with plenty of soap and water. Wash contaminated clothing before reuse. If skin irritation or rash occurs: Get medical attention. IF IN EYES: Rinse cautiously with water for several minutes. Remove contact lenses, if present and easy to do. Continue rinsing. If eye irritation persists: Get medical attention.
Storage :	Store locked up. Store in a well-ventilated place. Keep cool.
Disposal :	Dispose of contents and container in accordance with all local, regional, national and international regulations.
Supplemental label elements :	None known.

2.3 Other hazards

Hazards not otherwise classified : None known.

SECTION 3: Composition/information on ingredients

Product definition : Mixture

Physical state : Liquid.

Product/ingredient name	Identifiers	%	GHS Classification
bis(4-(1,2-bis(ethoxycarbonyl)ethylamino)-3-methylcyclohexyl)methane copolymer with acidic groups fumaric acid diethyl ester	136210-32-7 623-91-6	≥25 - ≤50 ≥1 - ≤3 ≥1 - ≤2.9	SKIN SENSITIZATION - Category 1 EYE IRRITATION - Category 2A ACUTE TOXICITY (oral) - Category 4 SKIN IRRITATION - Category 2 SERIOUS EYE DAMAGE - Category 1 SPECIFIC TARGET ORGAN TOXICITY (SINGLE EXPOSURE) (Respiratory tract irritation) - Category 3
solvent naphtha (petroleum), light arom.	64742-95-6	≥1 - ≤3	FLAMMABLE LIQUIDS - Category 3 SPECIFIC TARGET ORGAN TOXICITY (SINGLE EXPOSURE) (Respiratory tract irritation) - Category 3 SPECIFIC TARGET ORGAN TOXICITY (SINGLE EXPOSURE) (Narcotic effects) - Category 3
1,3,3-trimethyl-N-(2-methylpropylidene)-5-[(2-methylpropylidene)amino] cyclohexanemethylamine	54914-37-3	≥1 - ≤3	ASPIRATION HAZARD - Category 1 SKIN CORROSION - Category 1C EYE IRRITATION - Category 2A
2-methoxy-1-methylethyl acetate	108-65-6	≥1 - ≤3	SKIN SENSITIZATION - Category 1A
bis (1,2,2,6,6-pentamethyl-4-piperidyl) sebacate	41556-26-7	≤0.3	FLAMMABLE LIQUIDS - Category 3
toluene	108-88-3	≤0.3	SKIN SENSITIZATION - Category 1 FLAMMABLE LIQUIDS - Category 2 SKIN IRRITATION - Category 2 TOXIC TO REPRODUCTION (Unborn child) - Category 2 SPECIFIC TARGET ORGAN TOXICITY (SINGLE EXPOSURE) (Narcotic effects) - Category 3 SPECIFIC TARGET ORGAN TOXICITY (REPEATED EXPOSURE) - Category 2 ASPIRATION HAZARD - Category 1

Any concentration shown as a range is to protect confidentiality or is due to batch variation.

There are no ingredients present which, within the current knowledge of the supplier and in the concentrations applicable, are classified as hazardous to health or the environment and hence require reporting in this section.



SECTION 4: First aid measures

4.1 Description of first aid measures

General :	In all cases of doubt, or when symptoms persist, seek medical attention. Never give anything by mouth to an unconscious person. If breathing is irregular, drowsiness, loss of consciousness or cramps: Call 911 and give immediate treatment (first aid).
Eye contact :	Check for and remove any contact lenses. Immediately flush eyes with plenty of water for at least 15 minutes, occasionally lifting the upper and lower eyelids. In all cases of doubt, or when symptoms persist, seek medical attention.
Inhalation :	Remove to fresh air. Keep person warm and at rest. If unconscious, place in recovery position and seek medical advice.
Skin contact :	Remove contaminated clothing and shoes. Wash skin thoroughly with soap and water or use recognized skin cleanser. Do NOT use solvents or thinners.
Ingestion :	If swallowed, seek medical advice immediately and show this container or label. Keep person warm and at rest. Do not induce vomiting unless directed to do so by medical personnel. Lower the head so that vomit will not re-enter the mouth and throat.
Protection of first-aiders :	No action shall be taken involving any personal risk or without suitable training. It may be dangerous to the person providing aid to give mouth-to-mouth resuscitation. Wash contaminated clothing thoroughly with water before removing it, or wear gloves.

4.2 Most important symptoms and effects, both acute and delayed

Potential acute health effects

Eye contact :	Causes serious eye irritation.
Inhalation :	No known significant effects or critical hazards.
Skin contact :	Causes skin irritation. May cause an allergic skin reaction.
Ingestion :	No known significant effects or critical hazards.

Over-exposure signs/symptoms

Eye contact :	Adverse symptoms may include the following: pain or irritation watering redness
Inhalation :	No specific data.
Skin contact :	Adverse symptoms may include the following: irritation redness
Ingestion :	No specific data.

4.3 Indication of any immediate medical attention and special treatment needed

Notes to physician :	If gasses have been inhaled, from the decomposition of the product, symptoms may be delayed.
Specific treatments :	No specific treatment.

SECTION 5: Firefighting measures

5.1 Extinguishing media

Extinguishing media :	Recommended: alcohol resistant foam, CO ₂ , powders, water spray. Not to be used: waterjet.
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5.2 Special hazards arising from the substance or mixture

Hazards from the substance or mixture :	In a fire or if heated, a pressure increase will occur and the container may burst. This material is harmful to aquatic life with long lasting effects. Fire water contaminated with this material must be contained and prevented from being discharged to any waterway, sewer or drain.
Hazardous combustion products :	Decomposition products may include the following materials: carbon oxides nitrogen oxides

5.3 Advice for firefighters



SECTION 5: Firefighting measures

Promptly isolate the scene by removing all persons from the vicinity of the incident if there is a fire. No action shall be taken involving any personal risk or without suitable training. Fire will produce dense black smoke. Exposure to decomposition products may cause a health hazard. Cool closed containers exposed to fire with water. Do not release runoff from fire to drains or watercourses. Fire-fighters should wear appropriate protective equipment and self-contained breathing apparatus (SCBA) with a full face-piece operated in positive pressure mode.

SECTION 6: Accidental release measures

6.1 Personal precautions, protective equipment and emergency procedures

Avoid breathing vapor or mist. Refer to protective measures listed in sections 7 and 8. No action shall be taken involving any personal risk or without suitable training. If the product contaminates lakes, rivers, or sewers, inform the appropriate authorities in accordance with local regulations.

6.2 Environmental precautions

Avoid dispersal of spilled material and runoff and contact with soil, waterways, drains and sewers. Inform the relevant authorities if the product has caused environmental pollution (sewers, waterways, soil or air). Water polluting material.

6.3 Methods and materials for containment and cleaning up

Stop leak if without risk. Move containers from spill area. Approach release from upwind. Prevent entry into sewers, water courses, basements or confined areas. Wash spillages into an effluent treatment plant or proceed as follows. Contain and collect spillage with non-combustible, absorbent material e.g. sand, earth, vermiculite or diatomaceous earth and place in container for disposal according to local regulations (see Section 13). Use spark-proof tools and explosion-proof equipment. Contaminated absorbent material may pose the same hazard as the spilled product.

6.4 Reference to other sections

See Section 1 for emergency contact information.
See Section 8 for information on appropriate personal protective equipment.
See Section 13 for additional waste treatment information.

SECTION 7: Handling and storage

7.1 Precautions for safe handling

Avoid inhalation of vapour, dust and spray mist. Avoid contact with skin and eyes. Eating, drinking and smoking should be prohibited in area where this material is handled, stored and processed. Appropriate personal protective equipment: see Section 8. Always keep in containers made from the same material as the original one.

7.2 Conditions for safe storage, including any incompatibilities

Store in accordance with local regulations. Store in a cool, well-ventilated area away from incompatible materials and ignition sources. Keep out of the reach of children. Keep away from: Oxidizing agents, strong alkalis, strong acids. No smoking. Prevent unauthorized access. Containers that are opened must be carefully resealed and kept upright to prevent leakage.

7.3 Specific end use(s)

See separate Product Data Sheet for recommendations or industrial sector specific solutions.

SECTION 8: Exposure controls/personal protection

8.1 Control parameters

Product/ingredient name	Exposure limit values
solvent naphtha (petroleum), light arom.	ACGIH TLV (United States). TWA Tentative: 25 ppm 8 hours. AIHA WEEL (United States, 10/2011). TWA: 50 ppm 8 hours. OSHA PEL Z2 (United States, 2/2013). TWA: 200 ppm 8 hours. CEIL: 300 ppm AMP: 500 ppm 10 minutes. NIOSH REL (United States, 10/2016). TWA: 100 ppm 10 hours.
2-methoxy-1-methylethyl acetate	
toluene	



SECTION 8: Exposure controls/personal protection

TWA: 375 mg/m³ 10 hours.
STEL: 150 ppm 15 minutes.
STEL: 560 mg/m³ 15 minutes.
ACGIH TLV (United States, 3/2017).
TWA: 20 ppm 8 hours.

Recommended monitoring procedures

If this product contains ingredients with exposure limits, personal, workplace atmosphere or biological monitoring may be required to determine the effectiveness of the ventilation or other control measures and/or the necessity to use respiratory protective equipment. Reference should be made to appropriate monitoring standards. Reference to national guidance documents for methods for the determination of hazardous substances will also be required.

8.2 Exposure controls

Appropriate engineering controls

Provide local exhaust and general ventilation systems to maintain airborne concentrations below OSHA, ACGIH, and manufacturer recommended exposure limits. Local exhaust ventilation is preferred because it prevents contaminant dispersion into work areas by controlling it at its source. Use local and general exhaust ventilation to effectively remove and prevent buildup of mists/vapors/fumes generated from the handling of this product.

Note: Local exhaust ventilation is designed to capture an emitted contaminant at or near its source, before the contaminant has a chance to disperse into the workplace air. General exhaust ventilation, also called dilution ventilation, is different from local exhaust ventilation because instead of capturing emissions at their source and removing them from the air, general exhaust ventilation allows the contaminant to be emitted into the workplace air and then dilutes the concentration of the contaminant to an acceptable level (e.g., to the PEL or below).

Individual protection measures

- General :** Gloves must be worn for all work that may result in soiling. Apron/coveralls/protective clothing must be worn when soiling is so great that regular work clothes do not adequately protect skin against contact with the product. Safety eyewear should be used when there is a likelihood of exposure.
- Hygiene measures :** Wash hands, forearms, and face thoroughly after handling compounds and before eating, smoking, using lavatory, and at the end of day.
- Eye/face protection :** Safety eyewear complying with an approved standard should be used when a risk assessment indicates this is necessary to avoid exposure to liquid splashes, mists, gases or dusts. If contact is possible, the following protection should be worn, unless the assessment indicates a higher degree of protection: chemical splash goggles.
- Hand protection :** Wear chemical-resistant gloves in combination with 'basic' employee training. The quality of the chemical-resistant protective gloves must be chosen as a function of the specific workplace concentrations and quantity of hazardous substances.
- Since the actual work situation is unknown. Supplier of gloves should be contacted in order to find the appropriate type. Below listed glove(s) should be regarded as generic advice:
- May be used: butyl rubber
Recommended: Silver Shield / Barrier / 4H gloves, polyvinyl alcohol (PVA), Viton®, nitrile rubber
Short term exposure: neoprene rubber, natural rubber (latex), polyvinyl chloride (PVC)
- Body protection :** Personal protective equipment for the body should be selected based on the task being performed and the risks involved handling this product.
- Respiratory protection :** If working areas have insufficient ventilation, wear half or totally covering mask equipped with gas filter of type Organic Vapor, when grinding use particle filter of type P95, P99 or P100. When spraying use a combined filter (organic vapor / HEPA or organic vapor / P100 type). Be sure to use approved/certified respirator or equivalent. Always wear an air-fed respirator when spraying in a continuous and prolonged work situation (e.g. hood with supply of fresh or compressed air or a full face, powered air purifying filter).
- Protective clothing (pictograms) :**



Note: Application of paint products by spraying requires additional safety precautions: Full body suit, Full face respirator with air supplied.

Environmental exposure controls

Emissions from ventilation or work process equipment should be checked to ensure they comply with the requirements of environmental protection legislation. In some cases, fume scrubbers, filters or engineering modifications to the process equipment will be necessary to reduce emissions to acceptable levels.



SECTION 9: Physical and chemical properties

9.1 Information on basic physical and chemical properties

Physical state :	Liquid.
Color :	Clear.
Odor :	Solvent-like
pH :	Testing not relevant or not possible due to nature of the product.
Melting point/freezing point :	Testing not relevant or not possible due to nature of the product.
Boiling point/boiling range :	Testing not relevant or not possible due to nature of the product.
Flash point :	Closed cup: 75°C (167°F)
Evaporation rate :	Testing not relevant or not possible due to nature of the product.
Flammability :	Not available.
Upper/lower flammability or explosive limits :	0.6 - 10.7 vol %
Vapor pressure :	0.01 kPa This is based on data for the following ingredient: bis(4-(1,2-bis(ethoxycarbonyl)ethylamino)-3-methylcyclohexyl)methane
Vapor density :	Testing not relevant or not possible due to nature of the product.
Relative density :	1.051 g/cm ³
Solubility(ies) :	Insoluble in the following materials: cold water and hot water.
Partition coefficient (LogKow) :	Testing not relevant or not possible due to nature of the product.
Auto-ignition temperature :	Testing not relevant or not possible due to nature of the product.
Decomposition temperature :	Testing not relevant or not possible due to nature of the product.
Viscosity :	Testing not relevant or not possible due to nature of the product.
Explosive properties :	Not available.
Oxidizing properties :	Testing not relevant or not possible due to nature of the product.

9.2 Other information

Solvent(s) % by weight (Included exempt solvent(s)):	17.2 % (w/w)
Water % by weight :	Weighted average: 0 %
VOC content (Coatings) :	1.51 lbs/gal (180.9 g/l)
VOC content (Regulatory) :	1.51 lbs/gal (181.1 g/l)
TOC Content (Volatile) :	Weighted average: 29 g/l
Solvent Gas :	Weighted average: 0.025 m ³ /l

SECTION 10: Stability and reactivity

10.1 Reactivity

No specific test data related to reactivity available for this product or its ingredients.

10.2 Chemical stability

The product is stable.

10.3 Possibility of hazardous reactions

Under normal conditions of storage and use, hazardous reactions will not occur.

10.4 Conditions to avoid

No specific data.

10.5 Incompatible materials



SECTION 10: Stability and reactivity

Reactive or incompatible with the following materials: oxidizing materials.

Slightly reactive or incompatible with the following materials: moisture.

10.6 Hazardous decomposition products

When exposed to high temperatures (i.e. in case of fire) harmful decomposition products may be formed:

Decomposition products may include the following materials: carbon oxides nitrogen oxides

SECTION 11: Toxicological information

11.1 Information on toxicological effects

Exposure to component solvent vapor concentrations may result in adverse health effects such as mucous membrane and respiratory system irritation and adverse effects on the kidneys, liver and central nervous system. Solvents may cause some of the above effects by absorption through the skin. Symptoms and signs include headaches, dizziness, fatigue, muscular weakness, drowsiness and, in extreme cases, loss of consciousness. Repeated or prolonged contact with the preparation may cause removal of natural fat from the skin, resulting in non-allergic contact dermatitis and absorption through the skin. If splashed in the eyes, the liquid may cause irritation and reversible damage. Accidental swallowing may cause stomach pain. Chemical lung inflammation may occur if the product is taken into the lungs via vomiting.

Acute toxicity

Product/ingredient name	Result	Species	Dose	Exposure
bis(4-(1,2-bis(ethoxycarbonyl)ethylamino)-3-methylcyclohexyl) methane	LC50 Inhalation Dusts and mists	Rat	4224 mg/m ³	4 hours
fumaric acid diethyl ester solvent naphtha (petroleum), light arom.	LD50 Dermal	Rat	>2000 mg/kg	-
	LD50 Oral	Rat	>2000 mg/kg	-
	LD50 Oral	Rat	1780 mg/kg	-
	LC50 Inhalation Vapor	Rat	6193 mg/m ³	4 hours
1,3,3-trimethyl-N-(2-methylpropylidene)-5-[(2-methylpropylidene)amino] cyclohexanemethylamine	LD50 Dermal	Rabbit	3160 mg/kg	-
	LD50 Oral	Rat	3492 mg/kg	-
	LD50 Dermal	Rat	>5000 mg/kg	-
2-methoxy-1-methylethyl acetate	LD50 Oral	Rat	4150 mg/kg	-
	LD50 Dermal	Rabbit	>5 g/kg	-
	LD50 Oral	Rat	8532 mg/kg	-
bis (1,2,2,6,6-pentamethyl-4-piperidyl) sebacate	LD50 Oral	Rat	>2000 mg/kg	-
	LD50 Dermal	Rat	>2000 mg/kg	-
toluene	LD50 Oral	Rat	>2000 mg/kg	-
	LC50 Inhalation Vapor	Rat	>20 mg/l	4 hours
	LD50 Oral	Rat	636 mg/kg	-

Acute toxicity estimates

Route	ATE value
Oral	65023.9 mg/kg
Dermal	232037.4 mg/kg

Irritation/Corrosion

Product/ingredient name	Result	Species	Score	Exposure
bis(4-(1,2-bis(ethoxycarbonyl)ethylamino)-3-methylcyclohexyl) methane	Skin - Mild irritant	Rabbit	-	-
solvent naphtha (petroleum), light arom.	Respiratory - Mild irritant	Rabbit	-	-
	Eyes - Mild irritant	Rabbit	-	24 hours 100 microliters
	Respiratory - Mild irritant	Rabbit	-	-
1,3,3-trimethyl-N-(2-methylpropylidene)-5-[(2-methylpropylidene)amino] cyclohexanemethylamine	Skin - Severe irritant	Rabbit	-	-
	Eyes - Severe irritant	Rabbit	-	-
	Respiratory - Mild irritant	Rabbit	-	-
2-methoxy-1-methylethyl acetate	Eyes - Mild irritant	Rabbit	-	-
	Eyes - Mild irritant	Rabbit	-	-



SECTION 11: Toxicological information

toluene	Eyes - Mild irritant Skin - Moderate irritant	Rabbit Rabbit	- -	0.5 minutes 100 milligrams 24 hours 20 milligrams
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Sensitizer

Product/ingredient name	Route of exposure	Species	Result
bis(4-(1,2-bis(ethoxycarbonyl)ethylamino)-3-methylcyclohexyl)methane	skin	Guinea pig	Sensitizing
1,3,3-trimethyl-N-(2-methylpropylidene)-5-[(2-methylpropylidene)amino]cyclohexanemethylamine	skin	Guinea pig	Sensitizing
bis (1,2,2,6,6-pentamethyl-4-piperidyl) sebacate	skin	Guinea pig	Sensitizing

Carcinogen Classification

Product/ingredient name	IARC	NTP	OSHA
toluene	3	-	-

Specific target organ toxicity (single exposure)

Product/ingredient name	Category	Route of exposure	Target organs
fumaric acid diethyl ester	Category 3	Not applicable.	Respiratory tract irritation
solvent naphtha (petroleum), light arom.	Category 3	Not applicable.	Respiratory tract irritation and Narcotic effects
toluene	Category 3	Not applicable.	Narcotic effects

Specific target organ toxicity (repeated exposure)

Product/ingredient name	Category	Route of exposure	Target organs
toluene	Category 2	Not determined	Not determined

Aspiration hazard

Product/ingredient name	Result
solvent naphtha (petroleum), light arom.	ASPIRATION HAZARD - Category 1
toluene	ASPIRATION HAZARD - Category 1

Information on the likely routes of exposure

Routes of entry anticipated: Oral, Dermal, Inhalation.

Potential chronic health effects

Sensitization : Contains bis(4-(1,2-bis(ethoxycarbonyl)ethylamino)-3-methylcyclohexyl)methane, 1,3,3-trimethyl-N-(2-methylpropylidene)-5-[(2-methylpropylidene)amino]cyclohexanemethylamine, bis (1,2,2,6,6-pentamethyl-4-piperidyl) sebacate. May produce an allergic reaction.

Other information : No additional known significant effects or critical hazards.

SECTION 12: Ecological information

12.1 Toxicity

Do not allow to enter drains or watercourses. Harmful to aquatic life with long lasting effects.

When spilled, this product may act as an oil, causing a film, sheen, emulsion, or sludge at or beneath the surface of a body of water. Oils of any kind can cause: (a) drowning of waterfowl due to lack of buoyancy, loss of insulating capacity of feathers, starvation and vulnerability to predators due to lack of mobility; (b) lethal effect on fish by coating gill surfaces, preventing respiration; (c) potential fish kills resulting from alteration in biochemical oxygen demand; (d) asphyxiation of benthic life forms when floating masses become engaged with surface debris and settle on the bottom; and (e) adverse aesthetic effects of fouled shoreline and beaches.



SECTION 12: Ecological information

Product/ingredient name	Result	Species	Exposure
bis(4-(1,2-bis(ethoxycarbonyl)ethylamino)-3-methylcyclohexyl) methane	EC50 88.6 mg/l	Daphnia	48 hours
solvent naphtha (petroleum), light arom.	IC50 113 mg/l LC50 66 mg/l NOEC 0.01 mg/l Acute EC50 2.6 mg/l	Algae Fish Daphnia Algae - Pseudokirchneriella subcapitata (green algae)	72 hours 96 hours 21 days 96 hours
bis (1,2,2,6,6-pentamethyl-4-piperidyl) sebacate	Acute EC50 6.14 mg/l Acute LC50 9.22 mg/l	Daphnia - Daphnia magna Fish - Oncorhynchus mykiss (rainbow trout)	48 hours 96 hours
toluene	Acute EC50 1.68 mg/l Acute LC50 0.97 mg/l Fresh water Chronic NOEC <500000 µg/l Fresh water Chronic NOEC 1000 µg/l Fresh water	Aquatic plants Fish - Lepomis macrochirus Algae - Pseudokirchneriella subcapitata Daphnia - Daphnia magna	72 hours 96 hours 96 hours 21 days

12.2 Persistence and degradability

Product/ingredient name	Test	Result	Dose	Inoculum
bis(4-(1,2-bis(ethoxycarbonyl)ethylamino)-3-methylcyclohexyl) methane	OECD 301 F 301F Ready Biodegradability - Manometric Respirometry Test	13 % - Not readily - 28 days	-	-
solvent naphtha (petroleum), light arom.	-	>70 % - Readily - 28 days	-	-

Product/ingredient name	Aquatic half-life	Photolysis	Biodegradability
bis(4-(1,2-bis(ethoxycarbonyl)ethylamino)-3-methylcyclohexyl) methane	-	-	Not readily
fumaric acid diethyl ester	-	-	Readily
solvent naphtha (petroleum), light arom.	-	-	Readily
2-methoxy-1-methylethyl acetate	-	-	Readily

12.3 Bioaccumulative potential

Product/ingredient name	LogP _{ow}	BCF	Potential
bis(4-(1,2-bis(ethoxycarbonyl)ethylamino)-3-methylcyclohexyl) methane	5.99	8228	high
solvent naphtha (petroleum), light arom.	-	10 - 2500	high
1,3,3-trimethyl-N-(2-methylpropylidene)-5-[(2-methylpropylidene) amino]cyclohexanemethylamine	5.2	-	high
2-methoxy-1-methylethyl acetate	1.2	-	low
toluene	2.73	90	low

12.4 Mobility in soil

Soil/water partition coefficient (K_{oc}) : No known data available in our database.

Mobility : No known data available in our database.

12.5 Other adverse effects

No known significant effects or critical hazards.



SECTION 13: Disposal considerations

13.1 Waste treatment methods

Disposal should be in accordance with applicable regional, national and local laws and regulations. Local regulations may be more stringent than regional or national requirements.

The information presented below only applies to the material as supplied. The identification based on characteristic(s) or listing may not apply if the material has been used or otherwise contaminated. It is the responsibility of the waste generator to determine the toxicity and physical properties of the material generated to determine the proper waste identification and disposal methods in compliance with applicable regulations.

Refer to Section 7 and Section 8 for additional handling information and protection of employees.

The generation of waste should be avoided or minimized wherever possible. Disposal of this product, solutions and any by-products should at all times comply with the requirements of environmental protection and waste disposal legislation and any regional local authority requirements. Dispose of surplus and non-recyclable products via a licensed waste disposal contractor. Waste should not be disposed of untreated to the sewer unless fully compliant with the requirements of all authorities with jurisdiction. Waste packaging should be recycled. Incineration or landfill should only be considered when recycling is not feasible. This material and its container must be disposed of in a safe way. Care should be taken when handling emptied containers that have not been cleaned or rinsed out. Empty containers or liners may retain some product residues. Vapor from product residues may create a highly flammable or explosive atmosphere inside the container. Do not cut, weld or grind used containers unless they have been cleaned thoroughly internally. Avoid dispersal of spilled material and runoff and contact with soil, waterways, drains and sewers.

SECTION 14: Transport information

Transport may take place according to national regulation or DOT for transport by road and by train, IMDG for transport by sea, IATA for Air shipment. Refer to specific Dangerous Goods Transport requirements under 49CFR, ICAO and IATA.

14.1 UN no.	14.2 Proper shipping name	14.3 Transport hazard class(es)	14.4 PG*	14.5 Env*	Additional information
DOT Code	Not regulated.				
TDG Code	Not regulated.				
SCT Code	Not regulated.				
IMDG Code	Not regulated.				
IATA Code	Not regulated.				

Code : Classification

PG* : Packing group

Env.* : Environmental hazards

14.6 Special precautions for user

Transport within user's premises: always transport in closed containers that are upright and secure. Ensure that persons transporting the product know what to do in the event of an accident or spillage.

14.7 Transport in bulk according to Annex II of MARPOL and the IBC Code

Not applicable.

SECTION 15: Regulatory information

15.1 Safety, health and environmental regulations/legislation specific for the substance or mixture

U.S. Federal regulations :

Not determined.

TSCA 8(a) PAIR: 2-methoxy-1-methylethyl acetate; acetonitrile

TSCA 8(a) CDR Exempt/Partial exemption: Not determined

United States inventory (TSCA 8b): Not determined.

Clean Water Act (CWA) 307: toluene; acetonitrile

Clean Water Act (CWA) 311: toluene; n-butyl acetate; acetic acid ; formaldehyde; phosphoric acid

Clean Air Act Section 112(b) Hazardous Air Pollutants (HAPs) : Listed



SECTION 15: Regulatory information

Product/ingredient name	CAS number	Concentration
2-(2-butoxyethoxy)ethyl acetate	124-17-4	11.873
toluene	108-88-3	0.11962
2-(2-butoxyethoxy)ethanol	112-34-5	0.090623
glass beads	65997-17-3	0.326
acetonitrile	75-05-8	0.0010573

Clean Air Act Section 602 Class I Substances : Not listed

Clean Air Act Section 602 Class II Substances : Not listed

DEA List I Chemicals (Precursor Chemicals) : Not listed

DEA List II Chemicals (Essential Chemicals) : Not listed

SARA 302/304 - SARA 311/312:

SARA 302/304: formaldehyde

SARA 311/312 Hazards identification: Fire hazard, Immediate (acute) health hazard, Delayed (chronic) health hazard

Product/ingredient name	%	Fire hazard	Sudden release of pressure	Reactive	Immediate (acute) health hazard	Delayed (chronic) health hazard
bis(4-(1,2-bis(ethoxycarbonyl)ethylamino)-3-methylcyclohexyl)methane	25 - 50	No.	No.	No.	Yes.	No.
copolymer with acidic groups	1 - 3	No.	No.	No.	Yes.	No.
fumaric acid diethyl ester	1 - 3	No.	No.	No.	Yes.	No.
solvent naphtha (petroleum), light arom.	1 - 3	Yes.	No.	No.	Yes.	No.
1,3,3-trimethyl-N-(2-methylpropylidene)-5-[(2-methylpropylidene)amino]cyclohexanemethylamine	1 - 3	No.	No.	No.	Yes.	Yes.
2-methoxy-1-methylethyl acetate	1 - 3	Yes.	No.	No.	No.	No.
bis (1,2,2,6,6-pentamethyl-4-piperidyl) sebacate	0.1 - 0.5	No.	No.	No.	Yes.	No.
toluene	0.1 - 0.5	Yes.	No.	No.	Yes.	Yes.

SARA 313 :

SARA 313 notifications must not be detached from the MSDS and any copying and redistribution of the MSDS shall include copying and redistribution of the notice attached to copies of the MSDS subsequently redistributed.

Form R - Reporting requirements :

Product/ingredient name	CAS number	Concentration
2-(2-butoxyethoxy)ethyl acetate	124-17-4	10 - 20

Supplier notification :

Product/ingredient name	CAS number	Concentration
2-(2-butoxyethoxy)ethyl acetate	124-17-4	10 - 20

State regulations :

Connecticut Carcinogen Reporting: None of the components are listed.

Connecticut Hazardous Material Survey: None of the components are listed.

Florida substances: None of the components are listed.

Illinois Chemical Safety Act: None of the components are listed.

Illinois Toxic Substances Disclosure to Employee Act: None of the components are listed.

Louisiana Reporting: None of the components are listed.

Louisiana Spill: None of the components are listed.

Massachusetts Spill: None of the components are listed.

Massachusetts Substances: None of the components are listed.

Michigan Critical Material: None of the components are listed.

Minnesota Hazardous Substances: None of the components are listed.

New Jersey Hazardous Substances: The following components are listed: GLYCOL ETHERS

New Jersey Spill: None of the components are listed.

New Jersey Toxic Catastrophe Prevention Act: None of the components are listed.

New York Acutely Hazardous Substances: None of the components are listed.

New York Toxic Chemical Release Reporting: None of the components are listed.

Pennsylvania RTK Hazardous Substances: None of the components are listed.

Rhode Island Hazardous Substances: None of the components are listed.

California Prop. 65 PFF :

WARNING: This product contains less than 0.1% of a chemical known to the State of California to cause cancer. **WARNING:** This product contains less than 1% of a chemical known to the State of California to cause birth defects or other reproductive harm.

Product/ingredient name	Cancer	Reproductive	No significant risk level	Maximum acceptable dosage level
toluene	No.	Yes.		Yes.
ethyl acrylate	Yes.	No.		
formaldehyde	Yes.	No.	Yes.	



SECTION 16: Other information

Remarks : Note: In USA, consult Code of Federal Regulations, Title 29, Labor, Parts 1910 and 1915 concerning occupational safety and health standards and regulations, as well as any other applicable Federal, State or local regulations that apply to safe practices in coating operations.

Warning! If you scrape, sand, or remove old paint, you may release lead dust. LEAD is TOXIC.

Validation : Validated by US - HSE Products Coordinator on 29 September 2018

GHS Classification

Procedure used to derive the classification.

Classification	Justification
FLAMMABLE LIQUIDS - Category 4 SKIN IRRITATION - Category 2 EYE IRRITATION - Category 2A SKIN SENSITIZATION - Category 1 TOXIC TO REPRODUCTION (Unborn child) - Category 2	On basis of test data Calculation method Calculation method Calculation method Calculation method

Hazardous Material Information System (U.S.A.)

Health	* 2
Fire hazard	2
Physical hazards	0
Personal protection	X

National Fire Protection Association (U.S.A.)

Health	2	0	Flammability
			Instability
			Special

Personal Protective Equipment (PPE) shown in this section is a suggestion. Since conditions vary from one work location to another consult the facility safety & health program. Customer or end user is responsible to evaluate worker exposure conditions at the site of application and determine the appropriate PPE suitable for workers at that particular facility or location.

Abbreviations and acronyms :

ANSI = American National Standards Institute

HCS = Hazardous Communication System

TSCA = Toxic Substances Control Act

CFR = Code of federal Regulations

GHS = Globally Harmonized System of Classification and Labelling of Chemicals

OSHA = United States Occupational Health and Safety Administration

NIOSH = National Institute for Occupational Safety and Health

ACGIH = American Conference of Industrial Hygienists

IARC = International Agency for Research on Cancer.

NTP = National Toxicology Program

ATE = Acute Toxicity Estimate

OECD = Organisation for Economic Co-operation and Development

BCF = Bioconcentration Factor

DOT = United States Department of Transportation

ERG = Emergency Response Guide

TDG = Transport of Dangerous Goods, Canada

SCT = Transportation & Communications Ministry, Mexico

IMDG = International Maritime Dangerous Goods

IATA = International Air Transport Association

SARA = Superfund Amendments Reauthorization Act

EPCRA = Emergency Planning and Community Right to Know Act

Notice to reader

Indicates information that has changed from previously issued version.

To the best of our knowledge, the information contained herein is accurate. However, neither the above named supplier nor any of its subsidiaries assumes any liability whatsoever for the accuracy or completeness of the information contained herein. Final determination of suitability of any material is the sole responsibility of the user. All materials may present unknown hazards and should be used with caution. Although certain hazards are described herein, we cannot guarantee that these are the only hazards that exist.

Safety Data Sheet

NEOGARD

70818 CHEMICAL RESISTANT URETHANE HS CURING AGENT



Conforms to ANSI Z400.1-2010 Standard - HCS 2012

Protective Clothing	General Hazard	DOT

SECTION 1: Identification of the substance/mixture and of the company/undertaking

1.1 Product identifier

Product name : NEOGARD
CHEMICAL RESISTANT URETHANE HS CURING AGENT

Product identity : 9505900000, 70818

Product type : Curing agent

1.2 Relevant identified uses of the substance or mixture and uses advised against

Field of application : used only as part of two- or multi component products.

Ready-for-use mixture : Used for: 57059

Identified uses : Industrial/Professional use

TSCA : **Unless otherwise stated. All components are listed or exempted.**

1.3 Details of the supplier of the safety data sheet

Company details : NEOGARD, a Division of Hempel (USA), Inc.
2728 Empire Central
Dallas, TX 75235
Phone number: 1-214-353-1600
E-mail: hempel@hempel.com

1.4 Emergency telephone number (with hours of operation)

For Transportation Emergencies : CHEMTREC: **1-800-424-9300** (Toll-free in the U.S., Canada and the U.S. Virgin Islands) **703-527-3887**
(24 hours)
For calls originating elsewhere (Collect calls are accepted). Contract number: CCN10384
To preserve the effectiveness of arrangements for providing accurate and timely emergency response information, the basic identifying information (shipper name or contract number) must be included on shipping papers.
If the purchaser of this product is going to be shipping this product to other locations, the purchaser must arrange for its own Emergency Information Provider to respond to transport incidents. Hempel's 24 hour response contract does not cover non-Hempel shipments.

For all other information : In USA toll free calling available: 1-800- 678-6641 or (936)-523-6000
(8 AM - 5 PM CST) See Section 4 of the safety data sheet (first aid measures).

SECTION 2: Hazards identification

2.1 Classification of the substance or mixture

OSHA/HCS status : This material is considered hazardous by the OSHA Hazard Communication Standard (29 CFR 1910.1200).

GHS Classification : ACUTE TOXICITY (inhalation) - Category 4
RESPIRATORY SENSITIZATION - Category 1
SKIN SENSITIZATION - Category 1
SPECIFIC TARGET ORGAN TOXICITY (SINGLE EXPOSURE) (Respiratory tract irritation) - Category 3
SPECIFIC TARGET ORGAN TOXICITY (REPEATED EXPOSURE) (lungs) - Category 2

2.2 Label elements

Hazard pictograms :





SECTION 2: Hazards identification

Signal word :	Danger
Hazard statements :	H332 - Harmful if inhaled. H334 - May cause allergy or asthma symptoms or breathing difficulties if inhaled. H317 - May cause an allergic skin reaction. H335 - May cause respiratory irritation. H373 - May cause damage to organs through prolonged or repeated exposure. (lungs)
Precautionary statements :	
Prevention :	Wear protective gloves. Wear respiratory protection. Use only outdoors or in a well-ventilated area. Do not breathe vapor. Contaminated work clothing must not be allowed out of the workplace.
Response :	Get medical attention if you feel unwell. IF INHALED: If breathing is difficult, remove person to fresh air and keep comfortable for breathing. Call a POISON CENTER or physician if you feel unwell. If experiencing respiratory symptoms: Call a POISON CENTER or physician. IF ON SKIN: Wash with plenty of soap and water. Wash contaminated clothing before reuse. If skin irritation or rash occurs: Get medical attention.
Storage :	Store locked up.
Disposal :	Dispose of contents and container in accordance with all local, regional, national and international regulations.
Supplemental label elements :	None known.

2.3 Other hazards

Hazards not otherwise classified : None known.

SECTION 3: Composition/information on ingredients

Product definition :	Mixture
Physical state :	Liquid.

Product/ingredient name	Identifiers	%	GHS Classification
homopolymer of hexamethylene diisocyanate	28182-81-2	≥90	ACUTE TOXICITY (inhalation) - Category 4 RESPIRATORY SENSITIZATION - Category 1 SKIN SENSITIZATION - Category 1 SPECIFIC TARGET ORGAN TOXICITY (SINGLE EXPOSURE) (Respiratory tract irritation) - Category 3 SPECIFIC TARGET ORGAN TOXICITY (REPEATED EXPOSURE) (lungs) (inhalation) - Category 2

Any concentration shown as a range is to protect confidentiality or is due to batch variation.

There are no ingredients present which, within the current knowledge of the supplier and in the concentrations applicable, are classified as hazardous to health or the environment and hence require reporting in this section.

SECTION 4: First aid measures

4.1 Description of first aid measures

General :	In all cases of doubt, or when symptoms persist, seek medical attention. Never give anything by mouth to an unconscious person. If breathing is irregular, drowsiness, loss of consciousness or cramps: Call 911 and give immediate treatment (first aid).
Eye contact :	Check for and remove any contact lenses. Immediately flush eyes with plenty of water for at least 15 minutes, occasionally lifting the upper and lower eyelids. In all cases of doubt, or when symptoms persist, seek medical attention.
Inhalation :	Remove to fresh air. Keep person warm and at rest. If not breathing, if breathing is irregular or if respiratory arrest occurs, provide artificial respiration or oxygen by trained personnel. Give nothing by mouth. If unconscious, place in recovery position and get medical attention immediately.
Skin contact :	Remove contaminated clothing and shoes. Wash skin thoroughly with soap and water or use recognized skin cleanser. Do NOT use solvents or thinners.
Ingestion :	If swallowed, seek medical advice immediately and show this container or label. Keep person warm and at rest. Do not induce vomiting unless directed to do so by medical personnel. Lower the head so that vomit will not re-enter the mouth and throat.



SECTION 4: First aid measures

Protection of first-aiders : No action shall be taken involving any personal risk or without suitable training. If it is suspected that fumes are still present, the rescuer should wear an appropriate mask or self-contained breathing apparatus. It may be dangerous to the person providing aid to give mouth-to-mouth resuscitation. Wash contaminated clothing thoroughly with water before removing it, or wear gloves.

4.2 Most important symptoms and effects, both acute and delayed

Potential acute health effects

Eye contact : No known significant effects or critical hazards.

Inhalation : Harmful if inhaled. May cause respiratory irritation. May cause allergy or asthma symptoms or breathing difficulties if inhaled.

Skin contact : May cause an allergic skin reaction.

Ingestion : No known significant effects or critical hazards.

Over-exposure signs/symptoms

Eye contact : No specific data.

Inhalation : Adverse symptoms may include the following:
respiratory tract irritation
coughing
wheezing and breathing difficulties
asthma

Skin contact : Adverse symptoms may include the following:
irritation
redness

Ingestion : No specific data.

4.3 Indication of any immediate medical attention and special treatment needed

Notes to physician : If gasses have been inhaled, from the decomposition of the product, symptoms may be delayed.

Specific treatments : No specific treatment.

SECTION 5: Firefighting measures

5.1 Extinguishing media

Extinguishing media : Recommended: alcohol resistant foam, CO₂, powders, water spray.
Not to be used: waterjet.

5.2 Special hazards arising from the substance or mixture

Hazards from the substance or mixture : In a fire or if heated, a pressure increase will occur and the container may burst.

Hazardous combustion products : Decomposition products may include the following materials: carbon oxides nitrogen oxides

5.3 Advice for firefighters

Promptly isolate the scene by removing all persons from the vicinity of the incident if there is a fire. No action shall be taken involving any personal risk or without suitable training. Fire will produce dense black smoke. Exposure to decomposition products may cause a health hazard. Cool closed containers exposed to fire with water. Do not release runoff from fire to drains or watercourses. Fire-fighters should wear appropriate protective equipment and self-contained breathing apparatus (SCBA) with a full face-piece operated in positive pressure mode.

SECTION 6: Accidental release measures

6.1 Personal precautions, protective equipment and emergency procedures

Avoid all direct contact with the spilled material. Refer to protective measures listed in sections 7 and 8. No action shall be taken involving any personal risk or without suitable training. If the product contaminates lakes, rivers, or sewers, inform the appropriate authorities in accordance with local regulations.

6.2 Environmental precautions

Avoid dispersal of spilled material and runoff and contact with soil, waterways, drains and sewers. Inform the relevant authorities if the product has caused environmental pollution (sewers, waterways, soil or air).



SECTION 6: Accidental release measures

6.3 Methods and materials for containment and cleaning up

Stop leak if without risk. Move containers from spill area. Approach release from upwind. Prevent entry into sewers, water courses, basements or confined areas. Wash spillages into an effluent treatment plant or proceed as follows. Contain and collect spillage with non-combustible, absorbent material e.g. sand, earth, vermiculite or diatomaceous earth and place in container for disposal according to local regulations (see Section 13). Contaminated absorbent material may pose the same hazard as the spilled product.

6.4 Reference to other sections

See Section 1 for emergency contact information.

See Section 8 for information on appropriate personal protective equipment.

See Section 13 for additional waste treatment information.

SECTION 7: Handling and storage

7.1 Precautions for safe handling

Contains isocyanates. Exposure to isocyanate may result in acute irritation and/or sensitisation when breathing.

Care should be taken when re-opening partly-used containers.

Avoid inhalation of vapour, dust and spray mist. Avoid contact with skin and eyes. Eating, drinking and smoking should be prohibited in area where this material is handled, stored and processed. Appropriate personal protective equipment: see Section 8. Always keep in containers made from the same material as the original one.

7.2 Conditions for safe storage, including any incompatibilities

Store in accordance with local regulations for flammable liquids. Store in a cool, well-ventilated area away from incompatible materials and ignition sources. Keep out of the reach of children. Keep away from: Oxidizing agents, strong alkalis, strong acids as well as of amines, alcohols and water. No smoking. Prevent unauthorized access. Containers that are opened must be carefully resealed and kept upright to prevent leakage.

7.3 Specific end use(s)

See separate Product Data Sheet for recommendations or industrial sector specific solutions.

SECTION 8: Exposure controls/personal protection

8.1 Control parameters

Product/ingredient name	Exposure limit values
No exposure limit value known.	

Recommended monitoring procedures

If this product contains ingredients with exposure limits, personal, workplace atmosphere or biological monitoring may be required to determine the effectiveness of the ventilation or other control measures and/or the necessity to use respiratory protective equipment. Reference should be made to appropriate monitoring standards. Reference to national guidance documents for methods for the determination of hazardous substances will also be required.

8.2 Exposure controls

Appropriate engineering controls

Provide local exhaust and general ventilation systems to maintain airborne concentrations below OSHA, ACGIH, and manufacturer recommended exposure limits. Local exhaust ventilation is preferred because it prevents contaminant dispersion into work areas by controlling it at its source. Use local and general exhaust ventilation to effectively remove and prevent buildup of mists/vapors/fumes generated from the handling of this product.

Note: Local exhaust ventilation is designed to capture an emitted contaminant at or near its source, before the contaminant has a chance to disperse into the workplace air. General exhaust ventilation, also called dilution ventilation, is different from local exhaust ventilation because instead of capturing emissions at their source and removing them from the air, general exhaust ventilation allows the contaminant to be emitted into the workplace air and then dilutes the concentration of the contaminant to an acceptable level (e.g., to the PEL or below).

Individual protection measures

General :

Gloves must be worn for all work that may result in soiling. Apron/coveralls/protective clothing must be worn when soiling is so great that regular work clothes do not adequately protect skin against contact with the product. Safety eyewear should be used when there is a likelihood of exposure.



SECTION 8: Exposure controls/personal protection

Hygiene measures :	Wash hands, forearms, and face thoroughly after handling compounds and before eating, smoking, using lavatory, and at the end of day.
Eye/face protection :	Safety eyewear complying with an approved standard should be used when a risk assessment indicates this is necessary to avoid exposure to liquid splashes, mists, gases or dusts. If contact is possible, the following protection should be worn, unless the assessment indicates a higher degree of protection: safety glasses with side-shields.
Hand protection :	Wear chemical-resistant gloves in combination with 'basic' employee training. The quality of the chemical-resistant protective gloves must be chosen as a function of the specific workplace concentrations and quantity of hazardous substances. Since the actual work situation is unknown. Supplier of gloves should be contacted in order to find the appropriate type.
Body protection :	Personal protective equipment for the body should be selected based on the task being performed and the risks involved handling this product.
Respiratory protection :	Use appropriate respiratory protection if there is a risk of exceeding any exposure limits.
Protective clothing (pictograms) :	



Note: Application of paint products by spraying requires additional safety precautions: Full body suit, Full face respirator with air supplied.

Environmental exposure controls

Emissions from ventilation or work process equipment should be checked to ensure they comply with the requirements of environmental protection legislation. In some cases, fume scrubbers, filters or engineering modifications to the process equipment will be necessary to reduce emissions to acceptable levels.

SECTION 9: Physical and chemical properties

9.1 Information on basic physical and chemical properties

Physical state :	Liquid.
Color :	Clear.
Odor :	isocyanate
pH :	Testing not relevant or not possible due to nature of the product.
Melting point/freezing point :	-39.85°C This is based on data for the following ingredient: homopolymer of hexamethylene diisocyanate
Boiling point/boiling range :	Testing not relevant or not possible due to nature of the product.
Flash point :	Closed cup: 157°C (314.6°F)
Evaporation rate :	Testing not relevant or not possible due to nature of the product.
Flammability :	Not available.
Upper/lower flammability or explosive limits :	No specific data.
Vapor pressure :	0 kPa This is based on data for the following ingredient: homopolymer of hexamethylene diisocyanate
Vapor density :	Testing not relevant or not possible due to nature of the product.
Relative density :	1.16 g/cm ³
Solubility(ies) :	
Partition coefficient (LogKow) :	Testing not relevant or not possible due to nature of the product.
Auto-ignition temperature :	Testing not relevant or not possible due to nature of the product.
Decomposition temperature :	Testing not relevant or not possible due to nature of the product.
Viscosity :	Testing not relevant or not possible due to nature of the product.
Explosive properties :	Not available.
Oxidizing properties :	Testing not relevant or not possible due to nature of the product.

9.2 Other information

Solvent(s) % by weight (Included exempt solvent(s)):	0 % (w/w)
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SECTION 9: Physical and chemical properties

Water % by weight :	Weighted average: 0 %
VOC content (Coatings) :	0 lbs/gal (0 g/l)
VOC content (Regulatory) :	0 lbs/gal (0 g/l)
TOC Content (Volatile) :	Weighted average: 0 g/l
Solvent Gas :	Weighted average: 0 m³/l

SECTION 10: Stability and reactivity

10.1 Reactivity

No specific test data related to reactivity available for this product or its ingredients.

10.2 Chemical stability

The product is stable.

10.3 Possibility of hazardous reactions

Under normal conditions of storage and use, hazardous reactions will not occur.

10.4 Conditions to avoid

No specific data.

10.5 Incompatible materials

10.6 Hazardous decomposition products

When exposed to high temperatures (i.e. in case of fire) harmful decomposition products may be formed:

Decomposition products may include the following materials: carbon oxides nitrogen oxides

SECTION 11: Toxicological information

11.1 Information on toxicological effects

Isocyanate containing products have characteristics that include producing acute irritation and/or sensitisation when breathing, subsequent asthmatic problems and lung contractions. Sensitised people can, as a result from this, show asthmatic symptoms with exposure to atmospheric concentrations far below the TLV. Repeated exposures will lead to permanent damage to the respiratory system.

Acute toxicity

Product/ingredient name	Result	Species	Dose	Exposure
homopolymer of hexamethylene diisocyanate	LC50 Inhalation Dusts and mists	Rat	18500 mg/m³	1 hours
	LC50 Inhalation Dusts and mists	Rat	1.5 mg/l	4 hours
	LD50 Dermal	Rat	>2000 mg/kg	-
	LD50 Oral	Rat	>5000 mg/kg	-

Acute toxicity estimates

Route	ATE value
Inhalation (dusts and mists)	1.504 mg/l

Irritation/Corrosion

Product/ingredient name	Result	Species	Score	Exposure
homopolymer of hexamethylene diisocyanate	Skin - Mild irritant	Rabbit	-	-
	Respiratory - Mild irritant	Rabbit	-	-

Sensitizer



SECTION 11: Toxicological information

Product/ingredient name	Route of exposure	Species	Result
homopolymer of hexamethylene diisocyanate	skin	Mouse	Sensitizing

Specific target organ toxicity (single exposure)

Product/ingredient name	Category	Route of exposure	Target organs
homopolymer of hexamethylene diisocyanate	Category 3	Not applicable.	Respiratory tract irritation

Specific target organ toxicity (repeated exposure)

Product/ingredient name	Category	Route of exposure	Target organs
homopolymer of hexamethylene diisocyanate	Category 2	Inhalation	lungs

Information on the likely routes of exposure

Routes of entry anticipated: Oral, Dermal, Inhalation.

Potential chronic health effects

Sensitization : Contains homopolymer of hexamethylene diisocyanate, hexamethylene-di-isocyanate. May produce an allergic reaction.

Other information : No additional known significant effects or critical hazards.

SECTION 12: Ecological information

12.1 Toxicity

Do not allow to enter drains or watercourses.

When spilled, this product may act as an oil, causing a film, sheen, emulsion, or sludge at or beneath the surface of a body of water. Oils of any kind can cause: (a) drowning of waterfowl due to lack of buoyancy, loss of insulating capacity of feathers, starvation and vulnerability to predators due to lack of mobility; (b) lethal effect on fish by coating gill surfaces, preventing respiration; (c) potential fish kills resulting from alteration in biochemical oxygen demand; (d) asphyxiation of benthic life forms when floating masses become engaged with surface debris and settle on the bottom; and (e) adverse aesthetic effects of fouled shoreline and beaches.

12.2 Persistence and degradability

Product/ingredient name	Test	Result	Dose	Inoculum
homopolymer of hexamethylene diisocyanate	OECD 301C Ready Biodegradability - Modified MITI Test (I)	2 % - Not readily - 28 days	-	-

Product/ingredient name	Aquatic half-life	Photolysis	Biodegradability
homopolymer of hexamethylene diisocyanate	-	-	Not readily

12.3 Bioaccumulative potential

Product/ingredient name	LogP _{ow}	BCF	Potential
homopolymer of hexamethylene diisocyanate	5.54	367.7	low

12.4 Mobility in soil

Soil/water partition coefficient (K_{oc}) : No known data available in our database.

Mobility : No known data available in our database.

12.5 Other adverse effects

No known significant effects or critical hazards.



SECTION 13: Disposal considerations

13.1 Waste treatment methods

Disposal should be in accordance with applicable regional, national and local laws and regulations. Local regulations may be more stringent than regional or national requirements.

The information presented below only applies to the material as supplied. The identification based on characteristic(s) or listing may not apply if the material has been used or otherwise contaminated. It is the responsibility of the waste generator to determine the toxicity and physical properties of the material generated to determine the proper waste identification and disposal methods in compliance with applicable regulations.

Refer to Section 7 and Section 8 for additional handling information and protection of employees.

The generation of waste should be avoided or minimized wherever possible. Disposal of this product, solutions and any by-products should at all times comply with the requirements of environmental protection and waste disposal legislation and any regional local authority requirements. Dispose of surplus and non-recyclable products via a licensed waste disposal contractor. Waste should not be disposed of untreated to the sewer unless fully compliant with the requirements of all authorities with jurisdiction. Waste packaging should be recycled. Incineration or landfill should only be considered when recycling is not feasible. This material and its container must be disposed of in a safe way. Care should be taken when handling emptied containers that have not been cleaned or rinsed out. Empty containers or liners may retain some product residues. Avoid dispersal of spilled material and runoff and contact with soil, waterways, drains and sewers.

SECTION 14: Transport information

Transport may take place according to national regulation or DOT for transport by road and by train, IMDG for transport by sea, IATA for Air shipment. Refer to specific Dangerous Goods Transport requirements under 49CFR, ICAO and IATA.

14.1 UN no.	14.2 Proper shipping name	14.3 Transport hazard class(es)	14.4 PG*	14.5 Env*	Additional information
DOT Code	Not regulated.				Reportable quantity (hexamethylene-di-isocyanate) 40000 lbs / 18160 kg [4135.7 gal / 15655.2 L] Package sizes shipped in quantities less than the product reportable quantity are not subject to the RQ (reportable quantity) transportation requirements.
TDG Code	Not regulated.				
SCT Code	Not regulated.				
IMDG Code	Not regulated.				
IATA Code	Not regulated.				

Code : Classification

PG* : Packing group

Env.* : Environmental hazards

14.6 Special precautions for user

Transport within user's premises: always transport in closed containers that are upright and secure. Ensure that persons transporting the product know what to do in the event of an accident or spillage.

14.7 Transport in bulk according to Annex II of MARPOL and the IBC Code

Not applicable.

SECTION 15: Regulatory information

15.1 Safety, health and environmental regulations/legislation specific for the substance or mixture

U.S. Federal regulations :

All components are listed or exempted.

TSCA 8(a) CDR Exempt/Partial exemption: Not determined

United States inventory (TSCA 8b): All components are listed or exempted.

TSCA 8(c) calls for record of SAR: hexamethylene-di-isocyanate

Clean Water Act (CWA) 307: hexamethylene-di-isocyanate

Clean Air Act Section 112(b) Hazardous Air Pollutants (HAPs) : Listed

Safety Data Sheet

NEOGARD

70818 CHEMICAL RESISTANT URETHANE HS CURING AGENT



SECTION 15: Regulatory information

Product/ingredient name	CAS number	Concentration
hexamethylene-di-isocyanate	822-06-0	0.25

Clean Air Act Section 602 Class I Substances : Not listed

Clean Air Act Section 602 Class II Substances : Not listed

DEA List I Chemicals (Precursor Chemicals) : Not listed

DEA List II Chemicals (Essential Chemicals) : Not listed

SARA 302/304 - SARA 311/312:

SARA 302/304: No products were found.

SARA 311/312 Hazards identification: Immediate (acute) health hazard

Product/ingredient name	%	Fire hazard	Sudden release of pressure	Reactive	Immediate (acute) health hazard	Delayed (chronic) health hazard
homopolymer of hexamethylene diisocyanate	75 - 100	No.	No.	No.	Yes.	Yes.

State regulations :

Connecticut Carcinogen Reporting: None of the components are listed.

Connecticut Hazardous Material Survey: None of the components are listed.

Florida substances: None of the components are listed.

Illinois Chemical Safety Act: None of the components are listed.

Illinois Toxic Substances Disclosure to Employee Act: None of the components are listed.

Louisiana Reporting: None of the components are listed.

Louisiana Spill: None of the components are listed.

Massachusetts Spill: None of the components are listed.

Massachusetts Substances: None of the components are listed.

Michigan Critical Material: None of the components are listed.

Minnesota Hazardous Substances: None of the components are listed.

New Jersey Hazardous Substances: None of the components are listed.

New Jersey Spill: None of the components are listed.

New Jersey Toxic Catastrophe Prevention Act: None of the components are listed.

New York Acutely Hazardous Substances: None of the components are listed.

New York Toxic Chemical Release Reporting: None of the components are listed.

Pennsylvania RTK Hazardous Substances: None of the components are listed.

Rhode Island Hazardous Substances: None of the components are listed.

SECTION 16: Other information

Remarks :

Note: In USA, consult Code of Federal Regulations, Title 29, Labor, Parts 1910 and 1915 concerning occupational safety and health standards and regulations, as well as any other applicable Federal, State or local regulations that apply to safe practices in coating operations.

Warning! If you scrape, sand, or remove old paint, you may release lead dust. LEAD is TOXIC.

Validation :

Validated by US - HSE Products Coordinator on 1 February 2018

GHS Classification

Procedure used to derive the classification.

Classification	Justification
ACUTE TOXICITY (inhalation) - Category 4 RESPIRATORY SENSITIZATION - Category 1 SKIN SENSITIZATION - Category 1 SPECIFIC TARGET ORGAN TOXICITY (SINGLE EXPOSURE) (Respiratory tract irritation) - Category 3 SPECIFIC TARGET ORGAN TOXICITY (REPEATED EXPOSURE) (lungs) - Category 2	Calculation method Calculation method Calculation method Calculation method Calculation method

Hazardous Material Information System (U.S.A.)

Health	2
Fire hazard	1
Physical hazards	0
Personal protection	X

National Fire Protection Association (U.S.A.)

Health	1	1	1
Flammability			
Instability			
Special			

Personal Protective Equipment (PPE) shown in this section is a suggestion. Since conditions vary from one work location to another consult the facility safety & health program. Customer or end user is responsible to evaluate worker exposure conditions at the site of application and determine the appropriate PPE suitable for workers at that particular facility or location.

Abbreviations and acronyms :



SECTION 16: Other information

ANSI = American National Standards Institute
HCS = Hazardous Communication System
TSCA = Toxic Substances Control Act
CFR = Code of Federal Regulations
GHS = Globally Harmonized System of Classification and Labelling of Chemicals
OSHA = United States Occupational Health and Safety Administration
NIOSH = National Institute for Occupational Safety and Health
ACGIH = American Conference of Industrial Hygienists
IARC = International Agency for Research on Cancer.
NTP = National Toxicology Program
ATE = Acute Toxicity Estimate

OECD = Organisation for Economic Co-operation and Development
BCF = Bioconcentration Factor
DOT = United States Department of Transportation
ERG = Emergency Response Guide
TDG = Transport of Dangerous Goods, Canada
SCT = Transportation & Communications Ministry, Mexico
IMDG = International Maritime Dangerous Goods
IATA = International Air Transport Association
SARA = Superfund Amendments Reauthorization Act
EPCRA = Emergency Planning and Community Right to Know Act

Notice to reader

▀ Indicates information that has changed from previously issued version.

To the best of our knowledge, the information contained herein is accurate. However, neither the above named supplier nor any of its subsidiaries assumes any liability whatsoever for the accuracy or completeness of the information contained herein. Final determination of suitability of any material is the sole responsibility of the user. All materials may present unknown hazards and should be used with caution. Although certain hazards are described herein, we cannot guarantee that these are the only hazards that exist.

Safety Data Sheet

NEOGARD

70800-02 NEOCRETE WATER BORNE POLYURETHANE

RESIN GRAY



Conforms to ANSI Z400.1-2010 Standard - HCS 2012

Protective Clothing	General Hazard	DOT

SECTION 1: Identification of the substance/mixture and of the company/undertaking

1.1 Product identifier

Product name : NEOGARD
NEOCRETE WATER BORNE POLYURETHANE RESIN GRAY

Product identity : 4801916650, 70800-02

Product type : polyurethane paint (base for multi-component product)

1.2 Relevant identified uses of the substance or mixture and uses advised against

Field of application : buildings and metal industry.

Ready-for-use mixture : Mixing Ratio:
48010 = 48019 1.56 Vol. / 98010 0.98 Vol. / Powder 66020
48011 = 48019 1.56 Vol. / 98010 0.98 Vol. / Powder 66021
48012 = 48019 1.56 Vol. / 98010 0.98 Vol. / Powder 66022

Identified uses : Industrial/Professional use

TSCA : **Unless otherwise stated. All components are listed or exempted.**

1.3 Details of the supplier of the safety data sheet

Company details : NEOGARD, a Division of Hempel (USA), Inc.
2728 Empire Central
Dallas, TX 75235
Phone number: 1-214-353-1600
E-mail: hempel@hempel.com

1.4 Emergency telephone number (with hours of operation)

For Transportation Emergencies : CHEMTREC: **1-800-424-9300** (Toll-free in the U.S., Canada and the U.S. Virgin Islands) **703-527-3887**
(24 hours)
For calls originating elsewhere (Collect calls are accepted). Contract number: CCN10384
To preserve the effectiveness of arrangements for providing accurate and timely emergency response information, the basic identifying information (shipper name or contract number) must be included on shipping papers.
If the purchaser of this product is going to be shipping this product to other locations, the purchaser must arrange for its own Emergency Information Provider to respond to transport incidents. Hempel's 24 hour response contract does not cover non-Hempel shipments.

For all other information : In USA toll free calling available: 1-800- 678-6641 or (936)-523-6000
(8 AM - 5 PM CST) See Section 4 of the safety data sheet (first aid measures).

SECTION 2: Hazards identification

2.1 Classification of the substance or mixture

OSHA/HCS status : This material is considered hazardous by the OSHA Hazard Communication Standard (29 CFR 1910.1200).

GHS Classification : EYE IRRITATION - Category 2B
SKIN SENSITIZATION - Category 1
SPECIFIC TARGET ORGAN TOXICITY (REPEATED EXPOSURE) (kidneys) - Category 2

2.2 Label elements

Hazard pictograms :



Safety Data Sheet

NEOGARD

70800-02 NEOCRETE WATER BORNE POLYURETHANE

RESIN GRAY



SECTION 2: Hazards identification

Signal word :	Warning
Hazard statements :	H320 - Causes eye irritation. H317 - May cause an allergic skin reaction. H373 - May cause damage to organs through prolonged or repeated exposure. (kidneys)
Precautionary statements :	
Prevention :	Wear protective gloves. Do not breathe vapor. Wash hands thoroughly after handling. Contaminated work clothing must not be allowed out of the workplace.
Response :	Get medical attention if you feel unwell. IF ON SKIN: Wash with plenty of soap and water. Wash contaminated clothing before reuse. If skin irritation or rash occurs: Get medical attention. IF IN EYES: Rinse cautiously with water for several minutes. Remove contact lenses, if present and easy to do. Continue rinsing. If eye irritation persists: Get medical attention.
Disposal :	Dispose of contents and container in accordance with all local, regional, national and international regulations.
Supplemental label elements :	None known.

2.3 Other hazards

Hazards not otherwise classified : None known.

SECTION 3: Composition/information on ingredients

Product definition : Mixture

Physical state : Liquid.

Product/ingredient name	Identifiers	%	GHS Classification
castor oil	8001-79-4	≥25 - ≤50	Not classified. EYE IRRITATION - Category 2B ASPIRATION HAZARD - Category 1
bis(2-ethylhexyl) terephthalate	6422-86-2	≥10 - ≤25	
hydrocarbons, C10-C13, n-alkanes, isoalkanes, cyclics, < 2% aromatics	64742-48-9	≥5 - ≤10	
titanium dioxide	13463-67-7	≥5 - ≤10	Not classified. ACUTE TOXICITY (oral) - Category 4 SPECIFIC TARGET ORGAN TOXICITY (REPEATED EXPOSURE) (kidneys) (oral) - Category 2 SKIN SENSITIZATION - Category 1
ethanediol	107-21-1	≥1 - ≤3	
4,4',4''-(1-methylpropanyl-3-ylidene)tris [6-tert-butyl-m-cresol]	1843-03-4	≤0.3	

Any concentration shown as a range is to protect confidentiality or is due to batch variation.

There are no ingredients present which, within the current knowledge of the supplier and in the concentrations applicable, are classified as hazardous to health or the environment and hence require reporting in this section.

SECTION 4: First aid measures

4.1 Description of first aid measures

General :	In all cases of doubt, or when symptoms persist, seek medical attention. Never give anything by mouth to an unconscious person. If breathing is irregular, drowsiness, loss of consciousness or cramps: Call 911 and give immediate treatment (first aid).
Eye contact :	Check for and remove any contact lenses. Immediately flush eyes with plenty of water for at least 15 minutes, occasionally lifting the upper and lower eyelids. In all cases of doubt, or when symptoms persist, seek medical attention.
Inhalation :	Remove to fresh air.
Skin contact :	Remove contaminated clothing and shoes. Wash skin thoroughly with soap and water or use recognized skin cleanser. Do NOT use solvents or thinners.
Ingestion :	If swallowed, seek medical advice immediately and show this container or label. Keep person warm and at rest. Do not induce vomiting unless directed to do so by medical personnel. Lower the head so that vomit will not re-enter the mouth and throat.
Protection of first-aiders :	No action shall be taken involving any personal risk or without suitable training. It may be dangerous to the person providing aid to give mouth-to-mouth resuscitation.

4.2 Most important symptoms and effects, both acute and delayed



SECTION 4: First aid measures

Potential acute health effects

Eye contact :	Causes serious eye irritation.
Inhalation :	No known significant effects or critical hazards.
Skin contact :	No known significant effects or critical hazards.
Ingestion :	No known significant effects or critical hazards.

Over-exposure signs/symptoms

Eye contact :	Adverse symptoms may include the following: pain or irritation watering redness
Inhalation :	No specific data.
Skin contact :	No specific data.
Ingestion :	No specific data.

4.3 Indication of any immediate medical attention and special treatment needed

Notes to physician :	If gasses have been inhaled, from the decomposition of the product, symptoms may be delayed.
Specific treatments :	No specific treatment.

SECTION 5: Firefighting measures

5.1 Extinguishing media

Extinguishing media :	Recommended: alcohol resistant foam, CO ₂ , powders, water spray. Not to be used: waterjet.
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5.2 Special hazards arising from the substance or mixture

Hazards from the substance or mixture :	In a fire or if heated, a pressure increase will occur and the container may burst.
Hazardous combustion products :	Decomposition products may include the following materials: carbon oxides nitrogen oxides metal oxide/oxides

5.3 Advice for firefighters

Promptly isolate the scene by removing all persons from the vicinity of the incident if there is a fire. No action shall be taken involving any personal risk or without suitable training. Fire will produce dense black smoke. Exposure to decomposition products may cause a health hazard. Cool closed containers exposed to fire with water. Do not release runoff from fire to drains or watercourses. Fire-fighters should wear appropriate protective equipment and self-contained breathing apparatus (SCBA) with a full face-piece operated in positive pressure mode.

SECTION 6: Accidental release measures

6.1 Personal precautions, protective equipment and emergency procedures

Refer to protective measures listed in sections 7 and 8. No action shall be taken involving any personal risk or without suitable training.

6.2 Environmental precautions

Avoid dispersal of spilled material and runoff and contact with soil, waterways, drains and sewers. Inform the relevant authorities if the product has caused environmental pollution (sewers, waterways, soil or air).

6.3 Methods and materials for containment and cleaning up

Stop leak if without risk. Move containers from spill area. Prevent entry into sewers, water courses, basements or confined areas. Wash spillages into an effluent treatment plant or proceed as follows. Contain and collect spillage with non-combustible, absorbent material e.g. sand, earth, vermiculite or diatomaceous earth and place in container for disposal according to local regulations (see Section 13).

6.4 Reference to other sections

See Section 1 for emergency contact information.
See Section 8 for information on appropriate personal protective equipment.
See Section 13 for additional waste treatment information.



SECTION 7: Handling and storage

7.1 Precautions for safe handling

Appropriate personal protective equipment: see Section 8. Always keep in containers made from the same material as the original one.

7.2 Conditions for safe storage, including any incompatibilities

Store in accordance with local regulations. Store in a cool, well-ventilated area away from incompatible materials and ignition sources.

7.3 Specific end use(s)

See separate Product Data Sheet for recommendations or industrial sector specific solutions.

This product may be applied using several application techniques and methods of handling may be different for each. Application techniques include [but are not limited to] brushing, rolling, and spray application [conventional, HPLV, airless, pleural component or aerosol can]. Avoid the breathing of vapors and, if spraying, do not breath spray mist or aerosols.

SECTION 8: Exposure controls/personal protection

8.1 Control parameters

Product/ingredient name	Exposure limit values
titanium dioxide	OSHA PEL (United States, 6/2016). TWA: 15 mg/m ³ 8 hours. Form: Total dust ACGIH TLV (United States, 3/2017). TWA: 10 mg/m ³ 8 hours.
ethanediol	ACGIH TLV (United States, 3/2017). STEL: 10 mg/m ³ 15 minutes. Form: Inhalable fraction. Aerosol only. STEL: 50 ppm 15 minutes. Form: Vapor fraction TWA: 25 ppm 8 hours. Form: Vapor fraction

Recommended monitoring procedures

If this product contains ingredients with exposure limits, personal, workplace atmosphere or biological monitoring may be required to determine the effectiveness of the ventilation or other control measures and/or the necessity to use respiratory protective equipment. Reference should be made to appropriate monitoring standards. Reference to national guidance documents for methods for the determination of hazardous substances will also be required.

8.2 Exposure controls

Appropriate engineering controls

Provide local exhaust and general ventilation systems to maintain airborne concentrations below OSHA, ACGIH, and manufacturer recommended exposure limits. Local exhaust ventilation is preferred because it prevents contaminant dispersion into work areas by controlling it at its source. Use local and general exhaust ventilation to effectively remove and prevent buildup of mists/vapors/fumes generated from the handling of this product.

Note: Local exhaust ventilation is designed to capture an emitted contaminant at or near its source, before the contaminant has a chance to disperse into the workplace air. General exhaust ventilation, also called dilution ventilation, is different from local exhaust ventilation because instead of capturing emissions at their source and removing them from the air, general exhaust ventilation allows the contaminant to be emitted into the workplace air and then dilutes the concentration of the contaminant to an acceptable level (e.g., to the PEL or below).

Individual protection measures

General :	Gloves must be worn for all work that may result in soiling. Apron/coveralls/protective clothing must be worn when soiling is so great that regular work clothes do not adequately protect skin against contact with the product. Safety eyewear should be used when there is a likelihood of exposure.
Hygiene measures :	Wash hands, forearms, and face thoroughly after handling compounds and before eating, smoking, using lavatory, and at the end of day.
Eye/face protection :	Safety eyewear complying with an approved standard should be used when a risk assessment indicates this is necessary to avoid exposure to liquid splashes, mists, gases or dusts. If contact is possible, the following protection should be worn, unless the assessment indicates a higher degree of protection: chemical splash goggles.
Hand protection :	Wear chemical-resistant gloves in combination with 'basic' employee training. The quality of the chemical-resistant protective gloves must be chosen as a function of the specific workplace concentrations and quantity of hazardous substances.



SECTION 8: Exposure controls/personal protection

Since the actual work situation is unknown. Supplier of gloves should be contacted in order to find the appropriate type. Below listed glove(s) should be regarded as generic advice:

Recommended: Silver Shield / Barrier / 4H gloves, nitrile rubber, Viton®, polyvinyl alcohol (PVA)

May be used: neoprene rubber, butyl rubber, natural rubber (latex), polyvinyl chloride (PVC)

Body protection :

Personal protective equipment for the body should be selected based on the task being performed and the risks involved handling this product.

Wear suitable protective clothing. Always wear protective clothing when spraying.

Respiratory protection :

If working areas have insufficient ventilation, wear half or totally covering mask equipped with gas filter of type Organic Vapor, when grinding use particle filter of type P95, P99 or P100. When spraying use a combined filter (organic vapor / HEPA or organic vapor / P100 type). Be sure to use approved/certified respirator or equivalent. Always wear an air-fed respirator when spraying in a continuous and prolonged work situation (e.g. hood with supply of fresh or compressed air or a full face, powered air purifying filter).

Protective clothing (pictograms) :



Note: Application of paint products by spraying requires additional safety precautions: Full body suit, Full face respirator with air supplied.

Environmental exposure controls

Emissions from ventilation or work process equipment should be checked to ensure they comply with the requirements of environmental protection legislation. In some cases, fume scrubbers, filters or engineering modifications to the process equipment will be necessary to reduce emissions to acceptable levels.

SECTION 9: Physical and chemical properties

9.1 Information on basic physical and chemical properties

Physical state :	Liquid.
Color :	Gray
Odor :	Non-characteristic.
pH :	Testing not relevant or not possible due to nature of the product.
Melting point/freezing point :	-14°C This is based on data for the following ingredient: castor oil
Boiling point/boiling range :	Testing not relevant or not possible due to nature of the product.
Flash point :	Closed cup: 100°C (212°F)
Evaporation rate :	Testing not relevant or not possible due to nature of the product.
Flammability :	Flammable in the presence of the following materials or conditions: open flames, sparks and static discharge, heat, oxidizing materials and reducing materials.
Upper/lower flammability or explosive limits :	1.4 - 15.3 vol %
Vapor pressure :	Testing not relevant or not possible due to nature of the product.
Vapor density :	Testing not relevant or not possible due to nature of the product.
Relative density :	1.003 g/cm ³
Solubility(ies) :	Easily soluble in the following materials: cold water and hot water.
Partition coefficient (LogKow) :	Testing not relevant or not possible due to nature of the product.
Auto-ignition temperature :	Testing not relevant or not possible due to nature of the product.
Decomposition temperature :	Testing not relevant or not possible due to nature of the product.
Viscosity :	Testing not relevant or not possible due to nature of the product.
Explosive properties :	Explosive in the presence of the following materials or conditions: open flames, sparks and static discharge.
Oxidizing properties :	Testing not relevant or not possible due to nature of the product.

9.2 Other information

Safety Data Sheet

NEOGARD

70800-02 NEOCRETE WATER BORNE POLYURETHANE

RESIN GRAY



SECTION 9: Physical and chemical properties

Solvent(s) % by weight (Included exempt solvent(s)):	9.3 % (w/w)
Water % by weight :	Weighted average: 24 %
VOC content (Coatings) :	0.776 lbs/gal (93 g/l)
VOC content (Regulatory) :	1.02 lbs/gal (122.1 g/l)
TOC Content (Volatile) :	Weighted average: 68 g/l
Solvent Gas :	Weighted average: 0.029 m³/l

SECTION 10: Stability and reactivity

10.1 Reactivity

No specific test data related to reactivity available for this product or its ingredients.

10.2 Chemical stability

The product is stable.

10.3 Possibility of hazardous reactions

Under normal conditions of storage and use, hazardous reactions will not occur.

10.4 Conditions to avoid

No specific data.

10.5 Incompatible materials

Highly reactive or incompatible with the following materials: oxidizing materials.

Reactive or incompatible with the following materials: reducing materials and acids.

10.6 Hazardous decomposition products

When exposed to high temperatures (i.e. in case of fire) harmful decomposition products may be formed:

Decomposition products may include the following materials: carbon oxides nitrogen oxides metal oxide/oxides

SECTION 11: Toxicological information

11.1 Information on toxicological effects

Exposure to component solvent vapor concentrations may result in adverse health effects such as mucous membrane and respiratory system irritation and adverse effects on the kidneys, liver and central nervous system. Solvents may cause some of the above effects by absorption through the skin. Symptoms and signs include headaches, dizziness, fatigue, muscular weakness, drowsiness and, in extreme cases, loss of consciousness. Repeated or prolonged contact with the preparation may cause removal of natural fat from the skin, resulting in non-allergic contact dermatitis and absorption through the skin. If splashed in the eyes, the liquid may cause irritation and reversible damage. Accidental swallowing may cause stomach pain. Chemical lung inflammation may occur if the product is taken into the lungs via vomiting.

Acute toxicity

Product/ingredient name	Result	Species	Dose	Exposure
hydrocarbons, C10-C13, n-alkanes, isoalkanes, cyclics, < 2% aromatics	LC50 Inhalation Vapor	Rat	8500 mg/m³	4 hours
	LD50 Dermal	Rabbit	>5000 mg/kg	-
	LD50 Oral	Rat	>6000 mg/kg	-
titanium dioxide	LC50 Inhalation Dusts and mists	Rat	>6.8 mg/l	4 hours
	LD50 Dermal	Rabbit	>5000 mg/kg	-
	LD50 Oral	Rat	>5000 mg/kg	-
ethanediol	LD50 Dermal	Rabbit	9530 uL/kg	-
	LD50 Oral	Rat	4700 mg/kg	-
	LD50 Oral	Rat	14000 mg/kg	-
4,4',4''-(1-methylpropanyl-3-ylidene) tris[6-tert-butyl-m-cresol]	LD50 Oral	Rat		

Acute toxicity estimates

Safety Data Sheet

NEOGARD

70800-02 NEOCRETE WATER BORNE POLYURETHANE

RESIN GRAY



SECTION 11: Toxicological information

Route	ATE value
Oral	19575.4 mg/kg

Irritation/Corrosion

Product/ingredient name	Result	Species	Score	Exposure
castor oil	Eyes - Mild irritant	Rabbit	-	500 milligrams
	Skin - Mild irritant	Rat	-	24 hours 100 milligrams
bis(2-ethylhexyl) terephthalate	Skin - Mild irritant	Human	-	504 hours 0.5 Percent Intermittent
titanium dioxide	Skin - Mild irritant	Human	-	72 hours 300 Micrograms Intermittent
ethanediol	Eyes - Mild irritant	Rabbit	-	24 hours 500 milligrams

Carcinogen Classification

Product/ingredient name	IARC	NTP	OSHA
titanium dioxide	2B	-	-

Specific target organ toxicity (repeated exposure)

Product/ingredient name	Category	Route of exposure	Target organs
ethanediol	Category 2	Oral	kidneys

Aspiration hazard

Product/ingredient name	Result
hydrocarbons, C10-C13, n-alkanes, isoalkanes, cyclics, < 2% aromatics	ASPIRATION HAZARD - Category 1

Information on the likely routes of exposure

Routes of entry anticipated: Oral, Dermal, Inhalation.

Potential chronic health effects

Other information : No additional known significant effects or critical hazards.

SECTION 12: Ecological information

12.1 Toxicity

Do not allow to enter drains or watercourses.

When spilled, this product may act as an oil, causing a film, sheen, emulsion, or sludge at or beneath the surface of a body of water. Oils of any kind can cause: (a) drowning of waterfowl due to lack of buoyancy, loss of insulating capacity of feathers, starvation and vulnerability to predators due to lack of mobility; (b) lethal effect on fish by coating gill surfaces, preventing respiration; (c) potential fish kills resulting from alteration in biochemical oxygen demand; (d) asphyxiation of benthic life forms when floating masses become engaged with surface debris and settle on the bottom; and (e) adverse aesthetic effects of fouled shoreline and beaches.

Product/ingredient name	Result	Species	Exposure
titanium dioxide	Acute LC50 >100 mg/l Acute LC50 >100 mg/l	Daphnia Fish	48 hours 96 hours

12.2 Persistence and degradability

Product/ingredient name	Test	Result	Dose	Inoculum
bis(2-ethylhexyl) terephthalate	301B Ready Biodegradability - CO ₂ Evolution Test	73.05 % - Readily - 28 days	-	-

Product/ingredient name	Aquatic half-life	Photolysis	Biodegradability
bis(2-ethylhexyl) terephthalate	-	-	Readily

12.3 Bioaccumulative potential

Safety Data Sheet

NEOGARD

70800-02 NEOCRETE WATER BORNE POLYURETHANE

RESIN GRAY



SECTION 12: Ecological information

Product/ingredient name	LogP _{ow}	BCF	Potential
bis(2-ethylhexyl) terephthalate	8.39	393	low
hydrocarbons, C10-C13, n-alkanes, isoalkanes, cyclics, < 2%	-	10 - 2500	high
aromatics			
ethanediol	-1.36	-	low

12.4 Mobility in soil

Soil/water partition coefficient (K_{oc}) : No known data available in our database.

Mobility : No known data available in our database.

12.5 Other adverse effects

No known significant effects or critical hazards.

SECTION 13: Disposal considerations

13.1 Waste treatment methods

Disposal should be in accordance with applicable regional, national and local laws and regulations. Local regulations may be more stringent than regional or national requirements.

The information presented below only applies to the material as supplied. The identification based on characteristic(s) or listing may not apply if the material has been used or otherwise contaminated. It is the responsibility of the waste generator to determine the toxicity and physical properties of the material generated to determine the proper waste identification and disposal methods in compliance with applicable regulations.

Refer to Section 7 and Section 8 for additional handling information and protection of employees.

The generation of waste should be avoided or minimized wherever possible. Disposal of this product, solutions and any by-products should at all times comply with the requirements of environmental protection and waste disposal legislation and any regional local authority requirements. Dispose of surplus and non-recyclable products via a licensed waste disposal contractor. Waste should not be disposed of untreated to the sewer unless fully compliant with the requirements of all authorities with jurisdiction. Waste packaging should be recycled. Incineration or landfill should only be considered when recycling is not feasible. This material and its container must be disposed of in a safe way. Care should be taken when handling emptied containers that have not been cleaned or rinsed out. Empty containers or liners may retain some product residues. Avoid dispersal of spilled material and runoff and contact with soil, waterways, drains and sewers.

SECTION 14: Transport information

Transport may take place according to national regulation or DOT for transport by road and by train, IMDG for transport by sea, IATA for Air shipment. Refer to specific Dangerous Goods Transport requirements under 49CFR, ICAO and IATA.

	14.1 UN no.	14.2 Proper shipping name	14.3 Transport hazard class(es)	14.4 PG*	14.5 Env*	Additional information
DOT Code	Not regulated.					
TDG Code	Not regulated.					
SCT Code	Not regulated.					
IMDG Code	Not regulated.					
IATA Code	Not regulated.					

Code : Classification

PG* : Packing group

Env.* : Environmental hazards

14.6 Special precautions for user

Transport within user's premises: always transport in closed containers that are upright and secure. Ensure that persons transporting the product know what to do in the event of an accident or spillage.

Safety Data Sheet

NEOGARD

70800-02 NEOCRETE WATER BORNE POLYURETHANE

RESIN GRAY



SECTION 14: Transport information

14.7 Transport in bulk according to Annex II of MARPOL and the IBC Code

Not applicable.

SECTION 15: Regulatory information

15.1 Safety, health and environmental regulations/legislation specific for the substance or mixture

U.S. Federal regulations :

All components are listed or exempted.

TSCA 8(a) PAIR: poly(dimethylsiloxane)

TSCA 8(a) CDR Exempt/Partial exemption: Not determined

United States inventory (TSCA 8b): All components are listed or exempted.

Clean Water Act (CWA) 307: di(2-ethyl hexyl)phthalate

Clean Air Act Section 112(b) Hazardous Air Pollutants (HAPs) : Listed

Product/ingredient name	CAS number	Concentration
ethanediol	107-21-1	2.5542
diethylene glycol	111-46-6	0.0258
di(2-ethyl hexyl)phthalate	117-81-7	0.012185

Clean Air Act Section 602 Class I Substances : Not listed

Clean Air Act Section 602 Class II Substances : Not listed

DEA List I Chemicals (Precursor Chemicals) : Not listed

DEA List II Chemicals (Essential Chemicals) : Not listed

SARA 302/304 - SARA 311/312:

SARA 302/304: No products were found.

SARA 311/312 Hazards identification: Immediate (acute) health hazard, Delayed (chronic) health hazard

Product/ingredient name	%	Fire hazard	Sudden release of pressure	Reactive	Immediate (acute) health hazard	Delayed (chronic) health hazard
bis(2-ethylhexyl) terephthalate	10 - 25	No.	No.	No.	Yes.	No.
ethanediol	1 - 3	No.	No.	No.	Yes.	Yes.
4,4'-(1-methylpropanyl-3-ylidene)tris [6-tert-butyl-m-cresol]	0.1 - 0.5	No.	No.	No.	Yes.	No.

SARA 313 :

SARA 313 notifications must not be detached from the MSDS and any copying and redistribution of the MSDS shall include copying and redistribution of the notice attached to copies of the MSDS subsequently redistributed.

Form R - Reporting requirements :

Product/ingredient name	CAS number	Concentration
ethanediol	107-21-1	1 - 3

Supplier notification :

Product/ingredient name	CAS number	Concentration
ethanediol	107-21-1	1 - 3

State regulations :

Connecticut Carcinogen Reporting: None of the components are listed.

Connecticut Hazardous Material Survey: None of the components are listed.

Florida substances: None of the components are listed.

Illinois Chemical Safety Act: None of the components are listed.

Illinois Toxic Substances Disclosure to Employee Act: None of the components are listed.

Louisiana Reporting: None of the components are listed.

Louisiana Spill: None of the components are listed.

Massachusetts Spill: None of the components are listed.

Massachusetts Substances: The following components are listed: ETHYLENE GLYCOL; 1, 2-DIHYDROXYETHANE; TITANIUM DIOXIDE; TIN DIOXIDE DUST

Michigan Critical Material: None of the components are listed.

Minnesota Hazardous Substances: None of the components are listed.

New Jersey Hazardous Substances: The following components are listed: ETHYLENE GLYCOL; 1, 2-ETHANEDIOL; TITANIUM DIOXIDE; TITANIUM OXIDE (TiO₂)

New Jersey Spill: None of the components are listed.

New Jersey Toxic Catastrophe Prevention Act: None of the components are listed.

New York Acutely Hazardous Substances: The following components are listed: Ethylene glycol

New York Toxic Chemical Release Reporting: None of the components are listed.

Pennsylvania RTK Hazardous Substances: The following components are listed: 1,2-ETHANEDIOL; TITANIUM OXIDE

Safety Data Sheet

NEOGARD

70800-02 NEOCRETE WATER BORNE POLYURETHANE

RESIN GRAY



SECTION 15: Regulatory information

Rhode Island Hazardous Substances: None of the components are listed.

California Prop. 65 PFF :

WARNING: This product contains a chemical known to the State of California to cause cancer and birth defects or other reproductive harm.

Product/ingredient name	Cancer	Reproductive	No significant risk level	Maximum acceptable dosage level
di(2-ethyl hexyl)phthalate	Yes.	Yes.	Yes.	Yes.
titanium dioxide	Yes.	No.		
carbonblack	Yes.	No.		
ethanediol	No.	Yes.		Yes.

SECTION 16: Other information

Remarks :

Note: In USA, consult Code of Federal Regulations, Title 29, Labor, Parts 1910 and 1915 concerning occupational safety and health standards and regulations, as well as any other applicable Federal, State or local regulations that apply to safe practices in coating operations.

Warning! If you scrape, sand, or remove old paint, you may release lead dust. LEAD is TOXIC.

Validation :

Validated by US - HSE Products Coordinator on 30 January 2018

GHS Classification

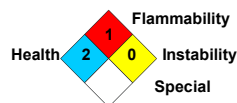
Procedure used to derive the classification.

Classification	Justification
EYE IRRITATION - Category 2B	Calculation method
SKIN SENSITIZATION - Category 1	Calculation method
SPECIFIC TARGET ORGAN TOXICITY (REPEATED EXPOSURE) (kidneys) - Category 2	Calculation method

Hazardous Material Information System (U.S.A.)



National Fire Protection Association (U.S.A.)



Personal Protective Equipment (PPE) shown in this section is a suggestion. Since conditions vary from one work location to another consult the facility safety & health program. Customer or end user is responsible to evaluate worker exposure conditions at the site of application and determine the appropriate PPE suitable for workers at that particular facility or location.

Abbreviations and acronyms :

ANSI = American National Standards Institute
HCS = Hazardous Communication System
TSCA = Toxic Substances Control Act
CFR = Code of federal Regulations
GHS = Globally Harmonized System of Classification and Labelling of Chemicals
OSHA = United States Occupational Health and Safety Administration
NIOSH = National Institute for Occupational Safety and Health
ACGIH = American Conference of Industrial Hygienists
IARC = International Agency for Research on Cancer.
NTP = National Toxicology Program
ATE = Acute Toxicity Estimate

OECD = Organisation for Economic Co-operation and Development
BCF = Bioconcentration Factor
DOT = United States Department of Transportation
ERG = Emergency Response Guide
TDG = Transport of Dangerous Goods, Canada
SCT = Transportation & Communications Ministry, Mexico
IMDG = International Maritime Dangerous Goods
IATA = International Air Transport Association
SARA = Superfund Amendments Reauthorization Act
EPCRA = Emergency Planning and Community Right to Know Act

Notice to reader

Indicates information that has changed from previously issued version.

To the best of our knowledge, the information contained herein is accurate. However, neither the above named supplier nor any of its subsidiaries assumes any liability whatsoever for the accuracy or completeness of the information contained herein. Final determination of suitability of any material is the sole responsibility of the user. All materials may present unknown hazards and should be used with caution. Although certain hazards are described herein, we cannot guarantee that these are the only hazards that exist.

Safety Data Sheet

NEOGARD

70804 NEOCRETE AGGREGATE SL



Conforms to ANSI Z400.1-2010 Standard - HCS 2012

Protective Clothing	General Hazard	DOT

SECTION 1: Identification of the substance/mixture and of the company/undertaking

1.1 Product identifier

Product name : NEOGARD
NEOCRETE AGGREGATE SL

Product identity : 6602209990, 70804

Product type : Sand.

1.2 Relevant identified uses of the substance or mixture and uses advised against

Field of application : buildings and metal industry.

Ready-for-use mixture : Used For: 48012

Identified uses : Industrial/Professional use

TSCA : **Unless otherwise stated. All components are listed or exempted.**

1.3 Details of the supplier of the safety data sheet

Company details : NEOGARD, a Division of Hempel (USA), Inc.
2728 Empire Central
Dallas, TX 75235
Phone number: 1-214-353-1600
E-mail: hempel@hempel.com

1.4 Emergency telephone number (with hours of operation)

For Transportation Emergencies : CHEMTREC: **1-800-424-9300** (Toll-free in the U.S., Canada and the U.S. Virgin Islands) **703-527-3887**
(24 hours)
For calls originating elsewhere (Collect calls are accepted). Contract number: CCN10384
To preserve the effectiveness of arrangements for providing accurate and timely emergency response information, the basic identifying information (shipper name or contract number) must be included on shipping papers.
If the purchaser of this product is going to be shipping this product to other locations, the purchaser must arrange for its own Emergency Information Provider to respond to transport incidents. Hempel's 24 hour response contract does not cover non-Hempel shipments.

For all other information : In USA toll free calling available: 1-800- 678-6641 or (936)-523-6000
(8 AM - 5 PM CST) See Section 4 of the safety data sheet (first aid measures).

SECTION 2: Hazards identification

2.1 Classification of the substance or mixture

OSHA/HCS status : This material is considered hazardous by the OSHA Hazard Communication Standard (29 CFR 1910.1200).

GHS Classification : SKIN IRRITATION - Category 2
SERIOUS EYE DAMAGE - Category 1
CARCINOGENICITY (inhalation) - Category 1A
SPECIFIC TARGET ORGAN TOXICITY (SINGLE EXPOSURE) (Respiratory tract irritation) - Category 3
SPECIFIC TARGET ORGAN TOXICITY (REPEATED EXPOSURE) (lungs) - Category 1

2.2 Label elements

Hazard pictograms :



Safety Data Sheet

NEOGARD

70804 NEOCRETE AGGREGATE SL



SECTION 2: Hazards identification

Signal word :	Danger
Hazard statements :	H318 - Causes serious eye damage. H315 - Causes skin irritation. H350 - May cause cancer if inhaled. H335 - May cause respiratory irritation. H372 - Causes damage to organs through prolonged or repeated exposure. (lungs)
Precautionary statements :	
Prevention :	Obtain special instructions before use. Do not handle until all safety precautions have been read and understood. Wear protective gloves. Wear eye or face protection. Wear protective clothing. Use only outdoors or in a well-ventilated area. Do not breathe dust. Do not eat, drink or smoke when using this product. Wash hands thoroughly after handling.
Response :	Get medical attention if you feel unwell. IF exposed or concerned: Get medical attention. IF INHALED: Remove person to fresh air and keep comfortable for breathing. Call a POISON CENTER or physician if you feel unwell. IF ON SKIN: Wash with plenty of soap and water. Take off contaminated clothing and wash it before reuse. If skin irritation occurs: Get medical attention. IF IN EYES: Rinse cautiously with water for several minutes. Remove contact lenses, if present and easy to do. Continue rinsing. Immediately call a POISON CENTER or physician.
Storage :	Store locked up.
Disposal :	Dispose of contents and container in accordance with all local, regional, national and international regulations.
Supplemental label elements :	None known.

2.3 Other hazards

Hazards not otherwise classified : None known.

SECTION 3: Composition/information on ingredients

Product definition :	Mixture
Physical state :	Solid. [Sand.]

Product/ingredient name	Identifiers	%	GHS Classification
Silica (vitreous) portland cement (with less than 2 mg water soluble chromate per kg)	60676-86-0 65997-15-1	≥50 - ≤75 ≥25 - ≤50	Not classified. SKIN IRRITATION - Category 2 SERIOUS EYE DAMAGE - Category 1 SPECIFIC TARGET ORGAN TOXICITY (SINGLE EXPOSURE) (Respiratory tract irritation) - Category 3
respirable quartz	14808-60-7	≤1	CARCINOGENICITY - Category 1A SPECIFIC TARGET ORGAN TOXICITY (REPEATED EXPOSURE) (lungs) (inhalation) - Category 1

Any concentration shown as a range is to protect confidentiality or is due to batch variation.

There are no ingredients present which, within the current knowledge of the supplier and in the concentrations applicable, are classified as hazardous to health or the environment and hence require reporting in this section.

SECTION 4: First aid measures

4.1 Description of first aid measures

General :	In all cases of doubt, or when symptoms persist, seek medical attention. Never give anything by mouth to an unconscious person. If breathing is irregular, drowsiness, loss of consciousness or cramps: Call 911 and give immediate treatment (first aid).
Eye contact :	Check for and remove any contact lenses. Immediately flush eyes with plenty of water for at least 15 minutes, occasionally lifting the upper and lower eyelids. Seek immediate medical attention.
Inhalation :	Remove to fresh air. Keep person warm and at rest. If unconscious, place in recovery position and seek medical advice.
Skin contact :	Remove contaminated clothing and shoes. Wash skin thoroughly with soap and water or use recognized skin cleanser. Do NOT use solvents or thinners.



SECTION 4: First aid measures

Ingestion :	If swallowed, seek medical advice immediately and show this container or label. Keep person warm and at rest. Do not induce vomiting unless directed to do so by medical personnel. Lower the head so that vomit will not re-enter the mouth and throat.
Protection of first-aiders :	No action shall be taken involving any personal risk or without suitable training. If it is suspected that fumes are still present, the rescuer should wear an appropriate mask or self-contained breathing apparatus. It may be dangerous to the person providing aid to give mouth-to-mouth resuscitation. Wash contaminated clothing thoroughly with water before removing it, or wear gloves.

4.2 Most important symptoms and effects, both acute and delayed

Potential acute health effects

Eye contact :	Causes serious eye damage.
Inhalation :	May cause respiratory irritation.
Skin contact :	Causes skin irritation.
Ingestion :	No known significant effects or critical hazards.

Over-exposure signs/symptoms

Eye contact :	Adverse symptoms may include the following: pain watering redness
Inhalation :	Adverse symptoms may include the following: respiratory tract irritation coughing
Skin contact :	Adverse symptoms may include the following: pain or irritation redness blistering may occur
Ingestion :	Adverse symptoms may include the following: stomach pains

4.3 Indication of any immediate medical attention and special treatment needed

Notes to physician :	Not applicable.
Specific treatments :	No specific treatment.

SECTION 5: Firefighting measures

5.1 Extinguishing media

Extinguishing media :	Recommended: alcohol resistant foam, CO ₂ , powders, water spray. Not to be used: waterjet.
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5.2 Special hazards arising from the substance or mixture

Hazards from the substance or mixture :	No specific fire or explosion hazard.
Hazardous combustion products :	Decomposition products may include the following materials: metal oxide/oxides

5.3 Advice for firefighters

Promptly isolate the scene by removing all persons from the vicinity of the incident if there is a fire. No action shall be taken involving any personal risk or without suitable training. Exposure to decomposition products may cause a health hazard. Cool closed containers exposed to fire with water. Do not release runoff from fire to drains or watercourses. Fire-fighters should wear appropriate protective equipment and self-contained breathing apparatus (SCBA) with a full face-piece operated in positive pressure mode.



SECTION 6: Accidental release measures

6.1 Personal precautions, protective equipment and emergency procedures

Refer to protective measures listed in sections 7 and 8. No action shall be taken involving any personal risk or without suitable training.

6.2 Environmental precautions

Avoid dispersal of spilled material and runoff and contact with soil, waterways, drains and sewers. Inform the relevant authorities if the product has caused environmental pollution (sewers, waterways, soil or air).

6.3 Methods and materials for containment and cleaning up

Move containers from spill area. Approach release from upwind. Prevent entry into sewers, water courses, basements or confined areas. Vacuum or sweep up material and place in a designated, labeled waste container.

6.4 Reference to other sections

See Section 1 for emergency contact information.

See Section 8 for information on appropriate personal protective equipment.

See Section 13 for additional waste treatment information.

SECTION 7: Handling and storage

7.1 Precautions for safe handling

Avoid inhalation of vapour, dust and spray mist. Avoid contact with skin and eyes. Eating, drinking and smoking should be prohibited in area where this material is handled, stored and processed. Appropriate personal protective equipment: see Section 8. Always keep in containers made from the same material as the original one.

7.2 Conditions for safe storage, including any incompatibilities

Store in accordance with local regulations. Store in a cool, well-ventilated area away from incompatible materials and ignition sources. Keep out of the reach of children. Keep away from: Oxidizing agents, strong alkalis, strong acids. No smoking. Prevent unauthorized access. Containers that are opened must be carefully resealed and kept upright to prevent leakage.

7.3 Specific end use(s)

See separate Product Data Sheet for recommendations or industrial sector specific solutions.

SECTION 8: Exposure controls/personal protection

8.1 Control parameters

Product/ingredient name	Exposure limit values
Silica (vitreous)	NIOSH REL (United States, 10/2016). TWA: 6 mg/m ³ 10 hours.
portland cement (with less than 2 mg water soluble chromate per kg)	ACGIH TLV (United States, 3/2018). TWA: 1 mg/m ³ 8 hours. Form: Respirable fraction
	NIOSH REL (United States, 10/2016). TWA: 5 mg/m ³ 10 hours. Form: Respirable fraction
	TWA: 10 mg/m ³ 10 hours. Form: Total
	OSHA PEL (United States, 5/2018). TWA: 5 mg/m ³ 8 hours. Form: Respirable fraction
	TWA: 15 mg/m ³ 8 hours. Form: Total dust
respirable quartz	OSHA PEL Z3 (United States, 6/2016). TWA: 250 mppcf / (%SiO ₂ +5) 8 hours. Form: Respirable
	TWA: 10 mg/m ³ / (%SiO ₂ +2) 8 hours. Form: Respirable
	OSHA PEL (United States, 5/2018). TWA: 50 µg/m ³ 8 hours. Form: Respirable dust
	ACGIH TLV (United States, 3/2018). TWA: 0.025 mg/m ³ 8 hours. Form: Respirable fraction
	NIOSH REL (United States, 10/2016). TWA: 0.05 mg/m ³ 10 hours. Form: respirable dust

Recommended monitoring procedures

SECTION 8: Exposure controls/personal protection

If this product contains ingredients with exposure limits, personal, workplace atmosphere or biological monitoring may be required to determine the effectiveness of the ventilation or other control measures and/or the necessity to use respiratory protective equipment. Reference should be made to appropriate monitoring standards. Reference to national guidance documents for methods for the determination of hazardous substances will also be required.

8.2 Exposure controls

Appropriate engineering controls

Provide local exhaust and general ventilation systems to maintain airborne concentrations below OSHA, ACGIH, and manufacturer recommended exposure limits. Local exhaust ventilation is preferred because it prevents contaminant dispersion into work areas by controlling it at its source. Use local and general exhaust ventilation to effectively remove and prevent buildup of mists/vapors/fumes generated from the handling of this product.

Note: Local exhaust ventilation is designed to capture an emitted contaminant at or near its source, before the contaminant has a chance to disperse into the workplace air. General exhaust ventilation, also called dilution ventilation, is different from local exhaust ventilation because instead of capturing emissions at their source and removing them from the air, general exhaust ventilation allows the contaminant to be emitted into the workplace air and then dilutes the concentration of the contaminant to an acceptable level (e.g., to the PEL or below).

Individual protection measures

- General : Gloves must be worn for all work that may result in soiling. Apron/coveralls/protective clothing must be worn when soiling is so great that regular work clothes do not adequately protect skin against contact with the product. Safety eyewear should be used when there is a likelihood of exposure.
- Hygiene measures : Wash hands, forearms, and face thoroughly after handling compounds and before eating, smoking, using lavatory, and at the end of day.
- Eye/face protection : Safety eyewear complying with an approved standard should be used when a risk assessment indicates this is necessary to avoid exposure to liquid splashes, mists, gases or dusts. If contact is possible, the following protection should be worn, unless the assessment indicates a higher degree of protection: chemical splash goggles and/or face shield. If inhalation hazards exist, a full-face respirator may be required instead.
- Hand protection : Wear chemical-resistant gloves in combination with 'basic' employee training. The quality of the chemical-resistant protective gloves must be chosen as a function of the specific workplace concentrations and quantity of hazardous substances.
- Since the actual work situation is unknown. Supplier of gloves should be contacted in order to find the appropriate type.
- Body protection : Personal protective equipment for the body should be selected based on the task being performed and the risks involved handling this product.
- Respiratory protection : Use appropriate respiratory protection if there is a risk of exceeding any exposure limits. Use dust protection mask, when there is a risk for dust.

Protective clothing (pictograms) :



Note: Application of paint products by spraying requires additional safety precautions: Full body suit, Full face respirator with air supplied.

Environmental exposure controls

Emissions from ventilation or work process equipment should be checked to ensure they comply with the requirements of environmental protection legislation. In some cases, fume scrubbers, filters or engineering modifications to the process equipment will be necessary to reduce emissions to acceptable levels.

SECTION 9: Physical and chemical properties

9.1 Information on basic physical and chemical properties

- Physical state : Sand.
- Color : Colorless.
- Odor : Non-characteristic.
- pH : Testing not relevant or not possible due to nature of the product.
- Melting point/freezing point : 1710°C This is based on data for the following ingredient: Silica (vitreous)
- Boiling point/boiling range : Testing not relevant or not possible due to nature of the product.
- Flash point : May be combustible at high temperature.
- Evaporation rate : Testing not relevant or not possible due to nature of the product.



SECTION 9: Physical and chemical properties

Flammability :	May be combustible at high temperature.
Upper/lower flammability or explosive limits :	No specific data.
Vapor pressure :	0.01 kPa This is based on data for the following ingredient: Silica (vitreous)
Vapor density :	Testing not relevant or not possible due to nature of the product.
Relative density :	2.76 g/cm ³
Solubility(ies) :	
Partition coefficient (LogKow) :	Testing not relevant or not possible due to nature of the product.
Auto-ignition temperature :	Testing not relevant or not possible due to nature of the product.
Decomposition temperature :	Testing not relevant or not possible due to nature of the product.
Viscosity :	Testing not relevant or not possible due to nature of the product.
Explosive properties :	Not available.
Oxidizing properties :	Testing not relevant or not possible due to nature of the product.

9.2 Other information

Solvent(s) % by weight (Included exempt solvent(s)):	0 % (w/w)
Water % by weight :	Weighted average: 0 %
VOC content (Coatings) :	0 lbs/gal (0 g/l)
VOC content (Regulatory) :	0 lbs/gal (0 g/l)
TOC Content (Volatile) :	Weighted average: 0 g/l
Solvent Gas :	Weighted average: 0 m ³ /l

SECTION 10: Stability and reactivity

10.1 Reactivity

No specific test data related to reactivity available for this product or its ingredients.

10.2 Chemical stability

The product is stable.

10.3 Possibility of hazardous reactions

Under normal conditions of storage and use, hazardous reactions will not occur.

10.4 Conditions to avoid

No specific data.

10.5 Incompatible materials

10.6 Hazardous decomposition products

When exposed to high temperatures (i.e. in case of fire) harmful decomposition products may be formed:

Decomposition products may include the following materials: metal oxide/oxides



SECTION 11: Toxicological information

11.1 Information on toxicological effects

Direct contact with the eyes can cause irreversible damage, including blindness.

Acute toxicity

Product/ingredient name	Result	Species	Dose	Exposure
Not available.				

Acute toxicity estimates

Route	ATE value
No known significant effects or critical hazards.	

Carcinogen Classification

Product/ingredient name	IARC	NTP	OSHA
Silica (vitreous) respirable quartz	3 1	- Known to be a human carcinogen.	- -

Specific target organ toxicity (single exposure)

Product/ingredient name	Category	Route of exposure	Target organs
portland cement (with less than 2 mg water soluble chromate per kg)	Category 3	Not applicable.	Respiratory tract irritation

Specific target organ toxicity (repeated exposure)

Product/ingredient name	Category	Route of exposure	Target organs
respirable quartz	Category 1	Inhalation	lungs

Information on the likely routes of exposure

Routes of entry anticipated: Oral, Dermal, Inhalation.

Potential chronic health effects

Other information : No additional known significant effects or critical hazards.

SECTION 12: Ecological information

12.1 Toxicity

Do not allow to enter drains or watercourses.

When spilled in water or drains, this product can cause: (a) contribute to suspended solid loading of the water body; (b) turbidity and reduce penetration of light into the water column; (c) alter water pH and/or alkalinity; (d) contribute to sediments at bottom of water column; (e) add colour to the sediment. When spilled to land surface with no runoff due to precipitation, this product can cause: (a) fines present may become air-borne and be transported by wind; (b) contribute to accumulation of surface "dirt"; (c) colour changes to surfaces on which it is spilled.

12.2 Persistence and degradability

12.3 Bioaccumulative potential

12.4 Mobility in soil

Soil/water partition coefficient (K_{oc}) : No known data available in our database.

Mobility : No known data available in our database.

12.5 Other adverse effects

No known significant effects or critical hazards.

SECTION 13: Disposal considerations

13.1 Waste treatment methods

Disposal should be in accordance with applicable regional, national and local laws and regulations. Local regulations may be more stringent than regional or national requirements.

The information presented below only applies to the material as supplied. The identification based on characteristic(s) or listing may not apply if the material has been used or otherwise contaminated. It is the responsibility of the waste generator to determine the toxicity and physical properties of the material generated to determine the proper waste identification and disposal methods in compliance with applicable regulations.

Refer to Section 7 and Section 8 for additional handling information and protection of employees.

The generation of waste should be avoided or minimized wherever possible. Disposal of this product, solutions and any by-products should at all times comply with the requirements of environmental protection and waste disposal legislation and any regional local authority requirements. Dispose of surplus and non-recyclable products via a licensed waste disposal contractor. Waste should not be disposed of untreated to the sewer unless fully compliant with the requirements of all authorities with jurisdiction. Waste packaging should be recycled. Incineration or landfill should only be considered when recycling is not feasible. This material and its container must be disposed of in a safe way. Care should be taken when handling emptied containers that have not been cleaned or rinsed out. Empty containers or liners may retain some product residues. Avoid dispersal of spilled material and runoff and contact with soil, waterways, drains and sewers.

SECTION 14: Transport information

Transport may take place according to national regulation or DOT for transport by road and by train, IMDG for transport by sea, IATA for Air shipment. Refer to specific Dangerous Goods Transport requirements under 49CFR, ICAO and IATA.

14.1 UN no.	14.2 Proper shipping name	14.3 Transport hazard class(es)	14.4 PG*	14.5 Env*	Additional information
DOT Code	Not regulated.				
TDG Code	Not regulated.				
SCT Code	Not regulated.				
IMDG Code	Not regulated.				
IATA Code	Not regulated.				

Code : Classification

PG* : Packing group

Env.* : Environmental hazards

14.6 Special precautions for user

Transport within user's premises: always transport in closed containers that are upright and secure. Ensure that persons transporting the product know what to do in the event of an accident or spillage.

14.7 Transport in bulk according to Annex II of MARPOL and the IBC Code

Not applicable.

SECTION 15: Regulatory information

15.1 Safety, health and environmental regulations/legislation specific for the substance or mixture

U.S. Federal regulations :

All components are listed or exempted.

TSCA 8(a) CDR Exempt/Partial exemption: Not determined

United States inventory (TSCA 8b): All components are listed or exempted.

Clean Air Act Section 112(b) Hazardous Air Pollutants (HAPs) : Not listed

Clean Air Act Section 602 Class I Substances : Not listed

Clean Air Act Section 602 Class II Substances : Not listed

DEA List I Chemicals (Precursor Chemicals) : Not listed

DEA List II Chemicals (Essential Chemicals) : Not listed



SECTION 15: Regulatory information

SARA 302/304 - SARA 311/312: SARA 302/304: No products were found.

SARA 311/312 Hazards identification: Delayed (chronic) health hazard

Product/ingredient name	%	Fire hazard	Sudden release of pressure	Reactive	Immediate (acute) health hazard	Delayed (chronic) health hazard
portland cement (with less than 2 mg water soluble chromate per kg)	25 - 50	No.	No.	No.	Yes.	No.
respirable quartz	0.1 - 0.5	No.	No.	No.	No.	Yes.

State regulations :

Connecticut Carcinogen Reporting: None of the components are listed.

Connecticut Hazardous Material Survey: None of the components are listed.

Florida substances: None of the components are listed.

Illinois Chemical Safety Act: None of the components are listed.

Illinois Toxic Substances Disclosure to Employee Act: None of the components are listed.

Louisiana Reporting: None of the components are listed.

Louisiana Spill: None of the components are listed.

Massachusetts Spill: None of the components are listed.

Massachusetts Substances: The following components are listed: SILICA, FUSED, DUST; PORTLAND CEMENT

Michigan Critical Material: None of the components are listed.

Minnesota Hazardous Substances: None of the components are listed.

New Jersey Hazardous Substances: The following components are listed: SILICA, FUSED; SILICA, VITREOUS; SILICATE, PORTLAND CEMENT; CEMENT, PORTLAND, CHEMICALS

New Jersey Spill: None of the components are listed.

New Jersey Toxic Catastrophe Prevention Act: None of the components are listed.

New York Acutely Hazardous Substances: None of the components are listed.

New York Toxic Chemical Release Reporting: None of the components are listed.

Pennsylvania RTK Hazardous Substances: The following components are listed: CEMENT, PORTLAND, CHEMICALS

Rhode Island Hazardous Substances: None of the components are listed.

California Prop. 65 PFF :

WARNING: This product contains a chemical known to the State of California to cause cancer.

Product/ingredient name	Cancer	Reproductive	No significant risk level	Maximum acceptable dosage level
respirable quartz	Yes.	No.		
titanium dioxide	Yes.	No.		

SECTION 16: Other information

Remarks :

Note: In USA, consult Code of Federal Regulations, Title 29, Labor, Parts 1910 and 1915 concerning occupational safety and health standards and regulations, as well as any other applicable Federal, State or local regulations that apply to safe practices in coating operations.

Warning! If you scrape, sand, or remove old paint, you may release lead dust. LEAD is TOXIC.

Validation :

Validated by US - HSE Products Coordinator on 29 October 2018

GHS Classification

Procedure used to derive the classification.

Classification	Justification
SKIN IRRITATION - Category 2 SERIOUS EYE DAMAGE - Category 1 CARCINOGENICITY (inhalation) - Category 1A SPECIFIC TARGET ORGAN TOXICITY (SINGLE EXPOSURE) (Respiratory tract irritation) - Category 3 SPECIFIC TARGET ORGAN TOXICITY (REPEATED EXPOSURE) (lungs) - Category 1	Calculation method Calculation method Expert judgment Calculation method Calculation method

Hazardous Material Information System (U.S.A.)

Health	3
Fire hazard	0
Physical hazards	0
Personal protection	X

National Fire Protection Association (U.S.A.)

Health	1	0	0	Flammability
				Instability
				Special

Personal Protective Equipment (PPE) shown in this section is a suggestion. Since conditions vary from one work location to another consult the facility safety & health program. Customer or end user is responsible to evaluate worker exposure conditions at the site of application and determine the appropriate PPE suitable for workers at that particular facility or location.

Abbreviations and acronyms :

Safety Data Sheet

NEOGARD

70804 NEOCRETE AGGREGATE SL



SECTION 16: Other information

ANSI = American National Standards Institute
HCS = Hazardous Communication System
TSCA = Toxic Substances Control Act
CFR = Code of federal Regulations
GHS = Globally Harmonized System of Classification and Labelling of Chemicals
OSHA = United States Occupational Health and Safety Administration
NIOSH = National Institute for Occupational Safety and Health
ACGIH = American Conference of Industrial Hygienists
IARC = International Agency for Research on Cancer.
NTP = National Toxicology Program
ATE = Acute Toxicity Estimate

OECD = Organisation for Economic Co-operation and Development
BCF = Bioconcentration Factor
DOT = United States Department of Transportation
ERG = Emergency Response Guide
TDG = Transport of Dangerous Goods, Canada
SCT = Transportation & Communications Ministry, Mexico
IMDG = International Maritime Dangerous Goods
IATA = International Air Transport Association
SARA = Superfund Amendments Reauthorization Act
EPCRA = Emergency Planning and Community Right to Know Act

Notice to reader

Indicates information that has changed from previously issued version.

To the best of our knowledge, the information contained herein is accurate. However, neither the above named supplier nor any of its subsidiaries assumes any liability whatsoever for the accuracy or completeness of the information contained herein. Final determination of suitability of any material is the sole responsibility of the user. All materials may present unknown hazards and should be used with caution. Although certain hazards are described herein, we cannot guarantee that these are the only hazards that exist.

Safety Data Sheet

Material: 60004600

HDK® N20
HYDROPHILIC PYROGENIC SILICA

Version: 2.1 (US)

Date of print: 11/11/2015

Date of last alteration: 06/24/2015

1. Product and company identification

1.1 Identification of the substance or preparation:

Commercial product name:

HDK® N20
HYDROPHILIC PYROGENIC SILICA

Use of substance / preparation

Industrial.
Auxiliary agent for: Plastics , Lacquer , Building materials , elastomer products , cosmetics .

1.2 Company/undertaking identification:

Manufacturer/distributor:

Wacker Chemie AG
Hanns-Seidel-Platz 4
81737 München
Germany

Customer information:

Wacker Chemical Corporation
3301 Sutton Road
Adrian, Michigan 49221-9397
USA
InfoLine:
Tel (517) 264-8240, Fax (517) 264-8740
Hours of operation:
Monday - Friday, 8 am to 5 pm (eastern standard time)
Corporate website: www.wacker.com**Emergency telephone no. (24h):**
Transportation emergency:(517) 264-8500
(800) 424-9300 (CHEMTREC, USA)
(703) 527-3887 (CHEMTREC, international)

This SDS was prepared by the Regulatory Affairs and Product Safety Department (RAPS) of Wacker Chemical Corporation.

2. Hazards identification

2.1 Classification of the substance or mixture

Classification (GHS):

Not a hazardous substance or mixture.

2.2 Label elements

Labelling (GHS):

No labeling according to GHS required.

2.3 Other hazards

No data available.

3. Composition/information on ingredients

3.1 Chemical characterization (substance)

CAS No.	Chemical characteristics
112945-52-5	Pyrogenic micro-dispersed silica, synthetic X-ray amorphous silicon dioxide (SiO ₂)

3.2 Information on ingredients:

This material does not contain any reportable hazardous ingredients.

Substances listed in the Subsections "HAPS" and "California Proposition 65 Carcinogens / Reproductive Toxins" that are not listed in this section are only present at quantities below 0.1% for California Proposition 65 listed toxins or below 1% for non-carcinogenic HAPS or they are inextricably bound in the product.

Safety Data Sheet

Material: 60004600

HDK® N20
HYDROPHILIC PYROGENIC SILICA

Version: 2.1 (US)

Date of print: 11/11/2015

Date of last alteration: 06/24/2015

4. First-aid measures

4.1 General information:

Get medical attention if irritation occurs or if breathing becomes difficult.

4.2 After inhalation

If inhaled, remove to fresh air.

4.3 After contact with the skin

If contact with skin, wash skin with plenty of water or with water and soap.

4.4 After contact with the eyes

If contact with eyes, immediately flush eyes with plenty of water.

4.5 After swallowing

Drink plenty of water. Get medical attention if symptoms occur. Show label if possible.

5. Fire-fighting measures

5.1 Flammable properties:

Property:	Value:	Method:
Flash point.....	not applicable	
Boiling point / boiling range	not applicable	
Lower explosion limit (LEL)	not applicable	
Ignition temperature	not applicable	

5.2 Fire and explosion hazards:

Material does not burn. Electrostatic charging is possible.

5.3 Recommended extinguishing media:

Use extinguishing measures appropriate to the source of fire.

5.4 Unsuitable extinguishing media:

not applicable

5.5 Special exposure hazards arising from the substance or preparation itself, combustion products, resulting gases

not applicable

5.6 Fire fighting procedures:

Fire fighters should wear full protective clothing including a self-contained breathing apparatus.

6. Accidental release measures

6.1 Precautions:

Avoid dust formation. Do not breathe dust. Wear personal protection equipment (see section 8).

HAZWOPER PPE Level: D

6.2 Containment:

Cover any spilled material in accordance with regulations to prevent dispersal by wind.

Spills of material which could reach surface waters must be reported to the United States Coast Guard National Response Center's toll free phone number (800) 424-8802.

6.3 Methods for cleaning up

Damp down dust and fill into containers.

7. Handling and storage

Safety Data Sheet

Material: 60004600

HDK® N20
HYDROPHILIC PYROGENIC SILICA

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Date of print: 11/11/2015

Date of last alteration: 06/24/2015

7.1 Handling

Precautions for safe handling:

Avoid dust formation.

Precautions against fire and explosion:

Electrostatic discharge possible during transport and processing. Take precautionary measures against electrostatic charging. Ensure all parts of equipment are well earthed. Use inert gas when working with combustible and explosive liquids. Avoid dust deposit, remove dust regularly.

7.2 Storage

Conditions for storage rooms and vessels:

none known

Advice for storage of incompatible materials:

not applicable

Further information for storage:

Keep container dry and tightly closed.

8. Exposure controls and personal protection

8.1 Engineering controls

Ventilation:

Use only with adequate ventilation.

Local exhaust:

In case of dust formation: (To maintain concentration below TLV.) Local exhaust ventilation which meets the requirements of ANSI Z9.2 is recommended to control airborne contaminants at the point of use.

8.2 Associate substances with specific control parameters such as limit values

Maximum airborne concentrations at the workplace:

CAS No.	Material	Type	mg/m ³	ppm	Dust fract.
7631-86-9	Silica, amorphous	OSHA PEL	0.8		
	Particulates not otherwise classified	ACGIH TWA	10.0		Inhalable dust/mist
	Particulates not otherwise classified	ACGIH TWA	3.0		Respirable dust/mist

Re Silica, amorphous: The exposure limits given for CAS-No. 7631-86-9 cover all types of synthetic amorphous silica; it appears, that there is an error in the OSHA PEL entry for amorphous silica. There should have been a footnote to the Z-3 Table indicating that the percent SiO₂ in formula (80 mg/m³ + % SiO₂) refers to crystalline silica content and not to amorphous silica. For evaluation of U.S. exposure measurements WACKER recommends use of the current NIOSH REL (6 mg/m³) instead of the old Z-3 Table entry.

Re Particulates not otherwise classified: The value is for particulate matter containing no asbestos and < 1% crystalline silica (ACGIH).

8.3 Personal protection equipment (PPE)

Respiratory protection:

In case of dust formation: A NIOSH approved particulate respirator with a P95 or higher rating.

Hand protection:

Recommendation: rubber gloves .

Eye protection:

Recommendation: Safety glasses with side shields or chemical safety goggles.

Other protective clothing or equipment:

Barrier cream may be used to prevent dryness of skin.

8.4 General hygiene and protection measures:

Do not breathe dust/vapor/mist/gas/aerosol. Wash thoroughly after handling.

Safety Data Sheet

Material: 60004600

HDK® N20
HYDROPHILIC PYROGENIC SILICA

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9. Physical and chemical properties

9.1 Appearance

Physical state / form : solid - powder
Colour : white
Odour : odourless

9.2 Safety parameters

Property:	Value:	Method:
Melting point / melting range	1700 °C (3,092 °F)	
Boiling point / boiling range	not applicable	
Flash point.....	not applicable	
Ignition temperature	not applicable	
Lower explosion limit (LEL)	not applicable	
Vapour pressure.....	not applicable	
Density	approx. 2.2 g/cm ³ at 20 °C (68 °F)	(DIN 51757)
Bulk density.....	20 - 130 kg/m ³	
Water solubility / miscibility.....	virtually insoluble at 20 °C (68 °F)	
pH-Value	3.6 - 4.5	(DIN EN ISO 787-9)
Viscosity (dynamic)	not applicable	

9.3 Further information

VOC..... : This material does not contain any reportable VOCs. 0 g/l (calculated value)

10. Stability and reactivity

10.1 General information:

If stored and handled in accordance with standard industrial practices no hazardous reactions are known.

10.2 Conditions to avoid

10.3 Materials to avoid

none known

10.4 Hazardous decomposition products

If stored and handled properly: none known .

10.5 Further information:

Hazardous polymerization cannot occur.

11. Toxicological information

11.1 Information on toxicological effects

11.1.1 Acute toxicity

Assessment:

Based on the available data acute toxic effects are not expected after single oral exposure. Based on the available data acute toxic effects are not expected after single dermal exposure. Based on the available data acute toxic effects are not expected after short-term inhalative exposure.

Product details:

Route of exposure	Result/Effect	Species/Test system	Source
oral	LD ₅₀ : > 5000 mg/kg	rat	literature
dermal	LD ₅₀ : > 5000 mg/kg	rabbit	literature

Safety Data Sheet

Material: 60004600

HDK® N20
HYDROPHILIC PYROGENIC SILICA

Version: 2.1 (US)

Date of print: 11/11/2015

Date of last alteration: 06/24/2015

by inhalation (dust)	LC ₅₀ : > 0.139 mg/l; 4 h At the technically highest possible concentration no mortality in animal test.	rat	literature
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11.1.2 Skin corrosion/irritation**Assessment:**

Based on the available data a clinically relevant skin irritation hazard is not expected.

Product details:

Result/Effect	Species/Test system	Source
not irritating	rabbit	literature

11.1.3 Serious eye damage / eye irritation**Assessment:**

Based on the available data a clinically relevant eye irritation hazard is not expected.

Product details:

Result/Effect	Species/Test system	Source
not irritating	rabbit	literature

11.1.4 Respiratory or skin sensitization**Assessment:**

By handling the product for many years no damage to health was observed.

11.1.5 Germ cell mutagenicity**Assessment:**

According to our present state of knowledge not mutagenic.

11.1.6 Carcinogenicity**Assessment:**

Animal tests have not revealed any carcinogenic effects.

11.1.7 Reproductive toxicity**Assessment:**

In animal experiments there have not been any indications of reproduction toxicity.

11.1.8 Specific target organ toxicity (single exposure)**Assessment:**

For this endpoint no toxicological test data is available for the whole product.

11.1.9 Specific target organ toxicity (repeated exposure)**Assessment:**

Changes in the lungs (inflammatory processes) observed in animal experiments after chronic inhalative exposure were reversible; no indication of silicosis.

11.1.10 Aspiration hazard**Assessment:**

For this endpoint no toxicological test data is available for the whole product.

11.1.11 Further toxicological information

No component of this product present at levels greater than or equal to 0.1% is identified as a known or anticipated carcinogen by NTP. No component of this product present at levels greater than or equal to 0.1% is identified as probable, possible or confirmed human carcinogen by IARC. No component of this product present at levels greater than or equal to 0.1% is identified as a carcinogen or potential carcinogen by OSHA.

Safety Data Sheet

Material: 60004600

HDK® N20
HYDROPHILIC PYROGENIC SILICA

Version: 2.1 (US)

Date of print: 11/11/2015

Date of last alteration: 06/24/2015

Other information: By handling the product for many years no damage to health was observed.

12. Ecological information

12.1 Toxicity

Assessment:

No expected damaging effects to aquatic organisms. According to current knowledge adverse effects on water purification plants are not expected.

Product details:

Result/Effect	Species/Test system	Source
LC ₅₀ : > 10000 mg/l	zebra fish (Danio rerio) (96 h)	literature
EC ₅₀ : > 10000 mg/l	Daphnia magna (24 h)	literature

12.2 Persistence and degradability

Assessment:

The substance is degradable in abiotic processes.

12.3 Bioaccumulative potential

Assessment:

No adverse effects expected.

12.4 Mobility in soil

Assessment:

No adverse effects expected.

12.5 Other adverse effects

none known

12.6 Additional information

Insoluble in water.

13. Disposal considerations

13.1 Product disposal

Recommendation:

Material that cannot be used, reprocessed or recycled should be disposed of in accordance with Federal, State, and local regulations at an approved facility. Depending on the regulations, waste treatment methods may include, e.g., landfill or incineration.

13.2 Packaging disposal

Recommendation:

Completely discharge containers (no tear drops, no powder rest, scraped carefully). Containers may be recycled or re-used. Observe local/state/federal regulations. Uncleaned packaging should be treated with the same precautions as the material.

14. Transport information

14.1 US DOT & CANADA TDG SURFACE

Valuation : Not regulated for transport

14.2 Transport by sea IMDG-Code

Valuation : Not regulated for transport

14.3 Air transport ICAO-TI/IATA-DGR

Valuation : Not regulated for transport

Safety Data Sheet

Material: 60004600

HDK® N20
HYDROPHILIC PYROGENIC SILICA

Version: 2.1 (US)

Date of print: 11/11/2015

Date of last alteration: 06/24/2015

15. Regulatory information

15.1 U.S. Federal regulations

TSCA inventory status and TSCA information:

This material or its components are listed on or are in compliance with the requirements of the TSCA Chemical Substance Inventory. This material is in compliance with TSCA under CAS Number 7631-86-9 (silica, amorphous).

TSCA 12(b) Export Notification:

This material does not contain any TSCA 12(b) regulated chemicals.

CERCLA Regulated Chemicals:

This material does not contain any CERCLA regulated chemicals.

SARA 302 EHS Chemicals:

This material does not contain any SARA extremely hazardous substances.

SARA 311/312 Hazard Class:

This product does not present any SARA 311/312 hazards.

SARA 313 Chemicals:

This material does not contain any SARA 313 chemicals above de minimus levels.

HAPS (Hazardous Air Pollutants):

This material does not contain any hazardous air pollutants.

15.2 U.S. State regulations

California Proposition 65 Carcinogens:

This material does not contain any chemicals known to the state of California to cause cancer.

California Proposition 65 Reproductive Toxins:

This material does not contain any chemicals known to the State of California to cause reproductive effects.

Massachusetts Substance List:

112945-52-5 Silica, amorphous, fumed

New Jersey Right-to-Know Hazardous Substance List:

112945-52-5 Silica, amorphous, fumed

Pennsylvania Right-to-Know Hazardous Substance List:

112945-52-5 Silica, amorphous, fumed

15.3 Canadian regulations

This product has been classified in accordance with the Hazard criteria of the CPR and the SDS contains all the information required by the CPR.

WHMIS Hazard Classes:

None.

15.4 Details of international registration status

Relevant information about individual substance inventories, where available, is given below.

South Korea (Republic of Korea).....	ECL (Existing Chemicals List): This product is listed in, or complies with, the substance inventory.
Japan.....	ENCS (Handbook of Existing and New Chemical Substances): This product is listed in, or complies with, the substance inventory.
Australia	AICS (Australian Inventory of Chemical Substances): This product is listed in, or complies with, the substance inventory.
People's Republic of China.....	IECSC (Inventory of Existing Chemical Substances in China): This product is listed in, or complies with, the substance inventory.
Canada.....	DSL (Domestic Substance List): This product is listed in, or complies with, the substance inventory.
Philippines	PICCS (Philippine Inventory of Chemicals and Chemical Substances): This product is listed in, or complies with, the substance inventory.

Safety Data Sheet

Material: 60004600
HDK® N20
HYDROPHILIC PYROGENIC SILICA

Version: 2.1 (US)
Date of print: 11/11/2015
Date of last alteration: 06/24/2015

United States of America (USA) : **TSCA** (Toxic Substance Control Act Chemical Substance Inventory):
This product is listed in, or complies with, the substance inventory.
European Economic Area (EEA) : **REACH** (Regulation (EC) No 1907/2006):
REACH registration number: 01-2119379499-16-0001
General note: the registration obligations for substances imported into the EEA or
manufactured within the EEA by the supplier mentioned in section 1 are fulfilled by
the said supplier. The registration obligations for substances imported into the EEA
by customers or other downstream users must be fulfilled by the latter.

16. Other information

16.1 Additional information:

This Safety Data Sheet (SDS) meets the requirements of the Federal OSHA Hazard Communication Standard (29 CFR 1910.1200). This product has been classified according to the hazard criteria of the Controlled Products Regulations (CPR) and the SDS contains all of the information required by the CPR. This information relates to the specific material designated and may not be valid for such material used in combination with any other materials or in any process. Such information is to the best of our knowledge and belief accurate and reliable as of the date compiled. However, no representation, warranty or guarantee expressed or implied, is made as to its accuracy, reliability or completeness. It is the user's responsibility to satisfy himself as to the suitability and completeness of such information for his own particular use. We do not accept liability for any loss or damage that may occur from the use of this information. Nothing herein shall be construed as a recommendation for uses which infringe valid patents or as extending a license under valid patents. This SDS provides selected regulatory information on this product, including its components. This is not intended to include all regulations. It is the responsibility of the user to know and comply with all applicable rules, regulations and laws relating to the product being used.

Vertical lines in the left-hand margin indicate changes compared with the previous version.

All deliveries are subject to the WACKER SILICONES Health Care Policy, which is available at www.wacker.com.

16.2 Glossary of Terms:

ACGIH - American Conference of Governmental Industrial Hygienists	ppm - Parts per Million
DOT - Department of Transportation	SARA - Superfund Amendments and Reauthorization Act
hPa - Hectopascals	STEL - Short Term Exposure Limit
mPa*s - Milli Pascal-Seconds	TSCA - Toxic Substances Control Act
OSHA - Occupational Safety and Health Administration	TWA - Time Weighted Average
PEL - Permissible Exposure Limit	WHMIS - Canadian Workplace Hazardous Materials Identification System
Flash point determination methods	Common name
ASTM D56.....	Tagliabue (Tag) closed cup
ASTM D92, DIN 51376, ISO 2592	Cleveland open cup
ASTM D93, DIN 51758, ISO 2719	Pensky-Martens closed cup
ASTM D3278, DIN 55680, ISO 3679	Setaflash or Rapid closed cup
DIN 51755.....	Abel-Pensky closed cup

16.3 Conversion table:

Pressure:.....: 1 hPa * 0.75 = 1 mm Hg = 1 torr; 1 bar = 1000 hPa
Viscosity:.....: 1 mPa*s = 1 centipoise (cP)

PRODUCT DATA

Product Data Sheet

Edition 8.20.2018

Sikaflex®-1a

Sikaflex®-1a

Description	Sikaflex-1a is a premium-grade, high-performance, moisture-cured, 1-component, polyurethane-based, non-sag elastomeric sealant. Sikaflex-1a can be used in green and damp concrete applications. Meets Federal Specification TT-S-00230C, Type II, Class A. Meets ASTM C-920, Type S, Grade NS, Class 35, use T, NT, O, M, G, I, A. Canadian standard CAN/CGSB 19.13-M87.
Where to Use	■ Designed for all types of joints where maximum depth of sealant will not exceed 1/2 in. ■ Excellent for small joints and fillets, windows, door frames, reglets, flashing, common roofing detail applications, and many construction adhesive applications. ■ Suitable for vertical and horizontal joints; readily placeable at 40°F ■ Has many applications as an elastic adhesive between materials with dissimilar coefficients of expansion. ■ Submerged conditions, such as canal and reservoir joints.
Advantages	■ Eliminates time, effort, and equipment for mixing, filling cartridges, pre-heating or thawing, and cleaning of equipment. ■ Fast tack-free and final cure times. ■ High elasticity - cures to a tough, durable, flexible consistency with exceptional cut and tear -resistance. ■ Stress relaxation. ■ Excellent adhesion - bonds to most construction materials without a primer. ■ Excellent resistance to aging, weathering. ■ Proven in tough climates around the world. ■ Can be applied to green concrete 24 hours after pour ■ Can be applied to damp concrete 1 hour after getting wet ■ Odorless, non-staining. ■ Jet fuel resistant. ■ Certified to the NSF/ANSI Standard 61 for potable water. ■ Urethane-based; suggested by EPA for radon reduction. ■ Paintable with water-, oil- and rubber-based paints. ■ Capable of ±35% joint movement.
Chemical Resistance	Good resistance to water, diluted acids, and diluted alkalines. Consult Technical Service for specific data.
Packaging	10.1 fl. oz. (300 mL) Cartridge, 20 fl. oz. uni-pac Sausages, 4.5 gal (17 L) in a 5 gal pail, 52 gal (197 L) in a 55 gal drum

Typical Data (Material and curing conditions @ 75°F (24°C) and 50% RH)

RESULTS MAY DIFFER BASED UPON STATISTICAL VARIATIONS DEPENDING UPON MIXING METHODS AND EQUIPMENT, TEMPERATURE, APPLICATION METHODS, TEST METHODS, ACTUAL SITE CONDITIONS AND CURING CONDITIONS.

Shelf Life:	Cartridge and Sausage: 12 months in original, unopened packaging. Pail and Drum: 6 months in original, unopened packaging.
Storage:	Store at 40°-95°F (4°-35°C).
Product Conditioning:	Condition material to 65°-75°F before using.
Colors:	White, colonial white, aluminum gray, limestone, black, dark bronze, capitol tan, stone and medium bronze. Special architectural colors on request.
Application Temperature:	40° to 100°F. Sealant should be installed when joint is at mid-range of its anticipated movement.
Service Range:	-40° to 170°F
Curing Rate:	Tack-free time 3 to 6 hours Tack-free to touch 3 hours
Final cure:	4 to 7 days
Tear Strength (ASTM D-624):	55 lb./in.
Shore A Hardness (ASTM C-661):	21 day 45±5
Movement Capability (ASTM C-719):	+/- 35%
Tensile Properties (ASTM D-412):	
21 day Tensile Stress	175 psi (1.21 MPa)
Elongation @ Break	550%
Stress @ 100%	85 psi (0.59 MPa)
Adhesion in Peel (TT-S-00230C, ASTM C 794):	
Substrate Peel Strength Adhesion Loss	Concrete 20 lb. 0% Aluminum 20 lb. 0% Glass 20 lb. 0%
Weathering Resistance	Excellent

PRIOR TO EACH USE OF ANY SIKA PRODUCT, THE USER MUST ALWAYS READ AND FOLLOW THE WARNINGS AND INSTRUCTIONS ON THE PRODUCT'S MOST CURRENT PRODUCT DATA SHEET, PRODUCT LABEL AND SAFETY DATA SHEET WHICH ARE AVAILABLE ONLINE AT [HTTP://USA.SIKA.COM/](http://usa.sika.com/) OR BY CALLING SIKA'S TECHNICAL SERVICE DEPARTMENT AT 800.933.7452 NOTHING CONTAINED IN ANY SIKA MATERIALS RELIEVES THE USER OF THE OBLIGATION TO READ AND FOLLOW THE WARNINGS AND INSTRUCTIONS FOR EACH SIKA PRODUCT AS SET FORTH IN THE CURRENT PRODUCT DATA SHEET, PRODUCT LABEL AND SAFETY DATA SHEET PRIOR TO PRODUCT USE.

Coverage

10.1 oz Cartridge: Yield in Linear feet				
Depth	1/4"	3/8"	1/2"	
Width	1/4"	24.3		
	3/8"	16.2	10.8	
	1/2"	12.1	8.1	6.1
	3/4"	8.1	5.4	4.0
	1"			3.0
	1.25"			2.4
	1.5"			2.0

20 oz Sausage: Yield in Linear feet				
Depth	1/4"	3/8"	1/2"	
Width	1/4"	48.1		
	3/8"	32.1	21.4	
	1/2"	24.1	16.0	12.0
	3/4"	16.0	10.7	8.0
	1"			6.0
	1.25"			4.8
	1.5"			4.0

1 gallon: Yield in Linear feet				
Depth	1/4"	3/8"	1/2"	
Width	1/4"	307.9		
	3/8"	205.3	136.8	
	1/2"	153.9	102.6	77.0
	3/4"	102.6	68.4	51.3
	1"			38.5
	1.25"			30.8
	1.5"			25.7

How to Use

Surface Preparation

Clean all surfaces. Joint walls must be sound, clean, frost-free, and free of oil and grease. Curing compound residues and any other foreign matter must be thoroughly removed. A roughened surface will also enhance bond. Install bond breaker tape or backer rod to prevent bond at base of joint. Priming is not usually necessary. Most substrates only require priming if testing indicates a need or where sealant will be subjected to water immersion after cure.

For green concrete applications control joints must be cut 8 hours prior to sealant installation and in expansion joint forms must be removed 4 hours prior to sealant installation. For wet concrete applications all excess or standing water must be displaced and concrete must then dry for a minimum of 60 min prior to sealant installation. Consult Sikaflex Primer Technical Data Sheet or Technical Service for additional information on priming.

Application

Recommended application temperatures: 40°-100°F. For cold weather application, condition units at approximately 70°F; remove prior to using. For best performance, Sikaflex-1a should be gunned into joint when joint slot is at mid-point of its designed expansion and contraction. Place nozzle of gun into bottom of the joint and fill entire joint. Keep the nozzle in the sealant, continue on with a steady flow of sealant preceding the nozzle to avoid air entrapment. Avoid overlapping of sealant to eliminate entrapment of air.

Sikaflex-1a can be applied on green concrete after the concrete has cured for a minimum of 24 hours at 75°F. Control joints must be cut and open for min of 8 hours prior to application. Expansion joints must have forms removed a minimum of 4 hours prior to application. For damp concrete applications Sikaflex-1a can be applied 60 minutes after any and all water has been displaced.

Tooling and Finishing

Tool sealant to ensure full contact with joint walls and remove air entrapment. Joint dimension should allow for 1/4 inch minimum and 1/2 inch maximum thickness for sealant. Proper design is 2:1 width to depth ratio. For use in horizontal joints in traffic areas, the absolute minimum depth of the sealant is 1/2 in. and closed cell backer rod is recommended.

Removal

Use personal protective equipment (chemical resistant gloves/goggles/clothing). Without direct contact, remove spilled or excess product and placed in suitable sealed container. Dispose of excess product and container in accordance with applicable environmental regulations.

Over Painting

Allow 1-week cure at standard conditions when using Sikaflex-1a in total water immersion situations and prior to painting.



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Limitations

- Allow 1 week cure at standard conditions when using Sikaflex-1a in total water immersion situations.
- When overcoating with water, oil and rubber based paints, compatibility and adhesion testing is essential.
- Sealant should be allowed to cure for 7 days prior to overcoating
- Avoid exposure to high levels of chlorine. (Maximum continuous level is 5 ppm of chlorine.)
- Maximum depth of sealant must not exceed 1/2 in.; minimum depth is 1/4 in.
- Maximum expansion and contraction should not exceed 35% of average joint width.
- Do not cure in the presence of curing silicone sealants.
- Avoid contact with alcohol and other solvent cleaners during cure.
- Do not apply when moisture-vapor-transmission condition exists from the substrate as this can cause bubbling within the sealant.
- Use opened cartridges and uni-pac sausages the same day.
- When applying sealant, avoid air-entrapment.
- Since system is moisture-cured, permit sufficient exposure to air.
- White color tends to yellow slightly when exposed to ultraviolet rays.
- Light colors can yellow if exposed to direct gas fired heating element.
- The ultimate performance of Sikaflex-1a depends on good joint design and proper application with joint surfaces properly prepared.
- The depth of sealant in horizontal joints subject to traffic is 1/2 in.
- Do not tool with detergent or soap solutions.
- Do not use in contact with bituminous/asphaltic materials.
- In green concrete applications sealing joints in poor or low strength concrete 24 hours after pour may impact ability of sealant to gain proper adhesion.
- In damp concrete applications all standing water and excess water must be eliminated prior to the 60 minute waiting time.

PRIOR TO EACH USE OF ANY SIKA PRODUCT, THE USER MUST ALWAYS READ AND FOLLOW THE WARNINGS AND INSTRUCTIONS ON THE PRODUCT'S MOST CURRENT PRODUCT DATA SHEET, PRODUCT LABEL AND SAFETY DATA SHEET WHICH ARE AVAILABLE ONLINE AT [HTTP://USA.SIKA.COM/](http://usa.sika.com/) OR BY CALLING SIKA'S TECHNICAL SERVICE DEPARTMENT AT 800.933.7452 NOTHING CONTAINED IN ANY SIKA MATERIALS RELIEVES THE USER OF THE OBLIGATION TO READ AND FOLLOW THE WARNINGS AND INSTRUCTIONS FOR EACH SIKA PRODUCT AS SET FORTH IN THE CURRENT PRODUCT DATA SHEET, PRODUCT LABEL AND SAFETY DATA SHEET PRIOR TO PRODUCT USE.

KEEP CONTAINER TIGHTLY CLOSED. KEEP OUT OF REACH OF CHILDREN. NOT FOR INTERNAL CONSUMPTION. FOR INDUSTRIAL USE ONLY. FOR PROFESSIONAL USE ONLY.

For further information and advice regarding transportation, handling, storage and disposal of chemical products, users should refer to the actual Safety Data Sheets containing physical, ecological, toxicological and other safety related data. Read the current actual Safety Data Sheet before using the product. In case of emergency, call CHEMTREC at 1-800-424-9300, International 703-527-3887.

Prior to each use of any Sika product, the user must always read and follow the warnings and instructions on the product's most current Product Data Sheet, product label and Safety Data Sheet which are available online at <http://usa.sika.com/> or by calling Sika's Technical Service Department at 800-933-7452. Nothing contained in any Sika materials relieves the user of the obligation to read and follow the warnings and instruction for each Sika product as set forth in the current Product Data Sheet, product label and Safety Data Sheet prior to product use.

SIKA warrants this product for one year from date of installation to be free from manufacturing defects and to meet the technical properties on the current Product Data Sheet if used as directed within shelf life. User determines suitability of product for intended use and assumes all risks. Buyer's sole remedy shall be limited to the purchase price or replacement of product exclusive of labor or cost of labor. NO OTHER WARRANTIES EXPRESS OR IMPLIED SHALL APPLY INCLUDING ANY WARRANTY OF MERCHANTABILITY OR FITNESS FOR A PARTICULAR PURPOSE. SIKI SHALL NOT BE LIABLE UNDER ANY LEGAL THEORY FOR SPECIAL OR CONSEQUENTIAL DAMAGES. SIKI SHALL NOT BE RESPONSIBLE FOR THE USE OF THIS PRODUCT IN A MANNER TO INFRINGE ON ANY PATENT OR ANY OTHER INTELLECTUAL PROPERTY RIGHTS HELD BY OTHERS.

SALE OF SIKI PRODUCTS ARE SUBJECT SIKI'S TERMS AND CONDITIONS OF SALE AVAILABLE AT [HTTP://USA.SIKA.COM/](http://usa.sika.com/) OR BY CALLING 201-933-8800.

Visit our website at usa.sika.com

1-800-933-SIKA NATIONWIDE

Regional Information and Sales Centers. For the location of your nearest Sika sales office, contact your regional center.

Sika Corporation
201 Polito Avenue
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Sika Mexicana S.A. de C.V.
Carretera Libre Celaya Km. 8.5
Fracc. Industrial Balvanera
Corregidora, Queretaro
C.P. 76920
Phone: 52 442 2385800
Fax: 52 442 2250537



SAFETY DATA

**1. Identification**

Product name : Sikaflex®-1A

Supplier : Sika Corporation
201 Polito Avenue
Lyndhurst, NJ 07071
USA
www.sikausa.com

Telephone : (201) 933-8800

Telefax : (201) 804-1076

E-mail address : ehs@sika-corp.com

Emergency telephone : CHEMTREC: 800-424-9300
INTERNATIONAL: 703-527-3887

Recommended use of the chemical and restrictions on use : For further information, refer to product data sheet.

2. Hazards identification**GHS Classification**

Respiratory sensitization, Category 1 H334: May cause allergy or asthma symptoms or breathing difficulties if inhaled.
Skin sensitization, Category 1 H317: May cause an allergic skin reaction.
Carcinogenicity, Category 1A (Inhalation) H350i: May cause cancer by inhalation.

GHS label elements

Hazard pictograms :



Signal Word : Danger

Hazard Statements : H317 May cause an allergic skin reaction.
H334 May cause allergy or asthma symptoms or breathing difficulties if inhaled.
H350i May cause cancer by inhalation.

Precautionary Statements : **Prevention:**
P201 Obtain special instructions before use.
P202 Do not handle until all safety precautions have been read and understood.
P261 Avoid breathing dust/ fume/ gas/ mist/ vapors/ spray.
P272 Contaminated work clothing must not be allowed out of the workplace.



P280 Wear protective gloves.
P281 Use personal protective equipment as required.
P285 In case of inadequate ventilation wear respiratory protection.
Response:
P302 + P352 IF ON SKIN: Wash with plenty of soap and water.
P304 + P341 IF INHALED: If breathing is difficult, remove person to fresh air and keep comfortable for breathing.
P308 + P313 IF exposed or concerned: Get medical advice/attention.
P333 + P313 If skin irritation or rash occurs: Get medical advice/attention.
P363 Wash contaminated clothing before reuse.
Storage:
P405 Store locked up.
Disposal:
P501 Dispose of contents/ container to an approved waste disposal plant.

See Section 11 for more detailed information on health effects and symptoms.
There are no hazards not otherwise classified that have been identified during the classification process.
There are no ingredients with unknown acute toxicity used in a mixture at a concentration $\geq 1\%$.

3. Composition/information on ingredients

Hazardous ingredients

Chemical name	CAS-No.	Concentration (%)
Quartz (SiO ₂) <5µm	14808-60-7	$\geq 0.1 - < 1\%$
aromatic polyisocyanate	53317-61-6	$\geq 0.1 - < 1\%$
4,4'-methylenediphenyl diisocyanate	101-68-8	$\geq 0.1 - < 1\%$

There are no additional ingredients present which, within the current knowledge of the supplier and in the concentrations applicable, are classified as hazardous to health or the environment and hence require reporting in this section.

4. First aid measures

If inhaled	: Move to fresh air. Consult a physician after significant exposure.
In case of skin contact	: Take off contaminated clothing and shoes immediately. Wash off with soap and plenty of water. If symptoms persist, call a physician.
In case of eye contact	: Remove contact lenses. Keep eye wide open while rinsing. If eye irritation persists, consult a specialist.
If swallowed	: Clean mouth with water and drink afterwards plenty of water. Do not induce vomiting without medical advice.



	Do not give milk or alcoholic beverages. Never give anything by mouth to an unconscious person. Obtain medical attention.
Most important symptoms and effects, both acute and delayed	: sensitizing effects carcinogenic effects Asthmatic appearance Allergic reactions See Section 11 for more detailed information on health effects and symptoms. May cause an allergic skin reaction. May cause allergy or asthma symptoms or breathing difficulties if inhaled. May cause cancer by inhalation.
Protection of first-aiders	: Move out of dangerous area. Consult a physician. Show this material safety data sheet to the doctor in attendance.
Notes to physician	: Treat symptomatically.

5. Fire-fighting measures

Suitable extinguishing media	: Use extinguishing measures that are appropriate to local circumstances and the surrounding environment.
Specific extinguishing methods	: Collect contaminated fire extinguishing water separately. This must not be discharged into drains. Fire residues and contaminated fire extinguishing water must be disposed of in accordance with local regulations.
Special protective equipment for fire-fighters	: In the event of fire, wear self-contained breathing apparatus.

6. Accidental release measures

Personal precautions, protective equipment and emergency procedures	: Use personal protective equipment. Deny access to unprotected persons.
Environmental precautions	: Do not flush into surface water or sanitary sewer system. If the product contaminates rivers and lakes or drains inform respective authorities. Local authorities should be advised if significant spillages cannot be contained.
Methods and materials for containment and cleaning up	: Soak up with inert absorbent material (e.g. sand, silica gel, acid binder, universal binder, sawdust). Keep in suitable, closed containers for disposal.



7. Handling and storage

- Advice on safe handling : Do not breathe vapors or spray mist.
 Avoid exceeding the given occupational exposure limits (see section 8).
 Do not get in eyes, on skin, or on clothing.
 For personal protection see section 8.
 Persons with a history of skin sensitization problems or asthma, allergies, chronic or recurrent respiratory disease should not be employed in any process in which this mixture is being used.
 Smoking, eating and drinking should be prohibited in the application area.
 Follow standard hygiene measures when handling chemical products.
- Conditions for safe storage : Prevent unauthorized access.
 Store in original container.
 Keep container tightly closed in a dry and well-ventilated place.
 Observe label precautions.
 Store in accordance with local regulations.
- Materials to avoid : No data available

8. Exposure controls/personal protection

Component	CAS-No.	Basis **	Value	Exposure limit(s)* / Form of exposure
Quartz (SiO ₂) <5µm	14808-60-7	OSHA Z-3	TWA	10 mg/m ³ / %SiO ₂ +2 respirable
		OSHA Z-3	TWA	250 mppcf / %SiO ₂ +5 respirable
		OSHA P0	TWA	0.1 mg/m ³ Respirable fraction
		ACGIH	TWA	0.025 mg/m ³ Respirable fraction
		OSHA Z-1	TWA	0.05 mg/m ³ Respirable dust
4,4'-methylenediphenyl diisocyanate	101-68-8	ACGIH	TWA	0.005 ppm
		OSHA Z-1	C	0.02 ppm 0.2 mg/m ³
		OSHA P0	C	0.02 ppm



				0.2 mg/m3
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*The above mentioned values are in accordance with the legislation in effect at the date of the release of this safety data sheet.

****Basis**

ACGIH. Threshold Limit Values (TLV)

OSHA P0. Table Z-1, Limit for Air Contaminant (1989 Vacated Values)

OSHA P1. Permissible Exposure Limits (PEL), Table Z-1, Limit for Air Contaminant

OSHA P2. Permissible Exposure Limits (PEL), Table Z-2

OSHA Z3. Table Z-3, Mineral Dust

Engineering measures : Use of adequate ventilation should be sufficient to control worker exposure to airborne contaminants. If the use of this product generates dust, fumes, gas, vapor or mist, use process enclosures, local exhaust ventilation or other engineering controls to keep worker exposure below any recommended or statutory limits.

Personal protective equipment

Respiratory protection : Use a properly fitted NIOSH approved air-purifying or air-fed respirator complying with an approved standard if a risk assessment indicates this is necessary.

The filter class for the respirator must be suitable for the maximum expected contaminant concentration (gas/vapor/aerosol/particulates) that may arise when handling the product. If this concentration is exceeded, self-contained breathing apparatus must be used.

Hand protection
Remarks : Chemical-resistant, impervious gloves complying with an approved standard should be worn at all times when handling chemical products if a risk assessment indicates this is necessary.

Eye protection : Safety eyewear complying with an approved standard should be used when a risk assessment indicates this is necessary.

Skin and body protection : Choose body protection in relation to its type, to the concentration and amount of dangerous substances, and to the specific work-place.

Hygiene measures : Avoid contact with skin, eyes and clothing.
Wash hands before breaks and immediately after handling the product.
Remove contaminated clothing and protective equipment before entering eating areas.
Wash thoroughly after handling.



9. Physical and chemical properties

Appearance	: paste
Color	: various
Odor	: characteristic
Odor Threshold	: No data available
Flash point	: Note: Not applicable
Ignition temperature	: No data available
Decomposition temperature	: No data available
Lower explosion limit (Vol%)	: No data available
Upper explosion limit (Vol%)	: No data available
Flammability (solid, gas)	: No data available
Oxidizing properties	: No data available
pH	: Note: Not applicable
Melting point/range / Freezing point	: No data available
Boiling point/boiling range	: No data available
Vapor pressure	: 0.01 mmHg (0.01 hpa)
Density	: ca.1.48 g/cm ³ at 73 °F (23 °C)
Water solubility	: Note: insoluble
Partition coefficient: n- octanol/water	: No data available
Viscosity, dynamic	: No data available
Viscosity, kinematic	: > 20.5 mm ² /s
Relative vapor density	: No data available
Evaporation rate	: No data available
Burning rate	: No data available
Volatile organic compounds (VOC) content	: 8 g/l

10. Stability and reactivity

Reactivity	: No dangerous reaction known under conditions of normal use.
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Chemical stability	: The product is chemically stable.
Possibility of hazardous reactions	: Stable under recommended storage conditions.
Conditions to avoid	: No data available
Incompatible materials	: No data available

11. Toxicological information

Acute toxicity

Not classified based on available information.

Ingredients:

aromatic polyisocyanate:

Acute oral toxicity : LD50 Oral (Rat): > 5,000 mg/kg

4,4'-methylenediphenyl diisocyanate:

Acute inhalation toxicity : Acute toxicity estimate: 1.5 mg/l
Test atmosphere: dust/mist
Method: Expert judgment

Skin corrosion/irritation

Not classified based on available information.

Serious eye damage/eye irritation

Not classified based on available information.

Respiratory or skin sensitization

Skin sensitization: May cause an allergic skin reaction.

Respiratory sensitization: May cause allergy or asthma symptoms or breathing difficulties if inhaled.

Germ cell mutagenicity

Not classified based on available information.

Reproductive toxicity

Not classified based on available information.

STOT-single exposure

Not classified based on available information.

STOT-repeated exposure

Once sensitized, a severe allergic reaction may occur when subsequently exposed to very low levels.

Aspiration toxicity

Not classified based on available information.

Carcinogenicity

May cause cancer by inhalation.

IARC

Group 1: Carcinogenic to humans

Quartz (SiO₂) <5µm

14808-60-7



Group 2B: Possibly carcinogenic to humans

NTP	titanium dioxide	13463-67-7
	Carbon black	1333-86-4
	Known to be human carcinogen	
	Quartz (SiO ₂) <5µm	14808-60-7
Carbon black (1333-86-4)		

Animal Toxicity:

Rat, oral, duration 2 year

Effect: no tumors

Mouse, oral, duration 2 years

Effect: no tumors

Mouse, dermal, duration 18 months

Effect: no skin tumors

Rat, inhalation, duration 2 years

Target organ: lungs

Effect: inflammation, fibrosis, tumors

Note: Tumors in the rat lung are considered to be related to the "particle overload phenomenon" rather than to a specific chemical effect of carbon black itself in the lung. These effects in rats have been reported in many studies on other poorly soluble inorganic particles and appear to be rat specific. Tumors have not been observed in other species (i.e., mouse and hamster) for carbon black or other poorly soluble particles under similar circumstances and study conditions.

Mortality studies (human data): A study on carbon black production workers in the UK (Sorahan, 2001) found an increased risk of lung cancer in two of the five plants studied; however, the increase was not related to the dose of carbon black. Thus, the authors did not consider the increased risk in lung cancer to be due to carbon black exposure. A German study of carbon black workers at one plant (Morfeld, 2006; Buechte, 2006) found a similar increase in lung cancer risk but, like the Sorahan, 2001 (UK study) found no association with carbon black exposure. A large US study of 18 plants showed a reduction in lung cancer risk in carbon black production workers (DEll, 2006). Based upon these studies, the February 2006 Working Group at the International Agency for Research on Cancer (IARC) concluded that the human evidence for carcinogenicity was inadequate (IARC, 2010).

Since the IARC evaluation of carbon black, Sorahan and Harrington (2007) have re-analyzed the UK study data using an alternative exposure hypothesis and found a positive association with carbon black exposure in two of the five plants. The same exposure hypothesis was applied by Morfeld and McCunney (2009) to the German cohort; in contrast, they found no association between carbon black exposure and lung cancer risk and, thus, no support for the alternative exposure hypothesis used by Sorahan and Harrington.

Overall, as a result of these detailed investigations, no causative link between carbon black exposure and cancer risk in humans has been demonstrated.

IARC CANCER CLASSIFICATION: In 2006 IARC re-affirmed its 1995 finding that there is "inadequate evidence" from human health studies to assess whether carbon black causes cancer in humans. IARC concluded that there is "sufficient evidence" in experimental animal studies for the carcinogenicity of carbon black. IARC's overall evaluation is that carbon black is "possibly carcinogenic to humans" (Group 2B)". This conclusion was based on IARC's guidelines, which generally require such a classification if one species exhibits carcinogenicity in



two or more animal studies (IARC, 2010).

Solvent extracts of carbon black were used in one study of rats in which skin tumors were found after dermal application and several studies of mice in which sarcomas were found following subcutaneous injection. IARC concluded that there was "sufficient evidence" that carbon black extracts can cause cancer in animals (Group 2B).

ICGIH CANCER CLASSIFICATION: Confirmed Animal Carcinogen with Unknown Relevance to Humans (Category A3 Carcinogen).

ASSESSMENT: Applying the guidelines of self-classification under the Globally Harmonized System of Classification and Labeling of Chemicals, carbon black is not classified as a carcinogen. Lung tumors are induced in rats as a result of repeated exposure to inert, poorly soluble particles like carbon black and other poorly soluble particles. Rats tumors are a result of a secondary non-genotoxic mechanism that has questionable relevance for classification in humans. In support of this opinion, the CLP Guidance for Specific Target Organ Toxicity - Repeated Exposure (STOT-RE), cites lung overload under mechanisms not relevant to humans. Human health studies show that exposure to carbon black does not increase the risk to carcinogenicity.

Titanium dioxide (13463-67-7)

In lifetime inhalation studies of rats, airborne respirable-size titanium dioxide particles have been shown to cause an increase in lung tumors at concentrations associated with substantial particle lung burdens and consequential pulmonary overload and inflammation. The potential for these adverse health effects appears to be closely related to the particle size and the amount of the exposed surface area that comes into contact with the lung. However, tests with other laboratory animals such as mice and hamsters, indicate that rats are significantly more susceptible to the pulmonary overload and inflammation that cause lung cancer. Epidemiology studies do not suggest an increased risk of cancer in humans from occupational exposure to titanium dioxide. Titanium dioxide has been characterized by IARC as possibly carcinogenic to humans (Group 2B) through inhalation (not ingestion). It has not been characterized as a potential carcinogen by either NTP or OSHA.

12. Ecological information

Other information	Do not empty into drains; dispose of this material and its container in a safe way. Avoid dispersal of spilled material and runoff and contact with soil, waterways, drains and sewers.
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13. Disposal considerations

Disposal methods

Waste from residues	: Disposal of this product, solutions and any by-products should at all times comply with the requirements of environmental protection and waste disposal legislation and any regional local authority requirements.
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Contaminated packaging : Empty containers should be taken to an approved waste handling site for recycling or disposal.

14. Transport information

DOT

Not dangerous goods

IATA

Not dangerous goods

IMDG

Not dangerous goods

Special precautions for user

No data available

Transport in bulk according to Annex II of MARPOL 73/78 and the IBC Code

Not applicable

15. Regulatory information

TSCA list : All chemical substances in this product are either listed on the TSCA Inventory or are in compliance with a TSCA Inventory exemption.

EPCRA - Emergency Planning and Community Right-to-Know

CERCLA Reportable Quantity

This material does not contain any components with a CERCLA RQ.

SARA304 Reportable Quantity

This material does not contain any components with a section 304 EHS RQ.

SARA 311/312 Hazards : Chronic Health Hazard
Acute Health Hazard

SARA 302 : This material does not contain any components with a section 302 EHS TPQ.

SARA 313 : This material does not contain any chemical components with known CAS numbers that exceed the threshold (De Minimis) reporting levels established by SARA Title III, Section 313.

Clean Air Act

**Ozone-Depletion Potential**

This product neither contains, nor was manufactured with a Class I or Class II ODS as defined by the U.S. Clean Air Act Section 602 (40 CFR 82, Subpt. A, App.A + B).

This product does not contain any hazardous air pollutants (HAP), as defined by the U.S. Clean Air Act Section 112 (40 CFR 61).

This product does not contain any chemicals listed under the U.S. Clean Air Act Section 112(r) for Accidental Release Prevention (40 CFR 68.130, Subpart F).

California Prop 65

WARNING: Cancer and Reproductive Harm - www.P65Warnings.ca.gov

16. Other information**HMIS Classification**

Health	*	3
Flammability		0
Physical Hazard		0
Personal Protection		X

Caution: HMIS® rating is based on a 0-4 rating scale, with 0 representing minimal hazards or risks, and 4 representing significant hazards or risks. Although HMIS® rating is not required on SDSs under 29 CFR 1910.1200, the preparer may choose to provide them. HMIS® rating is to be used with a fully implemented HMIS® program. HMIS® is a registered mark of the National Paint & Coatings Association (NPCA). Please note HMIS® attempts to convey full health warning information to all employees.

Notes to Reader

The information contained in this Safety Data Sheet applies only to the actual Sika Corporation ("Sika") product identified and described herein. This information is not intended to address, nor does it address the use or application of the identified Sika product in combination with any other material, product or process. All of the information set forth herein is based on technical data regarding the identified product that Sika believes to be reliable as of the date hereof. Prior to each use of any Sika product, the user must always read and follow the warnings and instructions on the product's current Product Data Sheet, product label and Safety Data Sheet for each Sika product, which are available at web site and/or telephone number listed in Section 1 of this SDS.

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All sales of Sika products are subject to its current terms and conditions of sale available at www.sikausa.com or 201-933-8800.

Revision Date 08/30/2017



Material number: 547580

PREVENTIVE MAINTENANCE DATA

Introduction

Neogard Floor Coating Systems are designed to be long lasting and provide years of trouble-free service. Good general maintenance practices along with preventive maintenance will ensure the floor coating system will continue to provide service for which it is intended. Neogard strongly encourages the original floor coating system applicator be retained for inspections, repairs and cleaning in order to simplify warranty issues involved in alterations or repairs.

Neogard flooring systems benefit from a regular, thorough maintenance program. Proper maintenance will increase the life and performance of your Neogard floor as well as maintain the desired aesthetic appearance. Dirt and dust particles are abrasive and may scratch the surface causing it to become dull in appearance. Liquids and chemicals such as oil and grease can also stain the surface and cause damage to your Neogard flooring system. The cleaner the floor is kept, the longer the floor will last. Note: A slip resistant finish is available with most Neogard flooring systems. However, as the degree of texture and friction increases, the ease of cleaning decreases.

Caring for your Floor: Dos and Don'ts

- Do clean up spills as soon as possible.
- Do place walk-off mats at entrance ways to eliminate tracking of dirt, stones and moisture.
- Do repair gouges or scratches as soon as possible.
- Don't slide or drag heavy equipment, drums, wooden pallets across the floor without protection.
- Don't drop heavy or pointed objects on the floor.
- Don't use stiff bristle brushes or abrasive pads when cleaning.

General Maintenance

Sweep the floor each day either with a good quality broom or mechanical sweeper.

As required, wash the floor with warm water and detergent, following detergent manufacturer's directions. Use neutral cleaning solutions; do not use strong acids, alkalis, or bleach. Note: Numerous cleaners and detergents are available. Select a reputable manufacturer and complete a test area. In areas subjected to spillage of food, oils, fats, etc., such as a meat packing plant, schedule frequent, thorough cleanings. A simple hot cleaning solution puts fatty residues into suspension, but after cooling, these fats re-solidify leaving a slippery residue film on the floor. To prevent this, use a free rinsing detergent and plenty of water. Follow this procedure:

1. Wet mop solution onto floor.
2. Scrub to loosen adhered dirt. Power scrubbing with a soft pad or soft bristle brush is recommended for large heavily soiled areas.
3. Flush off wash water with mop, squeegee or wet vacuum and discard.
4. Rinse thoroughly with clean water and hose if area permits and floor is sloped to a floor drain.

Removal of Forklift Wheel Marks

1. Broom sweep area for debris and heavy build up.
2. Apply a mild solvent (mineral spirits or xylene) on wheel mark.
3. Spread solvent and allow to soften wheel mark.
4. Agitate with scrubbing device by hand or motorized device.
5. Remove solution with wet vacuum or rags.
6. Detergent scrub and rinse with water.

Preventive Maintenance

Preventive Maintenance procedures should include:

- Physical Inspections
- Periodic Cleaning
- Floor System Repairs

Physical Inspections

The building owner should establish semi-annual flooring inspection of the facility. Inspections should be conducted by qualified personnel (ideally the original floor coating applicator). It is important to report any visible damage to the applicator (cracks due to movement of the building, gouges, solvent or chemical attack).

Floor coating systems are subject to a wide range of abusive conditions from chemical attack to physical damage. Some more common causes of damage to floor coating systems are:

- Opening the facility for traffic prior to proper cure of the system.
- Failure to remove and clean chemical spills within a reasonable time.
- Dragging heavy, pointed metal or concrete objects across the floor coating system.
- Failure to repair damaged areas in a timely manner.
- Gouges from extreme Impact
- Excessive cracking due to movement in concrete slab or at expansion joints.
- Excessive moisture drive through concrete slab.
- Dragging pallets rather than lifting and moving.

Physical inspections should be conducted as defined by use of the area. More heavily used areas should be inspected more frequently than lesser used areas of the floor.

Periodic Cleaning

Improper cleaning procedures can damage the floor coating system. Please refer to the General Maintenance section above for recommendations.

Floor System Repairs

In order to protect the integrity and aesthetics of your Neogard floor coating system, contact a Neogard representative or Neogard applicator to make any necessary repairs. Immediately notify Neogard personnel or the original floor coating applicator if damage occurs. Damage will only get worse if it is left unattended. Depending upon the use of the floor coating system (traffic, mechanical abuse, chemical exposure, etc.), a new finish coat may be required after repairs have been made.

Issued by: Hempel (USA) – Preventive Maintenance Manual, Floor Coating Systems

This document is intended for professional use and provides generic advice in respect of the subject matter hereof only. It is not intended to be used as a comprehensive guide. The buyer/applicator should always read the relevant Product Data Sheets ("PDS"), Safety Data Sheets and Guide Specification relating to the applicable products/system. If in doubt, please contact your local Neogard representative for further advice. To the extent relevant, the disclaimer set forth in the relevant PDS or Guide Specification applies to this document.

Preventive Maintenance Flooring ksk 07092021.docx

Neogard®, a part of Hempel
2728 Empire Central - Dallas, Texas 75235 - Phone (214) 353-1600 - Fax (214) 357-7532 - www.neogard.com

WARRANTY LETTERS



Painting, Wallcovering, & Industrial

91-310 Komohana Street
Kapolei • Hawaii • 96707

Phone: (808) 682-1321
Fax: (808) 682-1535

www.zelinskycompany.com
Contractor Lic. #C20440

Certified Women Owned Business

Date

Client Name
Address

E-Mail:

RE: **LETTER OF WARRANTY**
Job Name

We hereby guarantee all materials provided and/or installed under the following sections of specifications for a period of One (1) year from the date of substantial completion (DATE) and/or beneficiary occupancy by the Owner/Tenant.

Section: 09900 - Painting

Should any failure occur during this warranty period due to defective materials and/or faulty workmanship, we will provide all necessary labor and materials to correct the problem within a reasonable time and at no cost to the Owner.

This warranty does not cover failures due to abuse, misuse, improper/lack of maintenance, and events beyond our control or liability.

Sincerely,

ZELINSKY COMPANY

James Wong
CFO

FLOORING -MATERIAL WARRANTY

Project Name:

Project Address:

Neogard System:

Applicator:

Completion Date:

Batch/Lot Number:

Owner:

Warranty No: (Neogard Use Only)

Substrate:

Area in Square Feet:

Our Promise - Neogard® warrants to the Owner that, for 1 year after the Completion Date, the NEOGARD products will be free from defects in material and will conform to the Neogard Guide Specifications in effect at the time of manufacture. This warranty is void unless Neogard products are installed in an institutional, commercial, industrial, or high-rise/multi-family residential building.

Our Obligations - If, upon inspection, Neogard determines that any Neogard product is defective, Neogard will, at its option, will either replace the product proven defective with an equivalent or grant a credit toward the purchase of different Neogard product. THE OBLIGATIONS OF NEOGARD DESCRIBED IN THE PREVIOUS SENTENCE ARE THE SOLE AND EXCLUSIVE REMEDIES OF THE OWNER FOR BREACH OF THIS WARRANTY.

- Exclusions and Limitations** - This Warranty does not cover:
1. Handling or installation of the Neogard products, or the workmanship of the applicator.
 2. Any portion of Neogard products that are not applied in accordance with the applicable Neogard Guide Specification or other written requirements provided by Neogard.
 3. Labor or equipment costs associated with repairing or accessing the Neogard products.
 4. Damage to the Neogard products caused by:
 - a. improper maintenance;
 - b. substrate construction or movement of greater than 1/16”;
 - c. exposure to ponding or standing water;
 - d. exposure to fatty acids, oils, greases, and other contaminants;
 - e. exposure to inclement weather or high winds
 - f. mechanical contact such as snow or ice removal, from falling objects, and contact with plant or animal life;
 - g. vapor drive, infiltration or condensation of moisture in, through or around walls, copings, the building structure or underlying or surrounding materials.
 5. Changes in appearance of the Neogard products that results from normal weathering or atmospheric conditions.
 6. Cleanliness or discoloration of the Neogard products that result from environmental conditions.
 7. Products not supplied by Neogard.

- Owner’s Obligations** - This Warranty is void unless:
1. Owner submits all warranty claims in writing within 30 days of discovery, but in any case, within the warranty period.
 2. Neogard is paid in full for the Neogard products.
 3. Owner uses reasonable care in maintaining the Neogard products.
 4. Owner only repairs the Neogard products with materials approved by Neogard.

NEOGARD MAKES NO OTHER WARRANTIES, WHETHER EXPRESS, IMPLIED, STATUTORY OR OTHERWISE, THAT EXTEND BEYOND THE WARRANTIES SET FORTH HEREIN, INCLUDING, WITHOUT LIMITATION, ANY IMPLIED WARRANTIES OF MERCHANTABILITY OR FITNESS FOR A PARTICULAR PURPOSE. NEOGARD SHALL NOT BE LIABLE FOR ANY INCIDENTAL OR CONSEQUENTIAL DAMAGES FOR BREACH OF ANY EXPRESS OR IMPLIED WARRANTY.

Neogard, a part of Hempel (USA), Inc.
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☒ **APPLICATOR CERTIFICATION:** Applicator hereby represents and warrants that all the information provided by Applicator is accurate and complete, and that the NEOGARD products have been applied and maintained in accordance with the applicable NEOGARD Guide Specification and NEOGARD written requirements.

BY: _____ TITLE: _____ DATE: _____